NEW EDITION

Based on New syllabus, Pokhara University

Calculus - II

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Bachelor of Engineering
3rd Semester

Toya Narayan Paudel
Youb Raj Gaire
Hem Raj Pandey

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Based on New Syllabus, Pokhara University

CALCULUS II

Bachelor of Engineering, 3rd Semester
Pokhara University

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Syllabus

Course Code: AHTH 310
Course title: Calculus II (3 + 1)
Nature of the Course: Theory
Level: Bachelor

Full Marks:
Pass Marks:
Total Lectures: 45 hrs
Program:

1. Course Description:
The Calculus II is designed to develop the competency of the students in the application of various mathematical concepts they learned in previous semesters. It is mainly equipped with Vector Calculus, Laplace transform, Multiple integral, Differential Equations, Fourier Series and with introduction of Partial differential equations. The pre-requisite for this course is Calculus I and Algebra & Geometry. The course will be delivered through lecture method, assignments on practice based engineering problems and class tests.

2. General Objectives:
The course is designed to acquaint the students with applications of mathematics in engineering.

3. Methods of Instruction:
Lecture, tutorials, discussions and assignments

4. Contents in Detail

Specific objectives	Contents
❑ Evaluate multiple integrals.	Unit I: Multiple Integrals (6 Hours) 1.1 Introduction 1.2 Double integrals in Cartesian and polar form, Fubini's theorem (statement only), change of order of integration, change of variable from in double integral Jacobian matrix and reduction into Polar. 1.3 Triple integrals in Cartesian form and Dirichlet's Integral, use of cylindrical and spherical coordinates to evaluate triple integral. 1.4 Application of double and triple integrals to find Area and volume.

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❑ Analyze the concept of solution of differential equations in terms of infinite series as power series.	**Unit II: Series Solution of Differential Equations and Special Functions** (6 Hours) 2.1 Power series method of solution of differential equations. 2.2 Legendre's Equation, Legendre's polynomials $P_n(x)$ of Graph of $P_1(x), P_2(x), P_3(x)$ 2.3 Frobenius method. Bessel's equation, Bessel's function $J_v(x)$ and its properties. Graph of $J_v(x)$ for $v = 1/2$
❑ Apply the integral transform in solving practical problems	**Unit III: Laplace Transform and Its Application (8Hours)** 3.1 Laplace Transform (LT), Inverse LT, Linearity of LT, LT of elementary functions, inverses and first shifting (s-shifting) theorem. Existence theorem of Laplace transform (without proof) and uniqueness. 3.2 Transform of Derivative and Integrals of a function. 3.3 Differentiation and Integration of Laplace transform. 3.4 Unit step function, periodic function and LT, second shifting-shifting) theorem. 3.5 Convolution theorem and its application to find inverse. 3.6 Application of Laplace transform to find the solutions of ordinary differential equations (IVP).
❑ Solve higher dimensional (multivariable) calculus problems	**Unit IV: Advanced Vector Calculus** (15 Hours) 4.1 Differentiation of vector function of scalar variable. 4.2 Point functions, Gradient, directional derivative, divergence and curl with properties (without proof) 4.3 Line integral with physical interpretation and evaluation of line integrals on various path 4.4 Line integral, potential function and independence of path 4.5 Green's theorem in plane (without proof) and its various applications 4.6 Surface integral and evaluation of surface integrals 4.7 Stoke's theorem (without proof) and its applications 4.8 Gauss Divergence theorem (without proof) and its applications

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❑ Illustrate periodic functions of practical importance by infinite trigonometric series	**Unit V: Fourier Series (5 Hours)** 5.1 Periodic Functions, odd and even functions 5.2 Fourier series of 2π periodic functions in the interval $(\alpha, \alpha + 2\pi)$. 5.3 Fourier series of $2l$ periodic functions. 5.4 Fourier series of odd and even functions, sine and cosine series
❑ Interpret physical phenomenon by partial differential equations	**Unit VI: Partial Differential Equations (5 Hours)** 6.1 Introduction 6.2 Linear constant coefficient equation 6.3 Applications in conservation laws, the breaking time, shock waves, nonlinear advection equations, and traffic flow.

Note: The figures in the parentheses indicate the approximate periods for the respective units.

Evaluation System

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation, etc. The tabular presentation of the internal evaluation is as follows.

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Attendance & Class Participation	10%	Semester End Board	50%	50
Assignments	20%			
Presentations/Quizzes 10%	10%			
Term exam	60%			
Total Internal	50%			
Full Marks: 50 + 50 = 100				

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Students' Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester- End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

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1

MULTIPLE INTEGRALS

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CHAPTER OUTLINE

After completion of this chapter, you will be able to gain the knowledge and understanding regarding:

1.1 Introduction

1.2 Double integrals in Cartesian and polar form, Fubini's theorem (statement only), change of order of integration, change of variable from in double integral Jacobian matrix and reduction into Polar.

1.3 Triple integrals in Cartesian form and Dirichlet's Integral, use of cylindrical and spherical coordinates to evaluate triple integral.

1.4 Application of double and triple integrals to find Area and volume.

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Introduction

Generally, the definite integral of a function of any single variable say x, arose from the area problem. Given a function $f : [a, b] \subseteq R \to R$ of any single variable, say x that is continuous and nonnegative on a closed bounded interval [a, b] on the x-axis, then the area of the plane region enclosed between the graph of or the curve $y = f(x)$ and the interval [a, b] is $\int_a^b f(x)\,dx$.

Integrals of functions of two variables arise from the problem of finding volumes under surfaces. Let $f : D \subseteq R^2 \to R$ be a function of any two variables, say x and y. Let it be that f is continuous and is nonnegative on a bounded region D in the xy-plane, then the volume of the solid E in space enclosed between the graph of the surface $z = f(x, y)$ and the xy-planar region D. This develops double integrals.

1.1 DOUBLE INTEGRALS

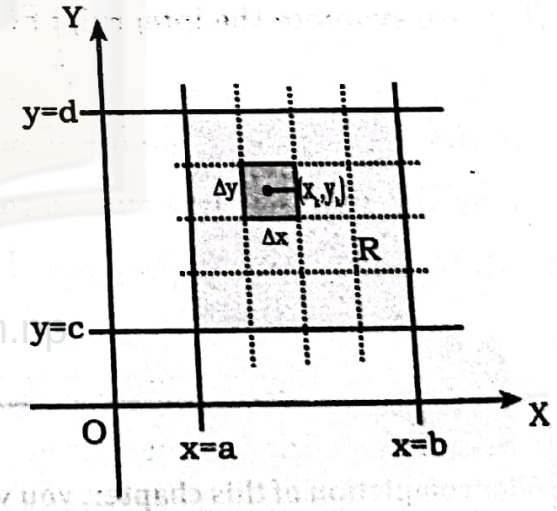
Let f(x, y) be a function of two variable x and y defined over a region R, and let the region R be divided into n elementary sub regions $R_1, R_2, R_3, \ldots, R_n$. Let $\Delta A_1, \Delta A_2, \ldots, \Delta A_n$ be the areas of regions $R_1, R_2, R_3, \ldots\ldots, R_n$, respectively. Let (x_k, y_k) be any point of region R_k having area ΔA_k. Then the total sum of the areas,

$$S_n = f(x_1, y_1)\Delta A_1 + f(x_2, y_2)\Delta A_2 + \ldots + f(x_k, y_k)\Delta A_k + \ldots + f(x_n, y_n)\Delta A_n$$

$$= \sum_{r=1}^{n} f(x_r, y_r)\Delta A_r \qquad \ldots\ldots (i)$$

If the limit of this sum as $n \to \infty$ (and consequently the largest of $\Delta A_r \to 0$) exists for all modes of subdivision, we say that the function is integrable over R; and the limit is called a double integral. It is represented by $\iint_R f(x, y)\,dA$. Thus

$$\iint_R f(x, y)\,dA = \lim_{\Delta A \to 0} \sum_{r=1}^{\infty} f(x_r, y_r)\,\Delta A_r$$



For elementary rectangular region in Cartesian form, we have dA = dxdy or dA = dydx.

Evaluation of Double Integrals

To evaluate a double integral we do it in stages,

- starting from the inside and then working outside,
- using our knowledge of the methods for single integrals.

The easiest kind of region R to work with is a rectangle.

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To evaluate $\iint_R f(x, y)\,dxdy$ proceed as follows:

- work out the limits of integration if they are not already known,
- work out the inner integral for a typical y,
- work out the outer integral.

The above idea develops the following process;

An integral of the form

$$\int_a^b dx \int_{g(x)}^{h(x)} f(x,y)\,dy \quad \text{or} \quad \int_a^b \left[\int_{g(x)}^{h(x)} f(x,y)\,dy\right] dx$$

where g(x) or h(x) or both are functions of x or constants, is called an iterated integral.

Here, the integral $\int_{g(x)}^{h(x)} f(x,y)\,dy$ is the inner integral.

First, we have to evaluate the inner integral, $\int_{g(x)}^{h(x)} f(x,y)\,dy$. Suppose,

$$F(x) = \int_{g(x)}^{h(x)} f(x,y)\,dy$$

Then, we evaluate the integral, $\int_a^b F(x)\,dx$.

Procedure for Finding Limits of Integration

Finding the limits of integration for double integral is the important task. The following steps are used to evaluate limits of integration of $\iint_R f(x, y)\,dA$ over region R.

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1. Identify the region of integration.
2. Sketch the region of integration and label the bounding curves.
3. Determine the limits of the integrals from the given region of integration, if the limits are not given in the problem.
4. Substitute the limits for x and y in the given integrals.
5. Evaluate the integrals.

Properties of Double Integrals

Let f(x, y) and g(x, y) are two functions of two variable x and y defined over a region R, then the following properties holds;

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1. $\iint_R f(x, y)\, dA \geq 0$ for $f(x, y) \geq 0$ in R.
2. $\iint_R k\, f(x, y)\, dA = k \iint_R f(x, y)\, dA$
3. $\iint_R [f(x, y) \pm g(x, y)]\, dA = \iint_R f(x, y)\, dA \pm \iint_R g(x, y)\, dA$
4. If $f(x, y) \geq g(x, y)$, then $\iint_R f(x, y)\, dA \geq \iint_R g(x, y)\, dA$
5. If R is the union of two non-overlapping rectangles R_1 and R_2

$$\iint_R f(x, y)\, dA = \iint_{R_1} f(x, y)\, dA + \iint_{R_2} f(x, y)\, dA.$$

1.2 FUBINI'S THEOREM ON DOUBLE INTEGRALS

When evaluating double integrals, it is very common not to be told the limits of integration but simply told that the integral is to be taken over a certain specified region R in xy–plane. In this case, we need to work out the limits of integration.

The integration can in principle be done in two ways:

a. integrating first with respect to x and then with respect to y, or

b. integrating first with respect to y and then with respect to x.

The limits of integration in the two approaches will in general be quite different, but both approaches must yield the same answer. Sometimes one way round is considerably harder than the other, and in some integrals one way works fine while the other leads to an integral that cannot be evaluated using the simple methods you have been taught. There are no simple rules for deciding which order to do the integration in. Italian mathematician Guido Fubini (1879 – 1943) studied such problems in double integrals, which now known as Fubini's Theorem.

Double integrals over rectangles can be calculated as iterated integrals and this can be done in two ways. In both produce, the double integral is evaluated by integrating with respect to one variable at a time. Thus the Fubini's Theorem says that, *we may calculate the double integral by integrating in either order.*

Fubini's Theorem (First Form) [Region is in Rectangular Form]

(When the region is bounded by the constant limits $a \leq x \leq b$ and $c \leq x \leq d$)

If a function f(x, y) is continuous over a rectangular region $R = \{(x, y): a \leq x \leq b, c \leq y \leq d\}$, where a, b, c and d are constants, then

$$\iint_R f(x, y)\, dx\, dy = \int_c^d \left[\int_a^b f(x,y)\, dx\right] dy = \int_a^b \left[\int_c^d f(x,y)\, dy\right] dx$$

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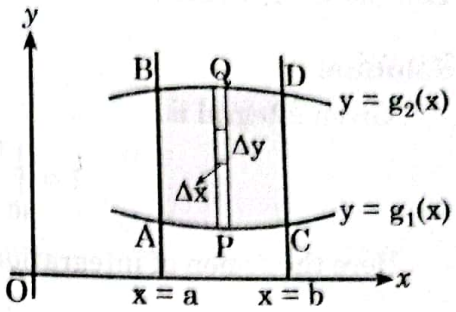
Fubini's Theorem (Second Form) [Region is in Non-Rectangular Form]

(When the region is bounded by the curves and lines)

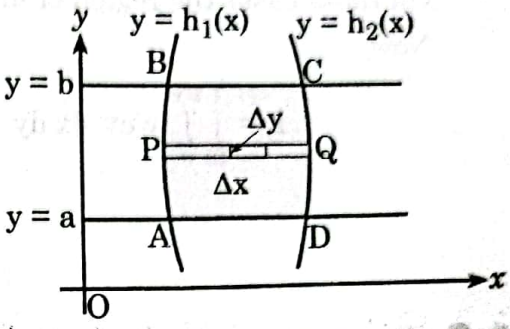
If a function f(x, y) is continuous in a region R and R is defined by $g_1(x) \le y \le g_2(x)$, $a \le x \le b$ in which $g_1(x)$ and $g_2(x)$ are continuous on [a, b] then

$$\iint_R f(x, y)\, dx\, dy = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x, y)\, dy \right] dx$$



Similarly, if a function f(x, y) is continuous in a region bounded by curves $x = h_1(y)$, $= h_2(y)$ where $h_1(y)$ and $h_2(y)$ are continuous and $(y) \le h_2(y)$, then

$$\iint_R f(x, y)\, dx\, dy = \int_a^b \left[\int_{h_1(y)}^{h_2(y)} f(x, y)\, dx \right] dy$$



mple 1: Evaluate the following integrals

(a) $\int_0^3 \int_0^2 (4 - y)^2\, dy\, dx$ (b) $\int_\pi^{2\pi} \int_0^\pi (\sin x + \cos y)\, dx\, dy$

ution:

Here,

$$\int_0^3 \int_0^2 (4 - y)^2\, dy\, dx = \int_0^3 \left[16y - 4y^2 + \frac{y^3}{3} \right]_0^2 dx$$

$$= \frac{56}{3} \int_0^3 dx = \frac{56}{3} [x]_0^3 = 56$$

Here,

$$\int_\pi^{2\pi} \int_0^\pi (\sin x + \cos y)\, dx\, dy = \int_\pi^{2\pi} [-\cos x + x \cos y]_0^\pi\, dy$$

$$= \int_\pi^{2\pi} (-\cos\pi + \pi\cos y + \cos 0 - 0)\, dy$$

$$= \int_\pi^{2\pi} (2 + \pi\cos y)\, dy = [2y + \pi \sin y]_\pi^{2\pi}$$

$$= 4\pi + 0 - 2\pi - 0 = 2\pi$$

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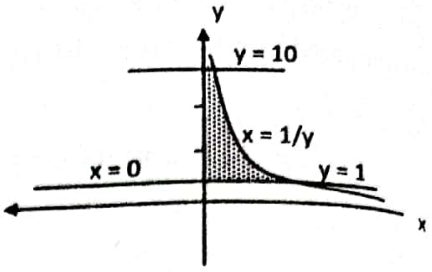
Example 2: Evaluate the integral $\int_{10}^{1}\int_{0}^{1/y} y\, e^{xy}\, dx\, dy$

Solution:

Given integral is

$$I = \int_{10}^{1}\int_{0}^{1/y} y\, e^{xy}\, dx\, dy$$



Here the region of integration is bounded by

$$10 \le y \le 1;\ 0 \le x \le \frac{1}{y}$$

On these bases the region of integration is as shown in figure.

Now,

$$I = \int_{10}^{1}\int_{0}^{1/y} y\, e^{xy}\, dx\, dy = \int_{10}^{1} y\left[\frac{e^{xy}}{y}\right]_0^{1/y} dy$$

$$= \int_{10}^{1} (e-1)\, dy = (e-1)\,[y]_{10}^{1} = (e-1)\,[1-10] = 9(1-e)$$

1.3 CHANGE OF ORDER OF INTEGRATION

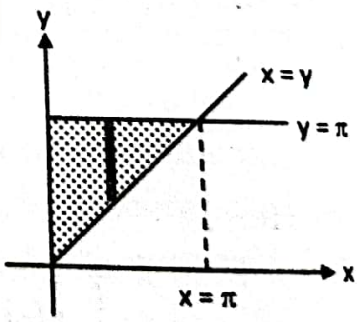
The double integration can be evaluated by integrating first with respect to x and then with respect to y or first with respect to y and then with respect to x. On changing the order of integration, the limits of integration is also changed. It is necessary to draw the rough sketch of the region of integration to find the new limits of integration.

Example 3: Evaluate the integral $\int_{0}^{\pi}\int_{x}^{\pi} \left(\frac{\sin y}{y}\right) dy\, dx$ by interchanging the equivalent integral obtained by reversing the order of integration if necessary.

Solution:

Given integral is

$$I = \int_{0}^{\pi}\int_{x}^{\pi} \left(\frac{\sin y}{y}\right) dy\, dx \quad \text{..... (1)}$$



(fig. 1)

Here the region of integration is bounded by $y = x$, and by $y = \pi$.

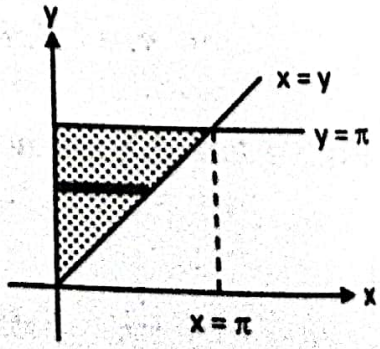
Since the line $y = x$ passes through the points (0, 0) and (1, 1).

And the line $y = \pi$ is a straight line that is parallel to x-axis.

Next, the line $x = 0$ is y-axis. And the line $x = \pi$ is a straight line that is parallel to y-axis.

Then, the region generated by (1) is the shaded portion that has vertical strip, as shown in figure-1.

Now, reversing the order of integration, we take the horizontal strip as in figure 2 for which x varies from $x = 0$ to $x = y$. Also, the strip moves from $y = 0$ to $y = \pi$.



(fig. 2)

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Therefore, after changing the order of integration of (1), it becomes,

$$I = \int_{y=0}^{\pi} \int_{x=0}^{y} \left(\frac{\sin y}{y}\right) dx\, dy$$

$$= \int_{y=0}^{\pi} \left(\frac{\sin y}{y}\right) \int_{x=0}^{y} dx\, dy$$

$$= \int_{y=0}^{\pi} \left(\frac{\sin y}{y}\right) \cdot [x]_0^y\, dy$$

$$= \int_{y=0}^{\pi} \left(\frac{\sin y}{y}\right) (y - 0)\, dy$$

$$= \int_{y=0}^{\pi} \sin y\, dy$$

$$= [-\cos y]_0^{\pi}$$

$$= -[\cos \pi - \cos 0]$$

$$= -[-1 - 1]$$

$$= 2$$

Thus, I = 2.

Example 4: Evaluate the integral $\int_0^2 \int_{y^2}^4 y \cos(x^2)\, dx\, dy$ by interchanging the equivalent integral obtained by reversing the order of integration.

Solution:

Given integral is

$$I = \int_0^2 \int_{y^2}^4 y \cos(x^2)\, dx\, dy \quad \ldots\ldots (1)$$

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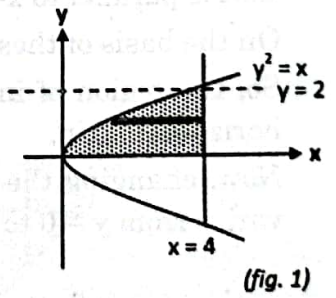
Here the region of integration is bounded by $x = y^2$, and by $x = 4$.

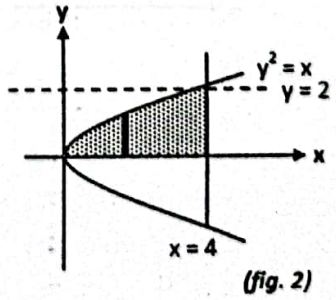
Since the curve $y^2 = x$ is a parabola that has vertex at (0, 0) and has line of symmetry y = 0. So, the parabola is right open ward. And, line x = 4 is a straight line that is parallel to y-axis.

Also, the region is bounded on below by y = 0 and on above by y = 2. On these bases the region of integration is as shown in figure.

Clearly, the region generated by (1) is the shaded portion that has horizontal strip, as shown in figure 1.



(fig. 1)



(fig. 2)

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Now interchanging the order of integration of (1), the integral takes the vertical strip as in figure 2 in which y varies from $y = 0$ to $y = \sqrt{x}$. Also, the strip moves from $x = 0$ to $x = 4$.

Therefore (1) reduces to

$$I = \int_0^4 \int_0^{\sqrt{x}} y \cos(x^2)\, dy\, dx = \int_0^4 \cos(x^2) \left[\frac{y^2}{2}\right]_0^{\sqrt{x}} dx$$

$$= \int_0^4 \cos(x^2) \frac{x}{2} dx$$

Put $x^2 = t$ then $2x\, dx = dt$. Also, $x = 0 \Rightarrow t = 0$, $x = 4 \Rightarrow t = 6$. Then,

$$I = \int_0^{16} \cos t \frac{dt}{4} = \frac{1}{4}[\sin t]_0^6 = \frac{\sin(6)}{4}$$

Thus, $I = \frac{\sin(16)}{4}$.

Example 5: Evaluate, the integral $\int_0^4 \int_y^4 \frac{x\, dx\, dy}{x^2 + y^2}$ by changing the order of integration.

Solution:

Given integral is

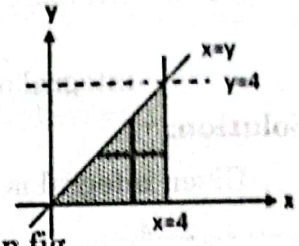
$$I = \int_0^4 \int_y^4 \frac{x\, dx\, dy}{x^2 + y^2} \quad \ldots\ldots (1)$$

Here the region of integration is bounded by $x = y$, and by $x = 4$.

Since the line $x = y$ passes through the points (0, 0) and (1, 1).

And the line $x = 4$ is a straight line that is parallel to y-axis.

Next, the line $y = 0$ is x-axis. And the line $y = 4$ is a straight line that is parallel to x-axis.



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On the basis of these boundaries the sketch of figure is shown as in fig.

So, the region of integration generated by integral (1) is the shaded portion that has horizontal strip.

Now, changing the order of integration, the region takes the vertical strip in which y varies from $y = 0$ to $y = x$. And, the strip moves from $x = 0$ to $x = 4$. Therefore,

$$I = \int_0^4 \int_y^x \frac{x}{x^2 + y^2} dy\, dx$$

$$= \int_0^4 x \cdot \left[\frac{1}{x} \tan^{-1}\left(\frac{y}{x}\right)\right]_0^x dx \quad \left[\because \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\frac{x}{a} + c\right]$$

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$$= \int_0^4 \tan^{-1}\left(\frac{x}{x}\right) dx \qquad [\because \tan^{-1} 0 = 0]$$

$$= \int_0^4 \tan^{-1}(1)\, dx$$

$$= \int_0^4 \frac{\pi}{4} dx \qquad \left[\because \tan\frac{\pi}{4} = 1 \Rightarrow \tan^{-1}(1) = \frac{\pi}{4}\right]$$

$$= \frac{\pi}{4}[x]_0^4$$

$$= \frac{\pi}{4}(4 - 0)$$

$$= \pi$$

Thus, $I = \pi$.

Example 6: Evaluate, the integral $\int_0^\infty \int_x^\infty \left(\frac{e^{-y}}{y}\right) dy dx$ by changing the order of integration.

Solution:

Given integral is

$$I = \int_0^\infty \int_x^\infty \left(\frac{e^{-y}}{y}\right) dy\, dx$$

Here, region of integration is $0 \le x < \infty$, $x \le y < \infty$.

Clearly, the region generated by I is the shaded portion in the figure that has vertical strip that moves from $x = 0$ to $x \to \infty$.

Now, changing the order of integration of I, the region takes horizontal strip in which x varies from $x = 0$ to $x = y$ that moves from $y = 0$ to $y \to \infty$. Then,

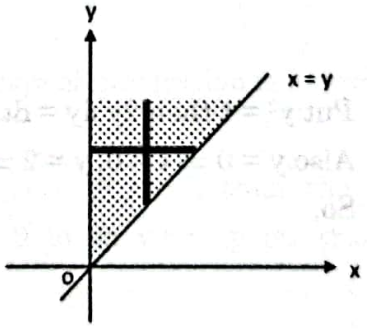
$$I = \int_0^\infty \int_0^y \left(\frac{e^{-y}}{y}\right) dx\, dy = \int_0^\infty \left(\frac{e^{-y}}{y}\right) \int_0^y dx\, dy$$

$$= \int_0^\infty \left(\frac{e^{-y}}{y}\right) [x]_0^y dy$$

$$= \int_0^\infty \frac{e^{-y}}{y}(y - 0)\, dy$$

$$= \int_0^\infty e^{-y} dy$$

$$= \left[\frac{e^{-y}}{-1}\right]_0^\infty$$



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$$= \left(\frac{e^{-\infty} - e^{0}}{-1}\right)$$

$$= \left(\frac{0-1}{-1}\right)$$

$$= 1$$

Thus, $I = 1$

Example 7: Sketch the region of integration and evaluate the integral by reversing the order of integration $\int_0^2 \int_x^2 y^2 \sin xy \, dy dx$.

Solution:

Given that,

$$I = \int_0^2 \int_x^2 y^2 \sin xy \, dy \, dx$$

Here the region of integration is bounded by $y = x$, and by $y = 2$.

Since the line $y = x$ passes through the points (0, 0) and (1, 1). And the line $y = 2$ is a straight line that is parallel to x–axis.

Next, the line $x = 0$ is y–axis. And the line $x = 2$ is a straight line that is parallel to y-axis.

On the basis of these boundaries the sketch of figure is shown as in fig-1.

Clearly, the required region generated by the integral (1) is the shaded portion that has vertical strip as shown in figure 1.

Now, reversing the order of integration, region has horizontal strip as in figure 2 in which x varies from $x = 0$ to $x = y$. Also, the strip moves from $y = 0$ to $y = 2$. Then (1) becomes,

$$I = \int_0^2 \int_x^y y^2 \sin xy \, dx \, dy = \int_0^2 y^2 \left[-\frac{\cos xy}{y}\right]_0^y dy$$

$$= -\int_0^2 y \left[\cos (y^2) - 1\right] dy$$

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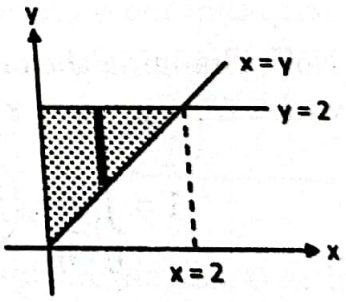


(fig. 1)

Put $y^2 = t$ then $2y \, dy = dt$.

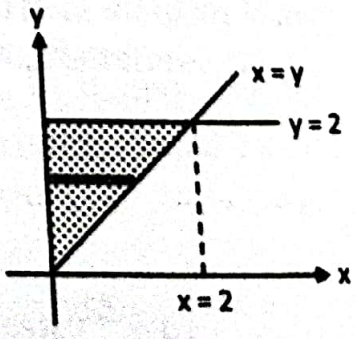
Also $y = 0 \Rightarrow t = 0$, $y = 2 \Rightarrow t = 4$.

So,

$$= \frac{1}{2} \int_0^4 [\cos t - 1] \, dt$$

$$= \frac{1}{2} [\sin t - t]_0^4$$

$$= \frac{\sin 4 - 4}{2}$$



(fig. 2)

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1.4 Change of Variables from Double Integral Using Jacobian Matrix

The double integrals in polar coordinates is defined as

$$\int_{\theta_1}^{\theta_2} \int_{f_1(\theta)}^{f_2(\theta)} f(r, \theta)\, dr\, d\theta$$

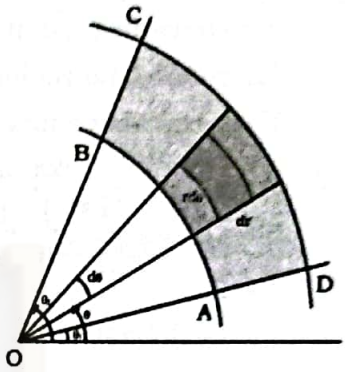
To Change the Integral $\iint_R f(x, y)\, dx\, dy$ into Polar Form

Let the given function is f(x, y). If x and y are transforming to u and v by the relation $x = \phi(u, v)$ and $y = \psi(u, v)$ then the Jacobian transformation for two variables is

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{pmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{pmatrix}$$

and is maintained its sign at all points of R via the relation

$$\iint_R f(x, y)\, dx\, dy = \iint_R f(\phi(u, v), \psi(u, v)) \left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\, dv$$



We put

$$x = r\cos\theta,\ y = r\sin\theta$$

Then

$$|J| = \left|\frac{\partial(x, y)}{\partial(r, \theta)}\right| = \begin{vmatrix} \partial x/\partial r & \partial x/\partial\theta \\ \partial y/\partial r & \partial y/\partial\theta \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$

Therefore, the double integral becomes

$$\iint_R f(x, y)\, dx\, dy = \iint_R f(r, \theta)\, r\, dr\, d\theta$$

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Note:

1. This type of transformation will be useful when the region of integration is a circle type or a part of the circle type.
2. If the region of integration is the complete circle $x^2 + y^2 = a^2$, then the radius r will vary from 0 to a and the angle θ vary from 0 to 2π when polar transform $x = r\cos\theta$, $y = r\sin\theta$ is used.
3. If the region r is the upper half of the circle $x^2 + y^2 = a^2$, then r will vary from 0 to a, and θ from 0 to π.
4. If the region R be the positive quadrant (First Quadrant) of the circle $x^2 + y^2 = a^2$, then r will vary from 0 to a, and θ from 0 to $\frac{\pi}{2}$.

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Example 8: Change the Cartesian integral into an equivalent polar integral and evaluat

the polar integral, $\int_{-a}^{a}\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dy\,dx$

Solution:

Given integral is,

$$I = \int_{-a}^{a}\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dy\,dx$$

Here, the region of integration is

$$-\sqrt{a^2-x^2} \le y \le \sqrt{a^2-x^2},\ -a \le x \le a.$$

That is the region of integration is the circle $x^2 + y^2 = a^2$ as shown in figure. For the radical strip in the region, r varies from $r = 0$ to $r = a$. In order to cover the region such type of radical strip varies from $\theta = 0$ to $\theta = 2\pi$.

Therefore, the region of integration is, $0 \le r \le a$ and $0 \le \theta \le 2\pi$.

Then the above integral change to

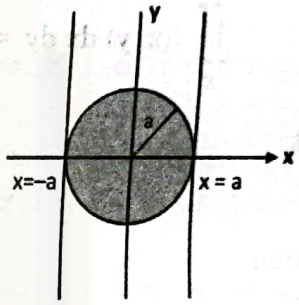
$$I = \int_0^{2\pi}\int_0^a r\,dr\,d\theta = \int_0^{2\pi}\left[\frac{r^2}{2}\right]_0^a d\theta$$

$$= \int_0^{2\pi}\frac{a^2}{2}\,d\theta$$

$$= \frac{a^2}{2}[\theta]_0^{2\pi}$$

$$= \frac{a^2}{2}\times 2\pi$$

$$= a^2\pi$$



Thus $I = a^2\pi$.

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Example 9: Evaluate, by changing the order of integration, $\int_{-a}^{a}\int_{0}^{\sqrt{a^2-x^2}} e^{-(x^2+y^2)}\,dy\,dx$

Solution:

Given integral is,

$$I = \int_{-a}^{a}\int_{0}^{\sqrt{a^2-x^2}} e^{-(x^2+y^2)}\,dy\,dx$$

Here, the variables y varies from $y = 0$ to $y = \sqrt{a^2-x^2}$ and the variable x varies from $x = -a$ to $x = a$.

Thus, the region of integration is the half circle that is in only the positive region of y.

Now, changing the region to polar we substitute

$x = r\cos\theta$ and $y = r\sin\theta$

Then, $dy\,dx = r\,dr\,d\theta$

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Clearly the half circle has radius r = 0 to r = a and the angular form moves from θ = 0 to θ = π. So, the region of integration is, $0 \le r \le a$; $0 \le \theta \le \pi$.

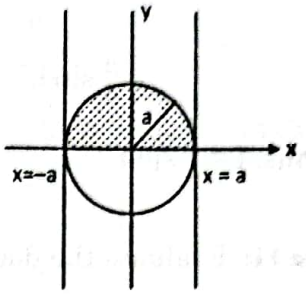
Here, $x^2 + y^2 = r^2(\cos^2\theta + \sin^2\theta) = r^2$

So that the above integral changes to,

$$I = \int_0^{\pi}\int_0^{a} e^{-r^2} r\, dr.\, d\theta$$

Put, $t = r^2$ then $\frac{dt}{dr} = 2r \Rightarrow \frac{dt}{2} = r\, dr$. Also, $r = 0 \Rightarrow t = 0$ and $r = a \Rightarrow t = a^2$. Then,

$$I = \int_0^{\pi}\int_0^{a^2} e^{-t}\frac{dt}{2}\, d\theta = \frac{1}{2}\int_0^{\pi}\left[\frac{e^{-t}}{-1}\right]_0^{a^2} d\theta$$



$$= \frac{1}{-2}\int_0^{\pi}(e^{-a^2} - 1)\, d\theta$$

$$= \frac{1}{2}(1 - e^{-a^2})\int_0^{\pi} d\theta$$

$$= \frac{1}{2}(1 - e^{-a^2})[\theta]_0^{\pi}$$

$$= \frac{\pi}{2}(1 - e^{-a^2}).$$

Thus, $I = \frac{\pi}{2}(1 - e^{-a^2})$.

Example 10: Evaluate, by changing the order of integration, $\int_0^{2}\int_0^{\sqrt{4-y^2}} \cos(x^2+y^2)dxdy$

Solution:

Given integral is,

$$I = \int_0^{2}\int_0^{\sqrt{4-y^2}} \cos(x^2 + y^2)\, dx\, dy$$

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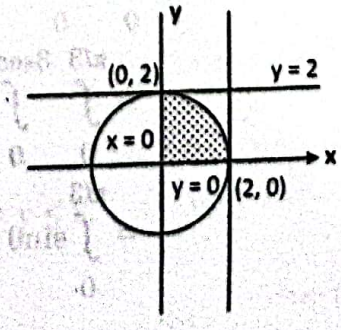
Here, the variables x varies from x = 0 to $x = \sqrt{4 - y^2}$ and the variable y varies from y = 0 to y = 2.

Thus, the region of integration is from the line x = 0 to the circle $x^2 + y^2 = 2^2$. Also, the region moves from origin to the length 2 toward y–axis.

Now, changing the region to polar we substitute $x = r\cos\theta$ and $y = r\sin\theta$.

Clearly the circle has radius r = 0 to r = 2 and the angular form moves from θ = 0 to θ = $\frac{\pi}{2}$. So, the region of integration is, $0 \le r \le a$; $0 \le \theta \le \frac{\pi}{2}$.



The above integral change to

$$I = \int_0^{\pi/2}\int_0^{2} \cos(r^2)\, r\, dr.\, d\theta$$

Put $r^2 = 1$ then $\frac{dt}{dr} = 2r \Rightarrow \frac{dt}{2} = r\, dr$.

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Also, $r = 0 \Rightarrow t = 0$ and $r = 2 \Rightarrow t = 4$. Then,

$$I = \int_0^{\pi/2}\int_0^4 \cos t \frac{dt}{2} d\theta$$

$$= \frac{1}{2}\int_0^{\pi/2} [\sin t]_0^4 d\theta$$

$$= \frac{\sin 4}{2}\int_0^{\pi/2} d\theta$$

$$= \frac{\pi}{4}\sin 4.$$

Thus, $I = \frac{\pi}{4}\sin 4$.

Example 11: Evaluate the double integral $\int_0^3\int_0^{x\sqrt{3}} \frac{y\,dy\,dx}{\sqrt{x^2+y^2}}$, by changing Cartesian integral t equivalent polar integral.

Solution:

Given integral is,

$$I = \int_0^3\int_0^{x\sqrt{3}} \frac{y\,dy\,dx}{\sqrt{x^2+y^2}}$$

Here, the variables x varies from $x = 0$ to $x = 3$ and the variable y varies from $y = 0$ to $y = x\sqrt{3}$.

Now, changing the region to polar we substitute $x = r\cos\theta$ and $y = r\sin\theta$. Then

$$x^2 + y^2 = r^2$$

And, to find r,

$$x = 0 \Rightarrow r\cos\theta = 0 \Rightarrow r = 0;$$
$$x = 3 \Rightarrow r\cos\theta = 3 \Rightarrow r = 3\sec\theta$$

Also, to find θ,

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$$\tan\theta = \frac{y}{x} = 0 \Rightarrow \theta = 0 \quad \text{when } y = 0.$$

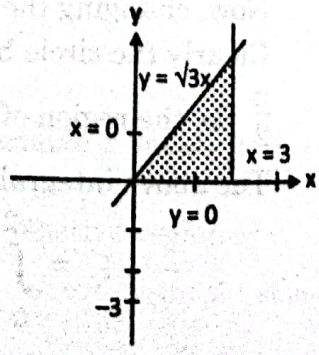
$$\tan\theta = \frac{y}{x} = \frac{x\sqrt{3}}{3} = \frac{3\sqrt{3}}{3} = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3} \text{ when } y = x\sqrt{3}$$

Now, the above integration reduces to,

$$I = \int_0^{\pi/3}\int_0^{3\sec\theta} \frac{r\sin\theta}{\sqrt{r^2}} r\,dr\,d\theta$$

$$= \int_0^{\pi/3}\int_0^{3\sec\theta} r\sin\theta\,dr\,d\theta$$

$$= \int_0^{\pi/3} \sin\theta\,[r]_0^{3\sec\theta} d\theta$$



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$$= \int_0^{\pi/3} \sin\theta\, 3\sec\theta\, d\theta$$

$$= 3\int_0^{\pi/3} \tan\theta\, d\theta$$

$$= 3\,[\log(\sec\theta)]_0^{\pi/3}$$

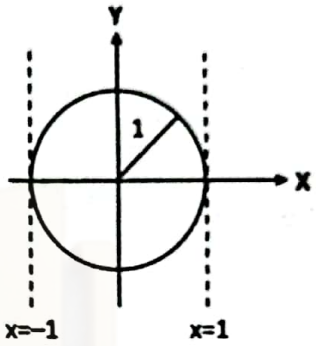
$$= 3\left[\log\left(\sec\frac{\pi}{3}\right) - \log(\sec 0)\right]$$

$$= 3\log(2).$$

Thus, $I = 3\log(2) = \log(2)^3 = \log(8)$.

Example 12: Change the following integral into polar form and evaluate:

$$\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2dy\,dx}{(1+x^2+y^2)^2}$$



Solution:

Given integral is

$$I = \int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2dy\,dx}{(1+x^2+y^2)^2}$$

Here, the variables x varies on $-1 < x < 1$ and y varies on $\sqrt{1-x^2} < y < \sqrt{1-x^2}$.

Now, changing the region to polar we substitute

$x = r\cos\theta$ and $y = r\sin\theta$.

Clearly the circle has radius $r = 0$ to $r = 1$ and the angular form moves from $\theta = 0$ to $\theta = 2\pi$. So, the region of integration is, $-1 \le r \le 1$; $0 \le \theta \le 2\pi$.

Here,

$(1+x^2+y^2)^2 = (1+1^2)^2 = 4$

Now,

$$I = \int_0^{2\pi}\int_0^1 \frac{2\,r\,d\theta\,dr}{4} = \frac{1}{2}\int_0^{2\pi}\left[\frac{r^2}{2}\right]_0^1 d\theta = \frac{1}{4}[\theta]_0^{2\pi} = \frac{\pi}{2}$$

Thus, $I = \frac{\pi}{2}$.

Example 13: Integrate $f(x, y) = \frac{1}{xy}$ over the rectangle R: $1 \le x \le 2$, $1 \le y \le 2$.

Solution:

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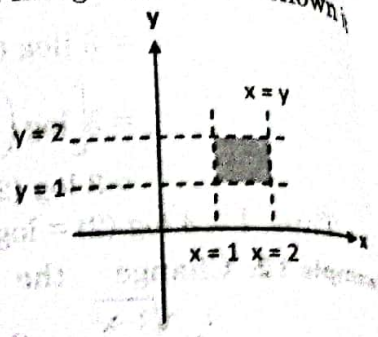
Given that,

$$f(x, y) = \frac{1}{xy}$$ over the region R: $1 \le x \le 2$, $1 \le y \le 2$.

Here the region of integration is bounded below by $y = 1$, above by $y = 2$, on the left by $x = 1$ and on the right by $x = 2$. On these bases the region of integration is as shown in figure.



Now,

$$I = \int_{y=1}^{2}\int_{x=1}^{2} f(x, y)\, dx\, dy = \int_{y=1}^{2}\int_{x=1}^{2}\left(\frac{1}{xy}\right) dx\, dy$$

$$= \int_{y=1}^{2} \frac{1}{y} [\log x]_1^2\, dy$$

$$= \log (2) \int_{y=1}^{2} \frac{dy}{y} \qquad [\because \log(1) = 0]$$

$$= \log (2) [\log (y)]_1^2$$

$$= \log (2) \log (2) \quad [\because \log (1) = 0]$$

$$= [\log (2)]^2$$

Thus, $I = [\log (2)]^2$

Exercise 1 [A]

1. Sketch the region and evaluate the following integrates,

a. $\int_0^{\log 2}\int_{e^y}^{2} dx\, dy$

b. $\int_0^3\int_0^2 (4 - y^2)\, dy\, dx$

c. $\int_0^3\int_{-2}^{0} (x^2y - 2xy)\, dy\, dx$

d. $\int_1^2\int_{-1}^{2} (12xy^2 - 8x^3)\, dy\, dx$

e. $\int_0^{\pi}\int_0^{x} x \sin y\, dy\, dx$

f. $\int_1^2\int_{1-x}^{\sqrt{x}} x^2y\, dy\, dx$

g. $\int_0^2\int_{y^2}^{2y} (4x - y)\, dx\, dy$

h. $\int_0^{\pi}\int_0^{\sin x} y\, dy\, dx$

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2. Evaluate, the integral by interchanging the equivalent integral obtained by reversing the order of integration if necessary.

a. $\int_0^2 \int_1^{e^x} dy\,dx$ b. $\int_0^{\sqrt{2}} \int_{-\sqrt{4-2y^2}}^{\sqrt{4-2y^2}} y\,dx\,dy$ c. $\int_0^1 \int_y^1 x^2 e^{xy}\,dx\,dy$

d. $\int_0^{\infty} \int_{x^{1/3}}^2 \left(\frac{1}{1+y^4}\right) dy\,dx$ e. $\int_0^1 \int_{2x}^2 e^{y^2}\,dy\,dx$ f. $\int_0^1 \int_{4y}^4 e^{x^2}\,dx\,dy$

g. $\int_0^{\infty} \int_0^x x\,e^{-x^2/y}\,dy\,dx$ h. $\int_0^a \int_{x^2/a}^{2a-x} xy\,dy\,dx$

3. Change the Cartesian integral into an equivalent polar integral and evaluate the polar integral.

a. $\int_0^2 \int_0^{4-y^2} y\,dx\,dy$ b. $\int_0^1 \int_0^{\sqrt{1-x^2}} y^2\,dy\,dx$ c. $\int_{a/\sqrt{2}}^{a} \int_{y}^{\sqrt{a^2-y^2}} x\,dx\,dy$

d. $\int_0^{a/\sqrt{2}} \int_y^{\sqrt{a^2-y^2}} \log(x^2+y^2)\,dx\,dy$ for $a > 0$ e. $\int_0^2 \int_0^{4-x^2} \frac{xe^{2y}}{4-y}\,dy\,dx$

f. $\int_0^2 \int_0^{\sqrt{4-x^2}} \frac{xy}{\sqrt{x^2+y^2}}\,dy\,dx$

4. Integrate f(x, y) over the region R:

a. If $f(x, y) = \frac{x}{y}$ over the region in the first quadrant bounded by the lines $y = x$, $y = 2x$, $x = 1$, $x = 2$.

b. If $f(x, y) = y - \sqrt{x}$ over the triangular region cut from the first quadrant by the lines $x + y = 1$.

c. Let R be the region in the xy plane bounded by the curves $y = x^2$ and $y = 2x$, evaluate: $\iint_R (x^2 + 4y)dA$.

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Answer Key

1. a. $2\log(2) - 1$ b. 16 c. 0 d. -36
   e. $\frac{4+\pi^2}{2}$ f. $\frac{168}{120}$ g. $\frac{36}{5}$ h. $\frac{\pi}{4}$
2. a. $\frac{e^2-3}{4}$ b. $\frac{8}{3}$ c. $\frac{e-2}{2}$ d. $\frac{1}{4}\log(17)$
   e. $\frac{e^{16}-1}{8}$ f. $\frac{1}{2}$ g. $\frac{3a^4}{8}$ h. $\frac{\pi a^4}{4}\left(\log a - \frac{1}{2}\right)$
3. a. 4 b. $\frac{\pi}{16}$ c. $\frac{8}{3\sqrt{2}}$ d. $\frac{\pi}{4}$
   e. $\frac{e^8-1}{4}$ f. $\frac{8}{3}$
4. a. $\frac{3}{2}\log 2$ b. $\frac{-1}{10}$ c. $\frac{248}{15}$

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1.5 APPLICATIONS OF DOUBLE INTEGRALS

Double Integrals can be used to find the area of a plane region and the volume of a solid

1.6 AREA OF A PLANE REGION BY DOUBLE INTEGRALS

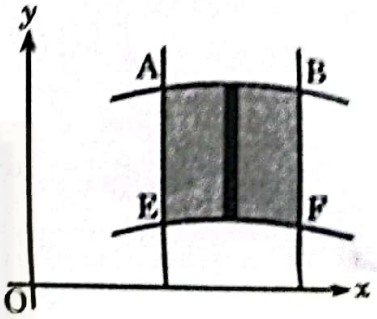
Let AB be any arc of a curve whose equation is $y = f(x)$, and let the coordinates of A and B be (a, f(a)) and (b, f(b)). Draw the ordinates AE and BF. Let P(x, y) and Q(x − Δx, y + Δy) be two neighboring points. Draw two lines PG and QH perpendiculars to x–axis.

Consider an elementary rectangle PRQS whose area is Δx Δy. Then

$$\text{Area of PRQS} = \lim_{\Delta y \to 0} (\Sigma \Delta x \Delta y) = \left(\int_0^{f(x)} dy \right) \delta x$$

Now, the area of the rectangle ABFE can be obtained by summing such elementary strips Therefore,

$$\text{area ABFE} = \int_a^b \left(\int_0^{f(x)} dy \right) dx$$



Thus, the area bounded by the curve $y = f(x)$ and the ordinates at $x = a$, $x = b$ is given by the iterated integral is

$$\int_a^b \left(\int_0^{f(x)} dy \right) dx$$

Similarly, the area bounded by the curves $y = \phi(x)$, $y = \psi(x)$, where $\phi(x) > \psi(x)$, and the ordinates $x = a$, $x = b$ is

$$\int_a^b \left[\int_{\psi(x)}^{\phi(x)} dy \right] dx$$

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1.7 VOLUME OF A REGION BY DOUBLE INTEGRALS

Let $z = f(x, y)$ be the equation of a surface S, and let R be the projection of S on the xy–plane.

Let the region R be divided into n elementary rectangles by drawing lines parallel to z–axis. Consider a typical rectangle whose area is $\delta x_i \, \delta y_i$; and erect an elementary prism on it. The volume of this elementary prism between R and the surface $z = f(x, y)$ is $z \, \delta x_i \, \delta y_i$ or $f(x_i, y_i) \, \delta x_i \, \delta y_i$.

Consider the sum

$$\Sigma f(x_i, y_i) \, \delta x_i \, \delta y_i$$

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If limit of this sum as $n \to \infty$ exists, we say that the function is integrable over R, and the limit is the volume of the solid then the volume is the given by

$$V = \iint_R f(x, y)\, dx\, dy \quad \text{or} \quad V = \iint_R z\, dx\, dy$$

where R is the region of integration.

Workout Examples

Example 1: Sketch the region bounded by the curves $y^2 = -x$, $x - y = 4$, $y = -1$, $y = 2$ and find its area using double integral.

Solution:

Given that the region bounded by the curves

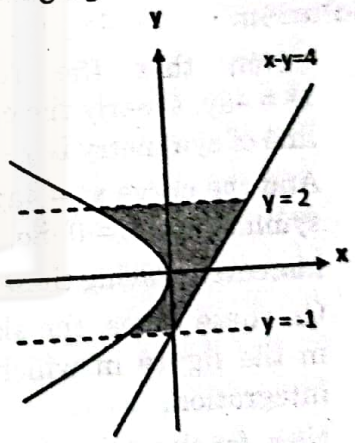
$$y^2 = -x,\ x - y = 4,\ y = -1,\ y = 2.$$

Since the curve $y^2 = -x$ is a parabola that has vertex at (0, 0) and has line of symmetry $y = 0$. So, the parabola is left openward. And, the line $x - y = 4$ passes through he point (4, 0) and (0, –4).

Also, the line $y = -1$ and $y = 2$ are parallel o the x-axis.

On the bases the region of integration is as shown in the figure.

Here the required region is the shaded part in the corresponding figure.



Taking horizontal strip

$$\text{Area} = \int_{-1}^{2}\int_{-y^2}^{y+4} dx\, dy = \int_{-1}^{2} [x]_{-y^2}^{y+4}\, dy$$

$$= \int_{-1}^{2} (y + 4 + y^2)\, dy$$

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$$= \left[\frac{y^2}{2} + 4y + \frac{y^3}{3}\right]_{-1}^{2}$$

$$= \left(\frac{4}{2} + 8 + \frac{8}{3}\right) - \left(\frac{1}{2} - 4 - \frac{1}{3}\right)$$

$$= \frac{76 + 23}{6}$$

$$= \frac{33}{2}$$

Thus, area of the region is $\frac{33}{2}$.

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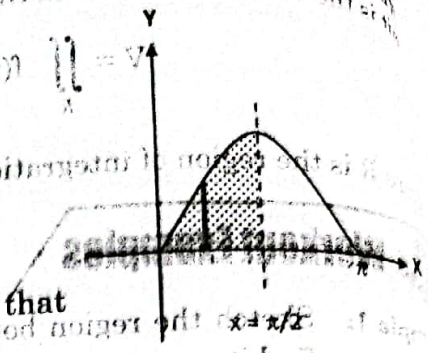
Example 2: Using the double integration find the area of the region bounded by $y = \sin x$ and the line $x = 0$ and $x = \frac{\pi}{2}$.

Solution:

Given that the region is bounded by curves

$$y = \sin x,\ x = 0 \text{ and } x = \frac{\pi}{2}.$$



Clearly the region is the shaded portion in the figure that is bounded by $0 \le y \le \sin x$ and $0 \le x \le \frac{\pi}{2}$.

Now,

$$I = \int_0^{\pi/2} \int_0^{\sin x} dy\, dx = \int_0^{\pi/2} [y]_0^{\sin x}\, dx$$

$$= \int_0^{\pi/2} \sin x\, dx = [-\cos x]_0^{\pi/2} = \cos\frac{\pi}{2} - \cos 0 = 1.$$

Thus, area of the region is 1 sq. unit.

Example 3: Using the double integrals find the area of the region, $y^2 = 4ax$ and the parabola $x^2 = 4ay$.

Solution:

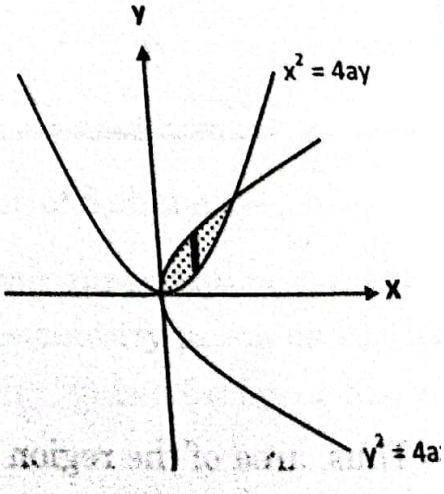
Given that the required region is bounded the curves $y^2 = 4ax$ and $x^2 = 4ay$. Clearly the curve $y^2 = 4ax$ is a parabola having vertex at (0, 0) and equation of line of symmetry is $y = 0$. So, it has right open ward.

And the curve $x^2 = 4ay$ is a parabola having vertex at (0, 0) and the equation of line of symmetry is $x = 0$. So, it has up open ward.

Moreover, solving these curves, the point of contact between them is (0, 0) and (4a, 4a).

On these bases, the sketch of the region of integration is as in the figure in which the shaded portion is the region of integration.

Now, for the area of the region, taking vertical strip we get,



$$A = \int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} dy\, dx = \int_0^{4a} [y]_{x^2/4a}^{2\sqrt{ax}}\, dx$$

$$= \int_0^{4a} \left(2\sqrt{ax} - \frac{x^2}{4a}\right) dx$$

$$= \left[2\sqrt{a}\,\frac{x^{3/2}}{3/2} - \frac{1}{4a}\cdot\frac{x^3}{3}\right]_0^{4a}$$

$$= \frac{4\sqrt{a}}{3}(4a)^{3/2} - \frac{(4a)^4}{12a} = \frac{32a^2}{3} - \frac{16a^2}{3} \Rightarrow A = \frac{16a^2}{3}$$

Thus, area of the region bounded by $x^2 = 4ay$ and $y^2 = 4ax$ is $\frac{16a^2}{3}$ sq. units.

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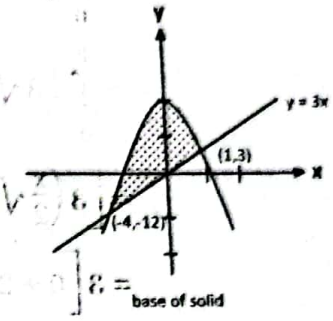
Example 4: Find the volume of the solid whose base is the region in the xy plane that is bounded by the parabola $y = 4 - x^2$ and the line $y = 3x$, while the top of the solid is bounded by the plane $z = x + 4$.

Solution:

Given that the parabola $y = 4 - x^2$ and the line $y = 3x$ made a solid in xy–plane which is bounded in the top by a plane $z = x + 4$. By solving $y = 4 - x^2$ and $y = 3x$, we get the base has common limits (1, 3) and (–4, –12).

Now, integrate the plane z over the region of base then,



$$V = \int_{-4}^{1}\int_{3x}^{4-x^2} (z)\, dy\, dx$$

$$= \int_{-4}^{1}\int_{3x}^{4-x^2} (x+4)\, dy\, dx$$

$$= \int_{-4}^{1} (x+4)\,[y]_{3x}^{4-x^2}\, dx$$

$$= \int_{-4}^{1} (x+4)(4-x^2-3x)\, dx$$

$$= \int_{-4}^{1} (-8x - x^3 - 7x^2 + 16)\, dx$$

$$= \left[-4x^2 - \frac{x^4}{4} - \frac{7x^3}{3} + 16x\right]_{-4}^{1}$$

$$= \left(-4 - \frac{1}{4} - \frac{7}{3} + 16\right) - \left(-64 - 64 + \frac{448}{3} - 64\right)$$

$$= \frac{144 - 31 + 2304 - 1792}{12}$$

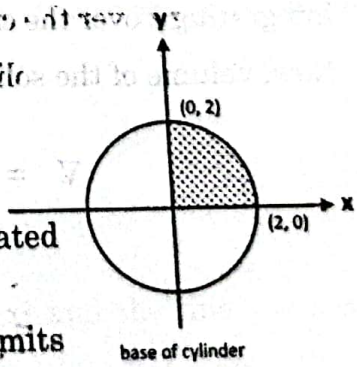
$$= \frac{625}{12}$$

Thus, volume of the solid is $\frac{625}{12}$ cubic units.

Example 5: Find the volume in the first octant bounded by the co–ordinate planes, the cylinder $x^2 + y^2 = 4$ and the plane $z + y = 3$.

Solution:

Given that, we have observe the volume in the first octant bounded by the coordinate planes, cylinder $x^2 + y^2 = 4$ and the plane $z + y = 3$.



From the figure it is clear that $z = 3 - y$ is to be integrated over the first quadrant of the circle $x^2 + y^2 = 4$.

For the base, limits for y are $y = 0$, and $y = \sqrt{4 - x^2}$ and limits for x are $x = 0$ and $x = 2$.

Now, volume of the solid is

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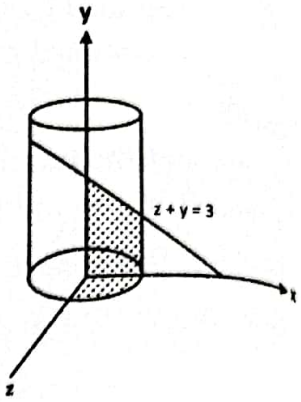


$$V = \int_0^2 \int_0^{\sqrt{4-x^2}} z \, dy \, dx$$

$$= \int_0^2 \int_0^{\sqrt{4-x^2}} (3 - y) \, dy \, dx$$

$$= \int_0^2 \left[3y - \frac{y^2}{2}\right]_0^{\sqrt{4-x^2}} dx$$

$$= \int_0^2 \left(3\sqrt{4-x^2} - \frac{4-x^2}{2}\right) dx$$

$$= \left[3\left(\frac{x}{2}\sqrt{4-x^2} + \frac{4}{2}\sin^{-1}\left(\frac{x}{2}\right) - \frac{1}{2}\left(4x - \frac{x^3}{3}\right)\right)\right]_0^2$$

$$= 3\left[0 + 2\sin^{-1}\left(\frac{2}{2}\right) - \frac{1}{2}\left(8 - \frac{8}{3}\right)\right] - 3[0 + 2\sin^{-1}0] + \frac{1}{2} \cdot 0$$

$$= 6\sin^{-1}(1) - \frac{4}{3} \qquad [\because \sin^{-1} 0 = 0]$$

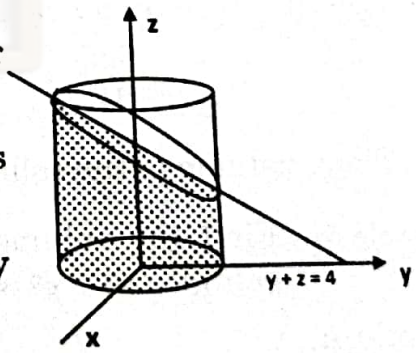
$$= 6 \cdot \frac{\pi}{2} - \frac{4}{3}$$

$$= 3\pi - \frac{4}{3}$$

Thus, volume of the solid is $3\pi - \frac{4}{3}$ cubic units.

Example 6: Find the volume bounded by circle cylinder $x^2 + y^2 = 4$ and the plan $y + z = 4$ and $z = 0$.

Solution:



Given that the solid is bounded by a circle cylinder $x^2 + y^2 = 4$, on the top by the plane $y + z = 4$ and by $z = 0$.

Clearly the plane $y + z = 4$ is parallel to x–axis and makes intercept 4 on y–axis and z–axis.

Then the volume generated by the solid is obtained by integrating z over the circle $x^2 + y^2 = 4$ that has radius 2.

Now, volume of the solid is

$$V = \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (z) \, dy \, dx$$

$$= \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4 - y) \, dy \, dx$$

Since $4 - y$ is an odd function. So,

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$$V = \int_0^2 \int_0^{\sqrt{4-x^2}} (4 - y)\, dy\, dx$$

$$= 16 \int_0^2 \int_0^{\sqrt{4-x^2}} dy\, dx - 4 \int_0^2 \int_0^{\sqrt{4-x^2}} y\, dy\, dx$$

$$= 16 \int_0^2 [y]_0^{\sqrt{4-x^2}}\, dx - 4 \int_0^2 \left[\frac{y^2}{2}\right]_0^{\sqrt{4-x^2}} dx$$

$$= 16 \int_0^2 \sqrt{4 - x^2}\, dx - 2 \int_0^2 (4 - x^2)\, dx$$

$$= 16 \left[\frac{x}{2}\sqrt{4 - x^2} + \frac{4}{2}\sin^{-1}\frac{x}{2}\right]_0^2 - 2\left[4x - \frac{x^3}{3}\right]_0^2$$

$$= 16\,[0 + 2\sin^{-1}(1)] - 2\left(8 - \frac{8}{3}\right)$$

$$= 16\left(2\frac{\pi}{2}\right) - 2\left(\frac{16}{3}\right)$$

$$= 16\pi - \frac{32}{\pi}$$

Thus, the volume of the solid is $\left(16\pi - \frac{32}{\pi}\right)$ cubic units.

Exercise 1 (B)

1. Sketch the region bounded by the graph of the equation and find its area using one or more double integral.

a. $y = \frac{1}{x^2}$, $y = -x^2$, $x = 1$, $x = 2$.

b. $y = x$, $y = 3x$, $x + y = 4$.

c. $y = e^x$, $y = \sin x$, $x = -\pi$, $x = \pi$.

2. Sketch the solid in the first octant bounded by the curves and find its volume.

a. $x^2 + z^2 = 9$, $y = 2x$, $y = 0$, $z = 0$

b. $2x + y + z = 4$, $x = 0$, $y = 0$, $z = 0$.

c. $z = x^2 + y^2$, $y = 4 - x^2$, $x = 0$, $y = 0$, $z = 0$.

d. $z = x^2 + 4$, $y = 4 - x^2$, $x + y = 2$, $z = 0$

3. a. Find the area lying between the parabola $y = 4x - x^2$ and the line $y = x$ by using double integral.

b. Find the area of the region bounded by the curve $y^2 + x = 0$ and the line $y = x + 2$ by using double integration.

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4. a. Find the volume of the solid in the first octant bounded by the co-ordinate planes the plane $x = 3$, and the parabolic cylinder $z = 4 - y^2$.

b. Find the volume of the prism whose base is the triangle in the xy–plane bounded by the x-axis and the line $y = x$, $x = 1$ and top lies in the plane $z = f(x, y) = 3 - x - y$.

c. Find the volume V of the solid that lies in the first octant and is bounded by the three co-ordinate planes and the cylinder $x^2 + y^2 = 9$ and $y^2 + z^2 = 9$.

d. Find the volume bounded by the xy–plane, the cylinder $x^2 + y^2 = 1$ and the plane $x + y + z = 3$.

e. Find the volume bounded by the xy–plane, the paraboloid $2z = x^2 + y^2$ and the cylinder $x^2 + y^2 = 4$.

f. Find the volume of the region bounded by $z = x^2 + y^2$, $z = 0$, $x = -a$, $x = a$ and $y = -a$, $y = a$.

g. Show that the volume enclosed between the cylinders $x^2 + y^2 = 2ax$ and $z^2 = 2ax$ is $\frac{64a^3}{15}$.

h. Find the volume of the solid that is bounded above by the cylinder $z = x^2$ and below by the region enclosed by the parabola $y = 2 - x^2$ and the line $y = x$ in the xy plane.

i. Find the volume of the solid whose base is the region in xy–plane that is bounded by the parabola $y = 3 - x^2$, $y = 2x$ while the top is bounded by the plane $z = x + 1$.

j. Find the volume of the tetrahedron bounded by the coordinate planes and the plane $x + 2y + z = 1$.

Answer Key

1. a. $\frac{17}{6}$ b. 2 c. $e^x - e^{-x}$

2. a. 18 b. $\frac{16}{3}$ c. $\frac{832}{35}$ d. $\frac{423}{20}$

3. a. $\frac{10}{3}$ b. 4.5

4. a. 16 b. 1 c. 18 d. 3π

e. 4π f. $\frac{8a^4}{3}$ h. $\frac{63}{20}$ i. 0

j. $\frac{1}{6}$

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1.8 TRIPLE INTEGRALS IN RECTANGULAR COORDINATES

We can define triple integrals for three variables just similar to double integrals of two variables.

Let f(x, y, z) be continuous function defined over the region of integration R and is closed and bounded in region D. We consider a rectangular region containing R and partition the rectangular region into rectangular cells by planes parallel to the coordinate planes. Let n be the number of cells that lie inside R and a typical cell have dimension Δx_k by Δy_k by Δz_k and volume ΔV_k.

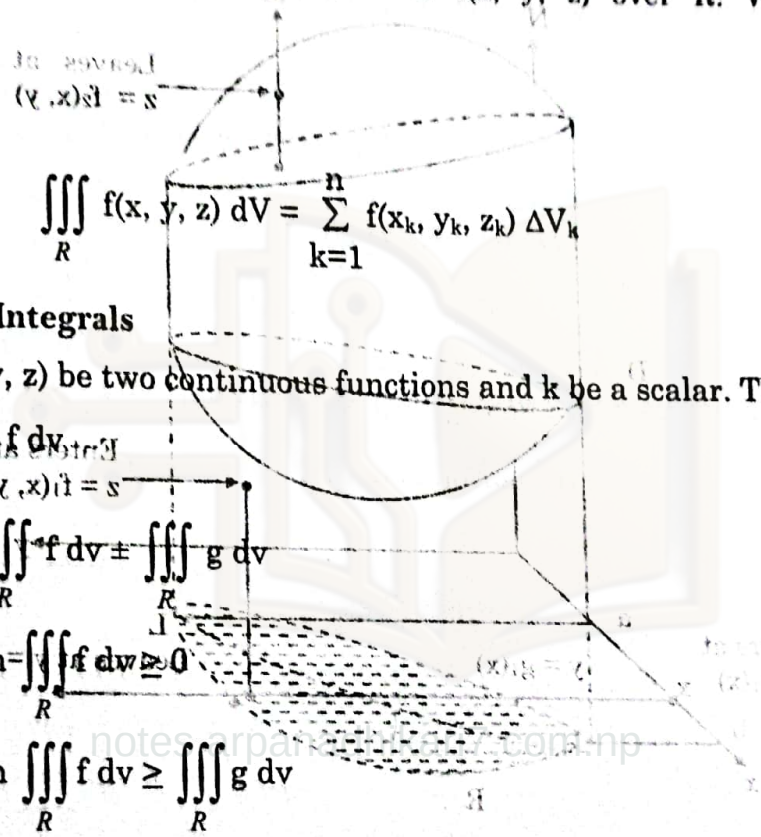
If R is made of smooth surfaces joined along continuous curves, then the

$$\lim_{n\to\infty} \sum_{k=1}^{n} f(x_k, y_k, z_k)\, \Delta V_k$$

exists and is called the **triple integral** of f(x, y, z) over R. We denote it by $\iiint_R f(x, y, z)\, dV$.

Hence,

$$\iiint_R f(x, y, z)\, dV = \sum_{k=1}^{n} f(x_k, y_k, z_k)\, \Delta V_k$$



Properties of Triple Integrals

Let f(x, y, z) and g(x, y, z) be two continuous functions and k be a scalar. Then

(1) $\iiint_R k f\, dv = k \iiint_R f\, dv$

(2) $\iiint_R (f \pm g)\, dv = \iiint_R f\, dv \pm \iiint_R g\, dv$

(3) If $f \geq 0$ on R, then $\iiint_R f\, dv \geq 0$

(4) If $f \geq g$ on R, then $\iiint_R f\, dv \geq \iiint_R g\, dv$

(5) If R is the union of two non-overlapping regions R_1 and R_2, then

$$\iiint_R f\, dv = \iiint_{R_1} f\, dv + \iiint_{R_2} f\, dv$$

1.9 VOLUME OF A REGION IN SPACE

If f(x, y, z) = 1 on the region R, then the volume of a closed and bounded region R is given by

$$\text{Volume (V)} = \iiint_R dV = \iiint_R dx\, dy\, dz$$

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Limits of Integration in Triple Integral

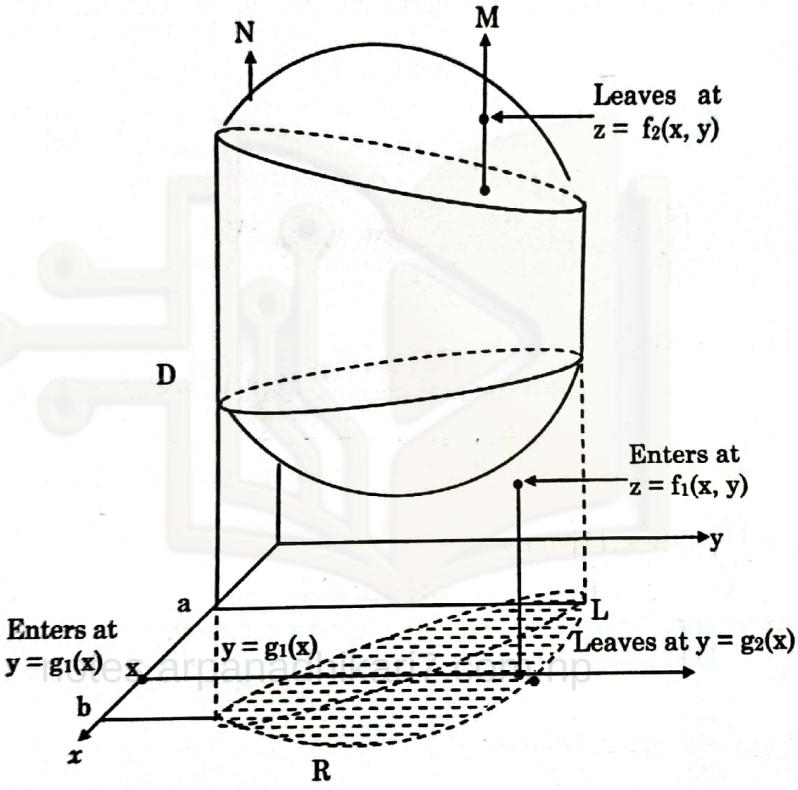
Step 1: **Sketch:** Sketch the projection of the region of integration R in the xy-pl
Label the upper and lower bounding surfaces of R and its upper and lo
bounding curves.

Step 2: **Find z-limits of integration:** Draw a line M passing through a typical p
(x, y) in R parallel to the z-axis. As z-increases, M at $z = f_1(x, y)$ and leaves
$= f_2(x, y)$. These are the z-limits of integration.

Step 3: **Find y-limits of integration:** Draw a line L through (x, y) parallel to th
axis. As y increases, L enters R at $y = g_1(x)$ and leaves at $y = g_2(x)$. These
the y-limits of integration.

Step 4: **Find x-limits of integration:** choose x-limits that include all lines throu
parallel to the y-axis (x = a and x = b). These are the x–limits of integration

These steps are explained diagrammatically as shown in the following figure.



Workout Examples

Example 1: Evaluate $\int_0^1 \int_0^1 \int_0^1 xyz \, dx \, dy \, dz$.

Solution:

Let,

$$I = \int_0^1 \int_0^1 \int_0^1 xyz \, dx \, dy \, dz = \int_0^1 \int_0^1 \left[\frac{x^2}{2}\right]_0^1 yz \, dy \, dz$$

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$$= \frac{1}{2}\int_0^1 \left[\frac{y^2}{2}\right]_0^1 dz$$

$$= \frac{1}{4}\int_0^1 dz$$

$$= \frac{1}{4}\left[\frac{z^2}{2}\right]_0^1$$

$$= \frac{1}{8}.$$

Example 2: Evaluate $\int_0^{2\pi}\int_0^1\int_{-1/2}^{1/2} (r^2\sin^2\theta + z^2)\, dz\, r\, dr\, d\theta$.

Solution:

Let,

$$I = \int_0^{2\pi}\int_0^1\int_{-1/2}^{1/2} (r^2\sin^2\theta + z^2)\, dz\, r\, dr\, d\theta$$

$$= \int_0^{2\pi}\int_0^1 \left[zr^2\sin^2\theta + \frac{z^3}{3}\right]_{-1/2}^{1/2} r\, dr\, d\theta$$

$$= \int_0^{2\pi}\int_0^1 \left[\left(\frac{1}{2}+\frac{1}{2}\right)r^2\sin^2\theta + \frac{1}{3}\left(\frac{1}{8}+\frac{1}{8}\right)\right] r\, dr\, d\theta$$

$$= \int_0^{2\pi}\int_0^1 \left[r^3\sin^2\theta + \frac{r}{12}\right] dr\, d\theta$$

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$$= \int_0^{2\pi} \left[\frac{r^4}{4}\sin^2\theta + \frac{r^2}{24}\right]_0^1 d\theta$$

$$= \int_0^{2\pi} \left[\frac{1}{4}\sin^2\theta + \frac{1}{24}\right] d\theta$$

$$= \int_0^{2\pi} \left[\frac{1}{4}\times\frac{1-\cos 2\theta}{2} + \frac{1}{24}\right] d\theta$$

$$= \left[\frac{1}{8}\times\left(\theta - \frac{\sin 2\theta}{2}\right) + \frac{\theta}{24}\right]_0^{2\pi}$$

$$= \left[\frac{1}{8}\times\left(2\pi - \frac{\sin 4\pi}{2}\right) + \frac{2\pi}{24}\right]$$

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$$= \left[\frac{1}{8} \times (2\pi - 0) + \frac{\pi}{12}\right]$$

$$= \frac{6\pi + 2\pi}{24}$$

$$= \frac{\pi}{3}$$

Example 3: Evaluate $\int_0^1 \int_0^y \int_0^{x+y} dz\, dx\, dy$.

Solution:

Here,

$$\int_0^1 \int_0^y \int_0^{x+y} dz\, dx\, dy = \int_0^1 \int_0^y [z]_0^{x+y} dx\, dy$$

$$= \int_0^1 \int_0^y [x + y]\, dx\, dy$$

$$= \int_0^1 \left[\frac{x^2}{2} + xy\right]_0^y dy$$

$$= \int_0^1 \left[\frac{y^2}{2} + y^2\right] dy$$

$$= \int_0^1 \left[\frac{3y^2}{2}\right] dy$$

$$= \left[\frac{y^3}{2}\right]_0^1$$

$$= \frac{1}{2}$$

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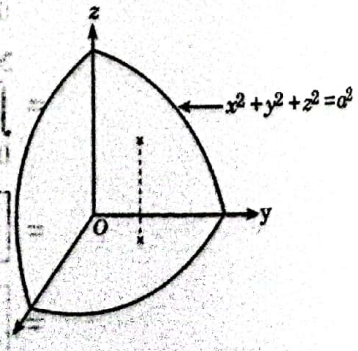
Example 4: Evaluate $\iiint \frac{dxdydz}{\sqrt{a^2 - x^2 - y^2 - z^2}}$ over the first octant of the sphere $x^2 + y^2 + z^2 = a^2$

Solution:

Divide the entire space $x^2 + y^2 + z^2 = a^2$ into 8 octants: xy, yz, xz plane.

Let,

$$I = \iiint_V \frac{1}{\sqrt{a^2 - x^2 - y^2 - z^2}}$$



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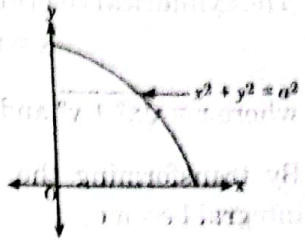
Since the limits lie in the first octant $x^2 + y^2 + z^2 = a^2$, this region of integration V is bounded by $x = 0$, $y = 0$, $z = 0$ and $x^2 + y^2 + z^2 = a^2$.

To find the limits for z, we drawn a line parallel to z-axis, we get the volume of integration in which z varies from 0 to $\sqrt{a^2 - x^2 - y^2}$.

After this we search the projection of the sphere in xy plane. As the process of double integral, we observe y varies from 0 to $\sqrt{a^2 - x^2}$ and x varies from 0 to a on the region. Then,



$$I = \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} \frac{1}{\sqrt{a^2 - x^2 - y^2 - z^2}} \, dz \, dy \, dx$$

Put, $a^2 - x^2 - y^2 = t^2$. Then

$$I = \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^t \frac{1}{\sqrt{t^2 - z^2}} \, dz \, dy \, dx$$

$$= \int_0^a \int_0^{\sqrt{a^2-x^2}} \left[\sin^{-1}\left(\frac{z}{t}\right)\right]_0 dy \, dx$$

$$= \int_0^a \int_0^{\sqrt{a^2-x^2}} [\sin^{-1}(1) - \sin^{-1}(0)] \, dy \, dx$$

$$= \frac{\pi}{2} \int_0^a \int_0^{\sqrt{a^2-x^2}} dy \, dx \qquad \left[\because \sin^{-1}(1) = \frac{\pi}{2}\right]$$

$$= \frac{\pi}{2} \int_0^a \sqrt{a^2 - x^2} \, dx$$

$$= \frac{\pi}{2}\left[\frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)\right]_0^a$$

$$= \frac{\pi}{2}\left[\left(0 + \frac{a^2}{2} \times \frac{\pi}{2}\right) - (0 + 0)\right]$$

$$= \frac{a^2\pi^2}{8}$$

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1.10 CHANGE OF VARIABLE IN TRIPLE INTEGRALS

(From Cartesian to Cylindrical and Spherical Polar Co-ordinates)

Case I: From Cartesian to Cylindrical Co-ordinates

The cylindrical co-ordinate of a point P(r, θ, z) is defined as

$$x = r\cos\theta,\ y = r\sin\theta \text{ and } z = z$$

where $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1}\left(\frac{y}{x}\right)$ with $0 \le \theta \le 2\pi$.

By transforming the rectangular co-ordinates to cylindrical co-ordinates, the triple integral becomes

$$\iiint_V f(x, y, z)\, dV = \iiint_V f(r\cos\theta, r\sin\theta, z)\, r\, dr\, d\theta\, dz$$

In particular, the volume of a region of space V is given by

$$\iiint_V dV = \iiint_V r\, dr\, d\theta\, dz$$

Case II: From Cartesian to Spherical Polar Co-ordinates

The cylindrical co-ordinates of a point P(r, θ, ϕ) is defined as

$$x = r\sin\phi\cos\theta;\quad y = r\sin\phi\sin\theta;\quad z = r\cos\phi$$

where $r = \sqrt{x^2 + y^2 + z^2}$ with $0 \le \theta \le \pi$ and $0 \le \phi \le 2\pi$.

By transforming the rectangular co-ordinates to spherical polar co-ordinates, the triple integral becomes

$$\iiint_V f(x, y, z)\, dV = \iiint_V f(r\sin\phi\cos\theta, r\sin\phi\sin\theta, r\cos\phi)\, r^2\sin\phi\, dr\, d\theta\, d\phi$$

In particular, the volume of a region of space V is given by

$$\iiint_V dV = \iiint_V r^2\sin\theta\, dr\, d\theta\, d\phi$$

Example 1: Change to spherical polar co-ordinates and then evaluate the integral $\iiint_V \frac{1}{x^2 + y^2 + z^2}\, dxdydz$ where v is the volume of the sphere $x^2 + y^2 + z^2 = a^2$.

Solution:

Given integral is

$$I = \iiint_V \frac{1}{x^2 + y^2 + z^2}\, dx\, dy\, dz \qquad \ldots (i)$$

In the spherical polar-co-ordinate system, let

$$x = r\sin\theta\cos\phi;\ y = r\sin\theta\sin\phi;\ z = r\cos\theta.$$

Also, $dx\, dy\, dz = r^2\sin\theta\, dr\, d\theta\, d\phi$, where $x^2 + y^2 + z^2 = a^2$.

The region of integration V is bounded by

$$r = 0,\ r = a,\ 0 \le \theta \le \pi \text{ and } 0 \le \phi \le 2\pi$$

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So, (i) becomes

$$I = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} \frac{1}{r^2} r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$= \int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta \, d\theta \int_0^{a} dr$$

$$= (2\pi)\left[-\cos\theta\right]_0^{\pi}(a)$$

$$= (-2a\pi)\left[\cos\pi - \cos 0\right]$$

$$= 4a\pi.$$

Example 2: Evaluate $\iiint_V z(x^2 + y^2 + z^2)\, dxdydz$ through the volume of the cylinder $x^2 + y^2 = a^2$ intercepted by the planes $z = 0$ and $z = b$.

Solution:

Let

$$I = \iiint_V z(x^2 + y^2 + z^2)\, dx\, dy\, dz \qquad \ldots (i)$$

where V is the volume of the cylinder $x^2 + y^2 = a^2$ intercepted by the planes $z = 0$ and $z = b$.

We change the co-ordinates to cylindrical by the substitution,

$$x = r\cos\theta,\ y = r\sin\theta \text{ and } z = z$$

then $$dx\, dy\, dz = r\, dr\, d\theta\, dz.$$

Clearly, z varies from $z = 0$ and $z = b$. Also, the base of the cylinder in xy plane, is $x^2 + y^2 = a^2$ in which r and θ varies; $0 \le r \le a$ and $0 \le \theta \le 2\pi$

Then, (i) becomes

$$I = \int_0^{b}\int_0^{2\pi}\int_0^{a} z(r^2 + z^2)\, r\, dr\, d\theta\, dz$$

$$= \int_0^{b}\int_0^{a} (zr^3 + rz^3)\left[\theta\right]_0^{2\pi} dr\, dz$$

$$= 2\pi \int_0^{b}\left[z\times\frac{r^4}{4} + z^3\times\frac{r^2}{2}\right]_0^{a} dz$$

$$= 2\pi \int_0^{b}\left[\frac{za^4}{4} + \frac{z^3a^2}{2}\right] dz$$

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$$= 2\pi\left[\frac{z^2a^4}{2\times4} + \frac{z^4a^2}{4\times2}\right]_0^b$$

$$= \pi\left[\frac{b^2a^4}{4} + \frac{b^4a^2}{4}\right]$$

$$= \frac{\pi}{4}(a^4b^2 + a^2b^4)$$

Example 3: Evaluate the following by changing to spherical co-ordinates;

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dz\, dy\, dx}{\sqrt{1-x^2-y^2-z^2}}$$

Solution:

Given integral is

$$I = \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dz\, dy\, dx}{\sqrt{1-x^2-y^2-z^2}} \quad \ldots (i)$$

Clearly, the region of integration is bounded by $z = 0$, $z = \sqrt{1-x^2-y^2}$, $y = 0$, $y = \sqrt{1-x^2}$, and $x = 0$, $x = 1$.

We change the co-ordinates to spherical co-ordinates by the substation:

$$x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta$$

where $r = \sqrt{x^2+y^2+z^2}$. Then, $dx\, dy\, dz = r^2\sin\theta\, dr\, d\theta\, d\phi$.

From (i), we observe that the sphere defines only in first octant. And, on the first octant i.e. on the positive octant, r varies from 0 to 1, θ varies from 0 to $\pi/2$ and ϕ varies from 0 to $\pi/2$.

Therefore, (i) becomes

$$I = \int_0^{\pi/2}\int_0^{\pi/2}\int_0^1 \frac{r^2\sin\theta\, dr\, d\theta\, d\phi}{\sqrt{1-r^2}}$$

$$= \int_0^{\pi/2}\int_0^{\pi/2}\int_0^1 \frac{[1-(1-r^2)]}{\sqrt{1-r^2}}\sin\theta\, dr\, d\theta\, d\phi$$

$$= \int_0^{\pi/2} d\phi \int_0^{\pi/2} \sin\theta\, d\theta \int_0^1 \left(\frac{1}{\sqrt{1-r^2}} - \sqrt{1-r^2}\right) dr$$

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$$= \left(\frac{\pi}{2}\right)[-\cos\theta]_0^{\pi/2}\left[\sin^{-1}(r) - \left(\frac{r\sqrt{1-r^2}}{2} + \frac{1}{2}\sin^{-1}(r)\right)\right]_0$$

$$= -\left(\frac{\pi}{2}\right)(0-1)\left[\left\{\sin^{-1}(1) - \left(0 + \frac{1}{2}\sin^{-1}(1)\right)\right\} - \left\{0 - \left(0 + \frac{1}{2} \times 0\right)\right\}\right]$$

$$= \frac{\pi}{2}\left[\frac{\pi}{2} - \frac{\pi}{4}\right]$$

$$= \frac{\pi^2}{8}$$

Example 4: Evaluate the volume of the sphere $x^2 + y^2 + z^2 = a^2$ by changing to spherical co-ordinates.

Solution:

Given sphere is

$$x^2 + y^2 + z^2 = a^2 \qquad \ldots (i)$$

We change the co-ordinates to spherical co-ordinates by the substitution;

$$x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta$$

where $r^2 = x^2 + y^2 + z^2$. Then, $dx\,dy\,dz = r^2\sin\theta\,dr\,d\theta\,d\phi$.

The volume of the sphere is eight times the volume of its portion in the positive octant for which r varies from 0 to a, θ varies from 0 to $\pi/2$ and ϕ varies from 0 to $\pi/2$. So the volume of the sphere (i) is

$$V = 8\int_0^a\int_0^{\pi/2}\int_0^{\pi/2} r^2\sin\theta\,d\theta\,d\phi\,dr = 8\int_0^a r^2\,dr\int_0^{\pi/2}\sin\theta\,d\theta\int_0^{\pi/2}d\phi$$

$$= 8\left[\frac{r^3}{3}\right]_0^a[-\cos\theta]_0^{\pi/2}\left(\frac{\pi}{2}\right)$$

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$$= 4\pi\left(\frac{a^3}{3}\right)(-0+1)$$

$$= \frac{4\pi a^3}{3}$$

Example 5: Find the volume of the portion of the sphere $x^2 + y^2 + z^2 = a^2$ lying inside the cylinder $x^2 + y^2 = ay$.

Solution:

Given sphere and cylinder be

$$x^2 + y^2 + z^2 = a^2 \qquad \ldots (i)$$

$$x^2 + y^2 = ay \qquad \ldots (ii)$$

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We change the co-ordinates to cylindrical by the substitution:

$$x = r\cos\theta,\ y = r\sin\theta \text{ and } z = z$$

then $$dx\,dy\,dz = r\,dr\,d\theta\,dz.$$

Since the cylinder (ii) is symmetrical through the line $x = 0$. Then the volume of the solid formed by the sphere (i) lying inside the cylinder (ii) is the twice of the volume of the solid formed in positive x octant. On the first octant, r varies from 0 to $a\sin\theta$ and θ varies from 0 to π. Also, $0 \le z \le \sqrt{a^2 - x^2 - y^2}$ i.e. $0 \le z \le \sqrt{a^2 - r^2}$. Then the volume of the solid is

$$V = 2\int_0^{\pi} \int_0^{a\sin\theta} \int_0^{\sqrt{a^2 - r^2}} r\,dz\,dr\,d\theta$$

$$= 2\int_0^{\pi} \int_0^{a\sin\theta} r\sqrt{a^2 - r^2}\,dr\,d\theta$$

$$= 2\int_0^{\pi} \left[\frac{-1}{2} \times \frac{(a^2 - r^2)^{3/2}}{3/2}\right]_0^{a\sin\theta} d\theta$$

$$= \frac{-2}{3}\int_0^{\pi} \left[(a^2 - a^2\sin^2\theta)^{3/2} - (a^2 - 0)^{3/2}\right] d\theta$$

$$= \frac{-2}{3}\int_0^{\pi} (a^3\cos^3\theta - a^3)\,d\theta$$

$$= \frac{2a^3}{3}\int_0^{\pi} (1 - \cos^3\theta)\,d\theta$$

$$= \frac{2a^3}{3}\int_0^{\pi} [1 - (1 - \sin^2\theta)\cos\theta]\,d\theta$$

$$= \frac{2a^3}{3}\left([\theta]_0^{\pi} - \int_0^0 (1 - t^2)\,dt\right)$$

$$= \frac{2\pi a^3}{3}$$

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1.11 VOLUME OF A SOLID BY TRIPLE INTEGRAL

Let P be (x, y, z). Draw a parallelepiped with edges parallel to the axes to the axes of lengths Δx, Δy and Δz. The element of volume ΔV is taken from this parallelopiped. Hence, the volume is

$$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0\\ \Delta z\to 0}} \sum\sum\sum \Delta x\Delta y\Delta z = \iiint_V dx\,dy\,dz$$

where V the limit of integration determined from the region within which the point must lie.

Example 6: Find the volume enclosed between the two surface $z = 8 - x^2 - 3y^2$ and $z = x^2 - 3y^2$.

Solution:

The volume V of the enclosed between the two surface $z = 8 - x^2 - 3y^2$ and $z = x^2 - 3y^2$ is

$$V = \iiint_V dx\,dy\,dz \qquad \ldots (i)$$

For V, the limits of z are from $z = x^2 + 3y^2$ to $z = 8 - x^2 - y^2$.

For limits of y, we have

$$8 - x^2 - y^2 = x^2 + 3y^2 \Rightarrow 2x^2 + 4y^2 = 8$$

$$\Rightarrow x^2 + 2y^2 = 4 \qquad \ldots (ii)$$

So, y varies from $-\frac{\sqrt{4-x^2}}{2}$ to $\frac{\sqrt{4-x^2}}{2}$.

Also, the limits of x are from -2 to 2.

So, (i) becomes

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$$V = \int_{-2}^{2} \int_{-\sqrt{1-x^2/4}}^{\sqrt{1-x^2/4}} \int_{x^2+3y^2}^{8-x^2-y^2} dz\,dy\,dx$$

$$= \int_{-2}^{2} \int_{-\sqrt{1-x^2/4}}^{\sqrt{1-x^2/4}} [8 - x^2 - y^2 - x^2 - 3y^2]\,dy\,dx$$

$$= \int_{-2}^{2} \int_{-\sqrt{1-x^2/4}}^{\sqrt{1-x^2/4}} [8 - 2x^2 - 4y^2]\,dy\,dx$$

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Since x and y are even. So,

$$V = 8\int_0^2 \int_0^{\sqrt{1-x^2/4}} [4 - x^2 - 2y^2]\, dy\, dx$$

Put $x = 2r\cos\theta$, $y = \sqrt{2}r\sin\theta$. Then $dxdy = 2\sqrt{2}\ rdrd\theta$.

And, $4 - x^2 - 2y^2 = 4 - 4r^2\cos^2\theta - 4r^2\sin^2\theta = 4 - 4r^2 = 4(1 - r^2)$

Also, r varies from 0 to 1 and θ varies from 0 to $\pi/2$.

Then,

$$V = 8\times4 \int_0^{\pi/2}\int_0^1 [1 - r^2]\, 2\sqrt{2}\ rdrd\theta$$

$$= 32\times2\sqrt{2} \int_0^{\pi/2}\left[\frac{r^2}{2} - \frac{r^4}{4}\right]_0^1 d\theta$$

$$= 64\sqrt{2} \int_0^{\pi/2}\left[\frac{1}{2} - \frac{1}{4}\right] d\theta$$

$$= 64\sqrt{2} \times\frac{1}{4}\times\frac{\pi}{2}$$

$$= 8\sqrt{2}\,\pi$$

Example 7: Find the volume of the tetrahedron bounded by the plane x = 0, y = z = 0 and x + y + z = a.

Solution:

Let V be the region bounded by four planes x = 0, y = 0, z = 0 and x + y + z = a. bounded below and above by z = 0 and z = a − x − y. The projection of the solid on plane is a triangle bounded by x = 0, y = 0, x + y = a.

Here x varies from 0 to a, y varies from 0 to a − x and z varies from 0 a − x − y. So the volume of the tetrahedron is

$$V = \int_0^a \int_0^{a-x} \int_0^{a-x-y} dz\, dy\, dx$$

$$= \int_0^a \int_0^{a-x} [a - x - y]\, dy\, dx$$

$$= \int_0^a \left[ay - xy - \frac{y^2}{2}\right]_0^{a-x} dx$$

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$$= \int_0^a \left[(a-x)[(a-x)-0] - \frac{[(a-x)-0]^2}{2} \right] dx$$

$$= \frac{1}{2} \int_0^a (a-x)^2 \, dx$$

$$= \frac{1}{2} \left[\frac{(a-x)^3}{-3} \right]_0^a$$

$$= \frac{a^3}{6}$$

1.12 DIRICHLET'S INTEGRALS

Let V be a closed region in the first octant in R^3, and is bounded by the surface $\left(\frac{x}{a}\right)^p + \left(\frac{y}{b}\right)^q + \left(\frac{z}{c}\right)^r = 1$ and the coordinate planes. Then, the integral of the form

$$\iiint_V (x^{\alpha-1} y^{\beta-1} z^{\gamma-1}) \, dx \, dy \, dz$$

is called a Dirichlet integral, where all a, b, c, p, q, r, α, β, γ are positive. Such integrals were named Dirichlet's integrals after a German mathematician Peter Gustav Lejeune Dirichlet (1805-1859)

Dirichlet Theorem:

If V be a region bounded by $x \geq 0, y \geq 0, z \geq 0, x + y + z \leq 1$ then

$$\iiint_V (x^{\alpha-1} y^{\beta-1} z^{\gamma-1}) \, dx \, dy \, dz = \frac{\Gamma(\alpha)\,\Gamma(\beta)\,\Gamma(\gamma)}{\Gamma(\alpha+\beta+\gamma+1)}.$$

Proof:

Let V be a region bounded by $x \geq 0, y \geq 0, z \geq 0, x + y + z \leq 1$. Then the volume of the region is

$$I = \iiint_V (x^{\alpha-1} y^{\beta-1} z^{\gamma-1}) \, dx \, dy \, dz$$

$$= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} (x^{\alpha-1} y^{\beta-1} z^{\gamma-1}) \, dz \, dy \, dx$$

$$= \int_0^1 \int_0^{1-x} (x^{\alpha-1} y^{\beta-1}) \left[\frac{z^\gamma}{\gamma} \right]_0^{1-x-y} dy \, dx$$

$$= \left(\frac{1}{\gamma}\right) \int_0^1 \int_0^{1-x} (x^{\alpha-1} y^{\beta-1})(1-x-y)^\gamma \, dy \, dx$$

Put $y = (1-x)t$ then $dy = (1-x)dt$. Then,

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$$I = \left(\frac{1}{\gamma}\right)\int_0^1\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}t^{\beta-1}(1-x-(1-x)t)^{\gamma}(1-x)\,dt\,dx$$

$$= \left(\frac{1}{\gamma}\right)\int_0^1\int_0^1 x^{\alpha-1}(1-x)^{\beta}t^{\beta-1}(1-x)^{\gamma}(1-t)^{\gamma}\,dt\,dx$$

$$= \left(\frac{1}{\gamma}\right)\int_0^1 x^{\alpha-1}(1-x)^{(\beta+\eta}\,dx\int_0^1 t^{\beta-1}(1-t)^{\gamma}\,dt$$

$$= \left(\frac{1}{\gamma}\right)B(\alpha, \beta+\gamma+1)\,B(\beta, \gamma+1)$$

$$= \left(\frac{1}{\gamma}\right)\times\frac{\Gamma(\alpha)\,\Gamma(\beta+\gamma+1)}{\Gamma(\alpha+\beta+\gamma+1)}\times\frac{\Gamma(\beta)\,\Gamma(\gamma+1)}{\Gamma(\beta+\gamma+1)}$$

$$= \left(\frac{1}{\gamma}\right)\times\frac{\Gamma(\alpha)\,\Gamma(\beta)\,\Gamma(\gamma+1)}{\Gamma(\alpha+\beta+\gamma+1)}$$

$$= \frac{\Gamma(\alpha)\,\Gamma(\beta)\,\Gamma(\gamma)}{\Gamma(\alpha+\beta+\gamma+1)} \qquad [\text{Being } \Gamma(\gamma+1) = \gamma\,\Gamma(\gamma)]$$

Example 8: The plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ meets the axes in A, B, C Dirichlet's integrals to fi the volume of the tetrahedron OABC.

Solution:

Substitute $\frac{x}{a}=u$, $\frac{y}{b}=v$, $\frac{z}{c}=w$. Then the tetrahedron OABC is bounded $u\geq 0, v\geq 0, w\geq 0$ and $u+v+w\leq 1$. So, the volume of the tetrahedron is

$$V = \iiint_V dx\,dy\,dz = \iiint_V abc\,du\,dv\,dw$$

$$= abc\iiint_V du\,dv\,dw$$

$$= abc\iiint_V (x^{1-1}y^{1-1}z^{1-1})\,du\,dv\,dw$$

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Using Dirichlet theorem, we get

$$V = abc\frac{\Gamma(1)\Gamma(1)\,\Gamma(1)}{\Gamma(1+1+1+1)} = abc\frac{1\times 1\times 1}{\Gamma(4)} = \frac{abc}{6}$$

Example 9: Evaluation the integral $\iiint_V x^{l-1}y^{m-1}z^{n-1}\,dx\,dy\,dz$ where x, y, z are all positi with the condition $(x/a)^p+(y/b)^q+(z/c)^r\leq 1$.

Solution:

Set, $(x/a)^p = u$, $(y/b)^q = v$, $(z/c)^r = w$. Then

$$dx = \frac{a}{p}u^{1/p-1}\,du; \quad dy = \frac{b}{q}v^{1/q-1}\,dv; \quad dw = \frac{c}{r}w^{1/r-1}\,dw$$

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Given that the region is bounded by $(x/a)^p + (y/b)^q + (z/c)^r \le 1$ with all are positive.

Now,

$$V = \iiint_V x^{l-1} y^{m-1} z^{n-1}\, dx\, dy\, dz$$

$$= \iiint_V (au^{1/p})^{l-1} (bv^{1/q})^{m-1} (cw^{1/r})^{n-1} \frac{a}{p} u^{1/p-1} du \frac{b}{q} v^{1/q-1} dv \frac{c}{r} w^{1/r-1} dw$$

$$= \frac{a^l b^m c^n}{pqr} \iiint_V u^{l/p-1} v^{m/q-1} w^{n/r-1}\, du\, dv\, dw$$

Using Dirichlet theorem, we get

$$V = \frac{a^l b^m c^n}{pqr} \times \frac{\Gamma(l/p)\Gamma(m/q)\,\Gamma(n/r)}{\Gamma[(l/p) + (m/q) + (n/r) + 1]}$$

Example 10: Evaluate $\iiint_V x^2yz\, dx\, dy\, dz$ throughout the volume bounded by the plane $x = 0$, $y = 0$, $z = 0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

Solution:

Put, $\frac{x}{a} = u$, $\frac{y}{b} = v$, $\frac{z}{c} = w$. Then $dx = adu$; $dy = bdv$; $dw = cdw$.

Given that the region is bounded by $x = 0$, $y = 0$, $z = 0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

Now,

$$V = \iiint_V x^2yz\, dx\, dy\, dz = \iiint_V a^2u^2\, bv\, cw\, adu\, bdv\, cdw$$

$$= a^3b^2c^2 \iiint_V u^2vw\, du\, dv\, dw$$

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Using Dirichlet theorem, we get

$$V = a^3b^2c^2 \times \frac{\Gamma(3)\Gamma(2)\,\Gamma(2)}{\Gamma(3+2+2+1)} = a^3b^2c^2 \times \frac{2!\,1!\,1!}{7!} = \frac{2a^3b^2c^2}{7!}$$

Example 11: Evaluate, $\iiint_V (x^2y^2 + y^2z^2 + z^2x^2)\, dx\, dy\, dz$ throughout the volume of the sphere $x^2 + y^2 + z^2 = a^2$.

Solution:

Let,

$$I = \iiint_V (x^2y^2 + y^2z^2 + z^2x^2)\, dx\, dy\, dz \quad \ldots (i)$$

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And the volume V given by the sphere,

$$x^2 + y^2 + z^2 = a^2 \quad \ldots (ii)$$

Clearly, the volume given by the sphere (ii) is same as 8 times positive octant sphere (ii).

Put, $\frac{x^2}{a^2} = u$, $\frac{y^2}{a^2} = v$, $\frac{z^2}{a^2} = w$. Then,

$$dx = \frac{a}{2} u^{-1/2} du, \quad dy = \frac{a}{2} v^{-1/2} dv, \; dz = \frac{a}{2} w^{-1/2} dw$$

As the sphere (ii) is 8 times positive octant of a sphere (ii), the equation (i) is

$$I = 8 \iiint_V (x^2y^2 + y^2z^2 + z^2x^2)\, dx\, dy\, dz \quad \ldots (iii)$$

in which the region taken by v is bounded with $x = 0$, $y = 0$, $z = 0$ a $x^2 + y^2 + z^2 \le a^2$.

Then,

$$I = 8 \iiint_V (a^2u\, a^2v + a^2v\, a^2w + a^2w\, a^2u) \frac{a}{2} u^{-1/2} du \frac{a}{2} v^{-1/2} dv \frac{a}{2} w^{-1/2} dw$$

$$= 8 \times a^4 \times \frac{a^3}{8} \iiint_V (uv + vw + wu)\, u^{-1/2} du\, v^{-1/2} dv\, w^{-1/2} dw$$

$$= a^7 \iiint_V (u^{1/2}v^{1/2}w^{-1/2} + u^{-1/2}v^{1/2}w^{1/2} + u^{1/2}v^{-1/2}w^{1/2})\, du\, dv\, dw$$

Using Dirichlet theorem, we get

$$= a^7 \left[\frac{\Gamma(3/2)\Gamma(3/2)\,\Gamma(1/2)}{\Gamma\left(\frac{3}{2}+\frac{3}{2}+\frac{1}{2}+1\right)} + \frac{\Gamma(1/2)\Gamma(3/2)\,\Gamma(3/2)}{\Gamma\left(\frac{1}{2}+\frac{3}{2}+\frac{3}{2}+1\right)} + \frac{\Gamma(3/2)\Gamma(1/2)\,\Gamma(3/2)}{\Gamma\left(\frac{3}{2}+\frac{1}{2}+\frac{3}{2}+1\right)} \right]$$

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$$= a^7 \left[3 \frac{\Gamma(3/2)\Gamma(3/2)\,\Gamma(1/2)}{\Gamma\left(\frac{3}{2}+\frac{3}{2}+\frac{1}{2}+1\right)} \right]$$

$$= a^7 \left[\frac{(\Gamma(3/2))^2\,\Gamma(1/2)}{\Gamma(9/2)} \right]$$

$$= 3a^7 \left[\frac{\left(\frac{1}{2} \times \Gamma\left(\frac{1}{2}\right)\right)^2 \Gamma\left(\frac{1}{2}\right)}{\frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \Gamma\left(\frac{1}{2}\right)} \right]$$

$$= 3a^7 \times \frac{4}{105} (\pi) \qquad \left[\text{Since, } \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}\right]$$

$$= \frac{4\pi a^7}{35}$$

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Exercise 1 (C)

1. **Evaluate the integrals:**

a. $\int_1^3 \int_2^3 \int_1^2 (x - y + z)\, dx\, dy\, dz$

c. $\int_0^1 \int_0^1 \int_0^1 (xy + yz + zx)\, dx\, dy\, dz$

c. $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2)\, dz\, dy\, dx$

d. $\int_0^1 \int_0^{3-3x} \int_0^{3-3x-y} dz\, dy\, dx$

e. $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} x\, dz\, dy\, dx$

f. $\int_0^1 \int_0^{1-x} \int_0^{x+y} e^z\, dx\, dy\, dz$

g. $\int_0^{\sqrt{2}} \int_0^{5y} \int_{x^2+3y^2}^{8-x^2-y^2} dz\, dx\, dy.$

2. Evaluate $\iiint_V (xy + yz + zx)\, dx\, dy\, dz$, where V is the region is bounded by $x = 0, x = 1, y = 0, y = 2, z = 0$ and $z = 3$.

3. Evaluate $\iiint_V (x^2 + y^3)\, dx\, dy\, dz$, where V is the region of space bounded by $x^2 + y^2 = 2z$ and the plane $z = 2$.

4. Evaluate $\iiint_V \frac{dxdydz}{(x + y + z + 1)^3}$ taken over the volume bounded by the planes $x = 0, y = 0, z = 0$, and $x + y + z = 1$.

5. Find the volume of the tetrahedron bounded by the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ and the co-ordinate planes.

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6. Find the volume of the cylinder $x^2 + y^2 = a^2$ above the xy-plane cut by the plane $x + y + z = a$.

7. Find the volume of the sphere $x^2 + y^2 + z^2 = 1$.

8. Evaluate $\iiint_V \frac{xyz}{\sqrt{x^2 + y^2 + z^2}}\, dxdydz$ taken throughout the first octant of the sphere $x^2 + y^2 + z^2 = a^2$.

9. Find the volume of region D enclosed by the surface $z = 8 - x^2 - y^2$ and $z = x^2 + y^2$.

10. Find the volume of the solid bounded by the surfaces $x^2 + y^2 = az$ and $x^2 + y^2 + z^2 = 2az$.

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Answer Key

1. a. 2	b. $\frac{3}{4}$	c. 1	d. $\frac{3}{2}$	e. $\frac{1}{3}$
f. $\frac{1}{2}$	g. $-\frac{190}{3}$			
2. $\frac{33}{2}$	3. 16π	4. $\frac{1}{2}\log 2 - \frac{5}{16}$	5. $\frac{abc}{6}$	6. πa^3
7. $\frac{4\pi}{3}$	8. $\frac{a^5}{48}$	9. 16π	10. $\frac{\pi a^3}{6}$	

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2

SERIES SOLUTION OF DIFFERENTIAL EQUATIONS AND SPECIAL FUNCTIONS

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CHAPTER OUTLINE

After completion of this chapter, you will be able to gain the knowledge and understanding regarding:

2.1 Power series method of solution of differential equations.

2.2 Legendre's Equation, Legendre's polynomials $P_n(x)$ of Graph of $P_1(x)$, $P_2(x)$, $P_3(x)$.

2.3 Frobenius method. Bessel's equation, Bessel's function $J_v(x)$ and its properties. Graph of $J_v(x)$ for $v = 1/2$

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Introduction

We have already discussed the solution of linear differential equation with const coefficients and its solution involves elementary continuous functions. However, the lin differential equations with variable coefficients, which arise from physical problems, do permit such solutions. Such equations can be solved by numerical method. There situations where we can find a solution in the form of an infinite convergent series. series helps to study the motion of wave in physics that arise in the field of heat conduct analysis etc. Also, the series solution of certain differential equation gives rise to spe functions such as Bessel's functions, Lagendre's polynomial, etc. These special functi have many applications in engineering.

2.1 SERIES SOLUTION OF A DIFFERENTIAL EQUATION

Consider a differential equation,

$$p_0(x)\frac{d^2y}{dx^2} + p_1(x)\frac{dy}{dx} + p_2(x)\,y = 0 \quad \ldots (1)$$

where $p_0(x)$, $p_1(x)$, $p_2(x)$ are polynomial in x in terms of infinite convergent series.

Dividing (1) by $p_0(x)$, we get

$$\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)\,y = 0 \quad \ldots (2)$$

where $p(x) = \frac{p_1(x)}{p_0(x)}$ and $q(x) = \frac{p_2(x)}{p_0(x)}$.

Then (2) have different forms like:

Case 1: If $p_0(a) \neq 0$ then the point $x = a$ is called an ordinary point of equation (2).

Case 2: If $p_0(a) = 0$ then the point $x = a$ is called a singular point of equation (2).

Remark: If $p_0(a) = 0$ and $xp(x)$, $x^2q(x)$ possess derivatives of all order in the neighborhood $x = 0$, then $x = 0$ is called the regular singularity of (2).

When $x = 0$ is a regular singularity of (2), at least one of its solution can be expresse as

$$y = x^k(a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots) = \sum_{m=0}^{\infty} a_m\, x^{m+k}$$

When $x = a$ is an ordinary point of (2), then its solution can be expressed as in th form

$$y = a_0 + a_1(x-a) + a_2(x-a)^2 + a_3(x-a)^3 + \ldots$$

$$= \sum_{m=0}^{\infty} a_m\,(x-a)^m$$

Note 1: Every differential equation of this form does not have series solution.

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Working Rule for Solving for which x = 0 is an Ordinary Point

1. Let $y = \sum_{m=0}^{\infty} a_m x^m$ be a series solution of (2).
2. Then, find y' and y" : $y' = \sum_{m=1}^{\infty} m a_m x^{m-1}$, $y'' = \sum_{m=2}^{\infty} m(m-1) a_m x^{m-2}$
3. Inserting these values into the given equation gives an identity.
4. Equate the like powers of x to zero, and get the coefficients.
5. These are then substituted in (2) give the required solution.

Example 1. Solve y" + 9y = 0, by power series method.

Solution:

Let

$$y'' + 9y = 0 \qquad \ldots (1)$$

Clearly x = 0 is an ordinary point of (1). So, let the solution of (1) is

$$y = \sum_{m=0}^{\infty} a_m x^m = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \ldots. \qquad \ldots (2)$$

Then

$$y' = \sum_{m=1}^{\infty} m a_m x^{m-1}, \quad y'' = \sum_{m=2}^{\infty} m(m-1) a_m x^{m-2}$$

Substituting in (1) we get

$$\sum_{m=2}^{\infty} m(m-1) a_m x^{m-2} + 9 \sum_{m=0}^{\infty} a_m x^m = 0$$

i.e. $$\sum_{m=0}^{\infty} (m+2)(m+1) a_{m+2} x^m + 9 \sum_{m=0}^{\infty} a_m x^m = 0$$

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i.e. $$\sum_{m=0}^{\infty} \left[(m+2)(m+1) a_{m+2} + 9 a_m\right] x^m = 0$$

This implies

$$(m + 2)(m + 1)a_{m+2} + 9a_m = 0 \qquad \text{for } m = 0, 1, 2, \ldots$$

$$\Rightarrow \qquad a_{m+2} = \frac{-9}{(m+2)(m+1)} a_m \qquad \ldots (3)$$

This gives

$$a_2 = \frac{-9}{2 \times 1} a_0 = \frac{-9}{2} a_0 \qquad a_3 = \frac{-9}{3 \times 2} a_1 = \frac{-3}{2} a_1$$

$$a_4 = \frac{-9}{4 \times 3} a_2 = \frac{-3}{4} \times \frac{-9}{2} a_0 = \frac{27}{8} a_0 \qquad a_5 = \frac{-9}{5 \times 4} a_3 = \frac{-9}{20} \times \frac{-3}{2} a_1 = \frac{27}{40} a_1$$

and so on.

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Substituting these values in (2), we get

$$y = a_0 + a_1x - \frac{9}{2}a_0x^2 - \frac{3}{2}a_1x^3 + \frac{27}{8}a_0x^4 + \frac{27}{40}a_1x^5 + \ldots$$

or, $$y = \left(1 - \frac{9}{2}x^2 + \frac{27}{8}x^4 + \ldots\right)a_0 + \left(x - \frac{3}{2}x^3 + \frac{27}{40}x^5 + \ldots\right)a_1$$

Alternative Method:

Let

$$y'' + 9y = 0 \qquad \ldots (1)$$

Clearly $x = 0$ is an ordinary point of (1). So, let the solution of (1) is

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \ldots. \qquad \ldots (2)$$

Then

$$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + \ldots$$

and $$y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + \ldots$$

Substituting these values in (1), we get

$$(2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + \ldots) + 9(a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots) = 0$$

Collecting like powers of x,

$$(2a_2 + 9a_0) + (6a_3 + 9a_1)x + (12a_4 + 9a_2)x^2 + (20a_5 + 9a_3)x^3 + \ldots = 0$$

Equating the coefficients of like terms we get

$$2a_2 + 9a_0 = 0 \quad \Rightarrow a_2 = -\frac{9}{2}a_0$$

$$6a_3 + 9a_1 = 0 \quad \Rightarrow a_3 = -\frac{3}{2}a_1$$

$$12a_4 + 9a_2 = 0 \Rightarrow a_4 = -\frac{3}{4}a_2 = -\frac{3}{4} \times \left(-\frac{9}{2}a_0\right) = \frac{27}{8}a_0$$

$$20a_5 + 9a_3 = 0 \Rightarrow a_5 = -\frac{9}{20}a_3 = -\frac{9}{20} \times \left(-\frac{3}{2}a_1\right) = \frac{27}{40}a_1$$

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Putting these values in (2), we get

$$y = a_0 + a_1x - \frac{9}{2}a_0x^2 - \frac{3}{2}a_1x^3 + \frac{27}{8}a_0x^4 + \frac{27}{40}a_1x^5 + \ldots$$

or, $$y = \left(1 - \frac{9}{2}x^2 + \frac{27}{8}x^4 + \ldots\right)a_0 + \left(x - \frac{3}{2}x^3 + \frac{27}{40}x^5 + \ldots\right)a_1$$

Example 2. Solve by power series method, $(x + 1)\frac{dy}{dx} - 3y = 0$.

Solution:

Let

$$(x + 1)\frac{dy}{dx} - 3y = 0 \qquad \ldots (1)$$

Clearly $x = 0$ is an ordinary point of (1). So, let the solution of (1) is

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots \qquad \ldots (2)$$

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Then

$$y' = \frac{dy}{dx} = a_1 + 2a_2x + 3a_3x^2 + \ldots$$

Substituting these values in (1), we get

$(x + 1)(a_1 + 2a_2x + 3a_3x^2 + \ldots) - 3(a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots) = 0$

$\Rightarrow$ $(a_1x + 2a_2x^2 + 3a_3x^3 + \ldots) + (a_1 + 2a_2x + 3a_3x^2 + \ldots) - 3(a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots) = 0$

$\Rightarrow$ $(a_1 - 3a_0) + (a_1 + 2a_2 - 3a_1)x + (2a_2 + 3a_3 - 3a_2)x^2 + \ldots = 0$

$\Rightarrow$ $(a_1 - 3a_0) + (2a_2 - 2a_1)x + (3a_3 - a_2)x^2 + \ldots = 0$

Equating the coefficients of various powers of x to zero,

$a_1 - 3a_0 = 0 \Rightarrow a_1 = 3a_0$

$2a_2 - 2a_1 = 0 \Rightarrow a_2 = a_1 = 3a_0$

$3a_3 - a_2 = 0 \Rightarrow a_3 = \frac{a_2}{3} = a_0$

Put the values in equation (2), which gives the solution.

$y = a_0 + 3a_0x + 3a_0x^2 + a_0x^3 + \ldots \ldots$

i.e. $y = a_0(1 + 3x + 3x^2 + x^2 + \ldots)$

This is the solution of equation (1).

Example 3. Solve by power series method, $y' = -2xy$.

Solution:

Given differential equation is,

$$y' = -2xy \qquad \ldots (1)$$

Let,

$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \ldots$ ... (2)

be solution of (1)

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Differentiating (2) w. r. t. x, then

$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \ldots$

Putting the value of y and y' in (1) then

$a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \ldots = -2x(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \ldots)$

i.e. $a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \ldots = -2a_0x - 2a_1x^2 - 2a_2x^3 - 2a_3x^4 - 2a_4x^5 - \ldots$

Equating coefficient of like terms from both sides then,

$a_1 = 0$

$2a_2 = -2a_0 \Rightarrow a_2 = -a_0$

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$3a_3 = -2a_1 \Rightarrow a_3 = \frac{-2}{3}a_1 = 0$

$4a_4 = -2a_2 \Rightarrow a_4 = \frac{-1}{2}a_2 = \frac{a_0}{2}$

and so on.

Putting the value of $a_1, a_2, a_3, a_4, \ldots$ in (2) then,

$$y = a_0 + 0 - a_0x^2 + 0 - \frac{a_0}{2}x^4 + \ldots$$

i.e. $$y = a_0\left(1 - x^2 - \frac{x^4}{2} - \ldots\ldots\right) = a_0e^{-x^2}$$

Example 4. Solve the differential equation $y' = 3x^2y$ by power series method.

Solution:

Given differential equation is,

$$y' = 3x^2y \quad \ldots (1)$$

Let,

$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \ldots$ ... (2)

be solution of (1)

Differentiating (2) w. r. t. x, then

$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + 6a_6x^5 + \ldots$

Putting the value of y and y' in (1) then

$a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_4x^4 + 6a_6x^5 + \ldots = 3x^2(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \ldots)$

Equating coefficient of like terms from both sides then,

$a_1 = 0$ $\quad 2a_2 = 0 \Rightarrow a_2 = 0$

$3a_3 = 3a_0 \Rightarrow a_3 = a_0$ $\quad 4a_4 = 3a_1 = 0$

$5a_5 = 3a_2 = 0$ $\quad 6a_6 = 3a_3 = \frac{1}{2}a_3 = \frac{1}{2}a_0$

and so on.

Putting the value of $a_1, a_2, a_3, a_4, \ldots$ in (2) then,

$$y = a_0 + a_0x^3 + \frac{1}{2}a_0x^6 + \ldots$$

i.e. $$y = a_0(1 + x^3 + \frac{1}{2}x^6 + \ldots) = a_0e^{x^3}$$

Example 5. Solve in series, the equation $(1 + x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$.

Solution:

Given differential equation is,

$$(1 + x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0 \quad \ldots (1)$$

Let,

$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \ldots$ ... (2)

be solution of (1)

Differentiating (2) w. r. t. x, then

$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + 6a_6x^5 + \ldots$

and $y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + \ldots$

Putting the value of y, y' and y'' in (1) then

$(1 + x^2)(2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + \ldots) + x(a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_4x^4 + 6a_6x^5 + \ldots) - (a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \ldots) = 0$

i.e. $(2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + \ldots) + (2a_2x^2 + 6a_3x^3 + 12a_4x^4 + 20a_5x^5 + 30a_6x^6 + \ldots) + (a_1x + 2a_2x^2 + 3a_3x^3 + 4a_4x^4 + 5a_4x^5 + 6a_6x^6 + \ldots) - (a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \ldots) = 0$

i.e. $(2a_2 - a_0) + (6a_3 + a_1 - a_1)x + (12a_4 + 2a_2 + 2a_2 - a_2)x^2 + (20a_5 + 6a_3 + 3a_3 - a_3)x^3 + (30a_6 + 12a_4 + 4a_4 - a_4)x^4 + \ldots = 0$

i.e. $(2a_2 - a_0) + (6a_3)x + (12a_4 + 3a_2)x^2 + (20a_5 + 8a_3)x^3 + (30a_6 + 15a_4)x^4 + \ldots = 0$

Equating coefficient of like terms from both sides then,

$2a_2 - a_0 = 0 \Rightarrow a_2 = \frac{1}{2}a_0$

$6a_3 = 0 \Rightarrow a_3 = 0$

$12a_4 + 3a_2 \Rightarrow a_4 = -\frac{1}{4}a_2 = -\frac{1}{8}a_0$

$20a_5 + 8a_3 \Rightarrow a_5 = -\frac{2}{5}a_3 = 0$

$30a_6 + 15a_4 \Rightarrow a_6 = -\frac{3}{6}a_4 = \frac{1}{16}a_0$

and so on.

Putting the value of $a_1, a_2, a_3, a_4, \ldots$ in (2) then,

$$y = a_0 + a_1x + \frac{1}{2}a_0x^2 + 0 - \frac{1}{8}a_0x^4 + 0 + \frac{1}{16}a_0x^6 + \ldots$$

i.e. $$y = a_0\left(1 + \frac{x^2}{2} - \frac{x^4}{8} + \frac{x^6}{16} - \frac{5x^8}{128} + \ldots\ldots\right) + a_1x$$

This is the general solution of (1).

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Exercise 2 (A)

Solve the following differential equation by the power series method.

1. $\frac{dy}{dx} - y = 0$
2. $(1 - x)\frac{dy}{dx} - y = 0$
3. $y' + y = 0$
4. $(x + 1)y' + y = 0$
5. $\frac{dy}{dx} + 2xy = 0$
6. $\frac{dy}{dx} - 3x^2y = 0$
7. $(x - 3)\frac{dy}{dx} - xy = 0$
8. $(x + 1)\frac{dy}{dx} = (2x + 3)y$
9. $y'' + 4y = 0$
10. $y'' - 9y = 0$
11. $y'' + xy = 0$
12. $y'' - y = x$
13. $\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$
14. $(1 - x^2)y'' - 2xy' + 2y = 0$
15. $\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + (4x^2 - 2)y = 0$
16. $y'' - xy' + x^2y = 0$

Answer Key

1. $y = a_0\left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots\dots\right)$
2. $y = a_0(1 + x + x^{2!} + x^3 + \dots\dots)$
3. $y = a_0\left(1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots\dots\right)$
4. $y = a_0(1 - x + x^2 - x^3 \dots\dots)$
5. $y = a_0\left(1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \frac{x^8}{4!} + \dots\dots\right)$
6. $y = \left(1 + \frac{x^3}{3!} + \frac{x^6}{2!} - \frac{x^9}{3!} + \dots\dots\right)$
7. $y = a_0\left(1 - \frac{x^2}{6} - \frac{x^3}{27} + \dots\dots\right)$
8. $y = a_0\left(1 + 3x + 4x^2 + \frac{10}{3}x^3 + 2x^4 + \dots\dots\right)$
9. $y = a_0\left(1 - 2x^2 + \frac{2x^4}{3} - \frac{4x^6}{45} + \dots\dots\right) + a_1\left(x - \frac{2x^3}{3} + \frac{2x^5}{15} + \dots\dots\right)$
10. $y = a_0\left(\frac{9x^2}{2!} - \frac{81x^4}{4!} + \frac{729x^6}{6!} + \dots\dots\right) + a_1\left(x + \frac{27x^3}{3!} + \frac{243x^5}{5!} + \dots\right)$
11. $y = a_0\left(1 - \frac{x^3}{3!} + \frac{1.4x^6}{6!} - \frac{1.4.7x^9}{9!} + \dots\dots\right) + a_1\left(x - \frac{2x^4}{4!} + \frac{2.5x^7}{7!} + \dots\dots\right)$
12. $y = a_0\left(1 + \frac{x^2}{2} + \frac{x^4}{4!} + \dots\dots\right) + a_1\left(x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots\dots\right)$
13. $y = a_0\left(1 - \frac{x^2}{2} + \frac{x^4}{2.4} - \frac{x^6}{2.4.6} + \dots\dots\right) + a_1\left(x - \frac{x^3}{3} + \frac{x^5}{3.5} - \frac{x^7}{3.5.7} + \dots\dots\right)$
14. $y = a_0\left(1 - x^2 - \frac{x^3}{3} - \frac{x^5}{5} \dots\dots\right)$
15. $y = a_0\left(1 + x^2 + \frac{x^4}{2} + \frac{x^6}{6} + \dots\dots\right) + a_1\left(x + x^3 + \frac{x^5}{2} + \dots\dots\right)$
16. $y = a_0\left(1 - \frac{x^4}{12} - \frac{x^6}{90} \dots\dots\right) + a_1\left(x + \frac{x^3}{6} - \frac{x^5}{40} - \frac{x^7}{144} \dots\dots\right)$

2.2 LEGENDRE'S EQUATION AND LEGENDRE'S POLYNOMIALS $P_N(x)$

The differential equation of the form

$$(1 - x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + n(n + 1)y = 0$$

i.e. $(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$ ... ... (1)

where n is a real number, is called Legendre's differential equation.

The solution of the equation (1) is known as Legendre's function.

Solution of the Legendre's differential equation

Consider the Legendre's differential equation,

$$(1 - x^2)y'' - 2xy' + n(n + 1)y = 0 \quad \dots\dots (1)$$

Here $x = 0$ is an ordinary point of the equation, since $1 - x^2 \neq 0$ when $x = 0$.

From (1), we have

$$[(1 - x^2)y'] + n(n + 1)y = 0 \quad \dots\dots (2)$$

Let

$$y = a_0 + a_1x + a_2x^2 + \dots\dots = \sum_{m=0}^{\infty} a_m x^m \quad \dots\dots (3)$$

be a solution in series of (1).

Clearly, the singular points of (3) are $x = \pm 1$. Therefore, the solution (3) is convergent in $|x| < 1$. Differentiating (3), we get

$$y' = \sum_{m=1}^{\infty} m\, a_m x^{m-1} \text{ and } y'' = \sum_{m=2}^{\infty} m(m-1)\, a_m x^{m-2}$$

Substituting the values in (1), we get

$$(1 - x^2)\sum_{m=2}^{\infty} m(m-1)\, a_m x^{m-2} - 2x\sum_{m=1}^{\infty} m\, a_m x^{m-1} + n(n+1)\sum_{m=0}^{\infty} a_m x^m = 0$$

i.e. $$\sum_{m=2}^{\infty} m(m-1)\, a_m x^{m-2} - \sum_{m=2}^{\infty} m(m-1)\, a_m x^m - 2x\sum_{m=1}^{\infty} m\, a_m x^{m-1} +$$

$$n(n+1)\sum_{m=0}^{\infty} a_m x^m = 0$$

i.e. $$\sum_{m=2}^{\infty} m(m-1)\, a_m x^{m-2} - 2\sum_{m=1}^{\infty} m\, a_m x^m + n(n+1)\sum_{m=0}^{\infty} a_m x^m -$$

$$\sum_{m=2}^{\infty} m(m-1)\, a_m x^m = 0$$

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Replacing m–2 by m in first term, we get

$$\sum_{m=0}^{\infty}(m+2)(m+1)\,a_{m+2}x^m - \sum_{m=1}^{\infty}2m\,a_m x^m + \sum_{m=0}^{\infty}n(n+1)\,a_m x^m -$$

$$\sum_{m=2}^{\infty}m(m-1)\,a_m x^m = 0$$

In general, equating the coefficients of x^r to zero, we get

$$(r+2)(r+1)a_{r+2} - 2ra_r + n(n+1)a_r - r(r-1)a_r = 0$$

i.e. $(r+2)(r+1)a_{r+2} = -(n^2+n-2r-r^2+r)a_r$

$$= -(n^2+n-r^2-r)a_r$$

$$\Rightarrow (r+2)(r+1)a_{r+2} = -(n-r)(n+r+1)a_r$$

$$\Rightarrow a_{r+2} = -\frac{(n-r)(n+r+1)}{(r+2)(r+1)}a_r \quad \ldots\ldots (4)$$

Putting r = 0, 1, 2, ... , we get

$$a_2 = -\frac{n(n+1)}{2\times 1}a_0 = -\frac{n(n+1)}{2!}a_0$$

$$a_3 = -\frac{(n-1)(n+2)}{3\times 2}a_1 = -\frac{(n-1)(n+2)}{3!}a_1$$

$$a_4 = -\frac{(n-2)(n+3)}{4\times 3}a_2 = \frac{(n-2)n(n+1)(n+3)}{4!}a_0$$

$$a_5 = -\frac{(n-3)(n+4)}{5\times 4}a_3 = \frac{(n-3)(n-1)(n+2)(n+4)}{5!}a_1$$

$$a_6 = -\frac{(n-4)(n+5)}{6\times 5}a_3 = \frac{(n-4)(n-2)n(n+1)(n+3)(n+5)}{6!}a_0$$

and so on.

Putting these values in (3), we get

$$y = a_0\left(1 - \frac{n(n+1)}{2!}x^2 + \frac{(n-2)(n+3)n(n+1)}{4!}x^4 - \ldots\right) +$$

$$a_1\left(x - \frac{(n-1)(n+2)}{3!} + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}x^5 - \ldots\right)$$

$$\Rightarrow y = a_0y_1(x) + a_2y_2(x) \quad \ldots (5)$$

where $y_1(x) = \left(1 - \frac{n(n+1)}{2!}x^2 + \frac{(n-2)(n+3)n(n+1)}{4!}x^4 + \ldots\right)$ ... (6)

$$y_2(x) = \left(x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}x^5 - \ldots\right) \ldots (7)$$

As the series in (3) converges for $|x| < 1$, the series $y_1(x)$ and $y_2(x)$ are convergent for $|x| < 1$.

Also, $y_1(x)$ and $y_2(x)$ are not in proportional, so the solutions are linearly independent solutions. Thus, the equation (5) with (6) and (7) be general solution of Legendre's equation on $-1 < x < 1$.

2.3 LEGENDRE'S POLYNOMIALS $P_N(x)$

From the solution of Legendre's equation we have

$$a_{r+2} = -\frac{(n-r)(n+r+1)}{(r+2)(r+1)}a_r \quad \ldots (1)$$

And, the Legendre's function of first kind, is given by Legendre's equation

$$y = a_0y_1(x) + a_2y_2(x) \quad \ldots (2)$$

where $y_1(x) = \left(1 - \frac{n(n+1)}{2!}x^2 + \frac{(n-2)(n+3)n(n+1)}{4!}x^4 - \ldots\right)$ ... (3)

$$y_2(x) = \left(x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}x^5 - \ldots\right) \ldots (4)$$

Set n = 0, 2, 4, ... we get

at n = 0, $y_1(x) = 1$

at n = 2, $y_1(x) = 1 - \frac{2(3)}{2!}x^2 = 1 - 3x^2$

at n = 4, $y_1(x) = 1 - \frac{4(5)}{2!}x^2 + \frac{(2)(7)4(5)}{4!}x^4 = 1 - 10x^2 + \frac{35}{3}x^4$

and so on.

Also, set n = 1, 3, 5, ... we get

at n = 1, $y_2(x) = x$

at n = 3, $y_2(x) = x - \frac{(2)(5)}{3!}x^3 = x - \frac{5}{3}x^3$

at n = 5, $y_2(x) = \left(x - \frac{(4)(7)}{3!}x^3 + \frac{(2)(4)(7)(9)}{5!}x^5\right) = x - \frac{14}{3}x^3 + \frac{21}{5}x^5$

and so on.

Thus, if n is even positive, the series $y_2(x)$ terminates and hence y_1 remains an infinite series. Similarly, if n is odd positive, then y_2 also becomes a polynomial.

This gives $a_0 = 1$ and $a_1 = 1$.

Let n is non-negative integer. If n = r then (1) gives

$$a_{n-2} = 0$$

So, if n is even positive integer then y_1 reduces to a polynomial of degree n. Likewise, if n is odd positive integer then y_2 reduces to a polynomial of degree n.

Choose a constant a_n as coefficient of x^n is given by

$$a_n = \frac{(2n)!}{2^n(n!)^2} \quad \text{when } n = 1, 2, 3, \ldots$$

At n = 0, $a_0 = 1$ and at n = 0, $a_1 = 1$.

Also, we have

$$a_r = \frac{(r+1)(r+2)}{(n-r)(m+r+1)}a_{r+2} \quad (\text{for } r \le n-2)$$

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At $r = n-2$,

$$a_{n-2} = \frac{(n-1)n}{-2(2n-1)} a_n = -\frac{n(n-1)}{2(2n-1)} \times \frac{(2n)!}{2^n (n!)^2}$$

$$= -\frac{n(n-1)}{2(2n-1)} \times \frac{(2n)(2n-1)(2n-2)!}{2^n n(n-1)!\, n(n-1)(n-2)!}$$

$$= -\frac{(2n-2)!}{2^n (n-1)!\, (n-2)!}$$

At $r = n-4$,

$$a_{n-4} = -\frac{(n-2)(n-3)}{4(2n-3)} a_{n-2}$$

$$= -\frac{(n-2)(n-3)}{4(2n-3)} \times (-1) \frac{(2n-2)!}{2^n (n-1)!\, (n-2)!}$$

$$= \frac{(n-2)(n-3)}{4(2n-3)} \times \frac{(2n-2)(2n-3)(2n-4)!}{2^n (n-1)(n-2)!\, (n-2)(n-3)(n-4)!}$$

$$= \frac{(2n-4)!}{2!\, 2^n (n-2)!\, (n-4)!}$$

At $r = n-6$,

$$a_{n-6} = -\frac{(n-4)(n-5)}{6(2n-5)} a_{n-4} = \frac{(2n-6)!}{3!\, 2^n (n-3)!\, (n-6)!}$$

and so on.

In general, at $r = n-2m$

$$a_{n-2m} = (-1)^m \frac{(2n-2m)!}{m!\, 2^n (n-m)!\, (n-2m)!}$$

Therefore, the Legendre polynomial of degree n is given by

$$P_n(x) = \sum_{m=0}^{M} (-1)^m \frac{(2n-2m)!}{m!\, 2^n (n-m)!\, (n-2m)!} x^{n-2m}$$

where, $M = \frac{n}{2}$ or $M = \frac{n-1}{2}$ whichever is an integer.

Note: Setting m is 0, 1, 2, 3, 4, 5, we get the initial some polynomial values

$P_0(x) = 1$, $P_1(x) = x$

$P_2(x) = \frac{1}{2}(3x^2 - 1)$, $P_3(x) = \frac{1}{2}(5x^3 - 3x)$

$P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3)$ $P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15x)$

2.4 GRAPH OF LEGENDRE'S POLYNOMIALS $P_1(x)$, $P_2(x)$, $P_3(x)$

From above article, we have the some initial Legendre's polynomials are

$P_0(x) = 1$, $P_1(x) = x$

$P_2(x) = \frac{1}{2}(3x^2 - 1)$, $P_3(x) = \frac{1}{2}(5x^3 - 3x)$

1. The polynomial has graph of $P_0(x)$.

   Clearly, $P_0(x) = 1$, like $y = 1$ in xy-plane.

2. The polynomial has graph of $P_1(x)$.

   Clearly, $P_1(x) = x$, like $y = x$ in xy-plane. Clearly, the curve $P_1(x) = x$ is a straight line passes through (0, 0) and (1, 1).

3. The polynomial has graph of $P_2(x)$.

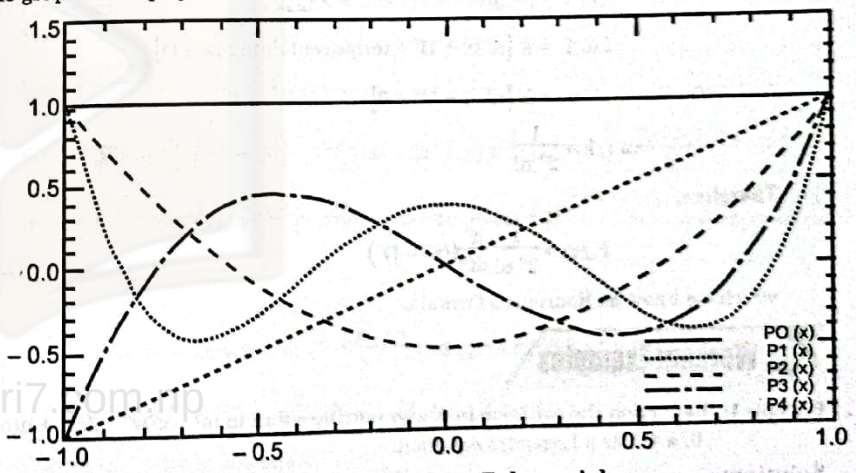
   Clearly, $P_2(x) = \frac{1}{2}(3x^2 - 1)$, like $x^2 = \frac{2}{3}\left(y + \frac{1}{2}\right)$ in xy-plane. Clearly, the curve $P_2(x) = \frac{1}{2}(3x^2 - 1)$ is a parabolic curve with vertex at $\left(0, \frac{-1}{2}\right)$, having symmetrical line $x = 0$, line of directrix at $y + \frac{1}{2} = -\frac{1}{6}$ i.e. $y = \frac{-2}{3}$ and focus at $\left(0, \frac{-7}{6}\right)$.

4. The polynomial has graph of $P_3(x)$.

   Clearly, $P_3(x) = \frac{1}{2}(5x^3 - 3x)$, like $5x^3 - 3x = 2y$ in xy-plane. Clearly, the curve $P_3(x) = \frac{1}{2}(5x^3 - 3x)$ is a cubic curve that has plots as:

x-value	0	0.25	–0.25	0.5	–0.5	0.75	–0.75	1	...
y-value	0	–0.34	0.34	–0.44	0.44	–0.19	0.07	1	...

The graph of the polynomials $P_0(x)$, $P_1(x)$, $P_2(x)$, $P_3(x)$



Note: Orthogonality Properties of Legendre's Polynomials

If $P_m(x)$ and $P_n(x)$ are Legendre's polynomials then

$$\int_{-1}^{1} P_m(x)\, P_n(x)\, dx = \begin{cases} 0 & \text{if } m \neq n \\ \frac{2}{2n+1} & \text{if } m = n \end{cases}$$

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2.5 RODRIQUE'S FORMULA

Put $t = (x^2 - 1)^n$ then

$$\frac{dt}{dx} = 2nx(x^2 - 1)^{n-1}$$

$$\Rightarrow (x^2 - 1)t_1 = 2nx(x^2 - 1)^n$$

$$\Rightarrow (1 - x^2)t_1 + 2nxt = 0$$

Applying Leibnitz rule,

$$(1 - x^2)t_{n+2} + (-2x)(n+1)t_{n+1} + (-2)\frac{n(n+1)}{2}t_n + 2nxt_{n+1} + 2n(n+1)(1)t_n = 0$$

$$\Rightarrow (1 - x^2)t_{n+2} - 2xt_{n+1} + n(n+1)t_n = 0 \quad \ldots (1)$$

which is Legendre's equation.

Let t_n be the solution of (1) then for some constant k,

$$P_n(x) = kt_n = k\frac{d^n}{dt^n}((x^2 - 1)^n)$$

$$= k\frac{d^n}{dt^n}((x+1)^n(x-1)^n) \quad \ldots (2)$$

Set, $x = 1$ then (2) gives

$$P_n(1) = \left[k\frac{d^n}{dt^n}((x+1)^n(x-1)^n)\right]_{at\ x=1}$$

i.e. $1 = k\left[n!(x+1)^n + \text{terms containing } (x-1)\right]_{at\ x=1}$

$$= k\left[n!(1+1)^n + 0\right]$$

$$\Rightarrow k = \frac{1}{2^n n!} \quad \ldots (3)$$

Therefore,

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dt^n}((x^2 - 1)^n)$$

which we know as Rodrique's formula.

Workout Examples

Example 1: Reduction the equation in y, z by putting $x = az$ in $(a^2 - x^2)y'' - 2xy' + n(n+1)y = 0$, $a = 0$ as a Legendre equation.

Solution:

The given differential equation is

$$(a^2 - x^2)y'' - 2xy' + n(n+1)y = 0 \quad \ldots (i)$$

Let,

$$az = x$$

Differentiating w.r.t. x, we get

$$\frac{dz}{dx} = \frac{1}{a}$$

Now,

$$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx} = \frac{1}{a}\frac{dy}{dz}$$

and

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{1}{a}\frac{d}{dx}\left(\frac{dy}{dz}\right) = \frac{1}{a}\left(\frac{d}{dz}\frac{dz}{dx}\right)\frac{dy}{dz} = \frac{1}{a^2}\frac{d^2y}{dz^2}$$

Substituting in (i), we get

$$(a^2 - a^2z^2)\frac{1}{a^2}\frac{d^2y}{dz^2} - 2az\frac{1}{a}\frac{dy}{dz} + n(n+1)y = 0$$

$$\Rightarrow (1 - z^2)\frac{d^2y}{dz^2} - 2z\frac{dy}{dz} + n(n+1)y = 0$$

which is Legendre's equation.

Example 2: Express $2x^2 - 4x + 2$ as a Legendre polynomial.

Solution:

By Rodrigue formulae, we have

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}\left[(x^2 - 1)^n\right] \quad \ldots (i)$$

Then (i) gives

$P_0(x) = 1$,

$$P_1(x) = \frac{1}{2}\frac{d}{dx}\left[(x^2 - 1)\right] = \frac{1}{2} \times 2x = x,$$

$$P_2(x) = \frac{1}{2^2 \times 2!}\frac{d^2}{dx^2}\left[(x^2 - 1)^2\right] = \frac{1}{8}\frac{d^2}{dx^2}(x^4 - 2x^2 + 1) = \frac{1}{8}(12x^2 - 4) = \frac{3x^2 - 1}{2}$$

$$\Rightarrow x^2 = \frac{2P_2(x) + 1}{3} = \frac{2P_2(x) + P_0(x)}{3} \quad [\because P_0(x) = 1]$$

Now,

$$f(x) = 2x^2 - 4x + 2 = 2x\frac{2P_2(x) + 1}{3} - 4P_1(x) + 2P_0(x)$$

$$= \frac{2}{3}\left[2P_2(x) - 6P_1(x) + 3P_0(x)\right]$$

which is the Legendre polynomials.

Example 3: Express $x^3 + 2x^2 - x - 3$ in terms of Legendre polynomials.

Solution:

By Rodrigue formulae, we have

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}\left[(x^2 - 1)^n\right] \quad \ldots (i)$$

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Then (i) gives

$P_0(x) = 1$,

$$P_1(x) = \frac{1}{2}\frac{d}{dx}[(x^2-1)] = \frac{1}{2} \times 2x = x,$$

$$P_2(x) = \frac{1}{2^2 \times 2!}\frac{d^2}{dx^2}[(x^2-1)^2] = \frac{1}{8}\frac{d^2}{dx^2}(x^4 - 2x^2 + 1) = \frac{1}{8}(12x^2 - 4) = \frac{3x^2-1}{2}$$

$$\Rightarrow \quad x^2 = \frac{2P_2(x)+1}{3} = \frac{2P_2(x)+P_0(x)}{3} \qquad [\because P_0(x) = 1]$$

$$P_3(x) = \frac{1}{2^3 3!}\frac{d^3}{dx^3}[(x^2-1)^3] = \frac{1}{48}\frac{d^3}{dx^3}(x^6 - 3x^4 + 3x^2 - 1)$$

$$= \frac{1}{48}(120x^3 - 72x)$$

$$= \frac{1}{2}(5x^3 - 3x)$$

$$\Rightarrow \quad x^3 = \frac{2P_3(x)+3x}{5} = \frac{1}{5}[2P_3(x) + 3P_1(x)]$$

Now,

$$f(x) = x^3 + 2x^2 - x - 3$$

$$= \frac{1}{5}[2P_3(x) + 3P_1(x)] + 2\left(\frac{2P_2(x)+P_0(x)}{3}\right) - P_1(x) - 3P_0(x)$$

$$= \frac{1}{15}[6P_3(x) + 9P_1(x) + 20P_2(x) + 10P_0(x) - 15P_1(x) - 45P_0(x)]$$

$$= \frac{1}{15}[6P_3(x) + 20P_2(x) - 6P_1(x) - 35P_0(x)]$$

which is the Legendre polynomial.

2.6 Frobenius Method

Consider a differential equation,

$$x^2\frac{d^2y}{dx^2} + x\,P_1(x)\frac{dy}{dx} + P_2(x)\,y = 0 \text{ (for } P_0 \neq 0) \qquad \ldots (1)$$

Clearly, the point $x = 0$ is a regular singular point of (1).

Let

$$y = x^k(a_0 + a_1x + a_2x^2 + \ldots) = x^k \sum_{m=0}^{\infty} a_m x^m = \sum_{m=0}^{\infty} a_m x^{m+k} \qquad \ldots\ldots (2)$$

be a solution in series of (1). Then

$$y' = \sum_{m=0}^{\infty} (m+k)\,a_m x^{m+k-1}$$

and

$$y'' = \sum_{m=0}^{\infty} (m+k)(m+k-1)\,a_m x^{m+k-2}$$

Substituting the values in (1) then

$$x^2 \sum_{m=0}^{\infty} (m+k)(m+k-1)\,a_m x^{m+k-2} + x\,P_1(x) \sum_{m=0}^{\infty} (m+k)\,a_m x^{m+k-1} + P_2(x) \sum_{m=0}^{\infty} a_m x^{m+k} = 0$$

i.e. $$\sum_{m=0}^{\infty} (m+k)(m+k-1)\,a_m x^{m+k} + P_1(x) \sum_{m=0}^{\infty} (m+k)\,a_m x^{m+k} + P_2(x) \sum_{m=0}^{\infty} a_m x^{m+k} = 0$$

$$\Rightarrow \quad \sum_{m=0}^{\infty} [(m+k)(m+k-1) + P_1(x)(m+k) + P_2(x)]\,a_m x^{m+k} = 0$$

$$\Rightarrow \quad [k(k-1) + P_1(x)k + P_2(x)]\,a_0x^k + \sum_{m=1}^{\infty} [(m+k)(m+k-1) + P_1(x)(m+k) + P_2(x)]$$

$$a_m x^{m+k} = 0$$

Here, the expression

$$k(k-1) + P_1(0)k + P_2(0) = I(k) \qquad \ldots (3)$$

is known as the indicial polynomial, is quadratic in k.

The roots of the equation (3) may have different cases:

Case 1: When the roots k_1, k_2 are distinct i.e. $k_1 \neq k_2$ and not differing by an integer. The complete solution is

$$y = a_1(y)_{k_1} + a_2(y)_{k_2}$$

Case 2: When the roots k_1, k_2 are equal i.e. $k_1 = k_2$. The complete solution is

$$y = a_1(y)_{k_1} + a_2\left(\frac{\partial y}{\partial m}\right)_{k_1}$$

Case 3: When the roots k_1, k_2 are distinct and differ by an integer with $k_1 < k_2$. The complete solution is

$$y = a_1(y)_{k_1} + a_2\left(\frac{\partial y}{\partial m}\right)_{k_1}$$

Case 4: When the roots k_1, k_2 are distinct and differ by an integer and making some coefficient indeterminate. The complete solution is

$$y = a_1(y)_{k_1} + a_2(y)_{k_2}$$

The solution will be finite if $k = k_2$.

[The roots are distinct and not differing by an integer]

Example 4: Find the solution of the equation in series form, $2x^2y'' + (2x^2 - x)y' + y = 0$.

Solution:

Given equation is

$$2x^2y'' + (2x^2 - x)y' + y = 0 \qquad \ldots (1)$$

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Here $x = 0$ is a regular singularity, so let the solution of (1) be

$$y = \sum_{m=0}^{\infty} a_m x^{m+k} \quad \ldots (2)$$

Then,

$$y' = \sum_{m=0}^{\infty} (m+k)\, a_m x^{m+k-1} \quad \text{and} \quad y'' = \sum_{m=0}^{\infty} (m+k)(m+k-1)\, a_m x^{m+k-2}$$

Substituting the values in (1) then

$$2x^2 \sum_{m=0}^{\infty} (m+k)(m+k-1)\, a_m x^{m+k-2} + (2x^2 - x) \sum_{m=0}^{\infty} (m+k) a_m x^{m+k-1} + \sum_{m=0}^{\infty} a_m x^{m+k} = 0$$

$$\text{i.e.} \quad \sum_{m=0}^{\infty} 2(m+k)(m+k-1)\, a_m x^{m+k} + \sum_{m=0}^{\infty} 2(m+k) a_m x^{m+k+1} -$$

$$\sum_{m=0}^{\infty} (m+k) a_m x^{m+k} + \sum_{m=0}^{\infty} a_m x^{m+k} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} [2(m+k)(m+k-1) - (m+k) + 1]\, a_m x^{m+k} + \sum_{m=0}^{\infty} 2(m+k) a_m x^{m+k+1} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} [2m^2 + 4mk + 2k^2 - 2m - 2k - m - k + 1]\, a_m x^{m+k} + \sum_{m=0}^{\infty} (2m + 2k) a_m x^{m+k+1} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} [2m^2 + 4mk + 2k^2 - 3m - 3k + 1] a_m x^{m+k} + \sum_{m=0}^{\infty} (2m + 2k)\, a_m x^{m+k+1} = 0 \quad \ldots (3)$$

Clearly, the lowest power of x in (3) is x^k and equating its coefficient to right part, i.e. zero, we get

$$2k^2 - 3k + 1 = 0$$

$$\Rightarrow (2k - 1)(k - 1) = 0$$

$$\Rightarrow k = \frac{1}{2}, 1$$

The roots are distinct and not differing by an integer. Equating the coefficients of x^{k+1} in (3), to zero, we get.

$$[2 + 4k + 2k^2 - 3 - 3k + 1] a_1 + (2k) a_0 = 0$$

$$\Rightarrow [2k^2 + k] a_1 + (2k) a_0 = 0$$

$$\Rightarrow a_1 = \frac{-2k\, a_0}{k(2k+1)} = \frac{-2\, a_0}{2k+1}$$

And, equating the coefficient of x^{m+2}, we get

$$[8 + 8k + 2k^2 - 6 - 3k + 1] a_2 + (2 + 2k) a_1 = 0$$

$$\text{i.e.} \quad [5k + 2k^2 + 3] a_2 + (2 + 2k) a_1 = 0$$

$$\Rightarrow (2k + 3)(k + 1) a_2 + 2(1 + k) a_1 = 0$$

$$\Rightarrow a_2 = \frac{-2}{2k+3} a_1 = \frac{4}{(2k+3)(2k+1)} a_0$$

and so on.

When $m = \frac{1}{2}$, $a_1 = -a_0$, $a_2 = \frac{1}{2} a_0$, and so on. Then

$$y_1 = x^{m_1} (a_0 + a_1 x + a_2 x^2 + \ldots)$$

$$\text{i.e.} \quad y_1 = a_0 x^{1/2} \left(1 - x + \frac{1}{2} x^2 + \ldots\right)$$

When $m = 1$, $a_1 = -\frac{2}{3} a_0$, $a_2 = \frac{4}{15} a_0$, and so on. Then

$$y_1 = x^{m_2} (a_0 + a_1 x + a_2 x^2 + \ldots)$$

$$\text{i.e.} \quad y_1 = a_0 x \left(1 - \frac{2}{3} x + \frac{4}{15} x^2 + \ldots\right)$$

Thus, the complete solution of (1) is

$$y = C_1 y_1 + C_2 y_2 = C_1 a_0 x^{1/2} \left(1 - x + \frac{1}{2} x^2 + \ldots\right) + C_2 a_0 x \left(1 - \frac{2}{3} x + \frac{4}{15} x^2 + \ldots\right)$$

Example 5: Find the solution of the equation in series form, $x(x-1)y'' + (3x-1)y' + y = 0$.

Solution:

Given equation is

$$x(x-1)y'' + (3x-1)y' + y = 0$$

$$\Rightarrow (x^2 - x)y'' + (3x-1)y' + y = 0 \quad \ldots (1)$$

Here $x = 0$ is a regular singularity, so let the solution of (1) be

$$y = \sum_{m=0}^{\infty} a_m x^{m+k} \quad \ldots (2)$$

Then,

$$y' = \sum_{m=0}^{\infty} (m+k)\, a_m x^{m+k-1} \quad \text{and} \quad y'' = \sum_{m=0}^{\infty} (m+k)(m+k-1)\, a_m x^{m+k-2}$$

Substituting the values in (1) then

$$(x^2 - x) \sum_{m=0}^{\infty} (m+k)(m+k-1)\, a_m x^{m+k-2} + (3x - 1) \sum_{m=0}^{\infty} (m+k) a_m x^{m+k-1} +$$

$$\sum_{m=0}^{\infty} a_m x^{m+k} = 0$$

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i.e. $\sum_{m=0}^{\infty}(m+k)(m+k-1)\,a_m x^{m+k} - \sum_{m=0}^{\infty}(m+k)(m+k-1)\,a_m x^{m+k-1} + \sum_{m=0}^{\infty}3(m+$

$k)a_m x^{m+k} - \sum_{m=0}^{\infty}(m+k)a_m x^{m+k-1} + \sum_{m=0}^{\infty}a_m x^{m+k} = 0$

$\Rightarrow \sum_{m=0}^{\infty}[(m+k)(m+k-1) + 3(m+k) + 1]\,a_m x^{m+k} + \sum_{m=0}^{\infty}[-(m+k)(m+k-1) - (m$

$+k)]\,a_m x^{m+k-1} = 0$

$\Rightarrow \sum_{m=0}^{\infty}[m^2 + 2mk + k^2 - m - k + 3m + 3k + 1]\,a_m x^{m+k} + \sum_{m=0}^{\infty}[-m^2 - 2mk - k^2 + m +$

$k - m - k]\,a_m x^{m+k-1} = 0$

$\Rightarrow \sum_{m=0}^{\infty}[m^2 + 2mk + k^2 + 2m + 2k + 1]\,a_m x^{m+k} + \sum_{m=0}^{\infty}[-m^2 - 2mk - k^2]\,a_m x^{m+k-1} = 0$

... (3)

Clearly, the lowest power of x in (3) is x^{k-1} and equating its coefficient to right part, i.e. zero, we get

$$a_0 k^2 = 0 \Rightarrow k = 0, 0 \text{ as } a_0 \neq 0$$

The roots are equal. Equating the coefficients of x^k (obtained by setting m = 0 in first term and m = 1 in second term of (3)) in (3), to zero, we get

$$[k^2 + 2k + 1]a_0 + [-1 - 2k - k^2]a_1 = 0$$

$$\Rightarrow (k+1)^2 a_0 - (k+1)^2 a_1 = 0$$

$$\Rightarrow a_1 = a_0$$

Equating the coefficients of x^{k+1} (obtained by setting m = 1 in first term and m = 2 in second term of (3)) in (3), to zero, we get

$$[1 + 2k + k^2 + 2 + 2k + 1]a_1 + [-4 - 4k - k^2]a_2 = 0$$

$$\Rightarrow [k^2 + 4k + 4]a_1 - [k^2 + 4k + 4]a_2 = 0$$

$$\Rightarrow (k+2)^2 a_0 - (k+2)^2 a_1 = 0$$

$$\Rightarrow a_2 = a_1 = a_0$$

and so on.

When m = 0, $a_1 = a_0$, $a_2 = a_0$, and so on. Then

$$y_1 = x^{m_1}(a_0 + a_1 x + a_2 x^2 + \dots)$$

i.e. $y_1 = a_0(1 + x + x^2 + \dots)$

Since m has equal roots, so for second independent solution is

$$y = a_0 x^m (1 + x + x^2 + \dots) \quad \dots (4)$$

Then, differentiating (4) partial w.r.t. m, we get

$$y_2 = \frac{\partial y}{\partial m} = a_0 x^m \log x\,(1 + x + x^2 + \dots) \text{ at } m = 0$$

i.e. $y_2 = a_0 \log x\,(1 + x + x^2 + \dots)$

Thus, the complete solution of (1) is

$$y = C_1 y_1 + C_2 y_2 = C_1 a_0 (1 + x + x^2 + \dots) + C_2 a_0 \log x\,(1 + x + x^2 + \dots).$$

Example 6: Find the series solution of the equation, $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - 4)y = 0$. **Solution:**

Given equation is

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - 4)y = 0 \quad \dots (1)$$

Here x = 0 is a regular singularity, so let the solution of (1) be

$$y = \sum_{m=0}^{\infty} a_m x^{m+k} \quad \dots (2)$$

Then,

$$y' = \sum_{m=0}^{\infty}(m+k)\,a_m x^{m+k-1} \quad \text{and} \quad y'' = \sum_{m=0}^{\infty}(m+k)(m+k-1)\,a_m x^{m+k-2}$$

Substituting the values in (1) then

$$x^2\sum_{m=0}^{\infty}(m+k)(m+k-1)\,a_m x^{m+k-2} + x\sum_{m=0}^{\infty}(m+k)a_m x^{m+k-1} +$$

$$(x^2 - 4)\sum_{m=0}^{\infty}a_m x^{m+k} = 0$$

i.e. $\sum_{m=0}^{\infty}(m+k)(m+k-1)\,a_m x^{m+k} + \sum_{m=0}^{\infty}(m+k)a_m x^{m+k} + \sum_{m=0}^{\infty}a_m x^{m+k+2} -$

$4a_m x^{m+k} = 0$

$\Rightarrow \sum_{m=0}^{\infty}[(m+k)(m+k-1) + (m+k) - 4]a_m x^{m+k} + \sum_{m=0}^{\infty}a_m x^{m+k+2} = 0$

$\Rightarrow \sum_{m=0}^{\infty}[m^2 + 2mk + k^2 - m - k + m + k - 4]a_m x^{m+k} + \sum_{m=0}^{\infty}a_m x^{m+k+2} = 0$

$\Rightarrow \sum_{m=0}^{\infty}[m^2 + 2mk + k^2 - 4]a_m x^{m+k} + \sum_{m=0}^{\infty}a_m x^{m+k+2} = 0 \quad \dots (3)$

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Clearly, the lowest power of x in (3) is x^k and equating its coefficient to right part, i.e. zero, we get

$$(k^2 - 4)a_0 = 0 \Rightarrow k = 2, -2 \text{ as } a_0 \neq 0$$

Equating the coefficients of x^{k+1} in (3), to zero, we get

$$[1 + 2k + k^2 - 4]a_1 = 0$$

$$\Rightarrow (k^2 + 2k - 3)a_1 = 0$$

$$\Rightarrow (k + 3)(k - 1)a_1 = 0$$

$$\Rightarrow a_1 = 0$$

Equating the coefficients of x^{k+2} in (3), to zero, we get

$$[4 + 4k + k^2 - 4]a_2 + a_0 = 0$$

$$\Rightarrow [4k + k^2]a_2 + a_0 = 0$$

$$\Rightarrow a_2 = \frac{-1}{k(k+4)}a_0$$

Equating the coefficients of x^{m+k+2} in (3), to zero, we get

$$[(m+2)^2 + 2(m+2)k + k^2 - 4]a_{m+2} + a_m = 0$$

$$\Rightarrow [m^2 + 4m + 4 + 2mk + 4k + k^2 - 4]a_{m+2} + a_m = 0$$

$$\Rightarrow [m^2 + 4m + 2mk + 4k + k^2]a_{m+2} + a_m = 0$$

$$\Rightarrow [m^2 + mk + mk + k^2 + 4m + 4k]a_{m+2} + a_m = 0$$

$$\Rightarrow (m + k)(m + k + 4)a_{m+2} = -a_m$$

$$\Rightarrow a_{m+2} = \frac{-a_m}{(m+k)(m+k+4)}$$

This implies $a_1 = 0 = a_3 = a_5 = \ldots$

And,

$$a_4 = \frac{-a_2}{(2+k)(2+k+4)} = \left(\frac{-1}{(k+2)(k+6)}\right)\left(\frac{-1}{k(k+4)}\right)a_0$$

and so on.

When k = 2,

$$y_1 = x^{k+1}(a_0 + a_1x + a_2x^2 + \ldots)$$

i.e. $$y_1 = a_0x^2\left(1 - \frac{x^2}{2(6)} + \frac{x^4}{2(4)(6)(8)} + \ldots\right)$$

i.e. $$y_1 = a_0x^2\left(1 - \frac{x^2}{12} + \frac{x^4}{384} + \ldots\right)$$

And, the coefficient of $x^4, x^6, \ldots$ are infinite at k = −2. Let $a_0 = b_0(k+2)$ then

$$y = b_0x^k\left((k+2) - \frac{(k+2)x^2}{k(k+4)} + \frac{x^4}{k(k+4)(k+6)} + \ldots\right) \quad \ldots (4)$$

Differentiating (4) partial w.r.t. k, we get

$$y_2 = \frac{\partial y}{\partial k} = b_0x^k \log x\left((k+2) - \frac{(k+2)x^2}{k(k+4)} + \frac{x^4}{k(k+4)(k+6)} + \ldots\right) +$$

$$b_0x^k\left[1 - \frac{k+2}{k(k+4)}\left(\frac{1}{k+2} - \frac{2(k+2)}{k(k+4)}x^2\right) + \frac{1}{k(k+4)(k+6)}\left(-\frac{1}{k} - \frac{1}{k+4} - \frac{1}{k+6}\right)x^4 + \ldots\right]$$

at k = −2

i.e. $$y_2 = \frac{\partial y}{\partial k} = b_0x^{-2}\log x\left(\frac{x^4}{-2(2)(4)} + \ldots\right) + b_0x^{-2}\left[1 - \left(\frac{1}{-2(2)} - 0\right)x^2 + \frac{1}{-2(2)(4)}\left(-\frac{1}{-2} - \frac{1}{2} - \frac{1}{4}\right)x^4 + \ldots\right]$$

i.e. $$y_2 = b_0x^{-2}\log x\left(-\frac{x^4}{16} + \ldots\right) + b_0x^{-2}\left[1 + \frac{x^2}{4} + \frac{x^4}{64} + \ldots\right]$$

Thus, the complete solution of (1) is

$$y = C_1y_1 + C_2y_2 = C_1a_0x^2\left(1 - \frac{x^2}{12} + \frac{x^4}{384} + \ldots\right) + C_2b_0x^{-2}\log x\left(-\frac{x^4}{16} + \ldots\right) + b_0x^{-2}\left[1 + \frac{x^2}{4} + \frac{x^4}{64} + \ldots\right]$$

Exercise 2 [B]

1. Show that $x^3 = \frac{2}{5}P_3(x) + \frac{3}{5}P_1(x)$ where $P_n(x)$ is Legendre polynomials.
2. a. Express $f(x) = x^4 + 3x^3 - x^2 + 5x - 2$ in terms of Legendre polynomial.
   b. Express $f(x) = x^3 - 5x^2 + x + 2$ in terms of Legendre polynomial.
3. Solve in power series the following differential equations.
   a. $\frac{d^2y}{dx^2} + \frac{1}{2x}\frac{dy}{dx} + \frac{1}{4x}y = 0$
   b. $x\frac{d^2y}{dx^2} + \frac{dy}{dx} + xy = 0$
   c. $3xy'' + 2y' + y = 0$
   d. $9x(1-x)y'' - 12y' + 4y = 0$

Answer Key

2. a. $\frac{8}{35}P_4(x) + \frac{6}{5}P_3(x) - \frac{2}{21}P_2(x) + \frac{34}{5}P_1(x) - \frac{224}{105}P_0(x)$
   b. $\frac{2}{5}P_3(x) - \frac{10}{3}P_2(x) + \frac{8}{3}P_1(x) + \frac{1}{3}(x)$
3. a. $y = C_1\left[1 - \frac{x}{2!} + \frac{x^2}{4!} - \frac{x^3}{6!} + \ldots\ldots\right] + C_2\left[1 - \frac{x}{3!} + \frac{x^2}{5!} - \frac{x^3}{7} + \ldots\ldots\right]$
   b. $y = C_1\left[1 - \frac{x^2}{2^2} - \frac{x^4}{2^2.4^2} - \frac{x^6}{2^2.4^2.6^2} + \ldots\ldots\right] + C_2\left[\log x\left(1 - \frac{x^2}{2^2} + \frac{x^4}{2^2.4^2} - \ldots\ldots\right) + \left(\frac{x^2}{2^2} - \frac{3x^4}{2^2.4^2} + \ldots\ldots\right)\right]$
   c. $y = C_1\left(1 - \frac{x}{2} + \frac{x^2}{20} - \frac{x^3}{480} + \ldots\ldots\right) + C_2x^{1/2}\left(1 - \frac{x}{4} + \frac{x^2}{56} - \frac{x^3}{1680} + \ldots\ldots\right)$
   d. $y = C_1\left(1 + \frac{x}{3} + \frac{2x^2}{9} + \frac{14x^3}{81} + \ldots\ldots\right) + C_2x^{7/3}\left(1 + \frac{4}{5}x + \frac{44}{65}x^2 + \frac{77}{130}x^3 + \ldots\ldots\right)$

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2.7 BESSEL'S EQUATION

The differential equation of the form

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - n^2)y = 0$$

is called the Bessel's equation of order n.

We solve the Bessel's equation by Frobenious method.

Solution of the Bessel's Equation

The Bessel's equation is

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - n^2)y = 0 \qquad \ldots (1)$$

Suppose the series

$$y = \sum_{m=0}^{\infty} a_m x^{m+k} \qquad \ldots (2)$$

Then,

$$y' = \sum_{m=0}^{\infty} (m+k)\, a_m x^{m+k-1} \quad \text{and} \quad y'' = \sum_{m=0}^{\infty} (m+k)(m+k-1)\, a_m x^{m+k-2}$$

Substituting the values in (1) then

$$x^2 \sum_{m=0}^{\infty} (m+k)(m+k-1) a_m x^{m+k-2} + x \sum_{m=0}^{\infty} (m+k) a_m x^{m+k-1} + (x^2 - n^2) \sum_{m=0}^{\infty} a_m x^{m+k} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} (m+k)(m+k-1) a_m x^{m+k} + \sum_{m=0}^{\infty} (m+k) a_m x^{m+k} + (x^2 - n^2) \sum_{m=0}^{\infty} a_m x^{m+k} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} [(m+k)(m+k-1) + (m+k) + (x^2 - n^2)]\, a_m x^{m+k} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} [(m+k)^2 - n^2]\, a_m x^{m+k} + \sum_{m=0}^{\infty} a_m x^{m+k+2} = 0 \qquad \ldots (3)$$

The lowest degree term in (3) is x^k corresponding to $m = 0$, equating its coefficient to zero, we get

$$(k^2 - n^2)a_0 = 0$$

$$\Rightarrow k = \pm n \quad \text{as} \quad a_0 \neq 0$$

And, equating the coefficient of x^{k+1} to zero, we get

$$[(k+1)^2 - n^2]a_1 = 0$$

$$\Rightarrow a_1 = 0 \text{ as } (k+1)^2 - n^2 \neq 0 \text{ for } k = \pm n.$$

Again, equating to zero the coefficient of x^{m+k+2}, we get.

$$[(m+k+2)^2 - n^2]a_{k+2} + a_k = 0$$

$$\Rightarrow a_{k+2} = -\frac{a_k}{(m+k+2)^2}$$

Putting $m = 1, 3, 5, \ldots$, we get

$$a_3 = a_5 = a_7 = \ldots = 0$$

Putting $m = 0, 2, 4, \ldots$, we get

$$a_2 = -\frac{a_0}{(k+2)^2 - n^2}$$

$$a_4 = -\frac{a_2}{(k+4)^2 - n^2} = \frac{a_0}{[(k+2)^2 - n^2]\,[(k+4)^2 - n^2]}$$

and so on.

The solution of (1) is

$$y = a_0 x^k \left[1 - \frac{1}{(k+2)^2 - n^2}x^2 + \frac{1}{[(k+2)^2 - n^2]\,[(k+4)^2 - n^2]}x^4 + \ldots\right]$$

For $k = n$, we get one independent solution

$$y_1 = a_0 x^n \left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2.4(2n+2)(2n+4)} \ldots \ldots\right]$$

$$= a_0 x^n \left[1 + (-1)^1 \frac{x^2}{2^2.1!(n+1)} + (-1)^2 \frac{x^4}{2^4 2!(n+1)(n+2)} + \ldots \ldots\right]$$

$$= a_0 x^n \sum_{m=0}^{\infty} \frac{(-1)^m}{2^{2m}\, m!(n+1)(n+2) \ldots \ldots (n+k)} x^{2m}$$

$$= a_0 x^n \sum_{m=0}^{\infty} \left(\frac{(-1)^m \Gamma(n+1)}{2^{2m}\, m!\, \Gamma(n+m+1)}\right) x^{2m}$$

The second independent solution corresponding to $k = -n$ is

$$y_2 = a_0 x^n \sum_{m=0}^{\infty} \left(\frac{(-1)^m \Gamma(-n+1)}{2^{2m}\, m!\, \Gamma(-n+m+1)}\right) x^{2m}$$

Therefore, the complete solution of (1) is

$$y = Ay_1 + By_2$$

The particular solution of the Bessel's equation is called the Bessel's functions of order n. The Bessel's functions are also called cylindrical and spherical functions. The Bessel's functions are applicable in the field of fluid motion, potential theory, waves (plane, cylindrical and spherical), Euler's theory of a circular membrane etc.

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2.8 BESSEL'S FUNCTION

One particular solution of the Bessel's equation

$$x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+(x^2-n^2)y=0 \quad \ldots (1)$$

is

$$y_1 = a_0x^n\sum_{k=0}^{\infty}\left(\frac{(-1)^k\Gamma(n+1)}{2^{2k}\,k!\,\Gamma(n+k+1)}\right)x^{2k} \quad \ldots (2)$$

If we choose the constant, $a_0=\frac{1}{2^n\Gamma(n+1)}$ then (2) becomes

$$y_1=\frac{x^n}{2^n\Gamma(n+1)}\sum_{k=0}^{\infty}\left(\frac{(-1)^k\Gamma(n+1)}{2^{2k}\,k!\,\Gamma(n+k+1)}\right)x^{2k}$$

$$=\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,(n+k)!}\left(\frac{x}{2}\right)^{2k+n}$$

which we called the Bessel's function of order n and is denoted by $J_n(x)$,

Some Elementary Properties of Bessel's Functions

1. $J_{-n}(x)=(-1)^n J_n(x)$

Proof:

Since we have

$$J_n(x)=\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,(n+k)!}\left(\frac{x}{2}\right)^{2k+n}$$

If we replace n by $-n$ in above, we get

$$J_{-n}(x)=\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,(-n+k)!}\left(\frac{x}{2}\right)^{2k-n}$$

Put $k-n=p$ then

$$J_{-n}(x)=\sum_{p=0}^{\infty}\frac{(-1)^{p+n}}{(p+n)!\,(p)!}\left(\frac{x}{2}\right)^{2p+n-n}$$

i.e. $J_{-n}(x)=(-1)^n\sum_{k=0}^{\infty}\frac{(-1)^p}{(p+n)!\,(p)!}\left(\frac{x}{2}\right)^{2p+n}$

$=(-1)^n J_n(x)$

2. $\frac{d}{dx}[x^n J_n(x)]=x^n J_{n-1}$

Proof:

Since we have

$$J_n(x)=\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,(n+k)!}\left(\frac{x}{2}\right)^{2k+n}$$

Then, $x^n J_n(x)=x^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,(n+k)!}\left(\frac{x}{2}\right)^{2k+n}$

Differentiate, we get

$$\frac{d}{dx}[x^n J_n(x)]=\frac{d}{dx}\left[x^n\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,(n+k)!}\left(\frac{x}{2}\right)^{2k+n}\right]$$

$$=\left[\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\,(n+k)!\,2^{2k+n}}\frac{d}{dx}x^{2n}\right]$$

$$=\left[\sum_{k=0}^{\infty}\frac{(-1)^k(2k+2n)\,x^{2k+2n-1}}{k!\,(n+k)!\,2^{2k+n}}\right]$$

$$=\left[x^n\sum_{k=0}^{\infty}\frac{(-1)^k\,x^{2k+n-1}}{k!\,(n+k-1)!\,2^{2k+n-1}}\right]=x^n J_{n-1}(x)$$

3. $\frac{d}{dx}[x^{-n}J_n(x)]=-x^{-n}J_{n+1}(x)$

Proof:

Since we have

$$\frac{d}{dx}[x^n J_n(x)]=x^n J_{n-1}(x) \text{ and } J_{-n}(x)=(-1)^n J_n(x)$$

Here,

$$\frac{d}{dx}[x^n J_n(x)]=x^n J_{n-1}(x)$$

Replacing n by $-n$ we get

$$\frac{d}{dx}[x^{-n}J_{-n}(x)]=x^{-n}J_{-n-1}(x)=x^{-n}J_{-(n+1)}(x)$$

i.e. $\frac{d}{dx}[x^{-n}(-1)^n J_n(x)]=x^{-n}(-1)^{n+1}J_{n+1}(x)$

i.e. $(-1)^n\frac{d}{dx}[x^{-n}J_n(x)]=(-1)^{n+1}x^{-n}J_{n+1}(x)$

i.e. $\frac{d}{dx}[x^{-n}J_n(x)]=-x^{-n}J_{n+1}(x)$

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Some Elementary Results based on Bessel's Function

Since, the Bessel's function of order n is,

$$J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,(n+k)!}\left(\frac{x}{2}\right)^{2k+n}$$

And, we have

$$\Gamma(m)\,\Gamma(1-m) = \frac{\pi}{\sin m\pi}$$

Set $m = \frac{1}{2}$ then

$$\Gamma\left(\frac{1}{2}\right)\Gamma\left(1-\frac{1}{2}\right) = \frac{\pi}{\sin \pi/2}$$

$$\Rightarrow \quad \left(\Gamma\left(\frac{1}{2}\right)\right)^2 = \frac{\pi}{1}$$

$$\Rightarrow \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

1. Bessel's function of order 0;

$$J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,(k)!}\left(\frac{x}{2}\right)^{2k}$$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2}\left(\frac{x}{2}\right)^{2k}$$

$$= 1 - \left(\frac{x}{2}\right)^2 + \frac{1}{(2!)^2}\times\left(\frac{x}{2}\right)^4 - \frac{1}{(3!)^2}\times\left(\frac{x}{2}\right)^6 + \ldots$$

$$= 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2\times 4^2} - \frac{x^6}{2^2\times 4^2\times 6^2} + \ldots$$

Note: $J_0(0) = 1$.

2. Bessel's function of order 1;

$$J_1(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,(k+1)!}\left(\frac{x}{2}\right)^{2k+1}$$

$$= \frac{x}{2} - \frac{1}{2!}\left(\frac{x}{2}\right)^3 + \frac{1}{2!\times 3!}\left(\frac{x}{2}\right)^5 - \frac{1}{3!\times 4!}\left(\frac{x}{2}\right)^7 + \ldots$$

$$= \frac{x}{2} - \frac{x^3}{2^2\times 4} + \frac{x^5}{2^2\times 4^2\times 6} - \frac{x^7}{2^2\times 4^2\times 6^2\times 8} + \ldots$$

Note: $J_1(0) = 0$.

3. Bessel's function of order $\frac{1}{2}$;

$$J_{1/2}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\left(k+\frac{1}{2}\right)!}\left(\frac{x}{2}\right)^{2k+1/2}$$

$$= \frac{(-1)^0}{0!\left(\frac{1}{2}\right)!}\left(\frac{x}{2}\right)^{1/2} + \frac{(-1)^1}{1!\left(\frac{3}{2}\right)!}\left(\frac{x}{2}\right)^{5/2} + \frac{(-1)^2}{2!\left(\frac{5}{2}\right)!}\left(\frac{x}{2}\right)^{9/2} + \ldots$$

$$= \left(\frac{x}{2}\right)^{1/2}\left[\frac{1}{\frac{1}{2}\Gamma\left(\frac{1}{2}\right)} - \frac{1}{\frac{3}{2}\times\frac{1}{2}\times\Gamma\left(\frac{1}{2}\right)}\left(\frac{x}{2}\right)^2 + \frac{1}{\frac{5}{2}\times\frac{3}{2}\times\frac{1}{2}\times\Gamma\left(\frac{1}{2}\right)}\left(\frac{x}{2}\right)^4\right] + \ldots$$

$$= \left(\frac{x}{2}\right)^{1/2}\left[\frac{2}{\sqrt{\pi}} - \frac{2x^2}{3\times 2\sqrt{\pi}} + \frac{2x^4}{5\times 4\times 3\times 2\times 1\sqrt{\pi}}\right] + \ldots$$

$$= 2\left(\frac{x}{2\pi}\right)^{1/2}\left[1 - \frac{x^2}{3!} + \frac{x^4}{5!} + \ldots\right]$$

$$= \sqrt{\frac{2}{\pi x}}\left[x - \frac{x^3}{3!} + \frac{x^5}{5!} + \ldots\right]$$

$$= \sqrt{\frac{2}{\pi x}}\sin x$$

4. Bessel's function of order $\frac{-1}{2}$;

Here,

$$J_{-1/2}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\left(k-\frac{1}{2}\right)!}\left(\frac{x}{2}\right)^{2k-1/2}$$

$$= \frac{(-1)^0}{0!\,\Gamma\left(\frac{1}{2}\right)}\left(\frac{x}{2}\right)^{-1/2} + \frac{(-1)^1}{1!\left(\frac{1}{2}\right)!}\left(\frac{x}{2}\right)^{3/2} + \frac{(-1)^2}{2!\left(\frac{3}{2}\right)!}\left(\frac{x}{2}\right)^{5/2} + \ldots$$

$$= \left(\frac{x}{2}\right)^{-1/2}\left[\frac{1}{\sqrt{\pi}} - \frac{2}{\sqrt{\pi}}\left(\frac{x}{2}\right)^2 + \frac{2}{3\sqrt{\pi}}\left(\frac{x}{2}\right)^4 - \ldots\right]$$

$$= \left(\frac{2}{\pi x}\right)^{1/2}\left[1 - \frac{x^2}{2} + \frac{x^4}{4!} - \ldots\right]$$

$$= \sqrt{\frac{2}{\pi x}}\cos x$$

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Recurrence Relations of Bessel's Function

1. $xJ_n'(x) = -nJ_n(x) + xJ_{n-1}(x)$

Solution:

We have

$$\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x) \quad \ldots (1)$$

Now,

$$\frac{d}{dx}[x^n J_n(x)] = n x^{n-1} J_n(x) + x^n \frac{d}{dx}(J_n(x))$$

$$\Rightarrow x^n J_{n-1}(x) = nx^{n-1} J_n(x) + x^n J_n'(x) \quad \text{[Using (1)]}$$

$$\Rightarrow J_{n-1}(x) = nx^{-1} J_n(x) + J_n'(x) \text{ [Taking out the common value } x^n\text{]}$$

$$\Rightarrow xJ_n'(x) = -nJ_n(x) + xJ_{n-1}(x)$$

2. $xJ_n'(x) = nJ_n(x) - xJ_{n+1}(x)$

Solution:

We have

$$\frac{d}{dx}[x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x) \quad \ldots (2)$$

Now,

$$\frac{d}{dx}[x^{-n} J_n(x)] = -n x^{n-1} J_n(x) + x^{-n} \frac{d}{dx}(J_n(x))$$

$$\Rightarrow -x^{-n} J_{n+1}(x) = -nx^{n-1} J_n(x) + x^{-n} J_n'(x) \quad \text{[Using (2)]}$$

$$\Rightarrow J_{n+1}(x) = nx^{-1} J_n(x) - J_n'(x) \text{ [Taking out the common value } -x^{-n}\text{]}$$

$$\Rightarrow xJ_n'(x) = nJ_n(x) - xJ_{n+1}(x)$$

3. $2J'_n(x) = J_{n-1}(x) - J_{n+1}(x)$

Solution:

We have

$$xJ_n'(x) = -nJ_n(x) + xJ_{n-1}(x) \quad \ldots (3)$$

$$xJ_n'(x) = nJ_n(x) - xJ_{n+1}(x) \quad \ldots (4)$$

Adding (3) and (4), we get

$$2xJ_n'(x) = xJ_{n-1}(x) - xJ_{n+1}(x)$$

$$\Rightarrow 2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$$

4. $2nJ_n(x) = x[J_{n-1}(x) + J_{n+1}(x)]$

Solution:

Subtracting (4) and (3), we get

$$0 = nJ_n(x) - xJ_{n+1}(x) + nJ_n(x) - xJ_{n-1}(x)$$

$$\Rightarrow 0 = 2nJ_n(x) - x[J_{n+1}(x) + J_{n-1}(x)]$$

$$\Rightarrow 2nJ_n(x) = x[J_{n+1}(x) + J_{n-1}(x)]$$

5. Show that $\int x^n J_{n-1}(x)\,dx = x^n J_n(x) + c$

Solution:

We have,

$$\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$$

$$\Rightarrow x^n J_{n-1}(x) = \frac{d}{dx}[x^n J_n(x)]$$

Integrating with respects to x, we get,

$$\int x^n J_{n-1}(x)\,dx = x^n J_n(x) + c.$$

6. Show that $\int x^{-n} J_{n+1}(x)\,dx = -x^{-n} J_n(x) + c$

Solution:

We have,

$$\frac{d}{dx}[x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$$

$$\Rightarrow x^{-n} J_{n+1}(x) = \frac{d}{dx}[-x^{-n} J_n(x)]$$

Integrating with respects to x, we get,

$$\int x^{-n} J_{n+1}(x)\,dx = -x^{-n} J_n(x) + c.$$

Workout Examples

Example 1: Prove that $\sqrt{\frac{\pi x}{2}} J_{3/2}(x) = \frac{1}{x}\sin x - \cos x$

Solution:

Since, the Bessel's function of order n is,

$$J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,(n+k)!}\left(\frac{x}{2}\right)^{2k+n}$$

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Set $n = \frac{3}{2}$,

$$J_{3/2}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\left(k+\frac{3}{2}\right)!}\left(\frac{x}{2}\right)^{2k+3/2}$$

$$= \frac{(-1)^0}{0!\left(\frac{3}{2}\right)!}\left(\frac{x}{2}\right)^{3/2} + \frac{(-1)^1}{1!\left(\frac{5}{2}\right)!}\left(\frac{x}{2}\right)^{7/2} + \frac{(-1)^2}{2!\left(\frac{9}{2}\right)!}\left(\frac{x}{2}\right)^{11/2} + \ldots$$

$$= \left(\frac{x}{2}\right)^{1/2}\left[\frac{x}{2\times\frac{3}{2}\times\frac{1}{2}\Gamma\frac{1}{2}} - \frac{x^3}{2^3\times\frac{5}{2}\times\frac{3}{2}\times\frac{1}{2}\Gamma\frac{1}{2}} + \frac{x^5}{2^5\times\frac{7}{2}\times\frac{5}{2}\times\frac{3}{2}\times\frac{1}{2}\Gamma\frac{1}{2}} + \ldots\right]$$

$$= \sqrt{\frac{2}{\pi x}}\left[\frac{x^2}{3} - \frac{x^4}{1\times3\times5} + \frac{x^6}{2\times3\times4\times5\times7} + \ldots\right]$$

$$= \sqrt{\frac{2}{\pi x}}\left[\left(\frac{x^2}{2!} - \frac{x^2}{3!}\right) - \left(\frac{x^4}{4!} - \frac{x^4}{5!}\right) + \left(\frac{x^6}{6!} - \frac{x^6}{7!}\right) + \ldots\right]$$

$$= \sqrt{\frac{2}{\pi x}}\left[\left(1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \ldots\right) - \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots\right)\right]$$

$$= \sqrt{\frac{2}{\pi x}}\left[\frac{1}{x}\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots\right) - \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots\right)\right]$$

$$= \sqrt{\frac{2}{\pi x}}\left[\frac{1}{x}\sin x - \cos x\right]$$

Thus, $\sqrt{\frac{\pi x}{2}}J_{3/2}(x) = \frac{1}{x}\sin x - \cos x$

Alternative Method;

We have

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}}\sin x \text{ and } J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}}\cos x$$

By recurrence relation of Bessel's function, we have

$$2nJ_n(x) = x\left[J_{n+1}(x) + J_{n-1}(x)\right]$$

$$\Rightarrow J_{n+1}(x) = \frac{2n}{x}J_n(x) - J_{n-1}(x)$$

At $n = \frac{1}{2}$

$$J_{3/2}(x) = \frac{1}{x}J_{1/2}(x) - J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}}\left[\frac{1}{x}\sin x - \cos x\right]$$

Thus, $\sqrt{\frac{\pi x}{2}}J_{3/2}(x) = \frac{1}{x}\sin x - \cos x$

Graph of $J_{1/2}(x)$

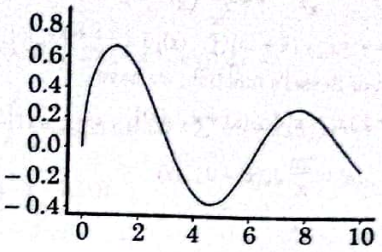

Figure: Graph of $J_{1/2}(x)$

Example 2: Prove that $J_{5/2}(x) = \sqrt{\frac{2}{\pi x}}\left(\frac{3-x^2}{x^2}\sin x - \frac{3}{x}\cos x\right)$

Solution:

We have

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}}\sin x;\ J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}}\cos x \quad \text{and} J_{-n}(x) = (-1)^n J_n(x)$$

By recurrence relation of Bessel's function, we have

$$2nJ_n(x) = x\left[J_{n+1}(x) + J_{n-1}(x)\right]$$

At $n = \frac{3}{2}$

$$3J_{3/2}(x) = x\left[J_{5/2}(x) + J_{1/2}(x)\right]$$

$$\Rightarrow J_{5/2}(x) = \frac{3}{x}J_{3/2}(x) - J_{1/2}(x) \quad \ldots (1)$$

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Again at $n = \frac{1}{2}$

$$J_{1/2}(x) = x\left[J_{3/2}(x) + J_{-1/2}(x)\right]$$

$$\Rightarrow J_{3/2}(x) = \frac{1}{x}J_{1/2}(x) - J_{-1/2}(x) \quad \ldots (2)$$

Now, (1) gives

$$J_{5/2}(x) = \frac{3}{x}\left(\frac{1}{x}J_{1/2}(x) - J_{-1/2}(x)\right) - J_{1/2}(x)$$

$$= \frac{3}{x^2}J_{1/2}(x) - \frac{3}{x}J_{-1/2}(x) - J_{1/2}(x)$$

$$= \left(\frac{3}{x^2} - 1\right)J_{1/2}(x) - \frac{3}{x}J_{-1/2}(x)$$

$$= \left(\frac{3-x^2}{x^2}\right)\sqrt{\frac{2}{\pi x}}\sin x - \frac{3}{x}\sqrt{\frac{2}{\pi x}}\cos x$$

$$= \sqrt{\frac{2}{\pi x}}\left(\frac{3-x^2}{x^2}\sin x - \frac{3}{x}\cos x\right)$$

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Example 3: Prove that $J_4(x) = \frac{8(6-x^2)}{x^3} J_1(x) - \left(\frac{24}{x^2} - 1\right) J_0(x)$.

Solution:

By recurrence relation of Bessel's function, we have

$$2nJ_n(x) = x[J_{n+1}(x) + J_{n-1}(x)]$$

$$\Rightarrow J_{n+1}(x) = \frac{2n}{x} J_n(x) - J_{n-1}(x) \quad \ldots (1)$$

At n = 3

$$J_4(x) = \frac{6}{x} J_3(x) - J_2(x)$$

$$= \frac{6}{x}\left[\frac{4}{x} J_2(x) - J_1(x)\right] - \left[\frac{2}{x} J_1(x) - J_0(x)\right] \quad \text{[Using (1)]}$$

$$= \frac{24}{x^2} J_2(x) - \frac{8}{x} J_1(x) + J_0(x)$$

$$= \frac{24}{x^2}\left(\frac{2}{x} J_1(x) - J_0(x)\right) - \frac{8}{x} J_1(x) + J_0(x) \quad \text{[Using (1)]}$$

$$= \frac{48}{x^3} J_1(x) - \frac{24}{x^2} J_0(x) - \frac{8}{x} J_1(x) + J_0(x)$$

$$= \frac{8(6-x^2)}{x^3} J_1(x) - \left(\frac{24}{x^2} - 1\right) J_0(x)$$

Example 4: Evaluate, $\int J_3(x)\, dx$.

Solution:

We have

$$\frac{d}{dx}\left[x^{-n} J_n(x)\right] = -x^{-n} J_{n+1}(x) \quad \ldots (1)$$

Integrating both sides, we get

$$x^{-n} J_n(x) = -\int x^{-n} J_{n+1}(x)\, dx + c$$

$$\Rightarrow \int x^{-n} J_{n+1}(x)\, dx = -x^{-n} J_n(x) + c \quad \ldots (2)$$

Putting n = 2 in (1), we get

$$\int x^{-2} J_3(x)\, dx = -x^{-2} J_2(x) + c$$

Here,

$$\int J_3(x)\, dx = \int x^2 x^{-2} J_3(x)\, dx$$

$$= x^2 \int x^{-2} J_3(x)\, dx - \int 2x \left(\int x^{-2} J_3(x)\right) dx$$

$$= x^2 \left(-x^{-2} J_2(x)\right) - \int 2x \left(-x^{-2} J_2(x)\right) dx$$

$$= -J_2(x) + 2\int x^{-1} J_2(x)\, dx$$

$$= -J_2(x) + 2\left(-x^{-1} J_1(x)\right) \quad \text{[Using (2)]}$$

$$= -J_2(x) - 2x^{-1} J_1(x)$$

Example 5: Show that $J_0'(x) = -J_1(x)$.

Solution:

We have,

$$J_n(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+n}}{2^{2m+n}\, m!\, (n+m)!}$$

At n = 1,

$$J_1(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+1}}{2^{2m+1}\, m!\, (m+1)!} = \frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} \quad \ldots (1)$$

At n = 0,

$$J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m}\, m!\, m!} = 1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{64 + 36} \quad \ldots (2)$$

Differentiating w. r. t. x, then

$$J_0'(x) = 0 - \frac{2x}{4} + \frac{4x^3}{64} - \frac{6x^5}{64 + 36} + \ldots$$

$$\Rightarrow J_0'(x) = -\left(\frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} \ldots\right)$$

$$\Rightarrow J_0'(x) = -J_1(x) \quad \text{(using (i)}$$

Alternative method:

Since we have,

$$xJ_n'(x) = nJ_n(x) - xJ_{n+1}(x)$$

Set n = 0 then,

$$xJ_0'(x) = 0 - xJ_1(x)$$

$$\Rightarrow J_0'(x) = -J_1(x)$$

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Example 6: Show that $\int_0^r xJ_0(ax)\,dx = \frac{r}{a}J_1(ar)$

Solution:

Set, $ax = u$ then

$$\int_0^r xJ_0(ax)\,dx = \frac{1}{a}\int_0^{ar}\frac{u}{a}J_0(u)\,du = \frac{1}{a^2}\int_0^{ar} u\,J_0(u)\,du \quad \ldots (i)$$

Since we have

$$\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$$

Putting $n = 1$,

$$\frac{d}{dx}[xJ_1(x)] = xJ_0(x)$$

Then (i) becomes

$$\int_0^r xJ_0(ax)\,dx = \frac{1}{a^2}\int_0^{ar}\frac{d}{du}(uJ_1(u))\,du = \frac{1}{a^2}[uJ_1(u)]_0^{ar} = \frac{1}{a^2}ar\,J_1(ar) = \frac{r}{a}J_1(ar)$$

Exercise 2 (C)

Prove the following relations (1 – 6) of Bessel's functions.

1. $J_2'(x) = \frac{1}{2}[J_1(x) - J_3(x)]$
2. $J_1'(x) = J_0(x) - x^{-1}J_1(x)$
3. $J_{-3/2}(x) = \sqrt{\frac{2}{\pi x}}\left(-\frac{\cos x}{x} - \sin x\right)$
4. $J_{n+3}(x) + J_{n+5}(x) = \frac{2}{x}(n+4)\,J_{n+4}(x)$
5. $J_{-1/2}(x) = J_{1/2}(x)\cot x$
6. $\int x^{n+1}J_n(x)\,dx = x^{n+1}J_{n+1}(x)$
7. Evaluate
   a. $\int J_3(x)\,dx$ b. $\int x^3 J_2(x)\,dx$ c. $\int J_5(x)\,dx$

Answer Key

a. $-2J_2(x) - J_0(x) + C$	b. $x^3J_3(x) + C$	c. $-2J_4(x) - 2J_2(x) - J_0(x) + C$

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3

LAPLACE TRANSFORM

CHAPTER OUTLINE

After completion of this chapter, you will be able to gain the knowledge and understanding regarding:

3.1 Laplace Transform (LT), Inverse LT, Linearity of LT, LT of elementary functions, inverses and first shifting (sshifting) theorem. Existence theorem of Laplace transform (without proof) and uniqueness.
3.2 Transform of Derivative and Integrals of a function.
3.3 Differentiation and Integration of Laplace transform.
3.4 Unit step function, periodic function and LT, second shifting (t-shifting) theorem.
3.5 Convolution theorem and its application to find inverse.
3.6 Application of Laplace transform to find the solutions of ordinary differential equations (IVP).

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Introduction

A transformation is a mathematical operation that converts one function to another [illegible] function. One of such transformation is Integral Transforms that transforms a function [illegible] another form by means of integral. An integral transforms is a relation

$$F(s) = \int_0^\infty k(s,t) f(t)\,dt \qquad \text{--- (1)}$$

where k(s, t) is a function of two variables which we called Kernel function.

As a particular form of (1), one transformation is Laplace transformation or Laplace transform, which was used for the first time by French mathematician Pierre Simon Marquis De Laplace (1749-1827) in 1799. Mostly, the transform is useful in solving linear differential equations with initial conditions. The Laplace transform is used to reduce differential equation to an algebraic problem.

3.1 Laplace Transform

One strong branch of integral transforms is Laplace transform. This transform is defined from (1) by putting the values: a = 0, b→∞, and choosing the kernel function $k(s, t) = e^{-st}$.

The Laplace transform of a piecewise continuous and absolutely integrable function f(t) over (0, ∞) is given by

$$F(s) = \int_0^\infty e^{-st} f(t)\,dt \qquad \text{--- (2)}$$

provided the integral converges. It is also denoted by L[f(t); s] or F(s) or L{f(t)}.

3.2 Function of Exponential Order

A function f(t) is said to be exponential order if there exists some constants k and M such that $|f(t)| \le Me^{kt}$ for all $t \ge 0$.

3.3 Sufficient Condition for Existence of Laplace Transform

Let a function f(t) be piece-wise continuous on every finite closed interval $0 \le t \le b$; $b > 0$ and f(t) is of exponential order, then the Laplace transform of f(t)

$$F(s) = \int_0^\infty e^{-st} f(t)\,dt$$

exists for all $s > k$.

Proof:

Let f(t) is of exponential order, then for given constants $M > 0$ and $k > 0$,

$$|f(t)| \le Me^{kt} \qquad \text{for } t \ge 0.$$

Let a function f(t) be piece-wise continuous on every finite closed interval $0 \le t \le b$; $b > 0$ and so $e^{-st}f(t)$ is integrable over any finite interval.

Now,

$$|F(s)| = \left|\int_0^\infty e^{-st} f(t)\,dt\right| \le \int_0^\infty e^{-st}|f(t)|\,dt \le \int_0^\infty e^{-st} Me^{kt}\,dt = M\int_0^\infty e^{-s+k}\,dt$$

Here,

$$\int_0^\infty e^{-s+k}\,dt = \lim_{T\to\infty}\int_0^T e^{-s+k}\,dt = \lim_{T\to\infty}\left[\frac{e^{-s+k}}{-(s-k)}\right]_0^T = \frac{1}{s-k} \quad (\text{for } s > k)$$

Therefore, for $s > k$

$$|F(s)| \le \frac{M}{s-k} \quad \text{which is finite.}$$

Thus, the Laplace transform of f(t) exists for $s > k$.

3.4 Some Elementary Properties of Laplace Transform

1. Linear Property

If f(t) and g(t) be two functions whose Laplace transform exist, then for any two constants a and b,

$$L\{af(t) + bg(t)\} = a\,L\{f(t)\} + b\,L\{g(t)\}$$

Proof:

By the definition,

$$L\{a f(t) + b g(t)\} = \int_0^\infty e^{-st}\{a f(t) + b g(t)\}\,dt$$

$$= a\int_0^\infty e^{-st} f(t)\,dt + b\int_0^\infty e^{-st} g(t)\,dt$$

$$= a\,L\{f(t)\} + b\,L\{g(t)\}$$

2. Change of Scale Property

If $L\{f(t)\} = F(s)$, then $L\{f(at)\} = \frac{1}{a}[F(s)]_{s\to s/a} = \frac{1}{a}F\left(\frac{s}{a}\right)$, $a > 0$.

Proof:

Let,

$$L\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\,dt$$

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Now,

$$L\{f(at)\} = \int_0^\infty e^{-st} f(at)\,dt = \int_0^\infty e^{-s(u/a)} f(u)\frac{du}{a} \quad [\text{Putting, } u = at]$$

$$= \frac{1}{a}\int_0^\infty e^{-s(u/a)} f(u)\,du$$

$$= \frac{1}{a}\left(\int_0^\infty e^{-su} f(u)\,du\right)_{s\to s/a} = \frac{1}{a}[F(s)]_{s\to s/a} = \frac{1}{a}F\left(\frac{s}{a}\right)$$

3. Multiplication of a Function by t

If $L\{f(t)\} = F(s)$, then $L\{t\,f(t)\} = -\frac{d}{ds}L\{f(t)\}$

Proof:

Let,

$$L\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\,dt$$

Then,

$$\frac{d}{ds}L\{f(t)\} = \frac{d}{ds}\left(\int_0^\infty e^{-st}\,t\,f(t)\,dt\right) = \int_0^\infty \frac{d}{ds}(e^{-st})f(t)dt = \int_0^\infty e^{-st}(-t)f(t)dt = -L\{t\,f(t)\}$$

$$\Rightarrow \quad L\{t\,f(t)\} = -\frac{d}{ds}L\{f(t)\}.$$

Note: a. $L\{t^2 f(t)\} = -\frac{d}{ds}(L\{t\,f(t)\}) = -\frac{d}{ds}\left(-\frac{d}{ds}(L\{f(t)\})\right) = (-1)^2\frac{d^2}{ds^2}(L\{f(t)\})$

b. $L\{t^3 f(t)\} = -\frac{d}{ds}(L\{t^2 f(t)\}) = -\frac{d}{ds}\left(-\frac{d^2}{ds^2}(L\{f(t)\})\right) = (-1)^3\frac{d^3}{ds^3}(L\{f(t)\})$

c. $L\{t^n f(t)\} = (-1)^n\frac{d^n}{ds^n}(L\{f(t)\})$ $\quad n = 1, 2, 3, \ldots$

Remark: (Derivative of Laplace Transform of a Function)

From above, we have

(a) $\frac{d}{ds}(L\{f(t)\}) = -L\{t\,f(t)\}$

(b) $\frac{d^n}{ds^n}(L\{f(t)\}) = (-1)^n L\{t^n f(t)\}$.

4. Division of a Function by t

If $L\{f(t)\} = F(s)$, then $L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds$.

Proof:

Let,

$$L\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\,dt$$

Now,

$$\int_s^\infty F(s)\,ds = \int_s^\infty\left(\int_0^\infty e^{-st}\,t\,f(t)\,dt\right)ds = \int_0^\infty\left(\int_s^\infty e^{-st}\,ds\right)f(t)\,dt$$

$$= \int_0^\infty\left[\frac{e^{-st}}{-t}\right]_s^\infty f(t)\,dt$$

$$= \int_0^\infty\left[\frac{0 - e^{-st}}{-t}\right]f(t)\,dt$$

$$= \int_0^\infty \frac{f(t)}{t}e^{-st}\,dt$$

$$= L\left\{\frac{f(t)}{t}\right\}.$$

Note: (Integration of Laplace Transform of a Function)

If the function f(t) is piecewise continuous and is of exponential order, then

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(\bar{s})\,d(\bar{s}), \text{ where } L\{f(t)\} = F(s).$$

Laplace Transform of Some Elementary Functions

Example 1: Find the Laplace transform of unity i.e. 1.

Solution:

Let $\quad f(t) = 1$

By the definition,

$$L\{f(t)\} = L\{1\} = \int_0^\infty e^{-st}(1)\,dt$$

$$= \int_0^\infty e^{-st}\,dt$$

$$= \left[\frac{e^{-st}}{-s}\right]_0^\infty = \frac{1}{s}$$

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Example 2: Find the Laplace transform of e^{at}, $s > a > 0$.

Solution:

Let $f(t) = e^{at}$, for $s > 0$ and $s > a$.

By the definition,

$$L\{f(t)\} = \int_0^\infty e^{-st} e^{at}\,dt = \int_0^\infty e^{-s+a)t}\,dt = \left[\frac{e^{-(s-a)t}}{-(s-a)}\right]_0^\infty = \frac{1}{s-a}$$

Example 3: Find the Laplace transform of (a) coswt (b) sinwt.

Solution:

(a) Let $f(t) = \cos wt$

By the definition,

$$L\{\cos wt\} = \int_0^\infty e^{-st}\cos wt\,dt$$

$$= \left[\frac{e^{-st}}{(-s)^2 + w^2}((-s)\cos wt + w\sin wt)\right]_0^\infty$$

$$= \left[0 - \frac{1}{s^2 + w^2}(-s\times 1 + w\times 0)\right]$$

$$= \frac{s}{s^2 + w^2}$$

(b) Let $f(t) = \sin wt$

By the definition,

$$L\{\sin wt\} = \int_0^\infty e^{-st}\sin wt\,dt$$

$$= \left[\frac{e^{-st}}{(-s)^2 + w^2}((-s)\sin wt - w\cos wt)\right]_0^\infty$$

$$= \left[0 - \frac{1}{s^2 + w^2}(-s\times 0 - w\times 1)\right]$$

$$= \frac{w}{s^2 + w^2}$$

Example 4: Find the Laplace transform of t^n.

Solution:

By Gamma function we have

$$\Gamma(n) = \int_0^\infty e^{-t} t^{n-1}\,dt$$

And, we have $\Gamma(n+1) = n!$

Now, by the definition of Laplace transform,

$$L\{t^n\} = \int_0^\infty e^{-st} t^n\,dt = \int_0^\infty e^{-x}\left(\frac{x}{s}\right)^n\left(\frac{dx}{s}\right) \quad [\text{Putting } st = x]$$

$$= \frac{1}{s^{n+1}}\int_0^\infty e^{-x} x^n\,dx$$

$$= \frac{1}{s^{n+1}}\Gamma(n+1)$$

$$= \frac{n!}{s^{n+1}}$$

Note: 1. $L\{1\} = \frac{1!}{s^2} = \frac{1}{s^2}$ 2. $L\{t^2\} = \frac{2!}{s^3} = \frac{2}{s^3}$ 3. $L\{t^3\} = \frac{3!}{s^4} = \frac{6}{s^4}$

and so on.

3.5 LAPLACE TRANSFORM OF DERIVATIVE OF A FUNCTION

Let $f(t)$ be continuous for all $t \geq 0$ and be of exponential order. If $f'(t)$ is piecewise continuous on every finite interval in the range $t \geq 0$, then the Laplace transform of the derivative $f'(t)$ exists for all $s > k$, and

$$L\{f'(t)\} = s\,L\{f(t)\} - f(0)$$

Proof:

Let $f'(t)$ be continuous for all $t \geq 0$. Then

$$L\{f'(t)\} = \int_0^\infty e^{-st} f'(t)\,dt$$

$$= [e^{-st} f(t)]_0^\infty + s\int_0^\infty e^{-st} f(t)\,dt \quad [\text{Integrating by parts}]$$

$$= [0 - f(0)] + s\int_0^\infty e^{-st} f(t)\,dt$$

$$= s\,L\{f(t)\} - f(0)$$

Note: The above theorem may be extended to piecewise continuous function $f(t)$.

$$L\{f''(t)\} = s\,L\{f'(t)\} - f'(0) = s\,[s\,L[f(t)] - f(0)] - f'(0)$$

i.e. $L\{f''(t)\} = s^2\,L\{f(t)\} - s\,f(0) - f'(0)$

Similarly,

$$L\{f'''(t)\} = s^2\,L\{f(t)\} - s\,f'(0) - f''(0)$$

$$= s^3\,L\{f(t)\} - s^2\,f(0) - s\,f'(0) - f''(0)$$

In general,

$$L\{f^n(t)\} = s^n\,L\{f(t)\} - s^{n-1}\,f(0) - s^{n-2}\,f'(0) - s^{n-3}f''(0) - \ldots - f^{n-1}(0)$$

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Finding Laplace Transform of a Function by Using the Laplace Transform of Derivative

Example A: Find the Laplace transform of unity i.e. 1, using differentiation property of Laplace transform.

Solution:

Let $f(t) = 1$ So, $f(0) = 1$

Then $f'(t) = 0$ So, $f'(0) = 0$

We know,

$$L\{f'(t)\} = s\,L\{f(t)\} - f(0)$$

$$\Rightarrow L\{0\} = s\,L\{f(t)\} - 1$$

$$\Rightarrow 0 = s\,L\{f(t)\} - 1$$

$$\Rightarrow L\{f(t)\} = \frac{1}{s}$$

Example B: Find the Laplace transform of e^{at}, $s > a > 0$, using differentiation property of Laplace transform.

Solution:

Let $f(t) = e^{at}$, So, $f(0) = 1$

Then $f'(t) = ae^{at}$, So, $f'(0) = a$

We know,

$$L\{f'(t)\} = s\,L\{f(t)\} - f(0)$$

$$\text{i.e. } L\{ae^{at}\} = s\,L\{e^{at}\} - 1$$

$$\Rightarrow a\,L\{e^{at}\} = s\,L\{e^{at}\} - 1$$

$$\Rightarrow 1 = (s - a)\,L\{e^{at}\}$$

$$\Rightarrow L\{e^{at}\} = \frac{1}{s - a}$$

Example C: Find the Laplace transform of coswt, using differentiation property of Laplace transform.

Solution:

Let $f(t) = \cos wt$ So, $f(0) = 1$

Then $f'(t) = -w \sin wt$ So, $f'(0) = 0$

$f''(t) = -w^2 \cos wt$ So, $f''(0) = -w^2$

We know,

$$L\{f''(t)\} = s^2\,L\{f(t)\} - s\,f(0) - f'(0)$$

$$\text{i.e. } L\{-w^2 \cos wt\} = s^2\,L\{\cos wt\} - s(1) - 0$$

$$\Rightarrow -w^2\,L\{\cos wt\} = s^2\,L\{\cos wt\} - s$$

$$\Rightarrow s = (s^2 + w^2)\,L\{\cos wt\}$$

$$\Rightarrow L\{\cos wt\} = \frac{s}{s^2 + w^2}$$

3.6 SHIFTING THEOREM ON LAPLACE TRANSFORM

Theorem 1: (First Shifting Theorem (First Shifting Property))

If the Laplace transform of $f(t)$ is $F(s)$, then the Laplace transform of $e^{at} f(t)$ is $F(s - a)$ for $s > a$. That is, if $L\{f(t)\} = F(s)$, then

$$L\{e^{at} f(t)\} = F(s - a) = [F(s)]_{s \to s - a} \text{ for } s > a.$$

Proof:

Let,

$$L\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\,dt$$

Now,

$$L\{e^{at} f(t)\} = \int_0^\infty e^{-st} e^{at} f(t)\,dt = \int_0^\infty e^{-(s-a)t} f(t)\,dt, \text{ for } s > a.$$

$$= \left[\int_0^\infty e^{-st} f(t)\,dt\right]_{s \to s - a}$$

$$= [F(s)]_{s \to s - a}$$

$$= F(s - a)$$

Thus, $L\{e^{at} f(t)\} = F(s - a) = [F(s)]_{s \to s - a} = [L\{f(t)\}]_{s \to s - a}$ for $s > a$.

Example 5: Find the Laplace transform of (a) coshwt (b) sinhwt.

Solution:

(a) We know,

$$\cosh wt = \frac{e^{wt} + e^{-wt}}{2} \text{ and } L\{e^{at}\} = \frac{1}{s - a}$$

Now,

$$L\{\cosh wt\} = L\left\{\frac{e^{wt} + e^{-wt}}{2}\right\} = \frac{1}{2}\left[L\{e^{wt}\} + L\{e^{-wt}\}\right]$$

$$= \frac{1}{2}\left[\frac{1}{s - w} + \frac{1}{s + w}\right]$$

$$= \frac{s}{s^2 - w^2}$$

Alternatively,

Let $f(t) = \cosh wt$ So, $f(0) = 1$

Then $f'(t) = w \sinh wt$ So, $f'(0) = 0$

$f''(t) = w^2 \cosh wt$ So, $f''(0) = w^2$

We know,

$$L\{f''(t)\} = s^2\,L\{f(t)\} - s\,f(0) - f'(0)$$

$$\text{i.e. } L\{w^2 \cosh wt\} = s^2\,L\{\cosh wt\} - s\,(1) - 0$$

$$\Rightarrow w^2\,L\{\cosh wt\} = s^2\,L\{\cosh wt\} - s$$

$$\Rightarrow s = (s^2 - w^2)\,L\{\cosh wt\}$$

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$\Rightarrow L\{\cosh wt\} = \frac{s}{s^2 - w^2}$

(b) We know,

$$\sinh wt = \frac{e^{wt} - e^{-wt}}{2} \text{ and } L\{e^{wt}\} = \frac{1}{s - a}$$

Now,

$$L\{\sinh wt\} = L\left\{\frac{e^{wt} - e^{-wt}}{2}\right\} = \frac{1}{2}[L\{e^{wt}\} - L\{e^{-wt}\}]$$
$$= \frac{1}{2}\left[\frac{1}{s - w} - \frac{1}{s + w}\right]$$
$$= \frac{w}{s^2 - w^2}$$

Example 6: Find the Laplace transform of $f(t) = e^{at} \sin wt$.

Solution:

We have

$$L\{\sin wt\} = \frac{s}{s^2 + w^2} \text{ and } L\{e^{at}\} = \frac{1}{s - a}$$

By first shifting property we have

$$L\{e^{at} f(t)\} = [L\{f(t)\}]_{s \to s - a}$$

Now,

$$L\{e^{at} \sin wt\} = [L\{\sin wt\}]_{s \to s - a} = \left[\frac{w}{s^2 + w^2}\right]_{s \to s - a} = \frac{w}{(s - a)^2 + w^2}$$

Example 7: Find the Laplace transform of $f(t) = e^{at} \cosh wt$.

Solution:

We have

$$L\{\cosh wt\} = \frac{w}{s^2 + w^2} \text{ and } L\{e^{at}\} = \frac{1}{s - a}$$

By first shifting property we have

$$L\{e^{at} f(t)\} = [L\{f(t)\}]_{s \to s - a}$$

Now,

$$L\{e^{at} \cosh wt\} = [L\{\cosh wt\}]_{s \to s - a} = \left[\frac{s}{s^2 - w^2}\right]_{s \to s - a} = \frac{s - a}{(s - a)^2 - w^2}$$

Example 8: Find the Laplace transform of $f(t) = e^{-5t} t$.

Solution:

We have

$$L\{t\} = \frac{1}{s^2} \text{ and } L\{e^{at}\} = \frac{1}{s - a}$$

By first shifting property we have

$$L\{e^{at} f(t)\} = [L\{f(t)\}]_{s \to s - a}$$

Now,

$$L\{e^{-5t} t\} = [L\{t\}]_{s \to s + 5} = \left[\frac{1}{s^2}\right]_{s \to s + 5} = \frac{1}{(s + 5)^2} = \frac{1}{s^2 + 10s + 25}$$

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Definition: (Unit Step Function)

A function u(t) that defined by

$$u(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$

is called unit step function. It also known as unit Heaviside function.

Note: 1. The unit step function $u(t - a)$ for $a > 0$, is given by

$$u_a(t) = u(t - a) = \begin{cases} 1 & \text{for } t - a \geq 0 \\ 0 & \text{for } t - a < 0 \end{cases}$$
$$= \begin{cases} 1 & \text{for } t \geq a \\ 0 & \text{for } t < a \end{cases}$$

2. The piecewise continuous function $f(t) = \begin{cases} k & \text{for } a < t < b \\ 0 & \text{for } t < a \text{ and } t > b \end{cases}$, where k is a constant, can be written as

$$f(t) = k[u(t - a) - u(t - b)].$$

Example 9: Find the Laplace transform of $f(t) = u(t - a)$.

Solution:

Let

$$f(t) = u(t - a) = \begin{cases} 1 & \text{for } t - a \geq 0 \\ 0 & \text{for } t - a < 0 \end{cases} = \begin{cases} 1 & \text{for } t \geq a \\ 0 & \text{for } t < a \end{cases}$$

Now,

$$L\{u(t - a)\} = \int_0^\infty e^{-st} u(t - a)\, dt = \int_a^\infty e^{-st}\, dt = \left[\frac{e^{-st}}{-s}\right]_a^\infty = \frac{e^{-as}}{s}$$

Theorem 2: (Second Shifting Theorem (Second Shifting Property))

If $L\{f(t)\} = F(s)$, then $L\{f(t - a)\, u(t - a)\} = e^{-as} F(s)$.

Proof:

By definition, we have

$$L\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\, dt \quad \ldots (1)$$

And, the unit step function is

$$u_a(t) = u(t - a) = \begin{cases} 1 & \text{for } t - a \geq 0 \\ 0 & \text{for } t - a < 0 \end{cases} = \begin{cases} 1 & \text{for } t \geq a \\ 0 & \text{for } t < a \end{cases}$$

Then,

$$f(t - a)\, u(t - a) = \begin{cases} f(t - a) & \text{for } t \geq a \\ 0 & \text{for } t < a \end{cases} \quad \ldots (2)$$

Now,

$$L\{f(t - a)\, u(t - a)\} = \int_0^\infty e^{-st} f(t - a)\, u(t - a)\, dt$$
$$= \int_a^\infty e^{-st} f(t - a)\, dt \quad \text{[Using (2)]}$$

Put $t - a = u$ then $dt = du$. Also, $t = a \Rightarrow u = 0$, $t \to \infty$, $u \to \infty$. Then

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$$L\{f(t-a)\,u(t-a)\} = \int_0^\infty e^{-s(a+u)} f(u)\,du$$

$$= e^{-as}\int_0^\infty e^{-su} f(u)\,du$$

$$= e^{-as} F(s)$$

$$= e^{-as} L\{f(t)\}$$

Note: If f(t) be a function of t and u(t – a) is unit step function then

$$L\{f(t)\,u(t-a)\} = e^{-as} L\{f(t+a)\}$$

Proof:

Here,

$$L\{f(t)\,u(t-a)\} = \int_0^\infty e^{-st} f(t)\,u(t-a)\,dt$$

$$= \int_a^\infty e^{-st} f(t)\,dt \qquad \text{[Using (2)]}$$

Put $t - a = u$ then $dt = du$. Also, $t = a \Rightarrow u = 0$, $t \to \infty$, $u \to \infty$. Then

$$L\{f(t)\,u(t-a)\} = \int_0^\infty e^{-s(a+u)} f(u+a)\,du$$

$$= e^{-as}\int_0^\infty e^{-su} f(u+a)\,du$$

$$= e^{-as}\int_0^\infty e^{-st} f(t+a)\,dt \qquad \text{[Being u is dummy,}$$

changing u to t]

$$= e^{-as} L\{f(t+a)\}$$

Example 10: Find the Laplace transform of $e^{-3t}\,u(t-2)$.

Solution:

By second shifting property, we get

$$L\{f(t)\,u(t-a)\} = e^{-as} L\{f(t+a)\}.$$

Now,

$$L\{e^{-3t}\,u(t-2)\} = e^{-2s} L\{e^{-3(t+2)}\}$$

$$= e^{-6} e^{-2s} L\{e^{-3t}\}$$

$$= e^{-2s-6}\left(\frac{1}{s+3}\right) \qquad \left[\because L\{e^{-at}\} = \frac{1}{s+a}\right]$$

$$= \frac{e^{-2s-6}}{s+3}$$

Alternatively,

By second shifting property, we get

$$L\{f(t-a)\,u(t-a)\} = e^{-as} L\{f(t)\}$$

Now,

$$L\{e^{-3t}\,u(t-2)\} = L\{e^{-3(t-2)} \times e^{-6}\,u(t-2)\}$$

$$= e^{-6} L\{e^{-3(t-2)}\,u(t-2)\}$$

$$= e^{-6} e^{-2s} L\{e^{-3t}\}$$

$$= e^{-2s-6}\left(\frac{1}{s+3}\right) \qquad \left[\because L\{e^{-at}\} = \frac{1}{s+a}\right]$$

$$= \frac{e^{-2s-6}}{s+3}$$

Example 11: Find the Laplace transform of $\sin 2t\,u(t-\pi)$.

Solution:

By second shifting property, we get

$$L\{f(t-a)\,u(t-a)\} = e^{-as} L\{f(t)\} \qquad \ldots (1)$$

Now,

$$L\{\sin 2t\,u(t-\pi)\} = L\{-\sin 2(t-\pi)\,u(t-\pi)\}$$

$$= -L\{\sin 2(t-\pi)\,u(t-\pi)\}$$

$$= -e^{-\pi s} L\{\sin 2t\}$$

$$= -e^{-\pi s}\left(\frac{2}{s^2+4}\right) \qquad \left[\because L\{\sin at\} = \frac{a}{s^2+a^2}\right]$$

$$= \frac{-2e^{-\pi s}}{s^2+4}$$

Example 12: Find the Laplace transform of $t \cos 2t$.

Solution:

Let $f(t) = t \cos 2t$.

Since we have,

$$L\{t\,f(t)\} = -\frac{d}{ds}(L\{f(t)\}) \text{ and } L\{\cos at\} = \frac{s}{s^2+a^2}$$

Now,

$$L\{t\cos 2t\} = -\frac{d}{ds}(L\{\cos 2t\}) = -\frac{d}{ds}\left(\frac{s}{s^2+4}\right)$$

$$= -\frac{s^2+4-2s^2}{(s^2+4)^2}$$

$$= \frac{s^2-4}{(s^2+4)^2}$$

Thus, $L\{t\cos 2t\} = \frac{s^2-4}{(s^2+4)^2}$.

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Example 13: Find the Laplace transform of $t^3 e^t$.

Solution:

Let $f(t) = t^3 e^t$.

Since we have,

$L[e^{at} f(t)] = [L\{f(t)\}]_{s \to s-a}$ and $L[t^n] = \frac{n!}{s^{n+1}}$

Now,

$$L[t^3 e^t] = [L[t^3]]_{s \to s-1} = \left(\frac{3!}{s^4}\right)_{s \to s-1} = \frac{3}{(s-1)^4}$$

Thus, $L[t^3 e^t] = \frac{3}{(s-1)^4}$.

Example 14: Find the Laplace transform of $te^{-t}\cosh t$

Solution:

Let $f(t) = te^{-t}\cosh t$.

Since we have,

$L[e^{at} f(t)] = [L\{f(t)\}]_{s \to s-a}$; $L[t\, f(t)] = -\frac{d}{ds}(L\{f(t)\})$; $L[\cosh at] = \frac{s}{s^2 - a^2}$

Now,

$$\begin{aligned} L[te^{-t}\cosh t] &= (L\{t\cosh t\})_{s \to s+1} \\ &= \left(-\frac{d}{ds}(L\{\cosh t\})\right)_{s \to s+1} \\ &= -\left(\frac{d}{ds}\left(\frac{s}{s^2-1}\right)\right)_{s \to s+1} \\ &= \left(\frac{s^2 - 1 - 2s^2}{(s^2-1)^2}\right)_{s \to s+1} \\ &= -\left(\frac{s^2+1}{(s^2-1)^2}\right)_{s \to s+1} \\ &= \frac{(s+1)^2 + 1}{\{(s+1)^2 - 1\}^2} \\ &= \frac{s^2 + 2s + 2}{(s^2+2s)^2} \\ &= \frac{s^2 + 2s + 2}{s^2(s+2)^2} \end{aligned}$$

Thus, $L[te^{-t}\cosh t] = \frac{s^2 + 2s + 2}{s^2(s+2)^2}$

3.7 LAPLACE TRANSFORM OF THE INTEGRAL OF A FUNCTION

If the function is piecewise continuous and is exponential order, and if

$L[f(t)] = F(s)$, then $L\left[\int_0^t f(u)\,du\right] = \frac{1}{s}F(s)$

Proof:

Let $G(t) = \int_0^t f(u)\,du$. Then $\frac{d}{dt}G(t) = f(t)$ as $f(0) = 0$ and $G(0) = 0$.

Now,

$$\begin{aligned} L[G(t)] &= \int_0^\infty e^{-st} G(t)\,dt \\ &= \left[\frac{e^{-st}}{-s}G(t)\right]_0^\infty - \int_0^\infty \left(\frac{e^{-st}}{-s}\frac{d}{dt}G(t)\right)dt \qquad \text{[Integrating by parts]} \\ &= 0 + \frac{1}{s}\int_0^\infty e^{-st} f(t)\,dt \end{aligned}$$

$$\Rightarrow L\left[\int_0^t f(u)\,du\right] = \frac{1}{s}F(s)$$

Example 15: Find the Laplace transform of $\frac{\cos 2t - \cos 3t}{t}$

Solution:

Let, $\frac{f(t)}{t} = \frac{\cos 2t - \cos 3t}{t}$

We have

$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(\bar{s})\,d(\bar{s})$ and $L[\cos at] = \frac{s}{s^2 + a^2}$

Now,

$$\begin{aligned} L\left\{\frac{\cos 2t - \cos 3t}{t}\right\} &= \int_s^\infty \left(\frac{\bar{s}}{(\bar{s})^2 + 4} - \frac{\bar{s}}{(\bar{s})^2 + 9}\right)d(\bar{s}) \\ &= \left[\frac{1}{2}\log\left((\bar{s})^2 + 4\right) - \frac{1}{2}\log\left((\bar{s})^2 + 9\right)\right]_s^\infty \\ &= \frac{1}{2}\log\left(\frac{(\bar{s})^2 + 4}{(\bar{s})^2 + 9}\right)_s^\infty \\ &= \frac{1}{2}\log\left(\frac{1 + 4/(\bar{s})^2}{1 + 9/(\bar{s})^2}\right)_s^\infty \end{aligned}$$

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$$= -\frac{1}{2}\log\left(\frac{1+4/s^2}{1+9/s^2}\right)$$

$$= -\frac{1}{2}\log\left(\frac{s^2+4}{s^2+9}\right)$$

$$= \frac{1}{2}\log\left(\frac{s^2+9}{s^2+4}\right)$$

Thus, $L\left\{\frac{\cos 2t - \cos 3t}{t}\right\} = \frac{1}{2}\log\left(\frac{s^2+9}{s^2+4}\right)$.

Example 16: Find the Laplace transform of $\frac{e^{-t}\sin t}{t}$

Solution:

Let, $\frac{f(t)}{t} = \frac{e^{-t}\sin t}{t}$

We have

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(\bar{s})\, d(\bar{s}) \quad \text{and} \quad L\{\sin at\} = \frac{a}{s^2+a^2}$$

Now,

$$L\left\{\frac{e^{-t}\sin t}{t}\right\} = \left[L\left\{\frac{\sin t}{t}\right\}\right]_{s\to s+1} = \left[\int_s^\infty \left(\frac{1}{\bar{s}^2+1}\right) d(\bar{s})\right]_{s\to s+1}$$

$$= \left[\left(\tan^{-1}(\bar{s})\right)_s^\infty\right]_{s\to s+1}$$

$$= \left[\frac{\pi}{2} - (\tan^{-1}(s))\right]_{s\to s+1}$$

$$= \frac{\pi}{2} - \tan^{-1}(s+1)$$

Thus, $L\left\{\frac{e^{-t}\sin t}{t}\right\} = \frac{\pi}{2} - \tan^{-1}(s+1)$.

3.8 Convolution on Laplace Transform

The convolution of two functions f(t) and g(t) for $t \geq 0$ is denoted by f*g and is defined as

$$f(t)*g(t) = \int_0^t f(T)\, g(t-T)\, dT = \int_0^t f(t-T)\, g(T)\, dT$$

Theorem 3: (Convolution Theorem for Laplace Transform)

Let f(t) and g(t) are piecewise continuous function in any finite interval for $t \geq 0$. Also, let $L\{f(t)\} = F(s)$ and $L\{g(t)\} = G(s)$. Then,

$$L\{f(t)*g(t)\} = F(s)\, G(s).$$

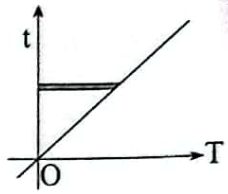
Proof:

Let $L\{f(t)\} = F(s)$ and $L\{g(t)\} = G(s)$.

Now,

$$L\{(f*g)(t)\} = \int_0^t (f*g)(t)\, e^{-st}\, dt$$

$$= \int_0^\infty \int_0^t f(T)\, g(t-T)\, e^{-st}\, dT\, dt$$

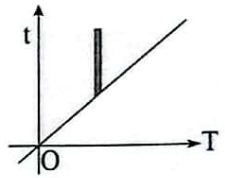


Now, reversing the order of integration we get

$$= \int_0^\infty \int_T^\infty f(T)\, g(t-T)\, e^{-st}\, dt\, dT$$

Put $t - T = x$ then $dt = dx$. Also,

$t = T \Rightarrow x = 0$, $t \to \infty \Rightarrow x \to \infty$.



Then,

$$L\{(f*g)(t)\} = \int_0^\infty \int_0^\infty f(T)\, g(x)\, e^{-s(x+T)}\, dx\, dT$$

$$= \int_0^\infty f(T)\, e^{-sT}\, dT \int_0^\infty g(x)\, e^{-sx}\, dx$$

$$= F(s)\, G(s).$$

Thus, $L\{(f*g)(t)\} = F(s)\, G(s)$.

List of Table

S.No.	f(t)	L{f(t)} = F(s)	S.N.	f(t)	L{f(t)} = F(s)
1.	1	$\frac{1}{s}$	6.	cos at	$\frac{s}{s^2+a^2}$
2.	t	$\frac{1}{s^2}$	7.	sin at	$\frac{a}{s^2+a^2}$
3.	t^n	$\frac{n!}{s^{n+1}}$	8.	cosh at	$\frac{s}{s^2-a^2}$
4.	e^{at}	$\frac{1}{s-a}$	9.	sinh at	$\frac{a}{s^2-a^2}$
5.	e^{-at}	$\frac{1}{s+a}$			

Example 17: Find the convolution of the function, $e^t * e^{-t}$.

Solution:

Let

$$f(t) = e^t, \quad g(t) = e^{-t}$$

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Then the convolution of f(t) and g(t) is

$$e^t * e^{-t} = f(t) * g(t) = \int_0^t f(t-T)\, g(T)\, dT$$

$$= \int_0^t e^{t-T} e^{-T}\, dT$$

$$= \int_0^t e^{t-2T}\, dT$$

$$= e^t \int_0^t e^{-2T}\, dT$$

$$= e^t \left[\frac{e^{-2T}}{-2}\right]_0^t$$

$$= e^t \left[\frac{e^{-2t}-1}{-2}\right]$$

$$= \frac{e^t - e^{-t}}{2}$$

$$= \sinh t$$

Example 18: Find the convolution of the function, sinwt * coswt

Solution:

Let

$$f(t) = \sin wt, \quad g(t) = \cos wt$$

Then the convolution of f(t) and g(t) is

$$\sin wt * \cos wt = f(t) * g(t) = \int_0^t f(t-T)\, g(T)\, dT$$

$$= \int_0^t \sin w(t-T) \cos wT\, dT$$

$$= \int_0^t (\sin wt \cos wT - \cos wt \sin wT) \cos wT\, dT$$

$$= \sin wt \int_0^t \cos^2 wT\, dT - \cos wt \int_0^t \sin wT \cos wT\, dT$$

$$= \sin wt \int_0^t \left(\frac{1+\cos 2wT}{2}\right) dT - \frac{\cos wt}{2} \int_0^t \sin 2wT\, dT$$

$$= \frac{\sin wt}{2}\left[T + \frac{\sin 2wT}{2w}\right]_0^t - \frac{\cos wt}{2}\left[\frac{-\cos 2wT}{2w}\right]_0^t$$

$$= \frac{\sin wt}{2}\left(t + \frac{\sin 2wt}{2w}\right) + \frac{\cos wt}{2}\left(\frac{\cos 2wt - 1}{2w}\right)$$

$$= \frac{t \sin wt}{2} - \frac{\cos wt}{4w} + \frac{\sin wt \sin 2wt + \cos wt \cos 2wt}{4w}$$

$$= \frac{t \sin wt}{2} - \frac{\cos wt}{4w} + \frac{\cos(2wt - wt)}{4w}$$

$$= \frac{t \sin wt}{2} - \frac{\cos wt}{4w} + \frac{\cos wt}{4w}$$

$$= \frac{t \sin wt}{2}$$

Example 19: Find the Laplace transform of convolution, $u(t-3) * e^{-2t}$.

Solution:

Let

$$f(t) = u(t-3), \quad g(t) = e^{-2t}$$

Then the convolution of f(t) and g(t) is

$$u(t-3) * e^{-2t} = f(t) * g(t) = \int_0^t f(T)\, g(t-T)\, dT$$

$$= \int_0^t u(t-3)\, e^{-2(t-T)}\, dT$$

$$= e^{-2t} \int_3^t e^{2T}\, dT = e^{-2t}\left[\frac{e^{2T}}{2}\right]_3^t = e^{-2t}\left[\frac{e^{2t} - e^6}{2}\right] = \frac{1 - e^{2(3-t)}}{2}$$

Now,

$$L\{u(t-3) * e^{-2t}\} = L\left\{\frac{1 - e^{2(3-t)}}{2}\right\} = L\left\{\frac{1 - e^6 e^{-2t}}{2}\right\}$$

$$= \frac{1}{2}\left[L\{1\} - e^6 L\{e^{-2t}\}\right] = \frac{1}{2}\left[\frac{1}{s} - e^6 \times \frac{1}{s+2}\right] = \left[\frac{s + 2 - se^6}{2s(s+2)}\right]$$

$$= \frac{s(1-e^6) + 2}{2s(s+2)}$$

Example 20: Find the Laplace transform of $f(t) = \begin{cases} 10\cos \pi t & \text{for } 0 < t \le 2 \\ 0 & \text{for other wise} \end{cases}$

Solution:

Let

$$f(t) = \begin{cases} 10\cos \pi t & \text{for } 0 < t \le 2 \\ 0 & \text{for other wise} \end{cases}$$

Now, Laplace transform of f(t) is

$$L\{f(t)\} = \int_0^\infty f(t)\, e^{-st}\, dt$$

$$= \int_0^2 10 \cos \pi t\, e^{-st}\, dt + 0$$

$$= 10\left[\frac{e^{-st}}{(-s)^2 + \pi^2}\{(-s)\cos \pi t + \pi \sin \pi t\}\right]_0^2$$

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$$= \frac{10}{s^2+\pi^2}\left[e^{-st}\{(-s)\cos\pi t + \pi\sin\pi t\}\right]_0^2$$

$$= \frac{10}{s^2+\pi^2}\left[e^{-2s}\{-s\cos 2\pi + \pi\sin 2\pi\} - e^0\{-s\cos 0 + \pi\sin 0\}\right]$$

$$= \frac{10}{s^2+\pi^2}\left[e^{-2s}\{-s+\pi\times 0\} - \{-s+\pi\times 0\}\right] = \frac{10}{s^2+\pi^2}(s - se^{-2s})$$

Thus, $L\{f(t)\} = \frac{10}{s^2+\pi^2}(s - se^{-2s})$.

Exercise 3 (A)

A. Find the Laplace transform of the following functions $f(t)$ where a, b, T, w, θ are constants:

1. $3t+4$ 2. t^2+at+b 3. $\sin\pi t$ 4. $\sin(wt+\theta)$
5. $\cos(wt+\theta)$ 6. e^{at+b} for $s > a$ 7. $4e^{5t} + 6t^2 - 4\cos 3t + 3\sin 4t$
8. $\cos^2 t$ 9. $\sin^2 t$

B. Find the Laplace transform of the following:

1. $e^{-2t}\cos t$ 2. $e^{-2t}\sin 4t$ 3. $e^{-2t}\sin n\pi t$ 4. $e^{at}\cos wt$
5. $e^{at}\sin wt$ 6. $e^{-at}(A\cos\theta t + B\sin\theta t)$
7. $e^{-3t}(2\cos 5t - 3\sin 5t)$ 8. $5e^{2t}\sinh 2t$ 9. $\sin at\sin bt$
10. $\sinh t\cos t$ 11. $\cosh at\cos at$ 12. $\sinh at\cos bt$

C. Find the Laplace transform of the following:

1. $e^{-3t}u_2(t)$ 2. $\sin 2t\, u_\pi(t)$ 3. $(t-1)u(t-1)$ 4. $t^2u(t-1)$
5. $4u(t-\pi)\cos t$ 6. $\cos(t-2)u_2(t)$

D. Find the Laplace transform of the functions

1. $t\cos 2t$ 2. $t\cos wt$ 3. $t\sin wt$
4. $\frac{1}{2w^2}(\sin wt - wt\cos wt)$ 5. $\frac{t\sin wt}{2w}$ 6. $t\cosh at$
7. $t\sinh at$ 8. $t\cosh t$ 9. $t\sinh 2t$ 10. $t^2\cos wt$
11. te^t 12. t^2e^t 13. t^2e^{-3t} 14. t^3e^{-3t}

E. Find the Laplace transform of the following functions:

1. $te^{-2t}\sin wt$ 2. $te^{-t}\sin t$ 3. $t^2e^{-2t}\cos t$ 4. $te^{-t}\cosh t$

F. Find the Laplace transform of the following functions:

1. $\frac{e^{-at}-e^{-bt}}{t}$ 2. $\frac{\sin^2 t}{t}$ 3. $\frac{\sinh t}{t}$

G. Calculate the following convolutions:

1. $1 * 1$ 2. $t * e^t$ 3. $t * 1$ 4. $\sin t * \sin t$

H. Find the Laplace transform of the following piecewise continuous functions

1. $f(t) = \begin{cases}\sin wt & \text{for } 0<t<\pi/w \\ 0 & \text{for other wise}\end{cases}$
2. $f(t) = \begin{cases}e^t & \text{for } 0<x<1 \\ 0 & \text{for otherwise}\end{cases}$
3. $f(t) = \begin{cases}1 & \text{if } 0<t<\pi \\ 0 & \text{if } \pi<1<2\pi \\ \sin t & \text{if } t>2\pi\end{cases}$

Answer Key

A.
1. $\frac{4}{s}+\frac{3}{s^2}$ 2. $\frac{2}{s^3}+\frac{a}{s^2}+\frac{b}{s}$ 3. $\frac{\pi}{s^2+\pi^2}$ 4. $\frac{w\cos\theta + s\sin\theta}{s^2+w^2}$
5. $\frac{s\cos\theta - w\sin\theta}{s^2+w^2}$ 6. $\frac{e^b}{s-a}$ 7. $\frac{4}{s-5}+\frac{12}{s^3}-\frac{4s}{s^2+9}+\frac{12}{s^2+16}$
8. $\frac{1}{2}\left(\frac{1}{s}+\frac{s}{s^2+4}\right)$ 9. $\frac{2}{s(s^2+4)}$

B.
1. $\frac{s+2}{(s+2)^2+1}$ 2. $\frac{4}{s^2+4s+20}$ 3. $\frac{n\pi}{(s+2)^2+n^2\pi^2}$ 4. $\frac{s-a}{(s-a)^2+w^2}$
5. $\frac{w}{(s-a)^2+w^2}$ 6. $\frac{A(s+a)+B\theta}{(s+a^2)+\theta^2}$ 7. $\frac{2s-9}{s^2+6s+34}$ 8. $\frac{10}{(s-2)^2-4}$
9. $\frac{2abs}{[s^2+(a+b)^2][s^2+(a-b)^2]}$ 10. $\frac{s^2-2}{s^4+4}$
11. $\frac{1}{2}\left[\frac{s-a}{(s-a)^2+a^2}+\frac{s+a}{(s+a)^2+a^2}\right]$ 12. $\frac{1}{2}\left[\frac{s-a}{(s-a)^2+b^2}-\frac{s+a}{(s+a)^2+b^2}\right]$

C.
1. $\frac{e^{-2(s+3)}}{s+3}$ 2. $-\frac{2e^{-\pi s}}{s^2+4}$ 3. $\frac{e^{-s}}{s^2}$ 4. $e^{-s}\left(\frac{2}{s^3}+\frac{2}{s^2}+\frac{1}{s}\right)$
5. $\frac{-4se^{-\pi s}}{s^2+1}$ 6. $\frac{se^{-2s}}{s^2+1}$

D.
1. $\frac{s^2-4}{(s^2+4)^2}$ 2. $\frac{s^2-w^2}{(s^2+w^2)^2}$ 3. $\frac{2ws}{(s^2+w^2)^2}$ 4. $\frac{1}{(s^2+w^2)^2}$
5. $\frac{s}{(s^2+w^2)^2}$ 6. $\frac{s^2+a^2}{(s^2-a^2)^2}$ 7. $\frac{2as}{(s^2-a^2)^2}$ 8. $\frac{s^2+1}{(s^2-1)^2}$
9. $\frac{4s}{(s^2-4)^2}$ 10. $\frac{2s(w^2-3s^2)}{(s^2+w^2)^3}$ 11. $\frac{1}{(s-1)^2}$ 12. $\frac{2}{(s-1)^3}$
13. $\frac{2}{(s+3)^3}$ 14. $\frac{6}{(s+3)^4}$

E.
1. $\frac{2w(s+2)}{((s+2)^2+w^2)^2}$ 2. $\frac{2(s+2)}{((s+2)^2+1)^2}$ 3. $\frac{2(s+2)\{1-3(s+2)^2\}}{\{(s+2)^2+1\}^3}$
4. $\frac{s^2+2s+2}{s^2(s+2)^2}$

F.
1. $\log\left(\frac{s+b}{s+a}\right)$ 2. $\frac{1}{4}\log\left(\frac{s^2+4}{s^2}\right)$ 3. $\frac{1}{2}\log\left(\frac{s+1}{s-1}\right)$

G.
1. t 2. $e^t - t - 1$ 3. $\frac{t^2}{2}$ 4. $-\frac{t\cos t}{2}+\frac{\sin t}{2}$

H.
1. $\frac{w}{s^2+w^2}(1+e^{-\pi s/w})$ 2. $\left(\frac{1-e^{1-s}}{s-1}\right)$ 3. $\frac{1-e^{-\pi s}}{s}+\frac{e^{-2\pi s}}{s^2+1}$

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3.9 INVERSE LAPLACE TRANSFORM

As stated earlier, we noted the Laplace transform of f(t) is F(s). Then the process to finding the value of f(t) with the help of given F(s), is called the inverse Laplace transform of f(t) and we written the as $L^{-1}\{F(s)\} = f(t)$.

That is, if the Laplace transform of f(t) is

$$F(s) = L\{f(t)\} = L[f(t); s] = \int_0^\infty e^{-st} f(t)\, dt.$$

Then the inverse Laplace transform of f(t) is

$$L^{-1}\{F(s)\} = \int_0^\infty e^{st} F(s)\, ds.$$

List of Some Formulae on Inverse Laplace Transform

The following result shows the relation of Laplace Transform and its Inverse form.

1. If $L\{1\} = \frac{1}{s}$ then $L^{-1}\left\{\frac{1}{s}\right\} = 1$
2. If $L\{e^{at}\} = \frac{1}{s-a}$ then $L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$
3. If $L\{\cos wt\} = \frac{s}{s^2+w^2}$ then $L^{-1}\left\{\frac{s}{s^2+w^2}\right\} = \cos wt$
4. If $L(\sin wt) = \frac{w}{s^2-w^2}$ then $L^{-1}\left\{\frac{w}{s^2-w^2}\right\} = \sin wt$
5. If $L(t^n) = \frac{n!}{s^{n+1}}$ then $L^{-1}\left\{\frac{n!}{s^{n+1}}\right\} = t^n$

Here, the lists the inverses Laplace transform of some standard function are

1. $L^{-1}\left\{\frac{1}{s}\right\} = 1$
2. $L^{-1}\left\{\frac{1}{s^2}\right\} = t$
3. $L^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}$
4. $L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$
5. $L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$
6. $L^{-1}\left\{\frac{a}{s^2+a^2}\right\} = \sin at$
7. $L^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos at$
8. $L^{-1}\left\{\frac{a}{s^2-a^2}\right\} = \sinh at$
9. $L^{-1}\left\{\frac{s}{s^2-a^2}\right\} = \cosh at$
10. $L^{-1}\{F(s-a)\} = \left[L^{-1}\{F(s)\}\right]_{s\to s-a} = e^{at} L^{-1}\{F(s)\} = e^{at} f(t)$
11. $L^{-1}\{e^{-as} F(s)\} = f(t-a)\, u(t-a)$
12. $L^{-1}\left\{\frac{d}{ds} F(s)\right\} = L^{-1}\{F'(s)\} = -t\, L^{-1}\{F(s)\} = -t\, f(t)$

13. $L^{-1}\left\{\frac{d^n}{ds^n} F(s)\right\} = L^{-1}\{F^n(s)\} = (-1)^n t^n f(t)$
14. $L^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t L^{-1}\{F(s)\}\, ds$
15. $L^{-1}\{s\, F(s)\} = \frac{d}{dt}\left(L^{-1}\{F(s)\}\right)$ as $f(0) = 0$
16. $L^{-1}\left[\int_s^\infty F(s)\, ds\right] = \frac{f(t)}{t} = \frac{1}{t} L^{-1}\{F(s)\}$
17. $L^{-1}\{F(s)\, G(s)\} = f(t) * g(t)$ where

$$f(t) * g(t) = \int_0^t f(T)\, g(t-T)\, dT = \int_0^t f(t-T)\, g(T)\, dT$$

Workout Examples

Example 1: Find inverse Laplace transform of $\frac{7}{s+3}$.

Solution:

Let,

$$F(s) = L\{f(t)\} = \frac{7}{s+3} = 7\left(\frac{1}{s}\right)_{s\to s+3} = 7(L\{1\})_{s\to s+3} = 7L\{e^{-3t}\} = L\{7e^{-3t}\}$$

Now, taking inverse Laplace transform on both sides then

$$f(t) = 7e^{-3t}$$

Example 2: If $L\{f(t)\} = \frac{1}{s^2+36}$ then find f(t).

Solution:

Let,

$$L\{f(t)\} = \frac{1}{s^2+36} = \frac{1}{6}\left\{\frac{6}{(s^2)+(6)^2}\right\}$$

Now, taking inverse Laplace transform on both sides we get

$$f(t) = \frac{1}{6} L^{-1}\left\{\frac{6}{(s^2)+(6)^2}\right\} = \frac{1}{6}\sin 6t \qquad \left[\because L\{\sin at\} = \frac{a}{s^2+a^2}\right]$$

Example 3: Find the inverse Laplace transform of $\frac{1}{s^4}$.

Solution:

Let,

$$L\{f(t)\} = \frac{1}{s^4}$$

Now, taking inverse Laplace transform on both sides then

$$f(t) = L^{-1}\left(\frac{1}{s^4}\right) = \frac{t^3}{3!} = \frac{t^3}{6} \qquad \left[\because L\{t^n\} = \frac{n!}{s^{n+1}} \text{ i.e. } L^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}\right]$$

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Example 4: Find the inverse Laplace transform of $\frac{s}{a^2s^2 + n^2\pi^2}$.

Solution:

Let,

$$L\{f(t)\} = \frac{s}{a^2s^2 + n^2\pi^2}$$

$$= \frac{1}{a^2}\left(\frac{s}{s^2 + (n\pi/a)^2}\right)$$

$$= \frac{1}{a^2} L\left\{\cos\frac{n\pi t}{a}\right\}$$

Now, taking inverse Laplace transform on both sides we get

$$f(t) = \frac{1}{a^2}\cos\frac{n\pi t}{a}$$

Example 5: Find f(t), where $L\{f(t)\} = \frac{8}{s^4 - 4s^2}$.

Solution:

Let,

$$L\{f(t)\} = \frac{8}{s^4 - 4s^2} \quad \ldots (i)$$

Let,

$$\frac{8}{s^4 - 4s^2} = \frac{8}{s^2(s^2 - 4)} = \frac{As + B}{s^2} + \frac{Cs + D}{s^2 - 4} \quad \ldots (ii)$$

$$\Rightarrow \frac{8}{s^4 - 4s^2} = \frac{(As + B)(s^2 - 4) + (Cs + d)s^2}{s^2(s^2 - 4)}$$

This gives,

$$8 = As^3 - 4As + Bs^2 - 4B + Cs^3 + Ds^2$$

$$\Rightarrow 8 = s^3(A + C) + s^2(B + D) - 4As - 4B$$

Equating the coefficient of like term on both sides then we get

$$A + C = 0, \quad B + D = 0,$$

$$-4A = 0, \quad -4B = 0$$

Solving we get,

$$A = 0, B = -2, C = 0, D = 2$$

Then, the equation (i) and (ii) gives,

$$L\{f(t)\} = \left(\frac{-2}{s^2} + \frac{2}{s^2 - 4}\right) = -2L\{t\} + L\{\sinh 2t\} = L\{-2t + \sinh 2t\}$$

Now, taking inverse Laplace transform we get

$$f(t) = \sinh 2t - 2t.$$

Example 6: Find f(t), where $L\{f(t)\} = \frac{n\pi}{(s + 2)^2 + n^2\pi^2}$.

Solution:

Let,

$$L\{f(t)\} = \frac{n\pi}{(s + 2)^2 + n^2\pi^2}$$

$$= \left(\frac{n\pi}{s^2 + n^2\pi^2}\right)_{s \to s+2}$$

$$= (L(\sin n\pi t))_{s \to s+2}$$

$$= L\{e^{-2t}\sin n\pi t\} \qquad [\because L\{e^{at} f(t)\} = (L\{f(t)\})_{s \to s-a}]$$

Now, taking inverse Laplace transform, we get,

$$f(t) = e^{-2t}\sin n\pi t.$$

Example 7: Using second shifting property, find f(t) where $L\{f(t)\} = \frac{e^{-\pi s}}{s^2 + 1}$.

Solution:

Let,

$$L\{f(t)\} = \frac{e^{-\pi s}}{s^2 + 1} = e^{-\pi s} L\{\sin t\} = L\{\sin(t - \pi)\, u_\pi(t)\}$$

$$[\because L\{f(t - a)\, u_a(t)\} = e^{-as} L\{f(t)\}]$$

Now, taking inverse Laplace transform we get

$$f(t) = \sin(t - \pi)\, u_\pi(t) = \sin(-(\pi - t))\, u_\pi(t)$$

$$= -\sin(\pi - t)\, u_\pi(t)$$

$$= -\sin t\, u_\pi(t)$$

Example 8: Find f(t), where $L\{f(t)\} = \frac{2s}{(s^2 - 4)^2}$.

Solution:

Since we have,

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(\bar{s})\, d\bar{s} = 2\int_s^\infty \frac{\bar{s}}{(\bar{s}^2 - 4)^2} d\bar{s}$$

Set, $\bar{s}^2 - 4 = u$ then $2\bar{s}\, d\bar{s} = du$.

Also, $\bar{s} = s \Rightarrow u = s^2 - 4$ and $\bar{s} \to \infty \Rightarrow u \to \infty$.

Now,

$$L\left\{\frac{f(t)}{t}\right\} = \int_{s^2-4}^\infty \frac{du}{u^2} = \left[-\frac{1}{u}\right]_{s^2-4}^\infty = \frac{1}{s^2 - 4}$$

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Taking inverse Laplace transform then,

$$\frac{f(t)}{t} = L^{-1}\left\{\frac{1}{s^2-4}\right\} = \frac{1}{2}L^{-1}\left\{\frac{2}{s^2-4}\right\} = \frac{1}{2}\sinh 2t$$

$$\Rightarrow f(t) = \frac{t}{2}\sinh 2t.$$

Example 9: Find inverse Laplace transform of $\log\left[1+\frac{w^2}{s^2}\right]$.

Solution:

Let,

$$L\{f(t)\} = F(s) = \log\left(1+\frac{w^2}{s^2}\right) = \log\left(\frac{s^2+w^2}{s^2}\right) = \log(s^2+w^2) - 2\log(s)$$

Then,

$$F'(s) = \frac{1}{s^2+w^2}(2s) - 2\frac{1}{s}$$

$$\Rightarrow -L\{t\,f(t)\} = 2\left(\frac{s}{s^2+w^2} - \frac{1}{s}\right) = 2\,L\{\cos wt - 1\}$$

$$\Rightarrow L\{t\,f(t)\} = L\{2(1-\cos wt)\}$$

Now taking inverse Laplace transform, we get,

$$t\,f(t) = 2(1-\cos wt)$$

$$\Rightarrow f(t) = \frac{2(1-\cos wt)}{t}.$$

Example 10: Find inverse Laplace transform of $\cot^{-1}\left(\frac{s}{w}\right)$.

Solution:

Let,

$$L\{f(t)\} = F(s) = \cot^{-1}\left(\frac{s}{w}\right)$$

Then,

$$F'(s) = -\frac{1}{\left(\frac{s^2}{w^2}+1\right)}\times\frac{1}{w} = -\frac{w}{s^2+w^2} = -L\{\sin wt\}$$

$$\Rightarrow -L[t\,f(t)] = L\{-\sin wt\} \quad \left[\because L\{t\,f(t)\} = -\frac{d}{ds}(L\{f(t)\}) = -F'(s)\right]$$

Now, taking inverse Laplace transform we get,

$$t\,f(t) = \sin wt$$

$$\Rightarrow f(t) = \frac{\sin wt}{t}$$

Example 11: Find the inverse Laplace transform by using convolution theorem

a. $\frac{1}{s(s^2+4)}$ b. $\frac{1}{(s^2+1)^2}$ c. $\frac{w}{s^2(s^2+w^2)}$

Solution:

a. Let

$$\frac{1}{s(s^2+4)} = \frac{1}{s}\times\frac{1}{s^2+4} = L\{1\} * \frac{1}{2}L\{\sin 2t\} = L\left\{1 * \frac{\sin 2t}{2}\right\}$$

Now, taking inverse Laplace transform we get,

$$L^{-1}\left\{\frac{1}{s(s^2+4)}\right\} = 1 * \frac{\sin 2t}{2}$$

$$= \int_0^t 1\times\frac{\sin 2u}{2}\,du$$

$$= \frac{1}{2}\left[-\frac{\cos 2u}{2}\right]_0^t$$

$$= \frac{1-\cos 2t}{4}$$

b. Let

$$\frac{1}{(s^2+1)^2} = \frac{1}{(s^2+1)}\times\frac{1}{(s^2+1)} = L\{\sin t\}\,L\{\sin t\} = L\{\sin t * \sin t\} \quad \ldots (1)$$

Now, taking inverse Laplace transform we get

$$L^{-1}\left\{\frac{1}{(s^2+1)^2}\right\} = \sin t * \sin t = \int_0^t \sin(t-u)\sin u\,du$$

$$= \int_0^t (\sin t\cos u - \cos t\sin u)\sin u\,du$$

$$= \sin t\int_0^t \cos u\sin u\,du - \cos t\int_0^t \sin 2u\,du$$

$$= \frac{\sin t}{2}\int_0^t \sin 2u\,du - \frac{\cos t}{2}\int_0^t (1-\cos 2u)\,du$$

$$= \frac{\sin t}{2}\left[\frac{-\cos 2u}{2}\right]_0^t - \frac{\cos t}{2}\left[u - \frac{\sin 2u}{2}\right]_0^t$$

$$= \frac{\sin t}{4}(1-\cos 2t) - \frac{\cos t}{2}\left(t - \frac{\sin 2t}{2}\right)$$

$$= \frac{\sin t}{4} - \frac{t\cos t}{2} + \frac{1}{4}(-\sin t\cos 2t + \cos t\sin 2t)$$

$$= \frac{\sin t}{4} - \frac{t\cos t}{2} + \frac{1}{4}\sin(2t-t)$$

$$= \frac{\sin t}{4} - \frac{t\cos t}{2} + \frac{\sin t}{4}$$

$$= -\frac{t\cos t}{2} + \frac{\sin t}{2}.$$

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c. Let $\frac{w}{s^2(s^2+w^2)} = \frac{1}{s^2}\cdot\frac{w}{s^2+w^2} = L\{t\}\,L\{\sin wt\} = L\{t * \sin wt\}$

Now taking inverse Laplace transform we get,

$$L^{-1}\left\{\frac{w}{s^2(s^2+w^2)}\right\} = t * \sin wt = \int_0^t (t-u)\sin wu\,du$$

$$= \left[(t-u)\left(-\frac{\cos wu}{w}\right) - (-1)\left(-\frac{\sin wu}{w^2}\right)\right]_0^t$$

[$\because$ applying successive integration]

$$= \left(0 - \frac{\sin wt}{w^2}\right) - \left(-\frac{t}{w} - 0\right)$$

$$= \frac{wt - \sin wt}{w^2}$$

Thus, $L^{-1}\left\{\frac{w}{s^2(s^2+w^2)}\right\} = \frac{wt - \sin wt}{w^2}$.

Exercise 3 (B)

1. Find f(t), it L{f(t)} equals the following:

a. $\frac{2s+6}{s^2+4}$ b. $\frac{4}{(s+1)(s+2)}$ c. $\frac{s+1}{s^2+2s}$

d. $\frac{-s-10}{s^2-s-2}$ e. $\frac{1}{(s-a)(s-b)}\,(a \neq b)$ f. $\frac{1}{4s+s^3}$

g. $\frac{1}{s^2}\left(\frac{s+1}{s^2+1}\right)$ h. $\frac{1}{s^3-s}$

2. Find f(t) if L {f(t)} equals:

a. $\frac{s}{(s+3)^2+1}$ b. $\frac{1}{(s+1)^2}$ c. $\frac{3}{s^2+6s+18}$

d. $\frac{4}{s^2-2s-3}$ e. $\frac{e^{-as}}{s}$ f. $\frac{e^{-\pi s}}{s^2}$

g. $\frac{e^{-2s}}{s-3}$ h. $\frac{e^{-3s}}{s^3}$ i. $\frac{e^{-3s}}{(s-1)^3}$

j. $\frac{se^{-2s}}{s^2+1}$ k. $\frac{se^{-2s}}{s^2+\pi^2}$

3. Find the inverse Laplace transform where the Laplace transform of the function is given by

a. $\frac{s}{(s^2+1)^2}$ b. $\frac{s}{(s^2+a^2)^2}$ c. $\frac{1}{(s-3)^3}$

d. $\frac{s^2-\pi^2}{(s^2+\pi^2)^2}$ e. $\frac{s}{(s^2+4)^2}$

4. Find f(t), where L{f(t)}:

a. $\log\left(\frac{s}{s-1}\right)$ b. $\log\left(\frac{1+s}{s}\right)$ c. $\log\left(1+\frac{1}{s^2}\right)$

d. $\log\left(1-\frac{a^2}{s^2}\right)$ e. $\log\left(\frac{s+a}{s+b}\right)$ f. $\log\left[\frac{s^2+1}{(s-1)^2}\right]$

g. $\cot^{-1}(1+s)$ h. $\cot^{-1}\left(\frac{s}{2}\right)$ i. $\tan^{-1}(1+s)$

j. $\tan^{-1}\left(\frac{2}{s}\right)$

5. Calculate the following inverse transform by convolution:

a. $\frac{1}{s^3}$ b. $\frac{1}{s^2(s^2+w^2)}$ c. $\frac{5}{(s^2+\pi^2)^2}$.

Answer Key

1. a. $2\cos 2t + 3\sin 2t$ b. $4(e^{-t} - e^{-4t})$ c. $\frac{1+e^{-2t}}{2}$ d. $3e^{-t} - 4e^{2t}$
   e. $\frac{e^{at}-e^{bt}}{a-b}$ f. $\frac{1-\cos 2t}{4}$ g. $1 + t - \cos t - \sin t$
   h. $\cosh t - 1$
2. a. $e^{-3t}(\cos t - 3\sin t)$ b. $t\,e^{-t}$ c. $e^{-3t}\sin 3t$ d. $e^{-t} - e^{3t}$
   e. $u_a(t)$ f. $(t-\pi)\,u_\pi(t)$ g. $e^{3(t-2)}\,u_2(t)$ h. $\frac{1}{2}(t-3)^2\,u(t-3)$
   i. $\frac{1}{2}\{e^{(t-3)}(t-3)^2\,u_3(t)\}$ j. $\cos(t-2)\,u_2(t)$ k. $\cos \pi t\,u(t-2)$
3. a. $\frac{t}{2}\sin t$ b. $\frac{t\sin at}{2a}$ c. $\frac{t^2 e^{3t}}{2}$ d. $t\cos \pi t$
   e. $\frac{t\sin 2t}{4}$
4. a. $\frac{e^t-1}{t}$ b. $\frac{1-e^{-t}}{t}$ c. $\frac{2(1-\cos t)}{t}$ d. $\frac{2(1-\cosh at)}{t}$
   e. $\frac{e^{-bt}-e^{-at}}{t}$ f. $\frac{2(e^t-\cos t)}{t}$ g. $\frac{e^{-t}\sin t}{t}$ h. $\frac{\sin 2t}{t}$
   i. $\frac{-e^{-t}\sin t}{t}$ j. $\frac{\sin 2t}{t}$
5. a. $\frac{t^2}{2}$ b. $\frac{1}{w^3}(wt - \sin wt)$ c. $\frac{\sin \pi t}{2\pi} - \frac{t\cos \pi t}{2}$

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3.10 APPLICATION OF LAPLACE TRANSFORM TO DIFFERENTIAL EQUATION

Laplace transform often use to solve ordinary and partial differential equations. Using ... transform, we can transform the linear differential equation to algebraic ... Normally, the process using Laplace transform is easier and shorter than the ... method to finding the solution of the ordinary differential equations with ... condition(s).

Steps to Solve the ODE with Initial Conditions

1. First we take Laplace transform on both sides of the given differential equation
2. Use the formulae of transform of derivatives with given conditions.
3. Finally, we find the inverse Laplace transform $L^{-1}(y)$ to get the solution

Review of Laplace Transform of the Derivative of a Function

Let f(t) be continuous for all $t \geq 0$ and be of exponential order. If f '(t) is piecewise continuous on every finite interval in the range $t \geq 0$, then the Laplace transform of the derivative f '(t) exists when $s > k$, and

$$L\{f'(t)\} = s\,L\{f(t)\} - f(0)$$

Moreover,
$$L\{f''(t)\} = s^2\,L\{f(t)\} - s\,f(0) - f'(0)$$
$$L\{f'''(t)\} = s^3\,L\{f(t)\} - s^2 f(0) - s\,f'(0) - f''(0)$$

In general,

$$L\{f^n(t)\} = s^n\,L\{f(t)\} - s^{n-1} f(0) - s^{n-2} f'(0) - s^{n-3} f''(0) - \ldots - f^{n-1}(0)$$

Workout Examples

Example 1: Solve the initial value problem by using the Laplace transformation;
$4y'' + \pi^2 y = 0,\ y(0) = 2,\ y'(0) = 0.$

Solution:

Given that,

$$4y'' + \pi^2 y = 0 \quad \ldots (i)$$

with $y(0) = 2,\ y'(0) = 0 \quad \ldots (ii)$

Taking Laplace transform both sides

$$4L\{y''\} + \pi^2 L\{y\} = L\{0\}$$
$$\Rightarrow \quad 4\{s^2 L\{y\} - sy(0) - y'(0) + \pi^2 L\{y\} = 0$$
$$\Rightarrow \quad 4\{s^2 L\{y\} - s\times 2 - 0\} + \pi^2 L\{y\} = 0 \quad \text{[using (ii)]}$$
$$\Rightarrow \quad 4s^2 L\{y\} - 8s + \pi^2 L\{y\} = 0$$
$$\Rightarrow \quad L\{y\}(4s^2 + \pi^2) = 8s$$
$$\Rightarrow \quad L\{y\} = \frac{8s}{4s^2 + \pi^2} = 2\frac{s}{s^2 + (\pi/2)^2} = L\left\{2\cos\left(\frac{\pi t}{2}\right)\right\}$$

Now, taking inverse Laplace transform, we get

$$y = 2\cos\frac{\pi t}{2}$$

Example 2: By using the Laplace transform, solve; $y' + 3y = 10\sin t,\ y(0) = 0.$

Solution:

Given that,

$$y' + 3y = 10\sin t \quad \ldots (i)$$

with $y(0) = 0 \quad \ldots (ii)$

Taking Laplace transform on both sides of (i) then

$$L\{y'\} + 3\,L\{y\} = 10\,L\{\sin t\}$$
$$\Rightarrow \quad s\,L\{y\} + 3\,L\{y\} = 10\frac{1}{s^2+1}$$
$$\Rightarrow \quad L\{y\}(s+3) = \frac{10}{s^2+1}$$
$$\Rightarrow \quad L\{y\} = \frac{10}{(s^2+1)(s+3)} \quad \ldots (iii)$$

Let,

$$\frac{10}{(s^2+1)(s+3)} = \frac{As+B}{s^2+1} + \frac{C}{s+3} = \frac{(As+B)(s+3) + C(s^2+1)}{(s^2+1)(s+3)}$$

This implies,

$$10 = As^2 + 3As + Bs + 3B + Cs^2 + C$$
$$\Rightarrow \quad 10 = s^2(A + C) + s(3A + B) + (3B + C)$$

Equating coefficient of like term on both sides then we get,

$$A + C = 0, \quad 3A + B = 0, \quad 3B + C = 10$$

Solving these equations we get,

$$A = -1,\ B = 3,\ C = 1.$$

Then (iii) becomes

$$L\{y\} = \frac{-s+3}{s^2+1} + \frac{1}{s+3} = -\frac{s}{s^2+1} + 3\frac{1}{s^2+1} + \frac{1}{s+3}$$
$$= -L\{\cos t\} + 3\,L\{\sin t\} + L\{e^{-3t}\}$$
$$= L\{-\cos t + 3\sin t + e^{-3t}\}$$

Now, taking inverse Laplace transform we get

$$y = -\cos t + 3\sin t + e^{-3t}$$

Example 3: Solve; $y'' + 2y' - 3y = 6e^{-2t},\ y(0) = 2,\ y'(0) = -14$, by using the Laplace transform.

Solution:

Given that,

$$y'' + 2y' - 3y = 6e^{-2t} \quad \ldots (i)$$

with $y(0) = 2,\ y'(0) = -14 \quad \ldots (ii)$

Taking Laplace transform of (i) then,

$$L\{y''\} + 2L\{y'\} - 3L\{y\} = 6L\{e^{-2t}\}$$
$$\Rightarrow \quad [s^2\,L\{y\} - sy(0) - y'(0)] + 2[sL\{y\} - y(0)] - 3L\{y\} = 6 \cdot \frac{1}{s+2}$$
$$\Rightarrow \quad L\{y\}[s^2 + 2s - 3] - 2s + 14 - 4 = \frac{6}{s+2} \quad \text{[using (ii)]}$$

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$$\Rightarrow L\{y\}(s^2+2s-3) = \frac{6}{s+2} + 2s - 10$$

$$= \frac{6 + 2s^2 - 10s + 4s - 20}{s+2}$$

$$= \frac{2s^2 - 6s - 14}{s+2}$$

$$\Rightarrow L\{y\} = \frac{2(s^2 - 3s - 14)}{(s+2)(s^2+2s-3)} \quad \ldots \text{(iii)}$$

Here, $\frac{s^2 - 3s - 7}{(s+2)(s^2+2s}$

$$= \frac{s^2 - 3s - 7}{(s+2)(s+3)(s-1)} = \frac{A}{s+2} + \frac{B}{s+3} + \frac{C}{s-1}$$

This gives,

$$s^2 - 3s - 7 = A(s+3)(s-1) + B(s+2)(s-1) + C(s+2)(s-3)$$
$$= A(s^2 + 2s - 3) + B(s^2 + s - 2) + C(s^2 - S - 6)$$
$$= (A + B + C)s^2 + (2A + B - C)s + (-3A - 2B - 6C)$$

Comparing the like terms from both sides we get,

$$A + B + C = 1, \quad 2A + B - C = -3, \quad -3A - 2B - 6C = -7$$

Solving we get,

$$A = -1, \quad B = \frac{1}{2}, \quad C = \frac{3}{2}$$

Then (iii) becomes

$$L\{y\} = -\frac{1}{s+2} + \frac{1}{2}\frac{1}{s+3} + \frac{3}{2}\frac{1}{s-1}$$

$$= -L\{e^{-2t}\} + \frac{1}{2}L\{e^{-3t}\} + \frac{3}{2}L\{e^{t}\}$$

$$= L\left\{\frac{e^{-3t} - 2e^{-2t} + e^{t}}{2}\right\}$$

Now, taking inverse Laplace transform we get

$$y = \frac{e^{-3t} - 2e^{-2t} + e^{t}}{2}$$

Example 4: Solve the initial value problem by using the Laplace transform;

$$y'' - 2y' + 10y = 0,\ y(0) = 3,\ y'(0) = 3$$

Solution:

Given that,

$$y'' - 2y' + 10y = 0 \quad \ldots \text{(i)}$$

with $y(0) = 3,\ y'(0) = 3 \quad \ldots \text{(ii)}$

Taking Laplace transform both side of (i) then,

$$L\{y''\} - 2L\{y'\} + 10L\{y\} = L\{0\}$$
$$\Rightarrow [s^2 L\{y\} - sy(0) - y'(0)] - 2[sL\{y\} + y(0)] + 10L\{y\} = 0$$
$$\Rightarrow s^2 L\{y\} - 3s - 3 - 2sL\{y\} + 6 + 10L\{y\} = 0 \quad \text{[using (ii)]}$$

$$\Rightarrow L\{y\}(s^2 - 2s + 10) = 3s - 3$$

$$\Rightarrow L\{y\} = \frac{3s-3}{s^2-2s+10} = \frac{3s-3}{(s-1)^2+3^2}$$

$$= \left(\frac{3s}{s^2+3^2}\right)_{s \to s-1}$$

$$= (3\,L\{\cos 3t\})_{s \to s-1}$$

$$= L\{3e^{t}\cos 3t\}$$

Now, taking inverse Laplace transform we get

$$y = 3e^{t}\cos 3t.$$

Example 5: By using the Laplace transform, solve; $x'' + 2x' + 5x = e^{-t}\sin t,\ x(0) = 0,\ x'(0) = 1.$

Solution:

Given that,

$$x'' + 2x' + 5x = e^{-t}\sin t \quad \ldots \text{(i)}$$

$$x(0) = x'(0) = 1 \quad \ldots \text{(ii)}$$

Taking Laplace transform both side of (i) then,

$$L\{x''\} + 2L\{x'\} + 5L\{x\} = L\{e^{-t}\sin t\}$$

$$\Rightarrow s^2 L\{x\} - sx(0) - x'(0) + 2sL\{x\} - 2x(0) + 5L\{x\} = [L\{\sin t\}]_{s \to s+1}$$

$$\Rightarrow s^2 L\{x\} - 0 - 1 + 2sL\{x\} - 0 + 5L\{x\} = \frac{1}{(s+1)^2+1}$$

$$\Rightarrow L\{x\}(s^2 + 2s + 5) = \frac{1}{s^2+2s+2} + 1$$

$$= \frac{1 + s^2 + 2s + 2}{(s^2 + 2s + 2)}$$

$$\Rightarrow L\{x\} = \frac{s^2 + 2s + 3}{(s^2+2s+2)(s^2+2s+5)} \quad \ldots \text{(iii)}$$

Here,

$$\frac{s^2 + 2s + 3}{(s^2+2s+2)(s^2+2s+5)} = \frac{As + B}{(s^2+2s+2)} + \frac{Cs + D}{(s^2+2s+5)}$$

$$= \frac{(As + B)(s^2 + 2s + 5) + (Cs + D)(s^2 + 2s + 2)}{(s^2+2s+2)(s^2+2s+5)}$$

$$\Rightarrow s^2 + 2s + 3 = As^3 + 2As^2 + 5As + Bs^2 + 2Bs + 5B + Cs^3 + 2Cs^2 + 2Cs + Ds^2 + 2Ds + 2D$$

$$= s^2(A + C) + s^2(2A + B + 2C + D) + s(5A + 2B + 2C + 2D) + (5B + 2D)$$

Equating coefficient of s and the constant term on both sides then we get,

$$A + C = 0, \quad 2A + B + 2C + D = 1,$$

$$5A + 2B + 2C + 2D = 2, \quad 5B + 2D = 3$$

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Solving we get,

$$A = 0, B = \frac{1}{3}, C = 0, D = \frac{2}{3}$$

Then (iii) becomes,

$$L\{x\} = \frac{1/3}{(s^2+2s+2)} + \frac{2/3}{(s^2+2s+5)}$$

$$= \frac{1/3}{(s+1)^2+1} + \frac{2/3}{(s+1)^2+4}$$

$$= \left(\frac{1/3}{s^2+1}\right)_{s\to s+1} + \left(\frac{2/3}{s^2+4}\right)_{s\to s+1}$$

$$= \frac{1}{3}(L\{\sin t\})_{s\to s+1} + \frac{1}{3}(L\{\sin 2t\})_{s\to s+1}$$

$$= L\left\{\frac{e^{-t}(\sin t + \sin 2t)}{3}\right\}$$

Now, taking inverse Laplace transform we get

$$x = \frac{e^{-t}(\sin t + \sin 2t)}{3}.$$

Example 6: By using the Laplace transform, solve the initial value problem;
$(D^2 - 2D + 2)x = 0, x(0) = 1, x'(0) = 1.$

Solution:

Given that,

$$(D^2 - 2D + 2)x = 0$$

i.e. $$\left(\frac{d^2}{dt^2} - 2\frac{d}{dt} + 2\right)x = 0$$

$$\Rightarrow \frac{d^2x}{dt^2} - 2\frac{dx}{dt} + 2x = 0$$

$$\Rightarrow x'' - 2x' + 2x = 0 \quad \ldots (i)$$

Satisfying the condition, $x(0) = 1, x'(0) = 1$ ... (ii)

Taking Laplace transform both side of (i) then,

$$L\{x''\} - 2L\{x'\} + 2L\{x\} = L\{0\}$$

$$\Rightarrow s^2L\{x\} - sx(0) - x'(0) - 2sL\{x\} + 2x(0) + 2L\{x\} = 0$$

$$\Rightarrow s^2L\{x\} - s - 1 - 2sL\{x\} + 2 + 2L\{x\} = 0$$

$$\Rightarrow L\{x\}(s^2 - 2s + 2) = s - 1$$

$$\Rightarrow L\{x\} = \frac{s-1}{s^2-2s+2} = \frac{s-1}{(s-1)^2+1}$$

$$= \left(\frac{s}{s^2+1}\right)_{s\to s-1}$$

$$= (L\{\cos t\})_{s\to s-1}$$

$$= L\{e^t \cos t\}$$

Now, taking inverse Laplace transform we get

$$x = e^t \cos t.$$

Example 7: Using the method of Laplace transformation solve the initial value problem:
$9y'' - 6y' + y = 0, y(0) = 3, y'(0) = 1.$

Solution:

Given equation is

$$9y'' - 6y' + y = 0 \quad \ldots (i)$$

with $y(0) = 3, \quad y'(0) = 1$ ... (ii)

Then taking Laplace transform on both sides of (i) then,

$$9L\{y''\} - 6L\{y'\} + L\{y\} = L\{0\}$$

$$\Rightarrow 9\{s^2\bar{y} - sy(0) - y'(0)\} - 6[s\bar{y} - y(0)] = \bar{y} = 0 \quad \text{for } \bar{y} = L\{y\}$$

[$\because$ Using the relation of L.T. of derivative of a function]

$$\Rightarrow 9\{s^2\bar{y} - 2s - 1) - 6(s\bar{y} - 3) + \bar{y} = 0 \quad [\because \text{using (ii)}]$$

$$\Rightarrow (9s^2 - 6s + 1)\bar{y} = 27s - 9$$

$$\Rightarrow \bar{y} = 9\left(\frac{3s-1}{9s^2-6s+1}\right) = 9\left(\frac{3s-1}{(3s-1)^2}\right) = \frac{9}{3s-1} = \frac{3}{s-1/3}$$

$$= 3L\{e^{t/3}\} \qquad \left[\because L\{e^{at}\} = \frac{1}{s-a}\right]$$

$$\Rightarrow L\{y\} = L\{3e^{t/3}\}$$

Taking inverse Laplace transform on both sides then,

$$y = 3e^{t/3}$$

Exercise 3 (C)

1. **Solve the following initial value problem using the Laplace transformation:**
   a. $y'' + 2y' - 8y = 0, y(0) = 1, y'(0) = 8.$
   b. $y'' - 4y' + 3y = 6t - 8, y(0) = 0, y'(0) = 0.$
   c. $y'' + w^2y = 0, y(0) = A, y'(0) = B.$
   d. $y'' - ky' = 0, y(0) = 2, y'(0) = k.$
   e. $y'' + ay' - 2a^2y = 0, y(0) = 6, y'(0) = 0.$
2. **Solve the following initial value problem:**
   a. $y'' + 2y' + 2y = 0, y(0) = 0, y'(0) = 1.$
   b. $y'' + 4y' + 5y = 0, y(0) = 1, y'(0) = 2.$
   c. $4y'' + 8y' + 5y = 0, \quad y(0) = 0, y'(0) = 1.$
   d. $y'' - 2y' + 2y = 0, \quad y(0) = 1, y'(0) = 1$.
   e. $y'' + 4y' + 3y = e^{-t}, y(0) = y'(0) = 1.$
   f. $y'' - 2y' + y = e^t, y(0) = 2, y'(0) = -1.$
   g. $y'' + 4y' + 3y = e^{-t}, \quad y(0) = y'(0) = 1.$
   h. $x'' - 3x' + 2x = 1 - e^{2t}, x(0) = 1, x'(0) = 0.$
   i. $y'' + y' - 2y = t, y(0) = 1, y'(0) = 0.$

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j. $x'' + 5x' + 6x = set$, $x(0) = 2$, $x'(0) = 1$.

k. $x'' - x = a\ cosht$, $x(0) = x'(0) = 0$.

l. $x'' + x = t\ cos2t$, $x(0) = x'(0) = 0$.

3. **Solve the following by the method of Laplace transformation,**

a. $y'' + 2y' + y = e^{-t}$, $y(0) = -1$, $y'(0) = 1$.

b. $y'' + 4y' + 4y = sint$, $y(0) = 1$, $y'(0) = 3$.

Answer Key

1. a. $y = 2e^{2t} - e^{-4t}$ b. $y = -e^{3t} + e^{t} + 2t$ c. $y = \frac{Aw\ coswt + B\ sinwt}{w}$

d. $y = 1 + e^{kt}$ e. $y = 2e^{-2at} + 4e^{at}$

2. a. $y = e^{-t}\ sint$ b. $y = e^{-2t}(\cos t + \sin t)$ c. $y = 2e^{-t} \sin \frac{t}{2}$

d. $y = e^{t} \cos t$ e. $y = \frac{-3e^{-3t} + 7e^{-t} + 2te^{-t}}{4}$ f. $y = e^{t}\left(2 - 3t + \frac{t^2}{2}\right)$

g. $y = \frac{-3e^{-3t} + 7e^{-t} + 2te^{-t}}{4}$ h. $y = \frac{1 - e^{2t} - te^{2t}}{2}$

i. $y = \frac{-1 - 2t + 4e^{t} + e^{-2t}}{4}$ j. $y = \frac{5e^{t} + 64e^{-2t} - 45e^{-3t}}{12}$

k. $x = \frac{at\ sinht}{2}$ l. $y = \frac{1}{9}[4\ sin2t - 5sint - 3t\ cos2t]$

3. a. $y = e^{-t}\left(\frac{t^2}{2} - 1\right)$ b. $y = \frac{1}{29}[e^{-2t}(25 + 165t) + 4cost - 3sint]$

□□□

4

DIFFERENTIATION AND INTEGRATION OF VECTOR FUNCTION

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CHAPTER OUTLINE

After completion of this chapter, you will be able to gain the knowledge and understanding regarding:

4.1 Differentiation of vector function of scalar variable.

4.2 Point functions, Gradient, directional derivative, divergence and curl with properties (without proof)

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Introduction

We are already studied the vector algebra on Calculus I and also familiar with the method of differentiating or integrating a scalar function. In this chapter, we will study about the differentiation and integration of a vector function and use the concept in some applications. In engineering, the differentiation and integration of a vector function helps to study dynamics, heat transfer, fluid mechanics etc.

4.1 Vector Function

If we associate a unique vector $\vec{r}$ with each element $t \in R$ by a rule $\vec{r} = \vec{r}(t)$, then this form we called a vector valued function of scalar variable t. For example, $\vec{r} = 2t^2\vec{i} + 2t\vec{j} + \sin t\vec{k}$ is a vector function of scalar variable t.

A vector function may be a function of one or more scalar variables. For instance, a vector function $\vec{r} = \vec{r}(x, y)$ is a vector function of two scalar variable x and y and a vector function $\vec{r} = \vec{r}(x, y, z) = x\vec{i} + y\vec{j} + z\vec{k}$ is a vector function of three scalar variable x, y and z.

We know every vector can be expressed uniquely as a linear combination of three fixed non-coplanar vectors as

$$\vec{f} = \vec{f}(t) = f_1(t)\vec{i} + f_2(t)\vec{j} + f_3(t)\vec{k}$$

where $\vec{i}$, $\vec{j}$ and $\vec{k}$ are unit vectors along x, y and z axes, respectively.
Example,

$$\vec{f} = \cos t\,\vec{i} + \sin t\,\vec{j} + t\vec{k}$$

4.2 Limit and Continuity of a Vector Function

Let $\vec{f}(t)$ be a vector function defined for every value $t \in (a, b)$. If there exists a vector $\vec{\ell}$ such that for any smallest positive number ε, there exists a positive number δ such that

$$|\vec{f}(t) - \vec{\ell}| < \varepsilon \text{ whenever } 0 < |t - t_0| < \delta,\ t_0 \in (a, b)$$

then the vector $\vec{\ell}$ is said to be the limit of $\vec{f}(t)$ as $t \to t_0$ and is denoted as

$$\lim_{t \to t_0} \vec{f}(t) = \vec{\ell}$$

A vector function $\vec{f}(t)$ is said to be continuous at $t = t_0$ if for every smallest positive number ε, there exists a positive number δ such that

$$|\vec{f}(t) - \vec{f}(t_0)| < \varepsilon \text{ whenever } 0 < |t - t_0| < \delta$$

i.e. $\lim_{t \to t_0} \vec{f}(t) = \vec{f}(t_0)$

Note: All the properties of continuity of vector function hold as in ordinary scalar calculus.

4.3 Derivative of a Vector Function

Let $\vec{f}(t)$ be a vector function of the scalar variable t, then we called $\vec{f}(t)$ is differentiable at a point t if $\lim_{\Delta t \to 0} \frac{\vec{f}(t + \Delta t) - \vec{f}(t)}{\Delta t}$ exists and is denoted by $\vec{f}\,'(t)$. Thus,

$$\frac{d\vec{f}}{dt} = \vec{f}\,'(t) = \lim_{\Delta t \to 0} \frac{\vec{f}(t + \Delta t) - \vec{f}(t)}{\Delta t}$$

Note: 1. Clearly, $\frac{d^2\vec{f}}{dt^2} = \frac{d}{dt}\left(\frac{d\vec{f}}{dt}\right)$ and so on.

2. All the basic properties of derivative for general function holds also by the vector function.

3. $\frac{d}{dt}(\vec{f}^{\,2}) = 2\vec{f} \cdot \frac{d\vec{f}}{dt}$

4. $\frac{d}{dt}(\vec{u} \times \vec{v}) = \vec{u} \times \frac{d\vec{v}}{dt} + \frac{d\vec{u}}{dt} \times \vec{v}$

Partial Derivatives of a Vector Function

If a vector $\vec{f} = \vec{f}(t) = f_1(t)\vec{i} + f_2(t)\vec{j} + f_3(t)\vec{k}$ be a vector valued then the partial derivative of $\vec{f}$ with respect to the scalar variable t is denoted by $\frac{\partial \vec{f}}{\partial t}$ is defined by

$$\frac{\partial \vec{f}}{\partial t} = \lim_{\Delta t \to 0} \frac{\vec{f}(t + \Delta t) - \vec{f}(t)}{\Delta t}$$

provided that the limit exists.

Theorem 1: The necessary and sufficient condition for the vector function $\vec{a}$ of the scalar variable t to have constant magnitude is $\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$.

Proof:

Necessary condition:

Let $\vec{a}$ has a constant magnitude. Then we have to show $\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$.

We know,

$$\vec{a} \cdot \vec{a} = (|\vec{a}|)^2$$

Differentiating with respect to t,

$$\Rightarrow \quad \frac{d}{dt}(\vec{a} \cdot \vec{a}) = \frac{d}{dt}[(|\vec{a}|)^2]$$

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$$\Rightarrow \vec{a}\cdot\frac{d\vec{a}}{dt}+\frac{d\vec{a}}{dt}\cdot\vec{a}=0 \qquad \text{since } |\vec{a}| \text{ is constant.}$$

$$\Rightarrow 2\vec{a}\cdot\frac{d\vec{a}}{dt}=0$$

$$\Rightarrow \vec{a}\cdot\frac{d\vec{a}}{dt}=0.$$

Sufficient condition:

Let $\vec{a}\cdot\frac{d\vec{a}}{dt}=0$, then we have to show that $\vec{a}$ has a constant magnitude.

We know, $\vec{a}\cdot\vec{a}=(|\vec{a}|)^2$.

Differentiating with respect to t

$$\frac{d}{dt}(\vec{a}\cdot\vec{a})=(|\vec{a}|)^2$$

$$\Rightarrow 2\left(\vec{a}\cdot\frac{d\vec{a}}{dt}\right)=2|\vec{a}|\frac{d}{dt}(|\vec{a}|)$$

$$\Rightarrow 0=|\vec{a}|\frac{d}{dt}(|\vec{a}|) \qquad [\text{Because of } \vec{a}\cdot\frac{d\vec{a}}{dt}=0]$$

Since $\vec{a}\neq 0$, so $|\vec{a}|\neq 0$. Therefore,

$$|\vec{a}|\frac{d}{dt}|\vec{a}|=0 \quad\Rightarrow\quad \frac{d}{dt}|\vec{a}|=0.$$

This means $|\vec{a}|$ is constant. That is, $\vec{a}$ has a constant magnitude.

Theorem 2: The necessary and sufficient condition for the vector valued function $\vec{a}$ of the scalar variable t to have a constant direction is $\vec{a}\times\frac{d\vec{a}}{dt}=\vec{0}$.

Proof:

Necessary condition:

Let $\vec{a}$ has a constant direction, then we have to show that $\vec{a}\times\frac{d\vec{a}}{dt}=\vec{0}$.

We know, $\vec{a}=a\hat{a}$ and $\hat{a}$ is a constant vector in this case where a is a magnitude of $\vec{a}$ and $\hat{a}$ be as unit vector along $\vec{a}$.

Since $\vec{a}$ has a constant direction, so $\frac{d\hat{a}}{dt}=\vec{0}$.

Here,

$$\vec{a}\times\frac{d\vec{a}}{dt}=(a\,\hat{a})\times\left(a\frac{d\hat{a}}{dt}+\frac{da}{dt}\hat{a}\right)$$

$$=a\,\hat{a}\times\left(\frac{da}{dt}\hat{a}\right)$$

$$=\left(a\frac{da}{dt}\right)(\hat{a}\times\hat{a}) \qquad \text{Since } \frac{d\hat{a}}{dt}=\vec{0}$$

$$=\left(a\frac{da}{dt}\right)\vec{0} \qquad [\text{Being } \hat{a}\times\hat{a}=0]$$

$$=\vec{0}.$$

Thus, $\vec{a}\times\frac{d\vec{a}}{dt}=\vec{0}$.

Sufficient Condition:

Let $\vec{a}\times\frac{d\vec{a}}{dt}=\vec{0}$, then we have to show that $\vec{a}$ has a constant direction.

Here,

$$\vec{a}\times\frac{d\vec{a}}{dt}=\vec{0} \quad\Rightarrow\quad (a\,\hat{a})\times\frac{d}{dt}(a\,\hat{a})=\vec{0}$$

$$\Rightarrow a^2\left(\hat{a}\times\frac{d\hat{a}}{dt}\right)=\vec{0}$$

$$\Rightarrow \hat{a}\times\frac{d\hat{a}}{dt}=\vec{0} \qquad \ldots\ldots (1) \qquad [\text{Since } a^2\neq 0]$$

Since we have, if a vector function $\vec{a}$ of the scalar variable t to have constant magnitude then $\vec{a}\cdot\frac{d\vec{a}}{dt}=0$. That is,

$$\vec{a}\cdot\frac{d\vec{a}}{dt}=0 \quad\Rightarrow\quad (a\hat{a})\cdot\frac{d}{dt}(a\hat{a})=0$$

$$\Rightarrow a^2\left(\hat{a}\cdot\frac{d\hat{a}}{dt}\right)=0$$

$$\Rightarrow \hat{a}\cdot\frac{d\hat{a}}{dt}=0$$

Thus, we have

$$\hat{a}\cdot\frac{d\hat{a}}{dt}=0 \text{ and } \hat{a}\times\frac{d\hat{a}}{dt}=\vec{0}$$

This will possible only if either $\hat{a}=\vec{0}$ or $\frac{d\hat{a}}{dt}=\vec{0}$.

Since $\hat{a}\neq\vec{0}$ then we have $\frac{d\hat{a}}{dt}=\vec{0}$.

This means $\hat{a}$ is constant. So $\vec{a}$ has a constant direction.

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Workout Examples

Example 1: If $\vec{r_1} = t^2\vec{i} - t\vec{j} + (2t+1)\vec{k}$, $\vec{r_2} = (2t-3)\vec{i} + \vec{j} - t\vec{k}$.

Find (i) $\frac{d}{dt}(\vec{r_1} \cdot \vec{r_2})$ at $t = 1$ (ii) $\frac{d}{dt}(\vec{r_1} \times \vec{r_2})$ at $t = 1$.

Solution:

Let $\vec{r_1} = t^2\vec{i} - t\vec{j} + (2t+1)\vec{k}$ and $\vec{r_2} = (2t-3)\vec{i} + \vec{j} - t\vec{k}$

Then,

$$\vec{r_1} \cdot \vec{r_2} = (t^2\vec{i} - t\vec{j} + (2t+1)\vec{k}) \cdot ((2t-3)\vec{i} + \vec{j} - t\vec{k})$$
$$= (t^2, -t, 2t+1) \cdot (2t-3, 1, -t)$$
$$= 2t^3 - 3t^2 - t - 2t^2 - t$$
$$= 2t^3 - 5t^2 - 2t$$

and

$$\vec{r_1} \times \vec{r_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -t & 2t+1 \\ 2t-3 & 1 & -t \end{vmatrix}$$
$$= (t^2 - 2t - 1)\vec{i} + ((2t-3)(2t+1) + t^3)\vec{j} + (t^2 + t(2t-3))\vec{k}$$
$$= (t^2 - 2t - 1)\vec{i} + (4t^2 - 4t - 3 + t^3)\vec{j} + (3t^2 - 3t)\vec{k}$$

Now,

(i) $\frac{d}{dt}(\vec{r_1} \cdot \vec{r_2}) = \frac{d}{dt}(2t^3 - 5t^2 - 2t) = 6t - 10t - 2$

At $t = 1$,

$$\frac{d}{dt}(\vec{r_1} \cdot \vec{r_2}) = 6 - 10 - 2 = -6$$

(ii) $\frac{d}{dt}(\vec{r_1} \times \vec{r_2}) = \frac{d}{dt}(t^2 - 2t - 1, t^3 + 4t^2 - 4t - 3, 3t^2 - 3t)$

$$= (2t - 2, 3t^2 + 8t - 4, 6t - 3)$$

At $t = 1$,

$$\frac{d}{dt}(\vec{r_1} \times \vec{r_2}) = (2 - 2, 3 + 8 - 4, 6 - 3) = (0, 7, 3)$$

Example 2: If $\vec{A} = t^2\vec{i} + (3t^2 - 2t)\vec{j} + \left(2t - \frac{1}{t}\right)\vec{k}$. Find $\left|\frac{d\vec{A}}{dt}\right|$ at $t = 1$.

Solution:

Let

$$\vec{A} = t^2\vec{i} + (3t^2 - 2t)\vec{j} + \left(2t - \frac{1}{t}\right)\vec{k}$$

Then,

$$\frac{d\vec{A}}{dt} = 2t\vec{i} + (6t-2)\vec{j} + (2 + t^{-2})\vec{k}$$

$$\left|\frac{d\vec{A}}{dt}\right| = \sqrt{(2t)^2 + (6t-2)^2 + (2 + t^{-2})^2}$$

So,

at $t = 1$,

$$\left|\frac{d\vec{A}}{dt}\right| = \sqrt{2^2 + (6-2)^2 + (2+1)^2}$$
$$= \sqrt{4 + 16 + 9}$$
$$= \sqrt{29}$$

Thus,

$$\left|\frac{d\vec{A}}{dt}\right|_{\text{at } t=1} = \sqrt{29}.$$

Example 3: If $\vec{r} = \vec{a}e^{nt} + \vec{b}e^{-nt}$, where $\vec{a}$ and $\vec{b}$ are constant vectors, show that $\frac{d^2\vec{r}}{dt^2} - n^2\vec{r} = 0$.

Solution:

Let $\vec{r} = \vec{a}e^{nt} + \vec{b}e^{-nt}$ for $\vec{a}$ and $\vec{b}$ are constant vectors.

Then,

$$\left|\frac{d\vec{r}}{dt}\right| = \vec{a}\,ne^{nt} + \vec{b}\,(-n)\,e^{-nt} = n[\vec{a}\,e^{nt} - \vec{b}\,e^{-nt}]$$

And $\frac{d^2\vec{r}}{dt^2} = n[\vec{a}\,ne^{nt} - (-n)\,\vec{b}\,e^{-nt}] = n^2[\vec{a}e^{nt} + \vec{b}e^{-nt}] = n^2\vec{r}$

$$\Rightarrow \frac{d^2\vec{r}}{dt^2} - n^2\vec{r} = 0.$$

Example 4: Find the unit tangent vector at any point on the curve $x = 3\cos t$, $y = 3\sin t$, $z = 4t$.

Solution:

Let, $x = 3\cos t$, $y = 3\sin t$ and $z = 4t$.

Then,

$$\vec{r} = x_i + y\vec{j} + z\vec{k} + (x, y, z) = (3\cos t, 3\sin t, 4t)$$

So, $\frac{d\vec{r}}{dt} = (-3\sin t, 3\cos t, 4)$

Therefore,

$$\left|\frac{d\vec{r}}{dt}\right| = \sqrt{(-3\sin t)^2 + (3\cos t)^2 + 4^2} = \sqrt{9(\sin^2 t + \cos^2 t) + 16} = \sqrt{9 + 16} = 5$$

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Now, the unit tangent vector be,

$$\left(\frac{d\hat{r}}{dt}\right) = \frac{\frac{d\vec{r}}{dt}}{\left|\frac{d\vec{r}}{dt}\right|} = \frac{1}{5}(-3\sin t, 3\cos t, 4).$$

Example 5: A particle moves along the curve $x = e^{-t}$, $y = 2\cos 3t$, $z = 2\sin 3t$, where t is time. Determine its velocity and acceleration vectors and also the magnitude velocity and acceleration at $t = 0$.

Solution:

Given curve is

$$x = e^{-t}, y = 2\cos 3t, z = 2\sin 3t$$

Then, $\vec{r} = (x, y, z) = (e^{-t}, 2\cos 3t, 2\sin 3t)$.

So that,

$$\frac{d\vec{r}}{dt} = (-e^{-t}, -6\sin 3t, 6\cos 3t)$$

and $$\frac{d^2\vec{r}}{dt^2} = (e^{-t}, -18\cos 3t, -18\sin 3t).$$

We know that, velocity of a curve is, $\vec{v} = \frac{d\vec{r}}{dt}$ and acceleration is $\vec{a} = \frac{d^2\vec{r}}{dt^2}$.

Therefore, velocity and acceleration vector are

$$\vec{v} = \frac{d\vec{r}}{dt} = (-e^{-t}, -6\sin 3t, 6\cos 3t)$$

and $$\vec{a} = \frac{d^2\vec{r}}{dt^2} = (e^{-t}, -18\cos 3t, -18\sin 3t)$$

Also, the velocity and acceleration vector at $t = 0$ is

$$\vec{v}_{at\ t=0} = (-e^{-0}, -6\sin 0, 6\cos 0) = (-1, 0, 6)$$

and $$\vec{a}_{at\ t=0} = (1, -18, 0)$$

Therefore, magnitude of velocity and acceleration at $t = 0$ is

$$|\vec{v}_{at\ t=0}| = \sqrt{(-1)^2 + 6^2} = \sqrt{1 + 36} = \sqrt{37}$$

and $$|\vec{a}_{at\ t=0}| = \sqrt{1^2 + (-18)^2} = \sqrt{1 + 324} = \sqrt{325} = 5\sqrt{13}$$

Example 6: A particle moves along the curve $x = t^3 + 1$, $y = t^2$, $z = 2t + 5$. Find the component of its velocity and acceleration at $t = 1$ in the direction $\vec{i} + \vec{j} + 3\vec{k}$.

Solution:

Given curve is

$$\vec{x} = t^3 + 1, \vec{y} = t^2, \vec{z} = 2t + 5.$$

Then the position vector of any point of the curve be,

$$\vec{r} = (x, y, z) = (t^3 + 1, t^2, 2t + 5)$$

Then, $$\frac{d\vec{r}}{dt} = (3t^2, 2t, 2) \quad \text{and} \quad \frac{d^2\vec{r}}{dt^2} = (6t, 2, 0)$$

at $t = 1$, $$\frac{d\vec{r}}{dt} = (3, 2, 2) \quad \text{and} \quad \frac{d^2\vec{r}}{dt^2} = (6, 2, 0)$$

Now, the velocity vector of $\vec{r}$ at $t = 1$ is

$$\vec{v} = \frac{d\vec{r}}{dt} \text{ at } t = 1 \quad \text{i.e. } \vec{v} = (3, 2, 2)$$

and the acceleration vector of $\vec{r}$ at $t = 1$ is

$$\vec{a} = \frac{d^2\vec{r}}{dt^2} \text{ at } t = 1 \quad \text{i.e. } \vec{a} = (6, 2, 0)$$

Also, given that a vector $\vec{i} + \vec{j} + 3\vec{k} = (1, 1, 3) = \vec{n}$ (say)

So, the unit vector along (1, 1, 3) is

$$\hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{(1, 1, 3)}{\sqrt{1 + 1 + 9}} = \frac{(1, 1, 3)}{\sqrt{11}}$$

Thus, the velocity component of $\vec{r}$ along $\vec{n}$ is

$$\vec{v} \cdot \hat{n} = (3, 2, 2) \cdot \frac{(1, 1, 3)}{\sqrt{11}} = \frac{3 + 2 + 6}{\sqrt{11}} = \frac{11}{\sqrt{11}} = \sqrt{11}$$

and the acceleration component of $\vec{r}$ along $\vec{n}$ is

$$\vec{a} \cdot \hat{n} = (6, 2, 0) \cdot \frac{(1, 1, 3)}{\sqrt{11}} = \frac{6 + 2 + 0}{\sqrt{11}} = \frac{8}{\sqrt{11}}$$

Example 7: If $\vec{r_1} = x^2yz\,\vec{i} - 2xz^2\,\vec{j} + xz^2\,\vec{k}$ and $\vec{r_2} = 2z\,\vec{i} - y\,\vec{j} + x^2\,\vec{k}$, then find the value of $\frac{\partial^2}{\partial y \partial x}(\vec{r_1} \times \vec{r_2})$.

Solution:

Given that,

$$\vec{r_1} = x^2yz\,\vec{i} - 2xz^2\,\vec{j} + xz^2\,\vec{k} \text{ and } \vec{r_2} = 2z\,\vec{i} - y\,\vec{j} + x^2\,\vec{k}$$

Then,

$$\vec{r_1} \times \vec{r_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x^2yz & -2xz^2 & xz^2 \\ 2z & -y & x^2 \end{vmatrix}$$

$$= (-2x^3z^2 + xyz^2)\,\vec{i} - (x^4yz - 2xz^3)\,\vec{j} + (-x^2y^2z + 4xz^3)\,\vec{k}$$

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Now,

$$\frac{\partial^2}{\partial y \partial x}(\vec{r}_1 \times \vec{r}_2) = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}(\vec{r}_1 \times \vec{r}_2)\right)$$

$$= \frac{\partial}{\partial y}\{(-6x^2z^2 + xz^2)\vec{i} + (2z^3 - 4x^3yz)\vec{j} + (4z^3 - 2xy^2z)\vec{k}\}$$

$$= 0\vec{i} - 4x^3z\vec{j} - 4xyz\vec{k}$$

$$= -4xz(x^2\vec{j} + y\vec{k}).$$

Example 8: For curve $x = 3t, y = 3t^2, z = 2t^3$, show that $[\vec{r}\ \vec{r}'\ \vec{r}''] = 36$ at $t = 1$.

Solution:

Given that,

$x = 3t, y = 3t^2, z = 2t^3$

Since

$\vec{r} = (x, y, z) = (3t, 3t^2, 2t^3)$

So, $\vec{r}' = (3, 6t, 6t^2)$ and $\vec{r}'' = (0, 6, 12t)$

At $t = 1$,

$\vec{r} = (3, 3, 2)$; $\vec{r}' = (3, 6, 6)$; $\vec{r}'' = (0, 6, 12)$

Now,

$$[\vec{r}\ \vec{r}'\ \vec{r}''] = \begin{vmatrix} 3 & 3 & 2 \\ 3 & 6 & 6 \\ 0 & 6 & 12 \end{vmatrix}$$

$$= (3\times3\times2)\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 2 & 6 \end{vmatrix}$$

$$= (18)[1(12-6) - 1(6-0) + 1(2-0)]$$

$$= (18)[6 - 6 + 2]$$

$$= 36$$

4.4 INTEGRATION OF VECTOR FUNCTION

Let $\vec{f}(t)$ be a vector valued function and $\vec{F}(t)$ be differentiation of $\vec{f}(t)$. Then,

$$\frac{d}{dt}\vec{F}(t) = \vec{f}(t)$$

Here, we called $\vec{F}(t)$ is an integral of $\vec{f}(t)$ with respect to t and we write as

$$\int \vec{f}(t)\,dt = \vec{F}(t) + \vec{C}$$

where $\vec{C}$ be any arbitrary constant vector independent of t.

The above integral is called the indefinite integral of $\vec{f}(t)$.

The definite integral of $\vec{f}(t)$ between the limits $t = a$ to $t = b$ is given by

$$\int_a^b \vec{f}(t)\,dt = \vec{F}(b) - \vec{F}(a).$$

Note: The basic properties of the integral, also satisfies by the vector function.

Example 9: If $\vec{v} = 3t^2\vec{i} + 3t\vec{j} - (3t+2)\vec{k}$, evaluate $\int_2^3 \vec{v}\,dt$.

Solution:

Let $\vec{v} = (3t^2, 3, -3t-2)$. Then,

$$\int_2^3 \vec{v}\,dt = 3\vec{i}\int_2^3 t^2\,dt + 3\vec{j}\int_2^3 t\,dt - \vec{k}\left[3\int_2^3 t\,dt - 2\int_2^3 dt\right]$$

$$= 3\vec{i}\left[\frac{t^3}{3}\right]_2^3 + 3\vec{j}\left[\frac{t^2}{2}\right]_2^3 - \vec{k}\left[3\frac{t^2}{2} - 2t\right]_2^3$$

$$= \vec{i}(27-8) + 3\vec{j}\left(\frac{9-4}{2}\right) - \vec{k}\left(\frac{27}{2} - 6 - 6 + 4\right)$$

$$= 19\vec{i} + \frac{15}{2}\vec{j} - \frac{11}{2}\vec{k}.$$

Example 10: If $\vec{r} = t\vec{i} - t^2\vec{j} + (t-1)\vec{k}$, $\vec{s} = 2t^2\vec{i} + 6t\vec{k}$ then evaluate, $\int_0^2 (\vec{r}\times\vec{s})\,dt$.

Solution:

Let $\vec{r} = t\vec{i} - t^2\vec{j} + (t-1)\vec{k}$ and $\vec{s} = 2t^2\vec{i} + 6t\vec{k}$

Then,

$$\vec{r}\times\vec{s} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ t & -t^2 & (t-1) \\ 2t^2 & 0 & 6t \end{vmatrix} = -6t^3\vec{i} - (2t^3 - 2t^2 - 6t^3)\vec{j} + 2t^4\vec{k}$$

$$= -t^3\vec{i} + (8t^2 - 2t^3)\vec{j} + 2t^4\vec{k}$$

Now,

$$\int_0^2 (\vec{r}\times\vec{s})\,dt = \int_0^2 \{-6t^3\vec{i} + (8t^2 - 2t^3)\vec{j} + 2t^4\vec{k}\}\,dt$$

$$= \left[-\frac{6t^4}{4}\vec{i} + \left(\frac{8t^3}{3} - \frac{2t^4}{4}\right)\vec{j} + \frac{2t^5}{5}\vec{k}\right]_0^2$$

$$= -24\vec{i} + \left(\frac{64}{3} - 8\right)\vec{j} + \frac{64}{5}\vec{k}$$

$$= -24\vec{i} + \frac{40}{3}\vec{j} + \frac{64}{5}\vec{k}$$

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Example 11: Let $\vec{a} = 12\cos 2t\,\vec{i} - 8\sin 2t\,\vec{j} + 16t\,\vec{k}$ be the acceleration of a particle at any time t. find the velocity $\vec{v}$ and displacement $\vec{r}$ at any time t and given $\vec{r} = \vec{v}$ at t = 0.

Solution:

Let $\vec{a} = 12\cos 2t\,\vec{i} - 8\sin 2t\,\vec{j} + 16t\,\vec{k}$ be acceleration at any time t. So,

$$\Rightarrow \vec{a} = 12\cos 2t\,\vec{i} - 8\sin 2t\,\vec{j} + 16t\,\vec{k} \quad \text{.....(i)}$$

Also, given that, $\vec{r} = 0 = \vec{v}$ at t = 0.

Since $\vec{v}$ be the velocity at any time t. Therefore,

$$\vec{r} = 0 = \frac{d\vec{r}}{dt} \text{ at } t = 0 \quad \text{.....(ii)}$$

Integrating (i) w.r.t. t then

$$\frac{d\vec{r}}{dt} = 6\sin 2t\,\vec{i} + 4\cos 2t\,\vec{j} + 8t^2\,\vec{k} + c \quad \text{.....(iii)}$$

at t = 0, (iii) gives,

$$0 = 0 + 4j + c \Rightarrow c = 4j \quad [\because \text{using (ii)}]$$

Then (iii) becomes,

$$\frac{d\vec{r}}{dt} = 6\sin 2t\,\vec{i} + 4\cos 2t\,\vec{j} + 8t^2\,\vec{k} - 4j \quad \text{.....(iv)}$$

Again integrating (iv) w.r.t. t then,

$$\vec{r} = -3\cos 2t\,\vec{i} - 2\sin 2t\,\vec{j} + \frac{8t^3}{3}\vec{k} - 4t\,\vec{j} + c \quad \text{.....(v)}$$

At t = 0, (v) gives,

$$0 = -3\vec{i} - 0 + 0 - 0 + c \Rightarrow c = 3\vec{i} \quad [\because \text{using (ii)}]$$

Then (v) becomes,

$$\vec{r} = -3\cos 2t\,\vec{i} - 2\sin 2t\,\vec{j} + \frac{8t^3}{3}\vec{k} - 4t\,\vec{j} + 3\vec{i}$$

$$\Rightarrow \vec{r} = (3 - 3\cos 2t)\vec{i} - (4t + 2\sin 2t)\,\vec{j} + \frac{8t^3}{3}\vec{k} \quad \text{..... (vi)}$$

Thus, (iv) be the velocity and (vi) be the displacement of the particle at time t.

4.5 EXACTNESS CONDITION

Let F_1, F_2, F_3 are continuous and have continuous first order partial derivatives exist in a domain D. Let an integral $\int_C (F_1dx + F_2dy + F_3dz)$ is integrable, then we called the value under the integral sign is exact if

$$\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x};\ \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x};\ \frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y}$$

are satisfied.

Since the given integral is independent of path and the integrand function (value under the integral sign) is exact, so the solution of the integral is given by

$$\int_C (F_1dx + F_2dy + F_3dz) = \int_C (F_1)\,dx + \int_C (\text{Term(s) of } F_2 \text{ free from x})\,dy + \int_C (\text{Term(s) of } F_3 \text{ free from x and y})\,dz$$

Example 12: Show that the value under the integral sign is exact and evaluate the integral $\int_{(4,0,3)}^{(-1,1,2)} [(yz + 1)dx + (xz + 1)dy + (xy + 1)dz]$

Solution:

Given integral is,

$$I = \int_{(4,0,3)}^{(-1,1,2)} [(yz + 1)dx + (xz + 1)dy + (xy + 1)dz] \quad \text{... (i)}$$

Comparing the value under the integral sign on (i) with $F_1dx + F_2dy + F_3dz$ then we get,

$$F_1 = yz + 1, \quad F_2 = xz + 1, \quad F_3 = xy + 1$$

Then,

$$\frac{\partial F_1}{\partial y} = z, \quad \frac{\partial F_1}{\partial z} = y, \quad \frac{\partial F_2}{\partial z} = x, \quad \frac{\partial F_2}{\partial x} = z, \quad \frac{\partial F_3}{\partial x} = y, \quad \frac{\partial F_3}{\partial y} = x$$

Here,

$$\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x}, \quad \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x}, \quad \frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y}$$

So, the value in (i) is exact.

Now,

$$\begin{aligned} I &= \int_{(4,0,3)}^{(-1,1,2)} [(yz + 1)dx + (xz + 1)dy + (xy + 1)dz] \\ &= \int_{(4,0,3)}^{(-1,1,2)} (yz + 1)\,dx + \int_{(4,0,3)}^{(-1,1,2)} (\text{Term(s) of } (xz + 1) \text{ free from x})\,dy + \int_{(4,0,3)}^{(-1,1,2)} (\text{Term(s) of } (xy + 1) \text{ free from x and y})\,dz \\ &= \int_{(4,0,3)}^{(-1,1,2)} (yz + 1)\,dx + \int_{(4,0,3)}^{(-1,1,2)} (1)\,dy + \int_{(4,0,3)}^{(-1,1,2)} (1)\,dz \\ &= [xyz + x]_{(4,0,3)}^{(-1,1,2)} + [y]_{(4,0,3)}^{(-1,1,2)} + [z]_{(4,0,3)}^{(-1,1,2)} \\ &= [(-2 - 1) - (0 + 4)] + [1 - 0] + [2 - 3] \\ &= -7 + 1 - 1 \\ &= -7 \end{aligned}$$

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Example 13: Show that the value under the integral sign is exact and evaluate the integral $\int_{(4,3/2)}^{(4,1/2)} (2x \sin\pi y\, dx + \pi x^2 \cos\pi y\, dy)$.

Solution:

Given integral is,

$$I = \int_{(4,3/2)}^{(4,1/2)} (2x \sin\pi y\, dx + \pi x^2 \cos\pi y\, dy) \quad \ldots (i)$$

Comparing the integrand value with $F_1 dx + F_2 dy$, we get

$F_1 = 2x \sin\pi y$ and $F_2 = \pi x^2 \cos\pi y$

Here,

$$\frac{dF_1}{dy} = 2\pi x \cos\pi y \quad \text{and} \quad \frac{dF_2}{dx} = 2\pi x \cos\pi y$$

This shows that $\frac{dF_1}{dy} = \frac{dF_2}{dx}$. So, the value under the integral sign of (i) is exact.

Therefore,

$$I = \int_{(4,3/2)}^{(4,1/2)} (2x \sin\pi y)\, dx + \int_{(4,3/2)}^{(4,1/2)} (\text{Term(s) of } \pi x^2 \cos\pi y \text{ free from x})\, dy$$

$$= \int_{(4,3/2)}^{(4,1/2)} (2x \sin\pi y)\, dx + \int_{(4,3/2)}^{(4,1/2)} (0)\, dy$$

$$= [x^2 \sin\pi y]_{(4,3/2)}^{(4,1/2)}$$

$$= \left(16 \sin\frac{\pi}{2} - 9 \sin\frac{3\pi}{2}\right)$$

$$= 16 \times 1 - 9 \times (-1)$$

$$= 25$$

Example 14: Determine, the value under the integral sign $\int_{(0,0,1)}^{(1,\pi/4,2)} [2xyz^2\, dx + (x^2z^2 + z\cos yz)\, dy + (2x^2yz + y\cos yz)\, dz]$ is exact or not. If exact, evaluate the integral.

Solution:

Given integral is,

$$I = \int_{(0,0,1)}^{(1,\pi/4,2)} [2xyz^2\, dx + (x^2z^2 + z\cos yz)\, dy + (2x^2yz + y\cos yz)\, dz] \quad \ldots (i)$$

Comparing the integrand value with $F_1 dx + F_2 dy + F_3 dz$, we get

$F_1 = 2xyz^2$, $F_2 = x^2z^2 + z\cos yz$, $F_3 = 2x^2yz + y\cos yz$

Then,

$$\frac{dF_1}{dy} = 2xz^2, \frac{dF_2}{dx} = 2xz^2, \frac{dF_1}{dz} = 4xyz, \quad \frac{dF_3}{dx} = 4xyz,$$

$$\frac{dF_2}{dz} = 2x^2z + \cos yz + yz\sin yz, \frac{dF_3}{dy} = 2x^2z + \cos yz + yz\sin yz$$

This shows that,

$$\frac{dF_1}{dy} = \frac{dF_2}{dx}, \frac{dF_1}{dz} = \frac{dF_3}{dx}, \quad \frac{dF_2}{dz} = \frac{dF_3}{dy}$$

This means the integrand value of (i) is exact.

Now, the solution of (i) is,

$$I = \int_{(0,0,1)}^{(1,\pi/4,2)} (2xyz^2)dx + \int_{(0,0,1)}^{(1,\pi/4,2)} (\text{Term(s) of } (x^2z^2 + z\cos yz) \text{ free from x})dy + \int_{(0,0,1)}^{(1,\pi/4,2)} (\text{Term(s) of } (2x^2yz + y\cos yz) \text{ free from x and y})\, dz$$

$$= \int_{(0,0,1)}^{(1,\pi/4,2)} (2xyz^2)dx + \int_{(0,0,1)}^{(1,\pi/4,2)} (z\cos yz)\, dy + \int_{(0,0,1)}^{(1,\pi/4,2)} (0)\, dz$$

$$= [x^2yz^2]_{(0,0,1)}^{(1,\pi/4,2)} + \left[z\frac{\sin yz}{z}\right]_{(0,0,1)}^{(1,\pi/4,2)} + [0]_{(0,0,1)}^{(1,\pi/4,2)}$$

$$= \left(\frac{\pi}{4} \times 4 - 0\right) + \left(\sin\frac{\pi}{2} - \sin 0\right) + 0$$

$$= \pi + 1$$

Exercise 4 (A)

1. a. Find $\frac{d}{dt}(\vec{r} \cdot \vec{r})$ where $\vec{r} = t\vec{i} + 3t^2\vec{j} + 4t^3\vec{k}$.

   b. If $\vec{r_1} = (2t+1)\vec{i} - t^2\vec{j} + 3t^3\vec{k}$ and $\vec{r_2} = t^2\vec{i} - t\vec{j} + (t-1)\vec{k}$. Find $\frac{d}{dt}(\vec{r_1} \times \vec{r_2})$.

   c. Verify $\frac{d}{dt}(\vec{a} \times \vec{b}) = \vec{a} \times \frac{d\vec{b}}{dt} + \frac{d\vec{a}}{dt} \times \vec{b}$ and $\frac{d}{dt}(\vec{a} \cdot \vec{b}) = \vec{a} \cdot \frac{d\vec{b}}{dt} + \frac{d\vec{a}}{dt} \cdot \vec{b}$ where $\vec{a} = 2t^3\vec{i} + t\vec{j} + 3t^2\vec{k}$ and $\vec{b} = 2t\vec{i} + 3\vec{j} + t^3\vec{k}$.

2. a. If $\vec{r} = \cos nt\,\vec{i} + \sin nt\,\vec{j}$, then show that $\vec{r} \times \frac{d\vec{r}}{dt} = n\vec{k}$.

   b. If $\vec{r} = \vec{a}\cos wt + \vec{b}\sin wt$, show that $\vec{r} \times \frac{d\vec{r}}{dt} = w(\vec{a} \times \vec{b})$ where $\vec{a}$ and $\vec{b}$ are constant vectors.

   c. If $\vec{r} = \vec{a}e^{mt} + \vec{b}e^{nt}$, where $\vec{a}$ and $\vec{b}$ are constant vectors, show that $\frac{d^2\vec{r}}{dt^2} - (m+n)\frac{d\vec{r}}{dt} + mn\vec{r} = 0$.

3. a. If $\vec{r} = t^2\vec{i} + (2t+1)\vec{j} + 3t\vec{k}$, then find $\left|\frac{d^2\vec{r}}{dt^2}\right|$.

   b. If $\frac{d\vec{a}}{dt} = \vec{c} \times \vec{a}$, $\frac{d\vec{b}}{dt} = \vec{c} \times \vec{b}$, show that $\frac{d}{dt}(\vec{a} \times \vec{b}) = \vec{c} \times (\vec{a} \times \vec{b})$.

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4. a. Find unit tangent vector to the curve $\vec{r} = t^2\vec{i} + 2t\vec{j} - t^3\vec{k}$ at $t = 1$.

b. Show that the unit tangent to the curve $\vec{r} = t\vec{i} + t^2\vec{j} + t^3\vec{k}$ at $t = 1$ is $\frac{1}{\sqrt{14}}(\vec{i} + 2\vec{j} + 3\vec{k})$.

c. A particle moves along the curve $x = t^3 + 1$, $y = t^2$, $z = 2t + 5$. Find the velocity and acceleration at $t = 1$.

d. A particle moves along the curves $x = 4\cos t$, $y = 4\sin t$, $z = 6t$. Find the velocity and acceleration at time $t = 0$ and $t = \frac{\pi}{2}$.

5. **Evaluate:**

a. $\int_0^1 \{t\vec{i} + (t^2 - 2t)\vec{j} + 3t^2\vec{k}\}\, dt$

b. $\int_0^1 \{t\vec{i} + e^t\vec{j} + e^{-2t}\vec{k}\}\, dt$

c. If $\vec{v} = 3t^2\vec{i} + 3t\vec{j} - (3t + 2)\vec{k}$, evaluate $\int_2^3 \vec{v}\, dt$.

d. If $\vec{r} = t\vec{i} - t^2\vec{j} + (t - 1)\vec{k}$, $\vec{s} = 2t^2\vec{i} + 6t\vec{k}$ then evaluate $\int_0^2 (\vec{r} \cdot \vec{s})\, dt$

e. If $\vec{r} = 5t^2\vec{i} + t\vec{j} - t^3\vec{k}$, find $\int_1^2 \left(\vec{r} \times \frac{d\vec{r}}{dt}\right) dt$.

6. a. Find the value of $\vec{r}$ satisfying the equation $\frac{d^2\vec{r}}{dt^2} = 6t\vec{i} - 24t^2\vec{j} + 4\sin t\vec{k}$ given that $\vec{r} = 2\vec{i} + \vec{j}$ and $\frac{d\vec{r}}{dt} = -\vec{i} - 3\vec{k}$ at $t = 0$.

b. If $\vec{r} \times \frac{d^2\vec{r}}{dt^2} = \vec{0}$, show that $\vec{r} \times \frac{d\vec{r}}{dt} = \vec{a}$, where $\vec{a}$ is a constant vector.

c. Solve $\frac{d^2\vec{r}}{dt^2} = t\vec{a} + \vec{b}$, where $\vec{a}$ and $\vec{b}$ are constant vectors, given that $t = 0$, $\vec{r} = 0$ and $\frac{d\vec{r}}{dt} = \vec{u}$.

7. Show that the value under integral sign is exact and then evaluate integral.

a. $\int_{(-1,5)}^{(4,3)} (3z^2\, dx + 6xz\, dz)$

b. $\int_{(0,1)}^{(2,3)} [(2x + y^3)\, dx + (3xy^2 + 4)\, dy]$

c. $\int_{(-1,2)}^{(3,1)} [(y^2 + 2xy)\, dx + (x + 2 + 2xy)\, dy]$

d. $\int_{(0,\pi)}^{(3,\pi/2)} [e^x \cos y\, dx - e^x \sin y\, dy]$

e. $\int_{(0,0,0)}^{(4,1,2)} (3ydx + 3xdy + 2z\, dz)$

f. $\int_{(0,0,0)}^{(a,b,c)} (2xy^2\, dx + 2x^2y\, dy + dz)$

g. $\int_{(1,0,2)}^{(-2,1,3)} [(6xy^3 + 2z^2)\, dx + 9x^2y^2\, dy + (4xz + 1)\, dz]$

h. $\int_{(0,1,1/2)}^{(\pi/2,3,2)} [y^2 \cos x\, dx + (2y \sin x + e^{2z})\, dy + 2ye^{2z}\, dz]$

i. $\int_{(0,2,3)}^{(1,1,1)} [yz \sinh(xz)\, dx + \cosh(xz)\, dy + xy \sinh(xz)\, dz]$

Answer Key

1. a. $\frac{d}{dt}(\vec{r} \cdot \vec{r}) = 2t + 36t^3 + 96t^5$

b. $\frac{d}{dt}(\vec{r_1} \times \vec{r_2}) = (2t - 3t^2 + 12t^3) - (4t - 1 - 15t^4)j + (4t3 - 4t - 1)k.$

3. a. 2

4. a. $\frac{1}{\sqrt{17}}(2, 2, 3)$

c. $\vec{v}$ $= (3, 2, 2)$, $\vec{a} = (6, 2, 0)$

d. at $t = 0$, $\vec{v} = (0, 4, 6)$; $\vec{a} = (-4, 0, 0)$; at $t = \frac{\pi}{2}$, $\vec{v} = (-4, 0, 6)$; $\vec{a} = (0, -4, 0)$

5. a. $\frac{\vec{i}}{2} - \frac{2\vec{j}}{3} + \vec{k}$ b. $\frac{\vec{i}}{2} + (e - 1)\vec{j} + \left(\frac{1 - e^{-2}}{2}\right)\vec{k}$ c. $19\vec{i} + \frac{15}{2}\vec{j} - \frac{11}{2}\vec{k}$

d. 12 e. $-\frac{15}{2}\vec{i} + 31\vec{j} - \frac{35}{3}\vec{k}$

6. a. $\vec{r} = (t^3 - t + 2)\vec{i} + (1 - 2t^4)\vec{j} + (t - 4\sin t)\vec{k}$ c. $\vec{r} = \frac{\vec{a}}{6}t^3 + \frac{\vec{b}}{2}t^2 + \vec{u}t$

7. a. 183 b. 66 c. 14
d. 1 e. 16 f. $a^2b^2 + c$
g. -31 h. $3e^4 - e + 9$ i. $\cosh 1 - 2$

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4.6 VECTOR FIELD AND VECTOR DIFFERENTIAL OPERATOR

Every physical quantity is a continuous function and has position coordinates in the given region. Such function assigned with direction form is a vector field.

The differential operator in vector field is called vector differential operator and is denoted by ∇ and is read as nabla and is defined by,

$$\nabla = \vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z} \left(\text{or } \Sigma\, \vec{i}\frac{\partial}{\partial x}\right)$$

It is useful in defining the gradient of a function, divergence and curl of a vector function. The gradient, divergence and curl are important in the field of physics and engineering.

4.7 GRADIENT OF A SCALAR FUNCTION

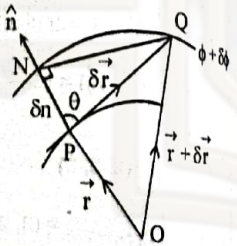
Let $\phi(x, y, z)$ be a scalar function of coordinates x, y, z. The gradient of ϕ is defined by grad ϕ or $\nabla\phi$. That is,

$$\text{grad}\,\phi = \nabla\phi = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)\phi = \left(\vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}\right)$$

Clearly, the components of gradient of a function are $\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z}$ along the direction $\vec{i}, \vec{j}, \vec{k}$. Thus, the gradient of a function is a vector function.

Geometrical Interpretation of Gradient of a Scalar Function

Let $\phi = \phi(x, y, z)$ be a surface. Let P(x, y, z) be a point and Q(x + δx, y + δy, z + δz) be a neighboring point of P, on surface ϕ. Let, $\vec{r}$ and $\vec{r} + \delta\vec{r}$ be the position vectors of P and Q, then $\vec{PQ} = \delta\vec{r}$ (i.e. $\delta x\,\vec{i} + \delta y\,\vec{j} + \delta z\,\vec{k}$).



Also, the gradient of a scalar function ϕ is

$$\nabla\phi = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}$$

Now,

$$\nabla\phi\cdot\delta\vec{r} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)\cdot\left(\delta x\,\vec{i} + \delta y\,\vec{j} + \delta z\,\vec{k}\right)$$

$$= \frac{\partial\phi}{\partial x}\delta x + \frac{\partial\phi}{\partial y}\delta y + \frac{\partial\phi}{\partial z}\delta z$$

$$= \delta\phi \quad \ldots\ldots (i)$$

If P and Q lie on the same surface, then $\delta\phi = 0$. Therefore,

$$\nabla\phi\cdot\delta\vec{r} = 0$$

i.e. $\nabla\phi$ is perpendicular to $\delta\vec{r}$ lying on the surface,

i.e. $\nabla\phi$ is normal to the surface ϕ.

Let $\nabla\phi = |\nabla\phi|\,\hat{n}$ where $\hat{n}$ is the unit vector normal to the surface ϕ. Let the distance PN is δn, along the direction of normal to the surface. Then the rate of change of ϕ in the direction of normal to the surface through P is

$$\frac{\partial\phi}{\partial n} = \lim_{\delta n\to 0}\frac{\delta\phi}{\delta n} = \lim_{\delta n\to 0}\frac{\nabla\phi\cdot\delta\vec{r}}{\delta n}$$

$$= \lim_{\delta n\to 0}\frac{|\nabla\phi|\,n\cdot\delta\vec{r}}{\delta n}$$

$$= |\nabla\phi|\lim_{\delta n\to 0}\frac{n\cdot\delta\vec{r}}{\delta n}$$

$$= |\nabla\phi|\lim_{\delta n\to 0}\frac{|n|\,|\delta\vec{r}|\cos\theta}{\delta n}$$

$$= |\nabla\phi|\lim_{\delta n\to 0}\frac{\delta n}{\delta n} \quad \text{[From figure]}$$

$$= |\nabla\phi|$$

Thus, the gradient of a scalar function is a vector function which is normal to the surface ϕ and whose magnitude is equal to the maximum rate of change of ϕ along this normal.

Theorem 3: The necessary and sufficient condition for a scalar function to be constant is that $\nabla\phi = 0$.

Proof: Necessary Condition:

Let the scalar function $\phi(x, y, z)$ is constant. Then,

$$\frac{\partial\phi}{\partial x} = \frac{\partial\phi}{\partial y} = \frac{\partial\phi}{\partial z} = 0$$

Therefore,

$$\nabla\phi = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z} = 0$$

Sufficient Condition:

Let $\phi(x, y, z)$ be a scalar function. Let, $\nabla\phi = 0$. That is,

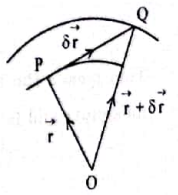
$$\vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z} = 0$$

This implies, $\frac{\partial\phi}{\partial x} = \frac{\partial\phi}{\partial y} = \frac{\partial\phi}{\partial z} = 0$.

This means the function ϕ must be independent of x, y, z, i.e. $\phi(x, y, z)$ is constant.

4.8 DIRECTIONAL DERIVATIVE OF A FUNCTION

Let $\phi(x, y, z)$ be a surface and P(x, y, z) be any point on the surface. Let Q(x + δx, y + δy, z + δz) be a neighboring point of P(x, y, z) on the surface ϕ. Let $\vec{r}$ and $\vec{r} + \delta\vec{r}$ be the position vector of P and Q. Then,



$$\frac{\partial\phi}{\partial r} = \lim_{\delta\vec{r}\to 0}\frac{\delta\phi}{\delta r}$$

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This limit is said to be the directional derivatives of ϕ at P along PQ. This means the rate of change of ϕ along PQ.

Note: We can compute a directional derivative in terms of the gradient as follows:

Let $\vec{r}$ be the position vector of any point on PQ is given by

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$
$$= \vec{P_0} + s\vec{a}$$

where $\vec{a}$ is unit vector along PQ. $\vec{P_0}$ be the position vector of P and s be along PQ. Then

$$d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k} = \vec{a} \text{ with } |\vec{a}| = 1$$

By the chain rule, the directional derivative of ϕ along PQ is

$$D_{PQ}\prod = \lim_{\delta\vec{r}\to 0} \frac{\delta\phi}{\delta r} = \frac{d\phi}{dr} = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz$$
$$= \left(\frac{\partial\phi}{\partial x}\vec{i} + \frac{\partial\phi}{\partial y}\vec{j} + \frac{\partial\phi}{\partial z}\vec{k}\right) \cdot \left(dx\vec{i} + dy\vec{j} + dz\vec{k}\right)$$
$$= \left(\frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{k}\right)\phi \cdot \left(d\vec{r}\right)$$
$$= \nabla\phi \cdot d\vec{r}$$

where $\vec{r}$ be the position vector of any point on PQ, so the tangential vector $d\vec{r}$ is unit vector. If $\hat{a} = d\vec{r}$ then

$$D_{PQ}\prod = \nabla\phi \cdot \hat{a} = \text{grad}\phi \cdot \hat{a}$$

If $\vec{a}$ is not unit vector then

$$D_{PQ}\prod = \nabla\phi \cdot \frac{\vec{a}}{|\vec{a}|} = \text{grad}\phi \cdot \frac{\vec{a}}{|\vec{a}|}$$

Theorem 4: Let ϕ be a level surface then the gradient of ϕ is maximum value of $\frac{d\phi}{ds}$ occurs in the direction of $\frac{d\phi}{ds}$.

Proof:

Let ϕ be a level surface, then the directional derivative of ϕ in the direction of $\hat{a}$ is

$$\frac{d\phi}{ds} = \hat{a} \cdot \text{grad}\phi = \hat{a} \cdot \vec{n}$$

where $\vec{n} = \text{grad}\phi = \nabla\phi$, normal vector to surface ϕ.

$$\frac{d\phi}{ds} = |\hat{a}|\,|\vec{n}|\cos\theta$$

This means the normal vector is fixed. That is, gradϕ is maximum when $\cos\theta$ is maximum and is possible when $\theta = 0°$. This implies, the vectors $\hat{a}$ and $\vec{n}$ are in same direction. Therefore, the gradient of ϕ is maximum value of $\frac{d\phi}{ds}$ occurs when $|\text{grad}\phi| = \frac{d\phi}{dn}$.

Properties of Directional Derivative

The function f increases most rapidly when $\cos\theta = 1$ i.e. $\theta = 0$. $D_{\vec{a}}f = |\nabla\phi|$.

The function f decreases most rapidly when $\cos\theta = -1$ i.e. $\theta = \pi$. $D_{\vec{a}}f = -|\nabla\phi|$.

The direction is 0 when $\cos\theta = 0$ i.e. $\theta = \frac{\pi}{2}$.

Workout Examples

Example 1: Find grad f, where $f = x^3 + y^3 + 3xyz$.

Solution:

Given that, $f = x^3 + y^3 + 3xyz$

Then,

$$\text{Grad (f)} = \nabla f = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)(x^3 + y^3 + 3xyz)$$
$$= (3x^2 + 3yz)\vec{i} + (3y^2 + 3zx)\vec{j} + 3xy\,\vec{k}.$$

Example 2: Find a unit normal to the surface $x^2y + 2xz - 4$.

Solution:

Let the given surface be,

$$f = x^2y + 2xz - 4$$

We know, the normal vector to the surface f is ∇f.

Here,

$$\nabla f = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)(x^2y + 2xz - 4)$$
$$= (2xy + 2z)\vec{i} + x^2\vec{j} + 2x\,\vec{k}$$

At point (2, –2, 3),

$$\nabla f = (-8 + 6)\vec{i} + 4\vec{j} + 4\vec{k}$$
$$= 2(-\vec{i} + 2\vec{j} + 2\vec{k})$$

Now, the unit vector of ∇f is

$$\hat{n} = \frac{\nabla f}{|\nabla f|} = \frac{2(\vec{i} + 2\vec{j} + 2\vec{k})}{2\sqrt{1+4+4}} = \frac{\vec{i} + 2\vec{j} + 2\vec{k}}{3}$$

Thus, the unit vector normal to f at (2, –2, 3) is $\frac{1}{3}(-\vec{i} + 2\vec{j} + 2\vec{k})$.

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Example 3: Find the directional derivatives of $f = xy^2 + yz^3$ at $P(2, -1, 1)$ along the direction of the normal to the surface $x \log z - y^2 + 4 = 0$ at $(-1, 2, 1)$.

Solution:

Given that, $f = xy^2 + yz^3$

And the surface is, $\phi = x \log(z) - y^2 + 4$

Then,

$$\nabla\phi = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{j}\frac{\partial}{\partial z}\right)(x \log(z) - y^2 + 4)$$

$$= \log(z)\,\vec{i} - 2y\,\vec{j} + \frac{x}{z}\vec{k}$$

at $(-1, 2, 1)$,

$$\nabla\phi = \log(1)\,\vec{i} - 4\vec{j} - \vec{k} = 0\vec{i} - 4\vec{j} - \vec{k}$$

Also,

$$\text{grad}(f) = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{j}\frac{\partial}{\partial z}\right)(xy^2 + yz^3)$$

$$= y^2\vec{i} + (2xy + z^3)\vec{j} + 3yz^2\vec{k}$$

at $(2, -1, 1)$,

$$\text{grad}(f) = \vec{i} + (-4 + 1)\vec{j} - 3\vec{k} = \vec{i} - 3\vec{j} - 3\vec{k}$$

Now, the directional derivative of f at $(2, -1, 1)$ is

$$D(f)|_{\text{at }(2,-1,1)} = (\text{grad}(f)|_{\text{at }(2,-1,1)}) \cdot \left(\frac{\nabla f}{|\nabla f|}\Big|_{\text{at }(-1,2,1)}\right)$$

$$= (\vec{i} - 3\vec{j} - 3\vec{k}) \cdot \left(\frac{0\vec{i} - 4\vec{j} - \vec{k}}{\sqrt{0 + 16 + 1}}\right)$$

$$= \frac{1}{\sqrt{17}}(0 + 12 + 3)$$

$$= \frac{15}{\sqrt{17}}$$

Example 4: The temperature at a point (x, y, z) in space is given by $T = x^2 + y^2 - z$. A mosquito located at $(1, 1, 2)$ desire to fly in such a direction that it wing get warm as soon as possible. In what direction should it fly?

Solution:

Given that, $T = x^2 + y^2 - z$

Then,

$$\nabla T = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)(x^2 + y^2 - z)$$

$$= 2x\vec{i} + 2y\vec{j} - \vec{k}$$

at point $(1, 1, 2)$,

$$\nabla T = 2\vec{i} + 2\vec{j} - \vec{k}.$$

The mosquito flies in the direction where it gets maximum temperature i.e. along the normal of T to get warm as soon as possible.

Thus, the mosquito should fly in the direction of $2\vec{i} + 2\vec{j} - \vec{k}$.

4.9 DIVERGENCE OF A VECTOR VALUED FUNCTION

Let $\vec{v}(x, y, z)$ be a vector valued function, defined and differentiable at each point (x, y, z) in space. Then the divergence of $\vec{v}$ written as div $\vec{v}$ (or $\nabla \cdot \vec{v}$) is defined by,

$$\text{div}\,\vec{v} = \nabla \cdot \vec{v} = \vec{i}\frac{\partial \vec{v}}{\partial x} + \vec{j}\frac{\partial \vec{v}}{\partial y} + \vec{k}\frac{\partial \vec{v}}{\partial z}$$

Clearly, the div $\vec{v}$ is a scalar function.

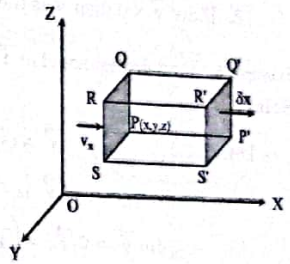
Note: If $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$, then

$$\text{div}\,\vec{v} = \nabla \cdot \vec{v}$$

$$= \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right) \cdot (v_1\vec{i} + v_2\vec{j} + v_3\vec{k}) = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

Physical Interpretation of Divergence of a Vector Function

Let $\vec{v} = v_x\vec{i} + v_y\vec{j} + v_z\vec{k}$ be the velocity of the fluid having density $\rho = \rho(x, y, z, t)$ at point P, at time t, where the components v_x, v_y, v_z are parallel to the axes.



Let the fluid enters through the faces PQRS and goes out from other faces P'Q'R'S'. Then the mass of fluid entering from face PQRS per unit time is (velocity × surface area) i.e. $(v_x\delta y\delta z)$ and outflow is $(v_{x+\delta x}\delta y\delta z)$.

Then the net preserving mass entering and outing from one face is

$$F_1 = (v_{x+\delta x}\delta y\delta z) - (v_x\delta y\delta z)$$

$$= \left(v_x + \frac{\partial v_x}{\partial x}\delta x\right)\delta y\delta z - (v_x\delta y\delta z)$$

$$= \frac{\partial v_x}{\partial x}\delta x\,\delta y\,\delta z$$

Similarly, the net preserving mass entering from RSR'S' and outing from one face PQQ'P' is

$$F_2 = \frac{\partial v_y}{\partial y}\delta x\,\delta y\,\delta z$$

Similarly, the net preserving mass entering from PSP'S' and outing from one face QRR'Q' is

$$F_3 = \frac{\partial v_z}{\partial z}\delta x\,\delta y\,\delta z$$

Total volume of fluid flowing through the parallelopiped per unit time is

$$F = F_1 + F_2 + F_3 = \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}\right)\delta x\,\delta y\,\delta z$$

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Dividing by $\delta x\delta y\delta z$ and then taking limit at origin, we get

$$\lim_{(\delta x,\,\delta y,\,\delta z)\to(0,\,0,\,0)}\left(\frac{F}{\delta x\,\delta y\,\delta z}\right) = \lim_{(\delta x,\,\delta y,\,\delta z)\to(0,\,0,\,0)}\left(\frac{\partial v_x}{\partial x}+\frac{\partial v_y}{\partial y}+\frac{\partial v_z}{\partial z}\right)$$

$$=\left(\frac{\partial v_x}{\partial x}+\frac{\partial v_y}{\partial y}+\frac{\partial v_z}{\partial z}\right)$$

$$=\left(\vec{i}\frac{\partial}{\partial x}+\vec{j}\frac{\partial}{\partial y}+\vec{k}\frac{\partial}{\partial z}\right)\cdot\left(v_x\vec{i}+v_y\vec{j}+v_z\vec{k}\right)$$

$$=\nabla\cdot\vec{v}$$

$$=\text{div}\,\vec{v}$$

Thus, the rate of flow of fluid through the parallelopiped per unit volume is div $\vec{v}$.

Hence, div $\vec{v}$ gives the rate of flow of fluid per unit volume at a point of the fluid.

Note: 1. If div $\vec{v}=0$ then $\vec{v}$ is called **Solenoidal.** That is, if div $\vec{v}=0$ then the given closed surface has no net flux across it.

2. If div $\vec{v}>0$ then $\vec{v}$ is the source of the field.

3. If div $\vec{v}<0$ then $\vec{v}$ is the sink of the field.

Example 5: Find divergence of $e^x\vec{i}+ye^{-x}\vec{j}+2z\sinh x\,\vec{k}$

Solution:

Let $\vec{v}=e^x\vec{i}+ye^{-x}\vec{j}+2z\sinh x\,\vec{k}$.

Then the divergence of $\vec{v}$ is,

$$\text{div}\,\vec{v}=\nabla\cdot\vec{v}=\left(\vec{i}\frac{\partial}{\partial x}+\vec{j}\frac{\partial}{\partial y}+\vec{k}\frac{\partial}{\partial z}\right)\cdot(e^x\vec{i}+ye^{-x}\vec{j}+2z\sinh x\,\vec{k})$$

$$=e^x+e^{-x}+2\sinh x$$

$$=e^x+e^{-x}+2\sinh x$$

$$=e^x+e^{-x}+2\left(\frac{e^x-e^{-x}}{2}\right)$$

$$=e^x+e^{-x}+e^x-e^{-x}$$

$$=2e^x.$$

Thus, divergence of $e^x\vec{i}+ye^{-x}\vec{j}+2z\sinh x\,\vec{k}$ is $2e^x$.

Example 6: Show that the vector $\vec{v}=(x+3y)\vec{i}+(y-3z)\vec{j}+(x-2z)\vec{k}$ is solenoidal.

Solution:

Let, $\vec{v}=(x+3y)\vec{i}+(y-3z)\vec{j}+(x-2z)\vec{k}$

Then,

$$\text{div}\,\vec{v}=\nabla\cdot\vec{v}=\left(\vec{i}\frac{\partial}{\partial x}+\vec{j}\frac{\partial}{\partial y}+\vec{k}\frac{\partial}{\partial z}\right)\cdot((x+3y)\vec{i}+(y-3z)\vec{j}+(x-2z)\vec{k})$$

$$=1+1-2=0.$$

This shows that $\vec{v}$ is solenoidal.

4.10 Curl of a Vector Function

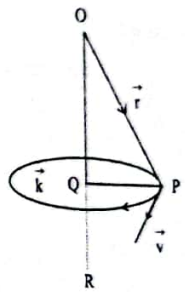
Let $\vec{v}=\vec{v}(x,y,z)$ be a vector valued function is continuous and differentiable at each point (x,y,z) in space, then the curl or rotation of $\vec{v}$ is denoted by curl$(\vec{v})$ or $\nabla\times\vec{v}$ and is defined by

$$\text{curl}\,\vec{v}=\nabla\times\vec{v}=\left(\vec{i}\frac{\partial}{\partial x}+\vec{j}\frac{\partial}{\partial y}+\vec{k}\frac{\partial}{\partial z}\right)\times\vec{v}$$

Note: If $\vec{v}=v_1\vec{i}+v_2\vec{j}+v_3\vec{k}$, then

$$\text{curl}\,\vec{v}=\nabla\times\vec{v}=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\ \partial/\partial x&\partial/\partial y&\partial/\partial z\\ v_1&v_2&v_3\end{vmatrix}$$

Physical Interpretation of Curl of a Vector Function



Consider a rigid body rotating about a fixed axis through origin O with an uniform angular velocity $\vec{\omega}=\omega_1\vec{i}+\omega_2\vec{j}+\omega_3\vec{k}$.

If $\vec{r}$ be the position vector of a particle P(x, y, z) on the rigid body, then the velocity $\vec{v}$ of the particle is given by

$$\vec{v}=\vec{\omega}\times\vec{r}=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\ \omega_1&\omega_2&\omega_3\\ x&y&z\end{vmatrix}$$

$$=(\omega_2z-\omega_3y)\vec{i}+(\omega_3x-\omega_1z)\vec{j}+(\omega_1y-\omega_2x)\vec{k}$$

Then,

$$\text{curl}\,\vec{v}=\nabla\times\vec{v}=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\ \partial/\partial x&\partial/\partial y&\partial/\partial z\\ \omega_2z-\omega_3y&\omega_3x-\omega_1z&\omega_1y-\omega_2x\end{vmatrix}$$

$$=(\omega_1+\omega_1)\vec{i}+(\omega_2+\omega_2)\vec{j}+(\omega_3-\omega_3)\vec{k}$$

$$=2\left(\omega_1\vec{i}+\omega_2\vec{j}+\omega_3\vec{k}\right)$$

$$=2\vec{\omega}.$$

Thus, the curl of the linear velocity of a rigid body is equal to twice of the angular velocity of the body. That is, the uniform velocity of a uniformly rotating body is half of the curl of the linear velocity. This means we often use rotation or *rot* for curl of a function.

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Note: (Irrotational Vector Function)

A vector function $\vec{v}$ is said to be irrotational if its curl is zero i.e. if $\nabla \times \vec{v} = 0$.

4.11 Some Properties of the Operator ∇

Let ϕ and ψ denotes scalar point function, $\vec{u}$ and $\vec{v}$ are vector point function, then,

a. $\text{grad}(\phi \pm \psi) = \text{grad}\phi \pm \text{grad}\psi$

b. $\text{div}(\vec{u} \pm \vec{v}) = \text{div}\vec{u} \pm \text{div}\vec{v}$

c. $\text{curl}(\vec{u} \pm \vec{v}) = \text{curl}\,\vec{u} \pm \text{curl}\,\vec{v}$

Theorem 1: Let f be a continuous and differentiable scalar valued function, then $\text{curl}(\text{grad}f) = \vec{0}$.

Proof:

Let f be a scalar valued function. Then

$$\text{grad}\, f = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right) f$$

$$= \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j} + \frac{\partial f}{\partial z}\vec{k}$$

Now,

$$\text{curl}(\text{grad}f) = \nabla \times \left(\frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j} + \frac{\partial f}{\partial z}\vec{k}\right)$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{i} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial f/\partial x & \partial f/\partial y & \partial f/\partial z \end{vmatrix}$$

$$= \vec{i}\left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y}\right) - \vec{j}\left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y}\right) + \vec{k}\left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y}\right)$$

$$= \vec{i}0 + \vec{j}0 + \vec{k}0$$

$$= \vec{0}$$

Thus, we get $\text{curl}(\text{grad}(f)) = \vec{0}$.

4.12 Conservative of a Function

A force $\vec{F}$ is said to be conservative if it is derivable from the potential function ϕ, i.e. $\vec{F} = \text{grad}\phi$.

Here,

$$\text{curl}\,\vec{F} = \text{curl}(\text{grad}\phi) = \vec{0}.$$

Thus, if $\vec{F}$ is conservative then $\text{curl}(\vec{F}) = \vec{0}$ and there exists a scalar potential function ϕ such that $\vec{F} = \text{grad}\phi = \nabla\phi$.

Theorem 2: Let $\vec{v}$ be a vector valued function, which is continuous and differentiable, then $\text{div}(\text{curl}\,\vec{v}) = 0$.

Proof:

Let $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$ be a vector valued function, which is continuous and differentiable. Then

$$\text{curl}\,\vec{v} = \nabla \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{i} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_1 & v_2 & v_1 \end{vmatrix}$$

$$= \vec{i}\left(\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z}\right) - \vec{j}\left(\frac{\partial v_3}{\partial x} - \frac{\partial v_1}{\partial z}\right) + \vec{k}\left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y}\right)$$

Now,

$$\text{div}(\text{curl}\,\vec{v}) = \nabla \cdot \left[\vec{i}\left(\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z}\right) - \vec{j}\left(\frac{\partial v_3}{\partial x} - \frac{\partial v_1}{\partial z}\right) + \vec{k}\left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y}\right)\right]$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z}\right) - \frac{\partial}{\partial y}\left(\frac{\partial v_3}{\partial x} - \frac{\partial v_1}{\partial z}\right) + \frac{\partial}{\partial z}\left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y}\right)$$

$$= \frac{\partial^2 v_3}{\partial x \partial y} - \frac{\partial^2 v_2}{\partial x \partial z} - \frac{\partial^2 v_3}{\partial y \partial x} + \frac{\partial^2 v_1}{\partial y \partial z} + \frac{\partial^2 v_2}{\partial z \partial x} - \frac{\partial^2 v_1}{\partial z \partial y}$$

$$= 0.$$

Thus, $\text{div}(\text{curl}\,\vec{v}) = 0$.

Example 7: Find the curl of vector, $xyz\vec{i} + 3x^2y\vec{j} + (xz^2 - y^2z)\vec{k}$

Solution:

Let $\vec{v} = xyz\,\vec{i} + 3x^2y\,\vec{j} + (xz^2 - y^2z)\,\vec{k}$

Then the curl of $\vec{v}$ is,

$$\text{curl}\,\vec{v} = \nabla \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xyz & 3x^2y & xy^2 - y^2z \end{vmatrix}$$

$$= -2yz\,\vec{i} - (z^2 - xy)\,\vec{j} + (6xy - xz)\vec{k}$$

Thus, curl of $\vec{v}$ is $-2yz\,\vec{i} + (xy - z^2)\,\vec{j} + x(6y - z)\vec{k}$.

Example 8: Show that the vector $\vec{v} = (\text{siny} + z)\vec{i} + (x\,\text{cosy} - z)\vec{j} + (x - y)\vec{k}$ is irrotational.

Solution:

Let,

$$\vec{v} = (\text{siny} + z)\vec{i} + (x\text{cosy} - z)\vec{j} + (x - y)\vec{k}$$

Then $\vec{v}$ is irretational if $\text{curl}\,\vec{v} = \vec{0}$.

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Here,

$$\text{curl } \vec{v} = \nabla \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \sin y + z & x\cos y - z & x - y \end{vmatrix}$$

$$= (-1+1)\vec{i} + (1-1)\vec{j} + (\cos y - \cos y)\vec{k}$$

$$= \vec{0}.$$

This shows that $\vec{v}$ is irrotational.

Example 9: If $\phi = \log(x^2 + y^2 + z^2)$, find div(grad ϕ) and curl(grad ϕ).

Solution:

Let,

$$\phi = \log(x^2 + y^2 + z^2)$$

Then,

$$\text{grad } \phi = \nabla\phi = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)(\log(x^2 + y^2 + z^2))$$

$$= \frac{2x}{x^2+y^2+z^2}\vec{i} + \frac{2y}{x^2+y^2+z^2}\vec{j} + \frac{2z}{x^2+y^2+z^2}\vec{k}$$

$$= \frac{2}{x^2+y^2+z^2}(x\vec{i} + y\vec{j} + z\vec{k})$$

Now,

$$\text{div (grad } \phi) = \nabla \cdot (\text{grad } \phi)$$

$$= \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right) \cdot \left(\frac{2(x\vec{i} + y\vec{j} + z\vec{k})}{x^2+y^2+z^2}\right)$$

$$= 2\left[\frac{(x^2+y^2+z^2) - 2x^2}{(x^2+y^2+z^2)}\right] + 2\left[\frac{(x^2+y^2+z^2) - 2y^2}{(x^2+y^2+z^2)}\right] + 2\left[\frac{(x^2+y^2+z^2) - 2z^2}{(x^2+y^2+z^2)}\right]$$

$$= \frac{2}{(x^2+y^2+z^2)^2}[y^2 + z^2 - x^2 + x^2 + z^2 - y^2 + x^2 + y^2 - z^2]$$

$$= \frac{2}{(x^2+y^2+z^2)^2}(x^2 + y^2 + z^2)$$

$$= \frac{2}{(x^2+y^2+z^2)}$$

And,

$$\text{curl(grad } \phi) = \nabla \times (\text{grad } \phi)$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \frac{2x}{x^2+y^2+z^2} & \frac{2y}{x^2+y^2+z^2} & \frac{2z}{x^2+y^2+z^2} \end{vmatrix}$$

$$= \left(\frac{-4yz}{(x^2+y^2+z^2)^2} - \frac{-4yz}{(x^2+y^2+z^2)^2}\right)\vec{i} - \left(\frac{-4xz}{(x^2+y^2+z^2)^2} - \frac{-4xz}{(x^2+y^2+z^2)^2}\right)\vec{j}$$

$$+ \left(\frac{-2xy}{(x^2+y^2+z^2)^2} - \frac{-2xy}{(x^2+y^2+z^2)^2}\right)\vec{k}$$

$$= \vec{0}.$$

Thus, div(grad ϕ) $= \frac{2}{(x^2+y^2+z^2)^2}$ and curl(grad ϕ) $= \vec{0}$.

Example 10: Show that the vector $\vec{F} = (2xy^2 + yz)\vec{i} + (2x^2y + xz + 2yz^2)\vec{j} + (2y^2z + xy)\vec{k}$ is conservative field.

Solution:

Given that,

$$\vec{F} = (2xy^2 + yz)\vec{i} + (2x^2y + xz + 2yz^2)\vec{j} + (2y^2z + xy)\vec{k}$$

Then, $\vec{F}$ is conservative is $\text{curl}(\vec{F}) = 0$

Here,

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2xy^2 + yz & 2x^2y + xz + 2yz^2 & 2y^2z + xy \end{vmatrix}$$

$$= (4yz + x - x - 4yz)\vec{i} - (y - y)\vec{j} + (4xy + z - 4xy - z)\vec{k}$$

$$= (0)\vec{i} + (0)\vec{j} + (0)\vec{k}$$

$$= \vec{0}$$

This shows that $\vec{F}$ is conservative.

Example 11: Show that the vector $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ is conservative and find the scalar potential function ϕ such that $\vec{F} = \nabla\phi$.

Solution:

Given that,

$$\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$$

Then, $\vec{F}$ is conservative if curl $\vec{F} = 0$.

Here,

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 - yz & y^2 - zx & z^2 - xy \end{vmatrix}$$

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$$= (-x + x)\vec{i} - (-y + y)\vec{j} + (-z + z)\vec{k}$$
$$= (0)\vec{i} + (0)\vec{j} + (0)\vec{k}$$
$$= \vec{0}$$

This shows that $\vec{F}$ is conservative.

Then, $\vec{F}$ can be written as $\vec{F} = \nabla\phi$. So,

$$\nabla\phi \cdot d\vec{r} = \left(\frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{j}\right) \cdot (dx\,\vec{i} + dy\,\vec{j} + dz\,\vec{k})$$
$$= \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz$$
$$= d\phi$$

i.e. $\vec{F} \cdot d\vec{r} = d\phi$

$$\Rightarrow d\phi = (x^2 - yz, y^2 - zx, z^2 - xy) \cdot (dx\,\vec{i} + dy\,\vec{j} + dz\,\vec{k})$$
$$= (x^2 - yz)dx + (y^2 - zx)dy + (z^2 - xy)dz$$

Integrating we get

$$\phi = \int (x^2 - yz)dx + \int (\text{free term of } (y^2 - zx) \text{ from x})dy +$$
$$\int (\text{free term of } (z^2 - xy) \text{ from x and y})dz$$

$$\Rightarrow \phi = \int (x^2 - yz)dx + \int (y^2)\,dy + \int (z^2)\,dz$$
$$= \frac{x^3}{3} - xyz + \frac{y^3}{3} + \frac{z^3}{3} + c$$
$$= \frac{x^3 + y^3 + z^3}{3} - xyz + c$$

Exercise 4 (B)

1. Find grad f, where
   a. $f = x^3 + yz$ b. $f = \log(x^2 + y^2 + z^2)$ c. $f = xy + yz + zx$.
2. Find a unit normal to the surface $xy^2z^2 = 4$ at $(-1, -1, 2)$.
3. Find divergence of
   a. $x\vec{i} + y\vec{j} + z\vec{k}$ b. $e^x(\cos y\,\vec{i} + \sin y\,\vec{j})$ c. $\left(-\frac{y\vec{i} + x\vec{j}}{x^2 + y^2}\right)$
   d. $(2x^2\vec{i} + 5xy^2\vec{j} + xyz^2\vec{k})$ at $(1, 2, 2)$.

   e. $(xy\sin\alpha\,\vec{i} + y^2\sin x\,\vec{j} + z^2\sin xy\,\vec{k})$ at $\left(0, \frac{\pi}{2}, \frac{\pi}{2}\right)$.
4. If the divergence of $\vec{F} = 2x\vec{i} + y\vec{j} + pz\vec{k}$ is zero, then find the value of p.
5. Find curl of
   a. $\vec{F} = 2y\vec{j} + 5x\vec{k}$ b. $\frac{1}{2}(x^2 + y^2 + z^2)(\vec{i} + \vec{j} + \vec{k})$
   c. $(x^3 + yz)\vec{i} + (y^2 + zx)\vec{j} + (z^2 + xy)\vec{k}$ d. $\vec{v} = x^2y\vec{i} + xz\vec{j} + 2yz\vec{k}$
6. a. If $u = x^2 + y^2 + z^2$ and $\vec{v} = x\vec{i} + y\vec{j} + z\vec{k}$. Show that $\text{div}(u\vec{v}) = 5u$
   b. If $\vec{A} = 3xz^2\vec{i} - yz\vec{j} + (x + 2z)\vec{k}$. Find curl (curl $\vec{A}$)
   c. If $\vec{v} = x^2yz\vec{i} + xy^2z\vec{j} + xyz^2\vec{k}$, find
      (i) div (curl $\vec{v}$) (ii) curl (curl $\vec{v}$).
   d. If $\phi = x^3 + y^3 + z^3 - 3xyz$. Find (i) div(grad ϕ), (ii) curl(grad ϕ).
   e. If $\vec{u} = y\vec{i} + z\vec{j} + x\vec{k}$ and $\vec{v} = yz\vec{i} + zx\vec{j} + xy\vec{k}$, then find
      (i) curl $(\vec{u} \times \vec{v})$ (ii) grad$(\vec{u} \cdot \vec{v})$.
7. Find the directional derivatives of f at P in the direction $\vec{a}$, where
   a. $f = x^2 + y^2$, P(1, 1), $\vec{a} = 2\vec{i} - 4\vec{j}$
   b. $f = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$, P(3, 0, 4); $\vec{a} = \vec{i} + \vec{j} + \vec{k}$.
   c. $f = xyz$, P(−1, 1, 3), $\vec{a} = \vec{i} - 2\vec{j} + 2\vec{k}$
   d. $f = 4xz^3 - 3x^2yz^2$ at (2, −1, 2) along z–axis.
   e. $f = x^2 - y^2 + 2z^2$ at the point A(1, 2, 3) in the direction of $\vec{a} = \vec{i} + \vec{j} + \vec{k}$.
8. a. In what direction from (3, 1, −2) is the directional derivative of $f = x^2y^2z^4$ maximum and what is its magnitude?
   b. What is the greatest rate of increase of $u = x^2 + yz^2$ at the point (1, −1, 3)?
9. Show that following vectors are conservative (or irrotational) vector field
   a. $\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$ b. $\vec{F} = \cos y\,\vec{i} - x\sin y\,\vec{j} - \cos z\,\vec{k}$
   c. $\vec{F} = (y + \sin z)\vec{i} + x\vec{j} + x\cos z\,\vec{k}$ d. $\vec{v} = 2xyz^3\vec{i} + x^2z^3\vec{j} + 3x^2yz^2\vec{k}$

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9. Show that following vectors are conservative and then find the scalar potential function ϕ such that $\vec{F} = \nabla\phi$.

a. $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$

b. $\vec{F} = yz\vec{i} + xz\vec{j} + xy\vec{k}$

c. $\vec{F} = (y\sin z - \sin x)\vec{i} + (x\sin z + 2yz)\vec{j} + (xy\cos z + y^2)\vec{k}$

Answer Key

1. a. $2x\vec{i} + z\vec{j} + y\vec{k}$ b. $\frac{2}{x^2+y^2+z^2}(x\vec{i} + y\vec{j} + z\vec{k})$
   c. $(y+z, x+z, x+y)$
2. $\frac{(-1, -3, 1)}{\sqrt{11}}$
3. a. 3 b. $2e^x \cos y$ c. 0 d. 80
   e. $\frac{\pi}{2}$ f. -3
4. a. $-5\vec{j}$ b. $(y-z)\vec{i} + (z-x)\vec{j} + (x-y)\vec{k}$ c. 0
   d. $(2z-x)\vec{i} + (z-x^2)\vec{k}$
5. b. $-6x\vec{i} + (6z-1)\vec{k}$ c. (i) 0 (ii) $4(yz\vec{i} + zx\vec{j} + xy\vec{k})$
   d. (i) $6(x+y+z)$ (ii) 0
   e. (i) $(xz - z^2 - xy)\vec{i} + (xy - x^2 - yz)\vec{j} + (yz - y^2 - zx)\vec{k}$
   (ii) $(z^2 + 2xy)\vec{i} + (x^2 + 2yz)\vec{j} + (y^2 + 2zx)\vec{k}$
6. a. $-\frac{2}{\sqrt{5}}$ b. $-\frac{14}{125\sqrt{3}}$ c. $\frac{7}{3}$
   d. 144 e. $\frac{1}{\sqrt{3}}(10)$
7. a. $96\sqrt{19}$ b. 11
9. a. $\phi = \frac{1}{2}(x^2 + y^2 + z^2) + C$ b. $\phi = xyz + C$
   c. $\phi = xy\sin z + \cos x + y^2 z + C$

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5

LINE AND SURFACE INTEGRAL

CHAPTER OUTLINE

After completion of this chapter, you will be able to gain the knowledge and understanding regarding:

5.1 Line integral with physical interpretation and evaluation of line integrals on various path

5.2 Line integral, potential function and independence of path

5.3 Surface integral and evaluation of surface integrals

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Introduction

A vector function of one variable is a function of the form $\vec{F}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$. This vector function is continuous at to if each component function is continuous at to. We may think of $\vec{F}(t)$ as the position vector of a curve in three dimensional space. For each t for which the vector is defined, let $\vec{F}(t)$ as an arrow from the origin to the point (x(t), y(t), z(t)). This arrow sweeps out a curve C as t varies. When thought of in this way, the coordinate functions are parametric equations of this curve.

In calculus, the derivative of a function gives the slope of the tangent to the graph at a point. In vector calculus, the derivative of the position vector of a curve gives the tangent vector to the curve at a point.

5.1 CURVE IN SPACE

If x, y and z are determined with the parametric value t by the relation, x = x(t), y = y(t), z = z(t). Let the space curve C represents by a parametric representation $\vec{r} = (x(t), y(t), z(t)) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$ with t is defines for $a \le t \le b$ (for fixed a, b). The space curve C is called the path of integration. The curve C has an orientation, since the object starts at the initial point (x(a), y(a), z(a)) at time t = a and ends at the terminal point (x(b), y(b),z(b)) at time t = b. We often indicate this orientation by putting arrows along the graph.

We call the curve C:

Continuous: If each coordinates function is continuous.

Differentiable: If each coordinates function is differentiable.

Smooth Curve: Let $\vec{r} = \vec{f}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$. If this curve f(t) is continuous and differentiable for all values within given interval $a \le t \le b$ which are not all zero for the same value of t, then the curve is called smooth.

Closed Curve: If a curve $\vec{r} = \vec{f}(t)$ has same initial and final points coincide, that is; (x(a), y(a), z(a)) = (x(b), y(b),z(b)) then it is called a closed curve.

A closed curve is called simple if $a < t_1 < t_2 < b$ implies that

$$(x(t_1), y)t_1),z(t_1)) \neq (x(t_2), y(t_2),z(t_2)).$$

This means a curve is simple if it does not intersect itself at different times.

Example 1:

(a) Let C have coordinate functions

$$x = 4\cos(t),\ y = 4\sin(t),\ z = 9 \quad \text{for } 0 \le t \le 2\pi.$$

Clearly, the graph of C is a circle with radius 4 about the origin in the plane z = 9. Also, the curve C is simple, closed and smooth.

(b) Let K be given by

$$x = 4\cos(t),\ y = 4\sin(t),\ z = 9 \quad \text{for } 0 \le t \le 4\pi.$$

Clearly, the graph of K is the same as the graph of K, except that a particle traversing K goes around this circle twice. This implies K is closed and smooth but not simple.

(c) Let L be the curve given by

$$x(t) = 4\cos(t),\ y = 4\sin(t),\ z = 9 \quad \text{for } 0 \le t \le 3\pi.$$

Clearly, the graph of L is again the circle with radius 4 about the origin in the plane z = 9. Also, the curve L is smooth and not simple, but L is also not closed, since the initial point of L is (4, 0, 9) and its terminal point is (–4, 0, 9).

5.2 LINE INTEGRAL

Suppose C is a smooth curve with coordinate functions x = x(t), y = y(t), z = z(t) for $a \le t \le b$. Let f, g and h be continuous at least at points on the graph of C. Then the line integral

$$\int_C (f\,dx + g\,dy + h\,dz)$$

is defined by

$$\int_C (f\,dx + g\,dy + h\,dz) = \int_a^b \left[f(x, y, z)\frac{dx}{dt} + g(x, y, z)\frac{dy}{dt} + h(x, y, z)\frac{dz}{dt} \right] dt$$

with f(x, y, z) = f(x(t), y(t), z(t)); g(x, y, z) = g(x(t), y(t), z(t)) and h(x, y, z) = h(x(t), y(t), z(t)).

Evaluation of Line Integrals

If a general point of a curve C is $\vec{r} = \vec{r}(t)$ and its vector function is $\vec{F} = \vec{F}(r)$ over a curve C. Then

$$\vec{r} = (x, y, z) = (x(t), y(t), z(t)) \quad \text{for } a \le t \le b$$

and,

$$\vec{F} = \vec{F}(r) = (f(r), g(r), h(r))$$

So that

$$\vec{F}\cdot d\vec{r} = (f(r), g(r), h(r))\cdot(dx, dy, dz)$$
$$= f(r)dx + g(r)dy + h(r)dz$$

Then the line integral of the vector function is $\vec{F} = \vec{F}(r)$ over C is defined by

$$\int_C \vec{F}\cdot d\vec{r} = \int_C \vec{F}(r)\cdot d\vec{r} = \int_a^b F(r(t))\cdot\frac{dr}{dt}\,dt = \int_a^b F(r(t))\cdot r'(t)\,dt$$

where r(t) is the parametric representation of C with $a \le t \le b$.

If the path of integration C is a closed curve, then instead of

$$\int_C \vec{F}\cdot d\vec{r} \quad \text{we also write} \quad \int_C \vec{F}\cdot d\vec{r}$$

Note that the integrand in above is a scalar, not a vector, because we take the dot product. We see that the integral in above is a definite integral of a function of t taken over the interval on the t-axis in the positive direction.

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Note:

The line integrals arise naturally in mechanics, where they give the work done by a force F in a displacement along the curve C. We may thus call the line integral as the work integral.

Properties of Line Integral

Some general properties of the line integral as follow:

1. For a constant k, $\int_C k\vec{F}\cdot d\vec{r} = k\int_C \vec{F}\cdot d\vec{r}$
2. $\int_C (\vec{F}+G)\cdot d\vec{r} = \int_C \vec{F}\cdot d\vec{r} + \int_C G\cdot d\vec{r}$
3. If the curve C is subdivided into two arcs C_1 and C_2 that have the same orientation as C then,

$$\int_C \vec{F}\cdot d\vec{r} = \int_{C_1} \vec{F}\cdot d\vec{r} + \int_{C_2} \vec{F}\cdot d\vec{r}$$

5.3 Line Integral of a Scalar Field with Respect to Arc Length

In some contexts it is useful to have a line integral with respect to arc length along C. If $\prod(x, y, z)$ is a scalar field and C is a smooth curve with coordinate functions $x = x(t)$, $y = y(t)$, $z = z(t)$ for $a \le t \le b$, we define

$$\int_C \prod(x, y, z)\, ds = \int_a^b \prod(x(t), y(t), z(t))\sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\, dt$$

5.4 Path Independence of Line Integrals

The path of integration of a curve along the curve is independent to choose of its path providing the curve is continuous. That is, the path of integration of a curve is depends only on the end points of the curve C. That is, the line integral $\int_C \vec{F}\cdot d\vec{r}$ is said to be path independent in a domain D in space if for every pair of endpoints A and B in domain D.

Theorem 1: Let $\vec{F}$ be a continuous vector valued function in a domain D, then the line integral $\int_C \vec{F}\cdot d\vec{r}$ is independent of the path C joining any two points in D if and only if there exists a scalar valued function ϕ such that $\vec{F} = \nabla\phi$.

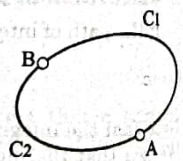
Proof:

Necessary condition

Let $\int_C \vec{F}\cdot d\vec{r}$ is independent of path C joining any two points $A(x_1, y_1, z_1)$ and $B(x, y, z)$. Let A be a fixed point and B be any point in C.

Let

$$\phi(x, y, z) = \int_A^B \vec{F}\cdot d\vec{r} = \int_A^B \left(\vec{F}\cdot\frac{d\vec{r}}{ds}\right) ds$$



Differentiating both sides with respect to s,

$$\frac{d\phi}{ds} = \vec{F}\cdot\frac{d\vec{r}}{ds}$$

i.e. $\vec{F}\cdot\frac{d\vec{r}}{ds} = \frac{d\phi}{ds} = \left(\frac{\partial\phi}{\partial x}\frac{dx}{ds} + \frac{\partial\phi}{\partial y}\frac{dy}{ds} + \frac{\partial\phi}{\partial z}\frac{dz}{ds}\right)$

$$= \left(\frac{\partial\phi}{\partial x}\vec{i} + \frac{\partial\phi}{\partial y}\vec{j} + \frac{\partial\phi}{\partial z}\vec{k}\right)\left(\vec{i}\frac{dx}{ds} + \vec{j}\frac{dy}{ds} + \vec{k}\frac{dz}{ds}\right)$$

$$= \left(\frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{k}\right)\phi\cdot\frac{d}{ds}\left(x\vec{i} + y\vec{j} + z\vec{k}\right)$$

$$= \nabla\phi\cdot\frac{d}{ds}\left(\vec{r}\right)$$

$$\Rightarrow (\vec{F} - \nabla\phi)\cdot\frac{d\vec{r}}{ds} = 0 \quad \ldots(1)$$

Fig.: Proof of Theorem 2

Since $\frac{d\vec{r}}{ds}$ is tangential vector, is non-zero. So,

$$\vec{F} = \nabla\phi.$$

Sufficient condition:

Let $\vec{F} = \nabla\phi$ and C be any path joining any two points A and B in domain D, then

$$\int_C \vec{F}\cdot d\vec{r} = \int_C \nabla\phi\cdot d\vec{r}$$

$$= \int_A^B \left(\frac{\partial\phi}{\partial x}\vec{i} + \frac{\partial\phi}{\partial y}\vec{j} + \frac{\partial\phi}{\partial z}\vec{k}\right)\cdot\left(\vec{i}\,dx + \vec{j}\,dy + \vec{k}\,dz\right)$$

$$= \int_A^B \left(\frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz\right)$$

$$= \int_A^B d\phi$$

$$= [\phi]_A^B$$

$$= \phi(B) - \phi(A)$$

This gives $\int_C \vec{F}\cdot d\vec{r}$ depends on the end points A and B and not on the path joining them. Hence, $\int_C \vec{F}\cdot d\vec{r}$ is independent of path C.

Remark 1:

If the line integral $\int_C \vec{F}\cdot d\vec{r}$ is independent of path joining any two points, then the force $\vec{F}$ is conservative i.e. curl $\vec{F} = \vec{0}$ i.e. $\nabla\times\vec{F} = \vec{0}$.

Proof:

Let the line integral $\int_C \vec{F}\cdot d\vec{r}$ is independent of path joining any two points, then above theorem implies us

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$$\vec{F} = \nabla\phi$$

Then

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \nabla \times \nabla\phi = (\nabla \times \nabla)\phi = \vec{0}$$

being $\nabla \times \nabla = \vec{0}$.

Remark 2:

If the line integral $\int_C \vec{F}\cdot d\vec{r}$ is independent of path joining any two points, then the force $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ is conservative. Therefore,

$$\text{curl } \vec{F} = \vec{0}$$

$$\Rightarrow \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_1 & F_2 & F_3 \end{vmatrix} = (0, 0, 0) = \left(0\vec{i} + 0\vec{j} + 0\vec{k}\right)$$

$$\Rightarrow \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\vec{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right)\vec{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\vec{k} = \left(0\vec{i} + 0\vec{j} + 0\vec{k}\right)$$

This implies

$$\Rightarrow \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} = 0;\ \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} = 0;\ \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = 0$$

i.e. $\frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z},\ \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x};\ \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$

Note that form the ODEs, the force $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ is an exact function when

$$\frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z},\ \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x};\ \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}.$$

The following definition leads the concepts.

Definition: (Criterion for Exactness and Path Independence)

Let C be a curve in a simply connected domain D in space. Let $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ be a continuous vector valued function having continuous first partial derivatives in D. Then we called the line integral $\int_C \vec{F}\cdot d\vec{r}$ is called exact if the following conditions hold;

$$\frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z},\ \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x};\ \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$$

Note that the solution of ODEs implies that if the function $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ is exact then its solution can be observe by the process

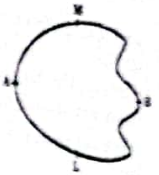
$$\int_C \vec{F}\cdot d\vec{r} = \int_{y,\,z \text{ are constant}} F_1\,dx + \int_{z \text{ is constant}} (\text{Term of } F_2 \text{ free from x})\,dy + \int_C (\text{Term of } F_3 \text{ free from x and y})\,dz.$$

Theorem 2: If $\int_A^B \vec{F}\cdot d\vec{r}$ is independent of the path joining any two points A and B in the region R, if an only if $\int_A^B \vec{F}\cdot d\vec{r} = 0$ for any simple closed curve C in domain D.

Proof:

Necessary condition

Let C be a simple closed curve joining the point ALBMA in a domain D as shown in figure.



We have, $\int_A^B \vec{F}\cdot d\vec{r}$ be independent of path ALB and AMB.

Then

$$\int_{ALB} \vec{F}\cdot d\vec{r} = \int_{AMB} \vec{F}\cdot d\vec{r} \quad \ldots (1)$$

Here,

$$\int_C \vec{F}\cdot d\vec{r} = \int_{ALB} \vec{F}\cdot d\vec{r} + \int_{BMA} \vec{F}\cdot d\vec{r}$$

$$= \int_{ALB} \vec{F}\cdot d\vec{r} - \int_{AMB} \vec{F}\cdot d\vec{r}$$

$$= 0 \qquad \text{[Using (1)]}$$

Thus, $\int_C \vec{F}\cdot d\vec{r} = 0$

Sufficient condition

For every closed curve C, let

$$\int_C \vec{F}\cdot d\vec{r} = 0,$$

then we have to show $\int_A^B \vec{F}\cdot d\vec{r}$ is independent of path

Here,

$$\int_C \vec{F}\cdot d\vec{r} = 0$$

$$\Rightarrow \int_{ALBMA} \vec{F}\cdot d\vec{r} = 0$$

$$\Rightarrow \int_{ALB} \vec{F}\cdot d\vec{r} + \int_{BMA} \vec{F}\cdot d\vec{r} = 0$$

$$\Rightarrow \int_{ALB} \vec{F}\cdot d\vec{r} - \int_{AMB} \vec{F}\cdot d\vec{r} = 0$$

$$\Rightarrow \int_{ALB} \vec{F}\cdot d\vec{r} = \int_{AMB} \vec{F}\cdot d\vec{r}$$

This implies the line integral is independent of the path joining the points A and B in D.

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Workout Examples

Example 1: Calculate $\int_C \vec{F}\cdot d\vec{r}$, where $\vec{F} = (xy, x^2y^2)$ and the displacement along C is the quarter circle from (2, 0) to (0, 2) with center (0, 0).

Solution:

Given that, $\vec{F} = (xy, x^2y^2) = xy\,\vec{i} + x^2y^2\,\vec{j}$

and the path of force $\vec{F}$ is the quarter circle from (2, 0) to (0, 2) with center at (0, 0).

Clearly, the length of (0, 0) to (2, 0) is 2. So, radius of the circle is 2. Therefore, the equation of path of curve is, $x^2 + y^2 = 4$.

Put $x = t$ then $y = \sqrt{4 - t^2}$. Also, t moves from $t = 2$ to $t = 0$

Then,

$$\vec{r} = x\,\vec{i} + y\,\vec{j}$$

$$\Rightarrow \quad \vec{r} = t\,\vec{i} + \sqrt{4 - t^2}\,\vec{j}$$

So, $$d\vec{r} = dt\,\vec{i} - \frac{t\,dt}{\sqrt{4 - t^2}}\,\vec{j} = \left(\vec{i} - \frac{t}{\sqrt{4 - t^2}}\,\vec{j}\right) dt$$

So that,

$$\vec{F}\cdot d\vec{r} = \left(t\sqrt{4 - t^2}\,\vec{i} + (4t^2 - t^4)\,\vec{j}\right)\cdot\left(\vec{i} - \frac{t}{\sqrt{4 - t^2}}\,\vec{j}\right) dt$$

$$= \left(t\sqrt{4 - t^2} - \frac{t(4t^2 - t^4)}{\sqrt{4 - t^2}}\right) dt$$

$$= \left(t\sqrt{4 - t^2} - \frac{t^3(4 - t^2)}{\sqrt{4 - t^2}}\right) dt$$

$$= \left(t\sqrt{4 - t^2} - t^3\sqrt{4 - t^2}\right) dt$$

$$= t\sqrt{4 - t^2}\,(1 - t^2)\,dt$$

Now,

$$\int_C \vec{F}\cdot d\vec{r} = \int_2^0 t\sqrt{4 - t^2}\,(1 - t^2)\,dt$$

Put $4 - t^2 = u^2$ then

$$-2t\,dt = 2u\,du \;\Rightarrow\; t\,dt = -u\,du.$$

Also, $t = 2 \Rightarrow u = 0$ and $t = 0 \Rightarrow u = 2$. So,

$$\int_C \vec{F}\cdot d\vec{r} = \int_0^2 u\,(4 - u^2)\,(-u\,du)$$

$$= -\int_0^2 (4u^2 - u^4)\,du$$

$$= -\left[4\frac{u^3}{3} - \frac{u^5}{5}\right]_0^2$$

$$= -\left(\frac{32}{3} - \frac{32}{5}\right)$$

$$= -\frac{64}{15}$$

Example 2: Calculate, $\int_C \vec{F}\cdot d\vec{r}$, where $\vec{F} = (\cosh x, \sinh y, e^z)$ and C be a path given by $\vec{r} = [t, t^2, t^3]$ from (0, 0, 0) to (2, 4, 8).

Solution:

Given that

$$\vec{F} = (\cosh x, \sinh y, e^z)$$

And, $\vec{r} = (t, t^2, t^3)$. Then, $d\vec{r} = (1, 2t, 3t^2)\,dt$.

Since we know $\vec{r} = (x, y, z)$. So, comparing it with $\vec{r} = (t, t^2, t^3)$ we get,

$$x = t,\ y = t^2,\ z = t^3$$

$$\vec{F} = (\cosh t, \sinh t^2, e^{t^3})$$

Then

and t moves from 0 to 2.

$$\vec{F}\cdot d\vec{r} = (\cosh t + 2t\sinh t^2 + 3t^2e^{t^3})\,dt$$

So,

Now,

$$\int_C \vec{F}\cdot d\vec{r} = \int_0^2 (\cosh t + 2t\sinh t^2 + 3t^2e^{t^3})\,dt$$

$$= \int_0^2 \cosh t\,dt + \int_0^2 2t\sinh t^2\,dt + \int_0^2 3t^2e^{t^3}\,dt$$

Put $t^2 = u$ and $t^3 = v$ then,

$$= \int_0^2 \cosh t\,dt + \int_0^4 \sinh u\,du + \int_0^8 e^v\,dv$$

$$= [\sinh t]_0^2 + [\cosh u]_0^4 + [e^v]_0^8$$

$$= \sinh 2 + \cosh 4 - 1 + e^8 - 1$$

$$= \sinh 2 + \cosh 4 + e^8 - 2.$$

Example 3: Evaluate, $\int_C \vec{F}\cdot d\vec{r}$ where $\vec{F} = x^2y^2\,\vec{i} + y\,\vec{j}$ and the curve C is $y^2 = 4x$ in the xy plane from (0, 0) to (4, 4).

Solution:

Given that the force is,

$$\vec{F} = x^2y^2\,\vec{i} + y\,\vec{j}$$

And the curve c is, $y^2 = 4x$

Then,

$$\vec{F} = \left(\frac{y^2}{4}\right)y^2\,\vec{i} + y\,\vec{j} = \frac{y^4}{4}\,\vec{i} + y\,\vec{j}$$

Since,

$$\vec{r} = x\,\vec{i} + y\,\vec{j} = \frac{y^2}{4}\,\vec{i} + y\,\vec{j}$$

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$$d\vec{r} = \left(\frac{y}{2}\vec{i} + \vec{j}\right)dy$$

Then,

Therefore,

$$\vec{F}\cdot d\vec{r} = \left(\frac{y^2}{4}\vec{i} + y\vec{j}\right)\cdot\left(\frac{y}{2}\vec{i} + \vec{j}\right)dy = \left(\frac{y^3}{8} + y\right)dy = \left(\frac{y^3 + 8y}{8}\right)dy.$$

Given that the force $\vec{F}$ moves along the curve c from (0, 0) to (4, 4), then

$$\int_C \vec{F}\cdot d\vec{r} = \int_0^4 \left(\frac{y^3 + 8y}{8}\right)dy$$

$$= \frac{1}{8}\left[\frac{y^4}{4} + 4y^2\right]_0^4$$

$$= \frac{1}{8}\left[\frac{(4)^4}{4} + 4.(4)^2\right]$$

$$= \frac{1}{8}[64 + 64]$$

$$= 16$$

Thus, $\int_C \vec{F}\cdot d\vec{r} = 16.$

Example 4: Evaluate, $\int_C \vec{F}\cdot d\vec{r}$ where $\vec{F} = \text{siny}\,\vec{i} + x(1 + \text{cosy})\vec{j}$ and c is the curve $x^2 + y^2 = a^2, z = 0$.

Solution:

Given that

$$\vec{F} = \text{siny}\,\vec{i} + x(1 + \text{cosy})\vec{j}$$

That moves along c:

$$x^2 + y^2 = a^2, z = 0$$

Since we have, $\vec{r} = \vec{i} + \vec{j}$

So, $d\vec{r} = dx\vec{i} + dy\vec{j}$

Then

$$\vec{F}\cdot d\vec{r} = (\text{siny}\,\vec{i} + x(1 + \text{cosy})\vec{j})\cdot(dx\vec{i} + dy\vec{j})$$
$$= \text{siny}\,dx + x(1 + \text{cosy})\,dy$$
$$= \text{siny}\,dx + x\,\text{cosy}\,dy + x\,dy$$
$$= d(x\,\text{siny}) + x\,dy$$

Now,

$$\int_C \vec{F}\cdot d\vec{r} = \int_C [d(x\,\text{siny}) + x\,dy]$$

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Since the path is circular curve. So, its parametric form is,

$$x = a\cos t, y = a\sin t$$

And t varies from t = 0 to t = 2π. Then

$$\int_C \vec{F}\cdot d\vec{r} = \int_0^{2\pi} [d(a\cos t \sin(a\sin t) + a\cos t\, a\, d(\sin t)]$$

$$= a[\cos t \sin(a\sin t)]_0^{2\pi} + a^2\int_0^{2\pi} \cos t \cos t\, dt$$

$$= a[\cos 2\pi \sin(a\sin 2\pi) - \cos 0 \sin(a\sin 0) +$$

$$a^2\int_0^{2\pi}\left(\frac{1 + \cos 2t}{2}\right)dt$$

$$= 0 + \frac{a^2}{2}\left[t + \frac{\sin 2t}{2}\right]_0^{2\pi}$$

$$= \frac{a^2}{2} \times 2\pi \qquad [\because \sin 2\pi = 0 = \sin 0]$$

$$= a^2\pi$$

Example 5: Evaluate, $\int_C \vec{F}\cdot d\vec{r}$ where $\vec{F} = \text{cosy}\,\vec{i} - x\,\text{siny}\,\vec{j}$ and c is the curve $y = \sqrt{1 - x^2}$ in the xy plane from (1, 0) to (0, 1).

Solution:

Let, $\vec{F} = \text{cosy}\,\vec{i} - x\,\text{siny}\,\vec{j}$.

And it moves along $y = \sqrt{1 - x^2}$ which is a half range circle having radius r = 1.

Since, $\vec{r} = x\vec{i} + y\vec{j}$.

So, $d\vec{r} = dx\vec{i} + dy\vec{j}$.

Then,

$$\int_C \vec{F}\cdot d\vec{r} = \int_C (\text{cosy}\,\vec{i} - x\sin\vec{j})\cdot(dx\vec{i} + dy\vec{j})$$

$$= \int_C (\text{cosy}\,dx - x\,\text{siny}\,dy)$$

$$= \int_C d(x\,\text{cosy})$$

$$= \int_1^0 d(x\cos\sqrt{1 - x^2})$$

$$= \left[x\cos\sqrt{1 - x^2}\right]_1^0$$

$$= -1$$

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Example 6: Evaluate, $\int_C \vec{F}\cdot d\vec{r}$ where $\vec{F} = yz\,\vec{i} + (xz + 1)\,\vec{j} + xy\,\vec{k}$, and c is the any path from (1, 0, 0) to (2, 1, 4).

Solution:

Given that,

$$\vec{F} = yz\,\vec{i} + (xz + 1)\,\vec{j} + xy\,\vec{k}$$

And the path is from (1, 0, 0) to (2, 1, 4).

Let,

$$\vec{r} = x\,\vec{i} + y\,\vec{j} + z\,\vec{k}$$

$$d\vec{r} = dx\,\vec{i} + dy\,\vec{j} + dz\,\vec{k}$$

Then,

Therefore,

$$\begin{aligned}\vec{F}\cdot d\vec{r} &= (yz\,\vec{i} + (xz + 1)\,\vec{j} + xy\,\vec{k})\cdot(dx\,\vec{i} + dy\,\vec{j} + dz\,\vec{k})\\ &= yz\,dx + (xz + 1)\,dy + xy\,dz\\ &= (yz\,dx + xz\,dy + xy\,dz) + dy\\ &= d(xyz) + dy\end{aligned}$$

Now,

$$\begin{aligned}\int_C \vec{F}\cdot d\vec{r} &= \int_{(1,0,0)}^{(2,1,4)} d(xyz) + \int_{(1,0,0)}^{(2,1,4)} dy\\ &= [xyz]_{(1,0,0)}^{(2,1,4)} + [y]_{(1,0,0)}^{(2,1,4)}\\ &= [8 - 0] + (1 - 0)\\ &= 9.\end{aligned}$$

Alternatively,

Here, the equation of the line passes through the points (1, 0, 0) and (2, 1, 4) is

$$\frac{x-1}{2-1} = \frac{y-0}{1-0} = \frac{z-0}{4-0}$$

i.e. $\frac{x-1}{1} = \frac{y}{1} = \frac{z}{4} = t$ (say)

i.e. $x = t + 1, y = t, z = 4t$

Then clearly, t moves from t = 0 to t = 1.

Here,

$$\begin{aligned}\vec{F}\cdot d\vec{r} &= (yz, xz + 1, xy)\cdot(dx, dy, dz)\\ &= (4t^2, 4t^2 + 4t + 1, t^2 + t)\cdot(dt, dt, 4dt)\\ &= (4t^2 + 4t^2 + 4t + 1 + 4t^2 + 4t)\,dt\\ &= (12t^2 + 8t + 1)\,dt\end{aligned}$$

Now,

$$\int_C \vec{F}\cdot d\vec{r} = \int_0^1 (12t^2 + 8t + 1)\,dt = [4t^3 + 4t^2 + t]_0^1 = 4 + 4 + 1 = 9.$$

Example 7: If $\vec{F} = (2xy - z)\,\vec{i} + yz\,\vec{j} + x\,\vec{k}$ then evaluate $\int_C \vec{F}\cdot d\vec{r}$ along the curve C, where

a. C is the curve $x = t, y = 2t, z = t^2 - 1$, with t increasing from 0 to 1.

b. C consists of two straight line from the origin to the point (1, 0, −1) and from (1, 0, −1) to the point (2, 3, −3).

Solution:

Given that,

$$\vec{F} = (2xy - z)\,\vec{i} + yz\,\vec{j} + x\,\vec{k}$$

(a) And the path of $\vec{F}\cdot d\vec{r}$ is, $x = t, y = 2t, z = t^2 - 1$. And, t moves from t = 0 to 1.

Since,

$$\vec{r} = x\,\vec{i} + y\,\vec{j} + z\,\vec{k} = t\,\vec{i} + 2t\,\vec{j} + (t^2 - 1)\,\vec{k}$$

$$d\vec{r} = dt\,\vec{i} + 2dt\,\vec{j} + 2t\,dt\,\vec{k}$$

So,

$$= (\vec{i} + 2\,\vec{j} + 2t\,\vec{k})\,dt$$

Then,

$$\begin{aligned}\vec{F}\cdot d\vec{r} &= [(2(t)(2t) - (t^2 - 1))\,\vec{i} + (2t(t^2 - 1))\,\vec{j} + t\,\vec{k}]\cdot(\vec{i} + 2\,\vec{j} + 2t\,\vec{k})dt\\ &= [(4t^2 - t^2 + 1)\,\vec{i} + (2t^3 - 2t)\,\vec{j} + t\,\vec{k}]\cdot(\vec{i} + 2\,\vec{j} + 2t\,\vec{k})\,dt\\ &= (4t^2 - t^2 + 1 + 4t^3 - 4t + 2t^2)\,dt\\ &= (4t^3 + 5t^2 - 4t + 1)\,dt\end{aligned}$$

Now,

$$\begin{aligned}\int_C \vec{F}\cdot d\vec{r} &= \int_0^1 (4t^3 + 5t^2 - 4t + 1)\,dt\\ &= \left[t^4 + \frac{5t^3}{3} - 2t^2 + t\right]_0^1\\ &= 1 + \frac{5}{3} - 2 + 1\\ &= \frac{5}{3}\end{aligned}$$

(b) Let C_1 be the line segment from (0, 0, 0) to (1, 0, −1). So, equation of C_1 is

$$\frac{x-1}{1-0} = \frac{y-0}{0-0} = \frac{z+1}{-1-0} = t\ (\text{say})$$

i.e. $x = t + 1, y = 0, z = -t - 1.$

So, $dx = dt, dy = 0, dz = -dt$

Since $\vec{r} = x\,\vec{i} + y\,\vec{j} + z\,\vec{k}$. Then $d\vec{r} = dx\,\vec{i} + dy\,\vec{j} + dz\,\vec{k}$.

Now,

$$\begin{aligned}\vec{F}\cdot d\vec{r} &= ((2xy - z)\,\vec{i} + yz\,\vec{j} + x\,\vec{k})\cdot(dx\,\vec{i} + dy\,\vec{j} + dz\,\vec{k})\\ &= (2xy - z)dx + yz\,dy + x\,dz\end{aligned}$$

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$$= [0 + (t + 1)]dt + 0 + (t + 1)(-dt)$$
$$= [(t + 1) - (t + 1)]\, dt$$
$$= 0\, dt$$

Here the integral along C_1 is,

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_1} 0\, dt = 0$$

Also, C_2 be the line segment from (1, 0, –1) to (2, 3, –3).
The equation of line C_2 is,

$$\frac{x-1}{2-1} = \frac{y-0}{3-0} = \frac{z+1}{-3+1} = u$$

i.e. $x = u + 1, y = 3u, z = -2u - 1$

Then

$$dx = du, dy = 3du \text{ and } dz = -2du.$$

Here,

$$\vec{F} \cdot d\vec{r} = (2xy - z, yz, x) \cdot (dx, dy, dz)$$
$$= (2xy - z)\, dx + yz\, dy + x\, dz$$
$$= [2(u + 1)(3u) + 2u + 1]\, du + 3u(-2u - 1)\, 3du + (u + 1)(-2\, du)$$
$$= [6u^2 + 6u + 2u + 1 - 18u^2 - 9u - 2u - 2]\, du$$
$$= [-12u^2 - 3u - 1]\, du$$

Also, y moves from 0 to 3. So, $y = 3u$ gives u moves from $u = 0$ to $u = 1$ along C_2. So,

$$\int_{C_2} \vec{F} \cdot d\vec{r} = \int_0^1 [-12u^2 - 3u - 1]\, du$$
$$= \left[-4u^3 - \frac{3u^2}{2} - u\right]_0^1$$
$$= -4 - \frac{3}{2} - 1$$
$$= -\frac{13}{2}$$

Now, C consists two lines C_1 and C_2. So,

$$\int_C \vec{F} \cdot d\vec{r} = \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r} = 0 - \frac{13}{2} = -\frac{13}{2}$$

Example 8: If $\vec{F} = 4xy\vec{i} + 8y\vec{j} + 3z\vec{k}$, find the line integral of $\vec{F}$ along the curve $y = 3x, z = 2x$ from (0, 0, 0) to (1, 3, 2).

Solution:

Given that $\vec{F} = 4xy\vec{i} + 8y\vec{j} + 3z\vec{k} = (4xy, 8y, 3z)$
And, along the curve, $y = 3x, z = 2x$.

Since we have, $\vec{r} = (x, y, z) = (x, 3x, 2x)$. So, $d\vec{r} = (1, 3, 2)\, dx$.

$$\vec{F} \cdot d\vec{r} = (4xy, 8y, 3z) \cdot (1, 3, 2)\, dx$$
$$= (12x^2, 24x, 6x) \cdot (1, 3, 2)\, dx$$
$$= (12x^2 + 72x + 12x)\, dx$$

Now, line integral of $\vec{F}$ along $y = 3x, z = 2x$ from (0, 0, 0) to (1, 3, 2) is

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 (12x^2 + 72x + 12x)\, dx$$
$$= [4x^3 + 36x^2 + 6x^2]_0^1 = 4 + 36 + 6 = 46.$$

Thus, $\int_C \vec{F} \cdot d\vec{r} = 46.$

Example 9: Find the work done by the force $\vec{F} = (2y + 3)\vec{i} + xz\vec{j} + (yz - x)\vec{k}$ when it moves a particle from the point (0, 0, 0) to the point (2, 1, 1) along the curve $x = 2t^2, y = t, z = t^3$.

Solution:

Given that,

$$\vec{F} = (2y + 3)\vec{i} + xz\vec{j} + (yz - x)\vec{k}$$

that moves from (0, 0, 0) to (2, 1, 1) along $x = 2t^2, y = t, z = t^3$. ... (i)

Then,

$$\vec{F} = (2t + 3)\vec{i} + 2t^5\vec{j} + (t^4 - 2t^2)\vec{k}$$

Since, y = t and y moves y = 0 to y = 1. So, t moves t = 0 to t = 1.
We know that,

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k} = 2t^2\vec{i} + t\vec{j} + t^3\vec{k}$$

Then,

$$d\vec{r} = (4t\vec{i} + \vec{j} + 3t^2\vec{k})\, dt$$

So that,

$$\vec{F} \cdot d\vec{r} = ((2t + 3)\vec{i} + 2t^5\vec{j} + (t^4 - 2t^2)\vec{k}) \cdot (2t\vec{i} + \vec{j} + 3t^2\vec{k})\, dt$$
$$= (8t^2 + 12t + 2t^5 + 3t^6 - 6t^4)\, dt$$

We have, the work done by the force $\vec{F}$ along the curve C: $\vec{r}$ is $\int_C \vec{F} \cdot d\vec{r}$.

Now,

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 (3t^6 + 2t^5 - 6t^4 + 8t^2 + 12t)\, dt$$
$$= \left[\frac{3t^7}{7} + \frac{2t^6}{6} - \frac{6t^5}{5} + \frac{8t^3}{3} + \frac{12t^2}{2}\right]_0^1$$
$$= \frac{3}{7} + \frac{2}{6} - \frac{6}{5} + \frac{8}{3} + \frac{12}{2}$$
$$= \frac{3}{7} + \frac{1}{3} - \frac{6}{5} + \frac{8}{3} + 6$$
$$= \frac{3}{7} - \frac{6}{5} + 3 + 6$$
$$= \frac{15 - 42}{35} + 9$$
$$= \frac{-27 + 315}{35}$$
$$= \frac{288}{35}$$

Thus, the work done by $\vec{F}$ is $\frac{288}{35}$.

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Example 10: The force at a point (x, y, z) in three dimensional is given by $\vec{F} = y\vec{i} + z\vec{j} + x\vec{k}$. Find the work done by $\vec{F}$ along the twisted cubic $x = t$, $y = t^2$, $z = t^3$ from $(0, 0, 0)$ to $(2, 4, 8)$.

Solution:

Given that, $\vec{F} = y\vec{i} + z\vec{j} + x\vec{k}$

And the force $\vec{F}$ works along $x = t$, $y = t^2$, $z = t^3$ from $(0, 0, 0)$ to $(2, 4, 8)$. Thus t varies from $t = 0$ to $t = 2$. Also, we have the position vector of the curve is,

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k} = (t\vec{i} + t^2\vec{j} + t^3\vec{k})$$

Then
$$d\vec{r} = (dt\,\vec{i} + 2t\,dt\,\vec{j} + 3t^2\,dt\,\vec{k})$$
$$= (\vec{i} + 2t\,\vec{j} + 3t^2\,\vec{k})\,dt$$

Then,
$$\vec{F}\cdot d\vec{r} = (y\vec{i} + z\vec{j} + x\vec{k})\cdot(\vec{i} + 2t\,\vec{j} + 3t^2\,\vec{k})\,dt$$
$$= (t^2\vec{i} + t^3\vec{j} + t\vec{k})\cdot(\vec{i} + 2t\,\vec{j} + 3t^2\,\vec{k})\,dt$$
$$= (t^2 + 2t^4 + 3t^5)\,dt$$

Now, the work done by $\vec{F}$ is,

$$\int_C \vec{F}\cdot d\vec{r} = \int_0^2 (t^2 + 2t^4 + 3t^5)\,dt$$
$$= \left[\frac{t^3}{3} + \frac{2t^5}{5} + \frac{3t^6}{4}\right]_0^2$$
$$= \frac{8}{3} + \frac{64}{5} + \frac{48}{4}$$
$$= \frac{40 + 192 + 180}{15}$$
$$= \frac{412}{15}$$

Thus, the work done by $\vec{F}$ along the given curve is $\frac{412}{15}$.

Exercise 5 (A)

1. Calculate $\int_C \vec{F}\cdot d\vec{r}$ for the following data. (If $\vec{F}$ is a force, this gives the work on the displacement along C).
   a. $\vec{F} = (y^2, -x^2)$, C is the straight line from $(0, 0)$ to $(1, 4)$.
   b. $\vec{F} = (2z, x, -y)$, C; $\vec{r} = (\cos t, \sin t, 2t)$ from $(0, 0, 0)$ to $(1, 0, 4\pi)$.
   c. $\vec{F} = (e^x, e^{-y}, e^z)$, C; $\vec{r} = (t, t^2, t)$ from $(0, 0, 0)$ to $(1, 1, 1)$.
2. Evaluate $\int_C \vec{F}\cdot d\vec{r}$
   a. $\vec{F} = [(x - y)^2, (y - x)^2]$, C; $xy = 1$, $1 \le x \le 4$.
   b. $\vec{F} = (x^2 + y^2)\vec{i} + (x^2 - y^2)\vec{j}$ and c is the curve $y^2 = x$ joining $(0, 0)$ and $(1, 1)$.
3. Evaluate the line integral along C
   a. $\int_C (6x^2y\,dx + xy\,dy)$, where C is the graph of $y = x^2 + 1$ from $(-1, 0)$ to $(1, 2)$.
   b. $\int_C [(x - y)\,dx + x\,dy]$, where C is the graph of $y^2 = x$ from $(4, -2)$ to $(4, 2)$.
4. Evaluate $\int_C \vec{F}\cdot d\vec{r}$
   a. $\vec{F} = (2x - y + z)\vec{i} + (x + y - z^2)\vec{j} + (3x - 2y + 4z)\vec{k}$, around the circle $x^2 + y^2 = a^2$, $z = 0$.
   b. $\vec{F} = z\vec{i} + x\vec{j} + y\vec{k}$ and C being the arc of the curve $\vec{r} = \cos t\,\vec{i} + \sin t\,\vec{j} + t\vec{k}$ from $t = 0$ to $t = 2\pi$.
5. a. Evaluate the integral, $\int_C [xz\,dx + (y - z)\,dy + x\,dz]$, if c is the graph of $x = e^t$, $y = e^{-t}$, $z = e^{2t}$, $0 \le t \le 1$.
   b. Find $\int_C \vec{F}\cdot d\vec{r}$ where, $\vec{F} = y^2\vec{i} + 2xy\vec{j}$ from O(0, 0) to P(1, 1) in each of the following cases:
      i. along the straight line OP.
      ii. along the parabola $y^2 = x$.
      iii. along the x–axis from $x = 0$ to $x = 1$ and then along the line $x = 1$, from $y = 0$ to $y = 1$.
   b. Evaluate $\int_C [(x + y + z)\,dx + (x - 2y + 3z)\,dy + (2x + y - z)\,dz]$, where C is the curve from (0, 0, 0) to (2, 3, 4) if
      i. C consists of three line segments the first parallel to the x–axis the second parallel to the y–axis and the third parallel to z–axis.
      ii. C consists of three line segments the first parallel to the z–axis the second parallel to the x–axis and the third is parallel to the y–axis.
      iii. C is the line segments.
6. Find the work done by the force $\vec{F}$ along the given curve C; where $\vec{F} = 3x^2\vec{i} + (2xz - y)\vec{j} + z\vec{k}$ along the curve defined by $x^2 = 4y$, $3x^3 = 8z$ from $x = 0$ to $x = 2$.

Answer Key

1.	a. 4	b. 9π	c. $2e - e^{-1} - 1$
2.	a. $\frac{891}{64}$	b. $\frac{7}{10}$	c. 228
3.	a. $\frac{34}{7}$	b. $\frac{16}{3}$	c. $\frac{7}{10}$
4.	a. $2\pi^2$	b. 3π	c. $\frac{1}{12}(3e^4 + 8e^3 - 12e + 6e^{-2} - 5)$
5.	a. i. 1;	ii. 1;	iii. 1
	b. i. 19;	ii. 35;	iii. 27
6.	16		

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5.5 SURFACE INTEGRAL

In many physical problems we come across functions defined on surface. In this section, we study integrals of the function defined over surfaces. An integral evaluated over a surface is called a surface integral. Before we introduce the surface integral, we present the equations of surfaces in different forms.

1. **A surface in space is represented by the equation**

$$z = f(x, y).$$

where $f(x, y)$ is defined in some region R in space. It is also represented by $f(x, y, z) = 0$, which is the implicit form of the equation of a surface. For example,

i. $x^2 + y^2 + z^2 = 0$ represents the equation of a sphere of radius a and center 0.

ii. $x^2 + y^2 = a^2, = -b \le z \le b$ represents the equation of circular cylinder of radius a and height 2b.

2. **A parametric equation of a curve C is generally given by**

$$\vec{r} = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}, \quad (a \le t \le b)$$

where t is a parameter.

Since surface is two dimensional, so we need two parameters. And, a parametric representation of a surface S is generally given by

$$x = f(u, v),\ y = g(u, v),\ z = h(u, v),\ (u, v \in R) \qquad \ldots (i)$$

where u, v are parameters, and f, g, h are defined on some region R of the uv plane.

The equation (i) can also be written as

$$\vec{r} = \vec{r}(u, v) \qquad \ldots (ii)$$

with $(x, y, z) = (x(u, v), y(u, v), z(u, v))$, which is the equation of the surface is vector form.

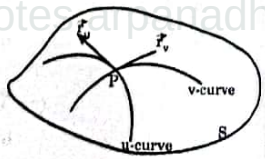
We assume here that u, v are continuous and x, y, z are single valued continuous functions of u and v

5.6 TANGENT PLANE AND UNIT NORMAL VECTOR

The vector equation of a surface S is

$$\vec{r} = \vec{r}(u, v) = (x(u, v), y(u, v), z(u, v))$$

Let $V = V_0$ (a constant), then the points on the surface describe a curve $\vec{r}(u, v_0)$, this curve is known as u-curve. By varying V_0, we get a family of v-curves. The collection of u-curves and v curves forms a set on S.



The vector $\frac{\partial \vec{r}}{\partial u} = \vec{r}_u$ is a tangent vector to u-curve and the vector $\frac{\partial \vec{r}}{\partial v} = \vec{r}_v$ is the tangent vector to v-curve.

If $\vec{r}_u$ and $\vec{r}_v$ are not parallel, then the locus of a point P is a surface. If $u = u(t)$, $v = v(t)$, then $\vec{r} = \vec{r}(u(t), v(t)) = \vec{r}(t)$ is a curve. Then the tangent vector is

$$\frac{d\vec{r}}{dt} = \frac{\partial \vec{r}}{\partial u}\cdot\frac{du}{dt} + \frac{\partial \vec{r}}{\partial v}\cdot\frac{dv}{dt}$$

At a fixed point P on S, the tangent vectors of all curves on S through P lie in the plane of the vector $\vec{r}_u$ and $\vec{r}_v$. This plane determined by $\vec{r}_u$ and $\vec{r}_v$, is the tangent plane to the surface at P. So, clearly the vector $\vec{r}_u \times \vec{r}_v$ is perpendicular to the tangent plane of S at the point P. Let $\vec{N}$ be a normal vector (i.e. perpendicular vector) of S at P, then

$$\vec{N} = \vec{r}_u \times \vec{r}_v$$

The corresponding unit normal vector $\vec{n}$ of S at P is

$$\vec{n} = \frac{\vec{N}}{|\vec{N}|} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}$$

Since $|\vec{r}_u \times \vec{r}_v|$ is the area of the parallelogram having its adjacent sides are $\vec{r}_u$ and $\vec{r}_v$. If ds denotes the element of surface area, then

$$ds = |\vec{r}_u \times \vec{r}_v|\, du\, dv$$

The vector element of surface area is defined by

$$d\vec{s} = (\vec{r}_u \times \vec{r}_v)\, du\, dv$$

Surface Integral

A surface integral is the integral that is defined over a surface.

Let $\vec{F}$ be a vector function, is defined over a closed surface S and let $\vec{n}$ be the unit normal vector drawn outward to the surface S at any point P. Then the component of $\vec{F}$ along $\vec{n}$ is $\vec{F}\cdot\vec{n}$, and the flux of vector $\vec{F}$ through the elementary area ds is $(\vec{F}\cdot\vec{n})ds$. Therefore, the integral of this flux taken over the surface S is the total flux or surface integral of $\vec{F}$ over S and is denoted by $\iint_S \vec{F}\cdot\vec{n}\, ds$. A surface integral is also called a flux integral.

Thus, the surface integral is denoted by any one;

$$\iint_S \vec{F}\cdot\vec{n}\, ds \quad \text{or} \quad \iint_S \vec{F}\cdot\vec{n}\, dA \quad \text{or} \quad \iint_S \vec{F}\cdot d\vec{s}$$

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Other Forms of Surface Integral

1. If R is the projection of S on xy-plane, then

$$\iint_S \vec{F}\cdot\vec{n}\,ds = \iint_R \left(\vec{F}\cdot\vec{n}\right)\frac{dx\,dy}{\vec{n}\cdot\vec{k}}, \text{ provided } \left|\vec{n}\cdot\vec{k}\right| \neq 0.$$

Similarly, if R is the projection of S on yz-plane, then

$$\iint_S \vec{F}\cdot\vec{n}\,ds = \iint_R \left(\vec{F}\cdot\vec{n}\right)\frac{dy\,dz}{\vec{n}\cdot\vec{i}}, \text{ provided } \left|\vec{n}\cdot\vec{i}\right| \neq 0.$$

and if R is the projection of S on zx-plane, then

$$\iint_S \vec{F}\cdot\vec{n}\,ds = \iint_R \left(\vec{F}\cdot\vec{n}\right)\frac{dx\,dz}{\vec{n}\cdot\vec{j}}, \text{ provided } \left|\vec{n}\cdot\vec{j}\right| \neq 0.$$

where $\vec{n}$ is unit normal vector to the surface at any point inside the element area ds of S.

2. If S is given in parametric representation $\vec{r}(u, v) = (x(u, v), y(u, v), z(u, v))$ in R then the surface S has a normal vector $\vec{N} = \left(\vec{r}_u \times \vec{r}_v\right)$ then the surface integral is

$$\iint_S \vec{F}\cdot\vec{n}\,ds = \iint_R \vec{F}\cdot\vec{N}\,dA$$

where $dA = du\,dv$ or $dA = dv\,du$; $\vec{N} = \vec{N}(u, v)$ and $\vec{F} = \vec{F}(u, v)$.

Some Important Surfaces

The parametric representations of some important surfaces are as;

1. **Cylinder:**

The equation of a circular cylinder of radius a and height 2h is given by

$$x^2 + y^2 = a^2, \quad -h \le z \le h$$

Its axis is a z-axis

The parametric representation of this cylinder is

$$x = a\cos u,\ y = a\sin u,\ z = v$$

In vector form,

$$\vec{r} = [a\cos u,\ a\sin u,\ v],$$

where u and v vary over a rectangle $R: 0 \le u \le 2\pi, -h \le v \le h$.

2. **Circular Cone**

A circular cone whose height is h is given by

$$z = \sqrt{x^2 + y^2}, \qquad 0 \le z \le h$$

Its equation in parametric form is

$$x = u\cos v,\ y = u\sin v,\ z = u.$$

3. **Sphere**

a. The equation of the sphere of radius a and centre 0 is

$$x^2 + y^2 + z^2 = a^2$$

A parametric representation of the sphere is

$$x = a\cos u\cos v,\ y = a\sin u\cos v,\ z = a\sin v$$

In vector form,

$$\vec{r} = (a\cos u\cos v,\ a\sin u\cos v,\ a\sin v)$$

where u and v vary over a rectangle $R: 0 \le u \le 2\pi, \frac{-\pi}{2} \le v \le \frac{\pi}{2}$

b. The upper part of the sphere

$$x^2 + y^2 + z^2 = a^2$$

is the hemisphere. Its equation is

$$z = +\sqrt{a^2 - x^2 - y^2},\ z \ge 0$$

And its parametric representation in vector form is

$$\vec{r} = (a\cos u\cos v,\ a\sin u\cos v,\ a\sin v)$$

where u and v vary over a rectangle $R: 0 \le u \le 2\pi, 0 \le v \le \frac{\pi}{2}$

c. The sphere $x^2 + y^2 + z^2 = a^2$ in the first octant, $x \ge 0, y \ge, z \ge 0$.

And its parametric representation in vector form is

$$\vec{r} = (a\cos u\cos v,\ a\sin u\cos v,\ a\sin v)$$

where u and v vary over a rectangle $R: 0 \le u \le \frac{\pi}{2}, 0 \le v \le \frac{\pi}{2}$

Workout Examples

Example 1: Evaluate $\iint_S \vec{F}\cdot\vec{n}\,dA$, where $\vec{F} = (3x^2, y^2, 0)$, S: $\vec{r} = (u, v, 2u + 3v)$, $0 \le u \le 2, -1 \le v \le 1$.

Solution:

Given that,

$$\vec{F} = (3x^2, y^2, 0) = 3x^2\vec{i} + y^2\vec{j} + 0\vec{k}$$

And $\vec{r} = (u, v, 2u + 3v) = u\vec{i} + v\vec{j} + (2u + 3v)\vec{k}$.

Then,

$$\vec{r}_u = (\vec{i} + 2\vec{k}) \quad \text{and} \quad \vec{r}_v = \vec{j} + 3\vec{k}.$$

Since we have,

$$\iint_S \vec{F}\cdot\vec{n}\,dA = \iint_R \vec{F}(\vec{r})\cdot\vec{N}\,du\,dv \qquad -(i)$$

where $\vec{N} = \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 2 \\ 0 & 1 & 3 \end{vmatrix} = -2\vec{i} - 3\vec{j} + \vec{k}.$

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Since, $\vec{r} = x\vec{i} - y\vec{j} + z\vec{k}$. And given that, $\vec{r} = u\vec{i} + v\vec{j} + (2u + 3v)\vec{k}$. From these we get

$$x = u,\ y = -v \text{ and } z = 2u + 3v.$$

So that, $\vec{F} = 3x^2\vec{i} + y^2\vec{j} = 3u^2\vec{i} + v^2\vec{j}$

This implies that, $\vec{F}(\vec{r})\cdot\vec{N} = (3u^2\vec{i} + v^2\vec{j})\cdot(-2\vec{i} - 3\vec{j} + \vec{k}) = -6u^2 - 3v^2$

Also, given that the region is $0 \le u \le 2,\ -1 \le v \le 1$.

Thus, (i) become,

$$\iint_S \vec{F}\cdot\vec{n}\,dA = \int_{-1}^{1}\int_{0}^{2}(-6u^2 - 3v^2)\,du\,dv$$

$$= \int_{-1}^{1}[-2u^3 - 3v^2u]_0^2\,dv$$

$$= \int_{-1}^{1}(-16 - 6v^2)\,dv$$

$$= [-16v - 2v^3]_{-1}^{1}$$

$$= (-16 - 2) - (16 + 2)$$

$$= -18 - 18$$

$$= -36$$

Thus, $\iint_S \vec{F}\cdot\vec{n}\,dA = -36$.

Example 2: Evaluate $\iint_S \vec{F}\cdot\vec{n}\,dA$, where $\vec{F} = (12x^2y, -3yz, 2z)$ and S is the portion of the plane $x + y + z = 1$ included in the first octant.

Solution:

Given that $\vec{F} = (12x^2y, -3yz, 2z)$.

And surface is $x + y + z = 1$ in first octant.

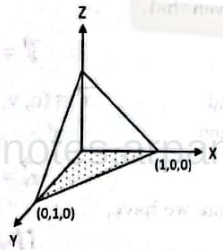
Set $x = u$ and $y = v$ then $z = 1 - u - v$

Here,

$$\vec{r} = (x, y, z) = (u, v, 1 - u - v)$$

So, $\vec{r}_u = (1, 0, -1)$ and $\vec{r}_v = (0, 1, -1)$

Then,



$$\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{vmatrix} = \vec{i} + \vec{j} + \vec{k} = (1, 1, 1)$$

By surface integral we have,

$$\iint_S \vec{F}\cdot\vec{n}\,dA = \iint_R \vec{F}\cdot\vec{N}\,dx\,dy \quad \ldots (i)$$

$$\vec{N} = \vec{r}_u \times \vec{r}_v = (1, 1, 1)$$

where,

Here, $\vec{F}\cdot\vec{N} = (12x^2y, -3yz, 2z)\cdot(1, 1, 1)$

$$= (12u^2v, -3v(1 - u - v), 2(1 - u - v))\cdot(1, 1, 1)$$

$$= 12u^2v - 3v(1 - u - v) + 2(1 - u - v)$$

$$= 12u^2v - 3v + 3uv + 3v^2 + 2 - 2u - 2v$$

$$= 12u^2v + 3v^2 + 3uv - 5v - 2u + 2$$

Also, the projection of the surface in xy–plane is $x + y = 1$. So, the surface in uv–plane is $u + v = 1$ in which u varies from $u = 0$ to $u = 1 - v$ and v moves from $v = 0$ to $v = 1$.

Also, $dx\,dy = du\,dv$.

Now, (i) becomes,

$$\iint_S \vec{F}\cdot\vec{n}\,dA = \int_0^1\int_0^{1-v}(12u^2v + 3v^2 + 3uv - 5v - 2u + 2)\,du\,dv$$

$$= \int_0^1\int_0^{1-v}[12u^2v + (3v - 2)u + (3v^2 - 5v + 2)]\,du\,dv$$

$$= \int_0^1\left[12v\frac{u^3}{3} + (3v - 2)\frac{u^2}{2} + (3v^2 - 5v + 2)u\right]_0^{1-v}dv$$

$$= \int_0^1\left[12v\frac{(1-v)^3}{3} + (3v - 2)\frac{(1-v)^2}{2} + (3v^2 - 5v + 2)(1 - v)\right]dv$$

$$= \frac{1}{6}\int_0^1(1 - v)[24v(1 - 2v + v^2) + 3(3v - 2)(1 - v) + 6(3v^2 - 5v + 2)]\,dv$$

$$= \frac{1}{6}\int_0^1[-24v^4 + 63v^3 - 48v^2 + 3v + 6]\,dv$$

$$= \frac{1}{2}\int_0^1(-8v^4 + 21v^3 - 16v^2 + v + 2)\,dv$$

$$= \frac{1}{2}\left[-\frac{8v^5}{5} + \frac{21v^4}{4} - \frac{16v^3}{3} + \frac{v^2}{2} + 2v\right]_0^1$$

$$= \frac{1}{2}\left[-\frac{8}{5} + \frac{21}{4} - \frac{16}{3} + \frac{1}{2} + 2\right]$$

$$= \frac{-96 + 315 - 320 + 30 + 120}{120}$$

$$= \frac{49}{120}$$

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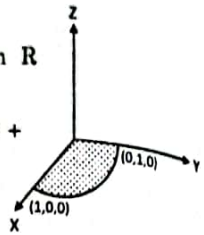
Thus, $\iint_S \vec{F}\cdot\vec{r}\, dA = \frac{49}{120}$.

Example 3: Evaluate $\iint_S \vec{F}\cdot\vec{n}\, dA$, where $\vec{F} = (yz, zx, xy)$ and S is the part of the surface $x^2 + y^2 + z^2 = 1$, which lies in the first octant.

Solution:

The projection of the surface on xy-plane is the region R bounded by x-axis, y-axis and the arc of circle $x^2 + y^2 = 1$.



Given that, $\vec{F} = (yz, zx, xy)$ and the sphere $x^2 + y^2 + z^2 = 1$.

Set $x = u$, $y = v$ then $z = \sqrt{1-u^2-v^2}$

Since, $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$= u\vec{i} + v\vec{j} + \sqrt{1-u^2-v^2}\,\vec{k}$.

Then,

$$\vec{F}(\vec{r}) = (v\sqrt{1-u^2-v^2},\ u\sqrt{1-u^2-v^2},\ uv)$$

Since we have,

$$\iint_S \vec{F}\cdot\vec{n}\, dA = \iint_R \vec{F}(\vec{r})\cdot\vec{N}\, du\, dv \quad \text{..... (i)}$$

So that $\vec{N} = \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -u/\sqrt{1-u^2-v} \\ 0 & 1 & -v/\sqrt{1-u^2-v^2} \end{vmatrix}$

$$= \left(\frac{u}{\sqrt{1-u^2-v^2}}, \frac{v}{\sqrt{1-u^2-v^2}}, 1\right)$$

Then, $\vec{F}\cdot\vec{N} = uv + uv + uv = 3uv$

And from figure, u varies on the region from $u = 0$ to the curve $u = \sqrt{1-v^2}$ and on the region, v moves from $v = 0$ to $v = 1$.

Then, (i) becomes,

$$\iint_S \vec{F}\cdot\vec{n}\, ds = \int_0^1 \int_0^{\sqrt{1-v^2}} 3uv\, du\, dv$$

$$= \frac{3}{2}\int_0^1 [u^2]_0^{\sqrt{1-v^2}}\, v\, dv$$

$$= \frac{3}{2}\int_0^1 (1-v^2)\, v\, dv$$

$$= \frac{3}{2}\int_0^1 (v - v^3)\, dv$$

$$= \frac{3}{2}\left[\frac{v^2}{2} - \frac{v^4}{4}\right]_0^1$$

$$= \frac{3}{2}\left(\frac{1}{2} - \frac{1}{4}\right)$$

$$= \frac{3}{8}$$

Thus, $\iint_S \vec{F}\cdot\vec{n}\, dA = \frac{3}{8}$.

Example 4: Let S be the part of the graph of $z = 9 - x^2 - y^2$ with $z \geq 0$. If $\vec{F} = 3x\vec{i} + 3y\vec{j} + z\vec{k}$. Find the flux of $\vec{F}$ through S.

Solution:

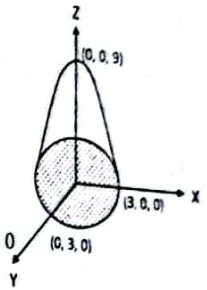
Given that $\vec{F} = (3x, 3y, z)$.

And S is part of $z = 9 - x^2 - y^2$ with $z \geq 0$.

Clearly, the projection of the parabolid in xy-plane is a circle $x^2 + y^2 = 9$.

By surface integral,

$$\iint_S \vec{F}\cdot\vec{n}\, dA = \iint_R (\vec{F}\cdot\vec{N})\, dx\, dy \quad \text{... (i)}$$



where, $\vec{N} = \vec{r}_x \times \vec{r}_y$

Since $\vec{r} = (x, y, z) = (x, y, 9 - x^2 - y^2)$

Then $\vec{r}_x = (1, 0, -2x)$, $\vec{r}_y = (0, 1, -2y)$.

For the circle, set $x = r\cos\theta$, $y = r\sin\theta$ then $z = 9 - r^2$.

On the circle, radius $r = 3$ and angular variation from $\theta = 0$ to $\theta = 2\pi$.

Here,

$$\vec{N} = \vec{r}_x \times \vec{r}_y = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -2x \\ 0 & 1 & -2y \end{vmatrix} = (2x, 2y, 1)$$

Then

$$\vec{F}\cdot\vec{N} = 6x^2 + 6y^2 + z$$
$$= 6(x^2 + y^2) + 9 - (x^2 + y^2)$$
$$= 6r^2 + 9 - r^2$$
$$= 9 + 5r^2$$

Now, (i) becomes,

$$\iint_S \vec{F}\cdot\vec{n}\, ds = \int_0^{2\pi}\int_0^3 (9 + 5r^2)\, r\, dr\, d\theta$$

$$= \int_0^{2\pi} \left[\frac{9r^2}{2} + \frac{5}{4}r^4\right]_0^3 d\theta$$

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$$= \int_0^{2\pi} \left(\frac{81}{2} + \frac{405}{4}\right) d\theta$$

$$= \frac{162 + 405}{4} \times 2\pi$$

$$= \frac{567\pi}{2}$$

Thus, $\iint_S \vec{F} \cdot \vec{n}\, ds = \frac{567\pi}{2}$.

Exercise 5 (B)

Evaluate $\iint_S \vec{F} \cdot \vec{n}\, dA$, where

1. $\vec{F} = (x, y)$, S: $z = 2x + 5y$, $0 \le x \le 2$, $-1 \le y \le 1$.
2. $\vec{F} = (0, x^2, -xz)$, S: $\vec{r} = (u, u^2, v)$, $0 \le u \le 1$, $-2 \le v \le 2$.
3. $\vec{F} = (x, y, z)$, S: $\vec{r} = (u \cos v, u \sin v, u^2)$, $0 \le u \le 4$, $-\pi \le v \le \pi$.
4. $\vec{F} = (x - z, y - x, z - y)$, S: $\vec{r} = (u \cos v, u \sin v, u)$, $0 \le u \le 3$, $0 \le v \le 2\pi$.
5. $\vec{F} = (e^{2y}, e^{-2x} \cdot 2x)$, S : $\vec{r} = (3 \cos u, 3 \sin u, v)$, $0 \le u \le \frac{\pi}{2}$, $0 \le v \le 2$.
6. $\vec{F} = (x^2, e^y, 1)$, S: $x + y + z = 1$ $x \ge 0$, $y \ge 0$, $z \ge 0$.
7. $\vec{F} = (0, x, 0)$, S: $x^2 + y^2 + z^2 = 1$, $x \ge 0$, $y \ge 0$, $z \ge 0$.
8. $\vec{F} = (x^2, 0, 3y^2)$ and S is the portion of the plane $x + y + z = 1$ in the first octant.
9. Find the flux of $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$ through the surface S is the first octant portion of the plane $2x + 3y + z = 6$.
10. $\vec{F} = (18z, -12, 3y)$, s is the surface of the plane $2x + 3y + 6z = 12$ in the first octant.
11. $\vec{F} = (yz, zx, xy)$ and S is the part of the surface $x^2 + y^2 + z^2 = 1$, which lies in the first octant.
12. $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$ and S is the upper half of the sphere $x^2 + y^2 + z^2 = a^2$.

Answer Key

1. -8	2. $\frac{-4}{3}$	3. -128π	4. 0
5. $e^6 - \frac{3}{2}e^{-4} + \frac{1}{2}$	6. $-e$	7. $\frac{1}{3}$	8. $\frac{1}{2}$
9. 11	10. 24	11. $\frac{3}{8}$	12. $2\pi a^3$

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6

INTEGRAL THEOREMS ON VECTOR CALCULUS

CHAPTER OUTLINE

After completion of this chapter, you will be able to gain the knowledge and understanding regarding:

6.1 Green's theorem in plane (without proof) and its various applications

6.2 Stoke's theorem (without proof) and its applications

6.3 Gauss Divergence theorem (without proof) and its applications.

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Introduction

Integral transforms i.e. integral transformation means the one type of forms or function is converting to another form or function by means of integrals. In this chapter we deal the transformation the line integrals into surface integrals and surface integrals into volume integrals. Here, we studied about the Green's theorem, Stoke's theorem and Gauss divergence theorem in which Green's theorem and Stoke's theorem transforms the line integral into surface integrals and Gauss divergence theorem transforms surface integrals into volume integrals. These theorems serve as tool in many engineering and physical problems.

6.1 GREEN'S THEOREM IN A PLANE

Green's theorem is a relationship between double integrals and line integrals around closed curves in the plane. It was formulated in 1828 independently by British philosopher George Green and the Ukrainian mathematician Michel Ostrogradsky. The Green's theorem is used in potential theory and partial differential equations.

A closed curve C in the xy-plane is positively oriented if a point on the curve moves counterclockwise as the parameter describing C increases. If the point moves clockwise, then C is negatively oriented. A simple closed curve C in the plane encloses a region, called the interior of C. The unbounded region that remains if the interior is cut out is the exterior of C.

We often denote a line integral over a closed path C as $\oint_C$ C, with a small oval on the integral sign. The Green's theorem connects the double integral of plane region D to curve line integral which bounded this surface.

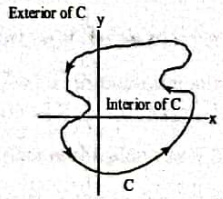

Fig.: Interior and exterior of a simple closed curve

Theorem: Let R be a closed bounded region in xy plane whose boundary C consists of finitely many smooth curves. Let $F_1(x, y)$ and $F_2(x, y)$ be functions that are continuous and have continuous partial derivatives $\frac{\partial F_1}{\partial y}$ and $\frac{\partial F_2}{\partial x}$ everywhere in some domain containing R. Then

$$\iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dA = \oint_C (F_1 dx + F_2 dy) = \oint_C \vec{F}.d\vec{r}$$

for $\vec{F} = (F_1, F_2)$. Here, the integration along the entire boundary C of R is in anticlockwise direction.

Proof:

Let us define a region R by

$a \le x \le b$, $u(x) \le y \le v(x)$ (fig. 1)

and $c \le y \le d$, $p(y) \le x \le q(y)$ (fig. 2)

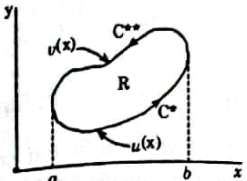

Fig.: Example of Special region

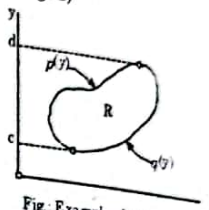

Fig.: Example of a Special region

Here,

$$\iint_R \frac{\partial F_1}{\partial y} dA = \int_a^b \int_{u(x)}^{v(x)} \frac{\partial F_1}{\partial y} dy\, dx$$

$$= \int_a^b \left[\int_{u(x)}^{v(x)} \frac{\partial F_1}{\partial y} dy\right] dx$$

$$= \int_a^b \left[F_1(x, y)\Big|_{u(x)}^{v(x)}\right] dx$$

$$= \int_a^b [F_1(x, v(x)) - F_1(x, u(x))]\, dx$$

$$= \int_a^b F_1[x, v(x)]\, dx - \int_a^b F_1[x, u(x)]\, dx$$

$$= -\int_b^a F_1(x, v(x))\, dx - \int_a^b F_1(x, u(x))\, dx$$

$$= -\int_a^b F_1[x, u(x)]\, dx - \int_b^a F_1[x, v(x)]\, dx \quad \ldots (1)$$

Here $y = u(x)$ represents the curve C_1 that moves from $x = a$ to $x = b$ and $y = v(x)$ represents C_2 that moves from $x = b$ to $x = a$. Therefore, (1) becomes

$$\iint_R \frac{\partial F_1}{\partial y} dA = -\int_{C_1} F_1[x, u(x)]\, dx - \int_{C_2} F_1[x, v(x)]\, dx \quad \ldots (2)$$

The combined form of C_1 and C_2 gives the closed curve C. So, (2) becomes

$$\iint_R \frac{\partial F_1}{\partial y} dA = -\oint_C F_1(x, y)\, dx \quad \ldots (3)$$

If $F_2 = 0$ then the proof is obvious. If $F_2 \neq 0$ then as above we get,

$$\iint_R \frac{\partial F_2}{\partial x} dA = \int_c^d \int_{p(y)}^{q(y)} \frac{\partial F_2}{\partial x} dx\, dy$$

$$= \int_c^d \left[\int_{p(y)}^{q(y)} \frac{\partial F_2}{\partial x} dx\right] dy$$

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$$= \int_c^d \left[F_2(x, y) \Big|_{p(y)}^{q(y)} \right] dy$$

$$= \int_c^d [F_2(q(y), y) - F_2(p(y), y)] \, dy$$

$$= \int_c^d F_2(q(y), y) \, dx - \int_c^d F_2(p(y), y) dx$$

$$= \int_c^d F_2(q(y), y) \, dx + \int_d^c F_2(p(y), y) dx \qquad \ldots (4)$$

Here $x = q(y)$ represents the curve that moves from $y = c$ to $y = d$ and then $x = p(y)$ represents the curve that moves from $y = d$ to $y = c$. This means the combined sum of the curves gives a closed curve C. Therefore, (4) gives

$$\iint_R \frac{\partial F_2}{\partial x} dA = \oint_c F_2(x, y) \, dy \qquad \ldots (5)$$

Thus adding (4) and (5) we get,

$$\iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA = \oint_c (F_1 dx + F_2 dy).$$

Note: If $\vec{F} = F_1 \vec{i} + F_2 \vec{j}$. Clearly, $F_1 = F_1(x, y)$ and $F_2 = F_2(x, y)$. Then

$$\text{curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_1 & F_2 & 0 \end{vmatrix} = \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \vec{k}$$

Then,

$$\text{curl } \vec{F} \cdot \vec{k} = \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)$$

Also,

$$\vec{F} \cdot d\vec{r} = \left(F_1 \vec{i} + F_2 \vec{j} \right) \cdot \left(dx \vec{i} + dy \vec{j} \right) = F_1 dx + F_2 dy$$

Therefore Green's Theorem can be written as,

$$\oint_c \vec{F} \cdot d\vec{r} = \oint_c (F_1 dx + F_2 dy) = \iint_R (\text{curl } \vec{F} \cdot \vec{k}) \, dA = \iint_R [\nabla \times \vec{F}) \cdot \vec{k}] \, dA$$

where $\vec{k}$ be unit vector along z–axis.

6.2 Application of Green's Theorem

(Area of the Region Enclosed by a Closed Curve)

If A be the area of the region bounded by a closed curve C is given by

$$A = \frac{1}{2} \oint_c (x \, dy - y \, dx).$$

Proof:

Let R be region that is bounded by a closed curve C. Then by Green's theorem we have

$$\oint_c (F_1 dx + F_2 dy) = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA$$

Replacing F_1 by $-y$ and F_2 by x then

$$\oint_c (-y dx + x dy) = \iint_R \left(\frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} \right) dx \, dy$$

$$= \iint_R (1 + 1) \, dx \, dy$$

$$= 2 \iint_R dx \, dy$$

$$= 2A$$

where A be the area of the closed bounded region.

Workout Examples

Example 1: Evaluate, $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = e^{x+y} \vec{i} + e^{x-y} \vec{j}$ where C is the boundary of the triangle region $x \le y \le 2x$, $0 \le x \le 1$.

Solution:

Given that the region is

$$x \le y \le 2x, \; 0 \le x \le 1$$

Clearly, the region is bounded.

Comparing the given integral $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = e^{x+y} \vec{i} + e^{x-y} \vec{j}$ with $\int_C (F_1 dx + F_2 dy)$ we get

$$F_1 = e^{x+y} \text{ and } F_2 = e^{x-y}$$

Then

$$\frac{\partial F_1}{\partial y} = e^{x+y} \text{ and } \frac{\partial F_2}{\partial x} = e^{x-y}$$

Therefore,

$$\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = e^{x-y} - e^{x+y} = e^x (e^{-y} - e^y)$$

Being the region is bounded, by Green's theorem

$$\oint_c (F_1 dx + F_2 dy) = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA$$

i.e. $$\int_C [e^{x+y} dx + e^{x-y} dy] = \int_0^1 \int_x^{2x} e^x (e^{-y} - e^y) \, dy \, dx$$

$$= \int_0^1 e^x \left[\frac{e^{-y}}{-1} - \frac{e^y}{1} \right]_x^{2x} dx$$

$$= \int_0^1 e^x \left[-e^{-2x} + e^{-x} - e^{2x} + e^x \right] dx$$

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$$= \int_0^1 [-e^{-x} + 1 - e^{3x} + e^{2x}]\, dx$$

$$= \left[-\frac{e^{-x}}{-1} + x - \frac{e^{3x}}{3} + \frac{e^{2x}}{2}\right]_0^1$$

$$= \left[e^{-x} + x - \frac{e^{3x}}{3} + \frac{e^{2x}}{2}\right]_0^1$$

$$= \left[e^{-1} - e^0 + 1 - 0 - \frac{e^3}{3} + \frac{e^0}{3} + \frac{e^2}{2} - \frac{e^0}{2}\right]$$

$$= \left[\frac{1}{e} - \frac{e^3}{3} + \frac{1}{3} + \frac{e^2}{2} - \frac{1}{2}\right]$$

$$= \frac{e^2}{2} - \frac{e^3}{3} - \frac{1}{e} - \frac{1}{6}$$

Alternative Method:

Given that,

$$\vec{F} = e^{x+y}\vec{i} + e^{x-y}\vec{j}$$

And, the boundaries of the region be $x \le y \le 2x$, $0 \le x \le 1$.

By Green's theorem we have,

$$\oint_c \vec{F}\cdot d\vec{r} = \iint_R (\text{curl } \vec{F}\cdot\vec{k})\, dA \qquad \ldots (i)$$

Here,

$$\text{curl } \vec{F} = \nabla\times\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ e^{x+y} & e^{x-y} & 0 \end{vmatrix} = (e^{x-y} - e^{x+y})\,\vec{k}$$

Then,

$$\text{curl } \vec{F}\cdot\vec{k} = e^{x-y} - e^{x+y}$$

Now, (i) becomes with boundaries $x \le y \le 2x$, $0 \le x \le 1$,

$$\oint_c \vec{F}\cdot d\vec{r} = \int_0^1\int_x^{2x} e^x(e^{-y} - e^y)\, dy\, dx$$

$$= \int_0^1 e^x \left[\frac{e^{-y}}{-1} - \frac{e^y}{1}\right]_x^{2x} dx$$

$$= \int_0^1 e^x [-e^{-2x} + e^{-x} - e^{2x} + e^x]\, dx$$

$$= \int_0^1 [-e^{-x} + 1 - e^{3x} + e^{2x}]\, dx$$

$$= \left[-\frac{e^{-x}}{-1} + x - \frac{e^{3x}}{3} + \frac{e^{2x}}{2}\right]_0^1$$

$$= \left[e^{-x} + x - \frac{e^{3x}}{3} + \frac{e^{2x}}{2}\right]_0^1$$

$$= \left[e^{-1} - e^0 + 1 - 0 - \frac{e^3}{3} + \frac{e^0}{3} + \frac{e^2}{2} - \frac{e^0}{2}\right]$$

$$= \left[\frac{1}{e} - \frac{e^3}{3} + \frac{1}{3} + \frac{e^2}{2} - \frac{1}{2}\right]$$

$$= \frac{e^2}{2} - \frac{e^3}{3} - \frac{1}{e} - \frac{1}{6}$$

Example 2: Evaluate, $\oint_c [(3x^2 - 8y^2)\, dx + (4y - 6xy)\, dy]$ where C is the boundary of the region defined by $y^2 = x$, $y = x^2$.

Solution:

Given that the region is

$$y^2 = x,\ y = x^2$$

Clearly, the region is bounded and solving the equations we get the point of intersections are (0, 0) and (1, 1).

On the region,

$$\sqrt{y} \le x \le y;\ 0 \le y \le 1$$

Comparing the given integral $\oint_c [(3x^2 - 8y^2)\, dx + (4y - 6xy)\, dy]$ with $\oint_c (F_1 dx + F_2 dy)$ we get

$$F_1 = 3x^2 - 8y^2 \text{ and } F_2 = 4y - 6xy$$

Then

$$\frac{\partial F_1}{\partial y} = -16y \text{ and } \frac{\partial F_2}{\partial x} = -6y$$

Therefore,

$$\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = -6y + 16y = 10y$$

Being the region is bounded, by Green's theorem

$$\oint_c (F_1 dx + F_2 dy) = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dA$$

i.e. $\oint_c [(3x^2 - 8y^2)\, dx + (4y - 6xy)\, dy] = \int_0^1\int_{\sqrt{y}}^{y} (10y)\, dx\, dy$

$$= \int_0^1 10y\, [x]_{\sqrt{y}}^{y}\, dy$$

$$= \int_0^1 10y(y - \sqrt{y})\, dy$$

$$= 10\int_0^1 (y^2 - y^{3/2})\, dy$$

$$= 10\left[\frac{y^3}{3} - \frac{y^{5/2}}{5/2}\right]_0^1$$

$$= 10\left[\frac{1}{3} - \frac{2}{5}\right] = \frac{-2}{3}$$

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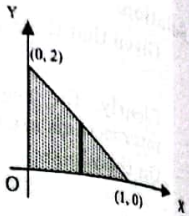
Example 3: Evaluate the integral $\oint_C [(3x^2 + y)\,dx + 4y^2\,dy]$ where C: the boundary of the triangle with vertices (0, 0), (1, 0), (0, 2); oriented in counterclockwise direction.

Solution:

Given that,

$$I = \oint_C [(3x^2 + y)\,dx + 4y^2\,dy] \quad \ldots (i)$$



And the region is the triangle having vertices (0, 0), (1, 0) and (0, 2) in counter wise direction.

Comparing the given integral I with the integral $\oint_C [F_1\,dx + F_2\,dy]$ then we get,

$$F_1 = 3x^2 + y \text{ and } F_2 = 4y^2$$

Then,

$$\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = 0 - 1 = -1$$

As given the region is bounded, so by Green's theorem we have,

$$\oint_C [F_1\,dx + F_2\,dy] = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dx\,dy \quad \ldots (ii)$$

In the figure, y varies from y = 0 to the line joining points (1, 0) and (0, 2). That is, y varies from y = 0 to y = – 2x + 2. And, x moves from x = 0 to x = 1. Then (i) becomes,

$$I = \iint_R [-1]\,dA = \int_0^1 \int_0^{2-2x} (-1)\,dy\,dx$$

$$= -\int_0^1 (2 - 2x)\,dx$$

$$= -[2 - 2x]_0^1$$

$$= -(2 - 1)$$

$$= -1$$

Thus, $\oint_C [(3x^2 + y)dx + 4y^2\,dy] = -1$.

Example 4: Evaluate the line integral $\oint_C \vec{F}\cdot d\vec{r}$ where $\vec{F} = (x^2e^y, y^2e^x)$ and the path C is the boundary of the rectangle with vertices (0, 0), (2, 0), (2, 3), (0, 3) in oriented counterclockwise direction.

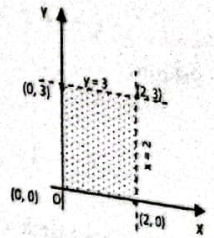
Solution:

Given that,

$$\vec{F} = (x^2e^y, y^2e^x)$$

Then,

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2e^y & y^2e^x & 0 \end{vmatrix} = (y^2e^x - x^2e^y)\vec{k}$$



So, $\text{curl } \vec{F}\cdot\vec{k} = (y^2e^x - x^2e^y)\vec{k}\cdot\vec{k} = y^2e^x - x^2e^y$

As given the path C is the boundary of the rectangle with vertices (0, 0), (2, 0), (2, 3), (0, 3) in oriented counterclockwise direction, then by Green's theorem, we have

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_R (\text{curl } \vec{F}\cdot\vec{k})\,dA = \iint_R (y^2e^x - x^2e^y)\,dA \quad \ldots (i)$$

Given that the path of $\vec{F}$ is C: the rectangle having vertices (0, 0), (2, 0), (2, 3) and (0, 3).

From the figure, y varies from y = 0 to y = 3 and x moves from x = 0 to x = 2

Therefore (i) becomes,

$$\oint_C \vec{F}\cdot d\vec{r} = \int_0^2 \int_0^3 (y^2e^x - x^2e^y)\,dy\,dx$$

$$= \int_0^2 \left[\frac{y^3e^x}{3} - x^2e^y\right]_0^3 dx$$

$$= \int_0^2 (9e^x - x^2e^3 + x^2)\,dx$$

$$= \left[9e^x - \frac{x^3}{3}e^3 + \frac{x^3}{3}\right]_0^2$$

$$= 9e^2 - \frac{8}{3}e^3 + \frac{8}{3} - 9$$

$$= 9(e^2 - 1) + \frac{8}{3}(1 - e^3)$$

Thus, $\oint_C \vec{F}\cdot d\vec{r} = 9(e^2 - 1) + \frac{8}{3}(1 - e^3)$.

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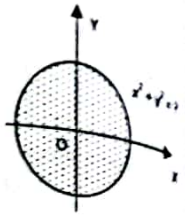
Example 5: Evaluate the line integral $\oint_c [xy\,dx + (y+x)\,dy]$, where C is the circle $x^2 + y^2 = 1$.

Solution:

Given that,

$$I = \oint_c [xy\,dx + (y+x)\,dy] \qquad \ldots (i)$$

where c is a circle $x^2 + y^2 = 1$



Comparing (i) with $\oint_c \vec{F}\cdot d\vec{r}$, then we get,

$$\vec{F} = xy\,\vec{i} + (y+x)\,\vec{j}$$

Then,

$$\text{curl}\,\vec{F} = \nabla\times\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy & y+x & 0 \end{vmatrix} = (1-x)\,\vec{k}$$

Then,

$$\text{curl}\,\vec{F}\cdot\vec{k} = (1-x)\,\vec{k}\cdot\vec{k} = 1-x$$

As given the path C is closed by a circle $x^2 + y^2 = 1$, having radius 1 and the angular movement from 0° to 2π, in oriented counterclockwise direction, then by Green's theorem, we have

$$\oint_c \vec{F}\cdot d\vec{r} = \iint_R (\text{curl}\,\vec{F}\cdot\vec{k})\,dA = \iint_R (1-x)\,dA \qquad \ldots (i)$$

Put $x = \cos\theta$, $y = \sin\theta$ then θ moves on C from $\theta = 0$ to $\theta = 2\pi$. Then (i) becomes

$$\oint_c \vec{F}\cdot d\vec{r} = \int_0^1\int_0^{2\pi} (1-\cos\theta)\,r\,d\theta\,dr$$

$$= \int_0^1 [\theta - \sin\theta]_0^{2\pi}\,r\,dr$$

$$= \int_0^1 [2\pi - \sin 2\pi]\,r\,dr$$

$$= 2\pi\int_0^1 r\,dr$$

$$= 2\pi\left[\frac{r^2}{2}\right]_0^1$$

$$= \pi$$

Example 6: Evaluate the line integral $\oint_c \vec{F}\cdot d\vec{r}$ where $\vec{F} = \frac{e^y}{x}\vec{i} + (e^y \log x + 2x)\vec{j}$ and the path C is the boundary of the region R: $1 + x^4 \le y \le 2$ in oriented counterclockwise direction.

Solution:

Given that,

$$\vec{F} = \left(\frac{e^y}{x}, e^y \log x + 2x\right)$$

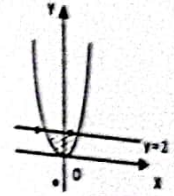
Then,

$$\text{curl}\,\vec{F} = \nabla\times\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ \frac{e^y}{x} & e^y\log x + 2x & 0 \end{vmatrix} = \left(\frac{e^y}{x} + 2 - \frac{e^y}{x}\right)\vec{k} = 2\vec{k}$$

$$\text{curl}\,\vec{F}\cdot\vec{k} = 2\vec{k}\cdot\vec{k} = 2$$

So,

As given the region R is closed by a path $1 + x^4 \le y \le 2$, in oriented counterclockwise direction, then by Green's theorem, we have



$$\oint_c \vec{F}\cdot d\vec{r} = \iint_R (\text{curl}\,\vec{F}\cdot\vec{k})\,dA = 2\iint_R dA \qquad \ldots (i)$$

From the figure, the region is bounded by $1 + x^2 \le y \le 2$ and solving the curves $y = 2$ and $y = x^2 + 1$ then we get $x = \pm 1$.

Now, (i) becomes,

$$\oint_c \vec{F}\cdot d\vec{r} = 2\int_{-1}^1\int_{1+x^4}^2 dy\,dx = 2\int_{-1}^1 [y]_{1+x^4}^2\,dx$$

$$= 2\int_{-1}^1 (2-1-x^4)\,dx$$

$$= 2\int_{-1}^1 (1-x^4)\,dx$$

$$= 2\left[x - \frac{x^5}{5}\right]_{-1}^1$$

$$= \left[\left(1-\frac{1}{5}\right) - \left(-1+\frac{1}{5}\right)\right]$$

$$= 2\left(2 - \frac{2}{5}\right)$$

$$= 4\left(5 - \frac{1}{5}\right)$$

$$= \frac{16}{5}.$$

Thus, $\oint_c \vec{F}\cdot d\vec{r} = \frac{16}{5}$.

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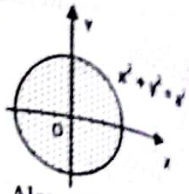
Example 7: Using Green's theorem, find the area of the circle, $x^2 + y^2 = a^2$.

Solution:

Given circle is,

$$x^2 + y^2 = a^2$$

Clearly, the region is bounded and has radius a. Then by Green's theorem the area of the circle is,



$$A = \frac{1}{2}\oint_c (x\,dy - y\,dx) \quad \ldots (i)$$

Set, $x = a\cos t$ and $y = a\sin t$. Then, $dx = -a\sin t\,dt$; $dy = a\cos t\,dt$. Also, radius of the circle is a and the angle rotates from $t = 0$ to $t = 2\pi$. Therefore, (i) becomes

$$A = \frac{1}{2}\int_0^{2\pi} (a\cos t \cdot a\cos t\,dt + a\sin t \cdot a\sin t\,dt)$$

$$= \frac{a^2}{2}\int_0^{2\pi} (\cos^2 t + \sin^2 t)\,dt$$

$$= \frac{a^2}{2}\int_0^{2\pi} (1)\,dt$$

$$= \frac{a^2}{2}\,2\pi$$

$$= \pi a^2$$

Example 8: Evaluate by using Green's theorem in the plane for $\oint_c [3x^2 - 8y^2)dx + (4y - 6xy)dy]$; where C is the boundary of the region $y = x^2$, $y = \sqrt{x}$.

Solution:

Given that,

$$I = \oint_c [3x^2 - 8y^2)dx + (4y - 6xy)dy] \quad \ldots (i)$$

where, the path c is determined by $y = x^2$, $y = \sqrt{x}$.

Comparing (i) with $\oint_c \vec{F}\cdot d\vec{r} = \oint_c (F_1 dx + F_2 dy)$ then we get

$$\vec{F} = (3x^2 - 8y^2)\vec{i} + (4y - 6xy)\vec{j}$$

Solving the curves $y = x^2$, $y = \sqrt{x}$ has point of intersections (0, 0) and (1, 1). Clearly, the region is bounded and the integrand function is continuous and is differentiable. Then by the Green's theorem, we have

$$\oint_c \vec{F}\cdot d\vec{r} = \iint_R (\text{curl}\,\vec{F}\cdot\vec{k})\,dA \quad \ldots (ii)$$

Here,

$$\text{curl}\,\vec{F} = \nabla\times\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 - 8y^2 & 4y - 6xy & 0 \end{vmatrix}$$

$$= (-6y + 16y)\vec{k}$$

$$= 10y\vec{k}$$

So,

$$\text{curl}\,\vec{F}\cdot\vec{k} = 10y\vec{k}\cdot\vec{k} = 10y$$

Then (ii) becomes,

$$\oint_c [3x^2 - 8y^2)dx + (4y - 6xy)dy] = \int_0^1 \int_{x^2}^{\sqrt{x}} (10y)\,dy\,dx$$

$$= \int_0^1 [5y^2]_{x^2}^{\sqrt{x}}\,dx$$

$$= 5\int_0^1 (x - x^4)\,dy$$

$$= 5\left[\frac{x^2}{2} - \frac{x^5}{5}\right]_0^1$$

$$= 5\left[\frac{1}{2} - \frac{1}{5}\right]$$

$$= \frac{3}{2}.$$

Thus, $I = \frac{3}{2}$.

Exercise 6 (A)

1. **Using Green's theorem, evaluate the following integrals:**

a. $\oint_c (ydx + 2xdy)$, C: the boundary of the square $0 \le x \le 1,\ 0 \le y \le 1$ (counterclockwise).

b. $\oint_c (x^2 + y^2)\,dy$, C: the boundary of the of the square $2 \le x \le 4,\ 2 \le y \le 4$

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c. $\oint_c [x^2 + y^2)dx + xy^2\, dy]$; where C is the closed curve determined by $y^2 = x$ and $y = x$ with $0 \le x \le 1$.

d. $\oint_c [15xy\ dx + x^3\ dy]$, where C is the closed curve consisting of the graphs of $y = x^2$ and $y = 2x$ between the points (0, 0) and (2, 4).

2. Using Greens theorem, evaluate the following integrals:

a. $\oint_c [2xy\ dx + (e^x + x^2)\ dy]$, C: the boundary of the triangle with vertices (0, 0), (1, 0), (1, 1) (clockwise).

b. $\oint_c [\sqrt{y}\, dx + \sqrt{x}\, dy]$ where C is the triangle with vertices (1, 1), (3, 1) and (2, 2).

c. $\oint_c [xy\ dx + siny\ dy]$, where C is the triangle with vertices (1, 1), (2, 2), (3, 0).

d. $\oint_c [(x^3 - 3y)\ dx + (x + siny)\ dy]$, C: the boundary of the triangle with vertices (0, 0), (1, 0), (0, 2).

e. $\oint_c [x^2y^2\ dx + (x^2 - y^2)\ dy]$; where C is the square with vertices (0, 0), (1, 0), (1, 1), (0, 1).

3. Use Green's theorem to evaluate the line integrals;

a. $\oint_c 2xy^3dx + 3x^2y^2dy)$, c: $x^2 + y^2 = 1$ counter clockwise.

b. $\oint_c [2xy\ dx + (x^2 + y^2)\ dy]$, where C is the ellipse $4x^2 + 9y^2 = 36$.

c. $\oint_c \vec{F}(r).d\vec{r}$ counterclockwise around the boundary C of the region R, where $\vec{F} = (y, -x)$, C the circle $x^2 + y^2 = \frac{1}{4}$

d. $\oint_c \vec{F}(r).d\vec{r}$ counterclockwise around the boundary C of the region R, where $\vec{F}$ = grad (sinx.cosy), C is the ellipse $25x^2 + 9y^2 = 225$.

e. $\oint_c \vec{F}(r).d\vec{r}$ counterclockwise around the boundary C of the region R, where $\vec{F} = (\tan 0.2x, x^5y)$, R: $x^2 + y^2 \le 25$, $y \ge 0$.

Answer Key

1. a. 1	b. 24	c. $\frac{13}{60}$	d. $\frac{28}{15}$	
2. a. −1	b. $\frac{\pi}{6}$	c. −3	d. 4	e. $\frac{2}{3}$
3. a.	b. 0	c. $\frac{-\pi}{2}$	d. 0	e. $\frac{2 \times 5^7}{7}$

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6.3 VOLUME INTEGRALS

Suppose V is a volume bounded by a surface S. Suppose f(x, y, z) be a single valued function of position defined over V. Then the volume integral of f(x, y, z) over V is denoted by $\iiint_V f(x, y, z)\, dV$. If we subdivide the volume V into small cuboids by drawing line parallel to the three coordinates axes then dV = dx dy dz. Then the volume of the function f(x, y, z) can be reintroduced as

$$\iiint_V f(x, y, z)\, dx\, dy\, dz.$$

If $\vec{F}$ be a vector valued function defined over V then the volume of the function $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ can be reintroduced as

$$\iiint_V \vec{F}\, dV.$$

The triple integrals can be transformed into surface integrals over the boundary surface of a region in space and conversely. It helps in establishing fundamental equations in fluid flow, heat conduction, etc. The transformation is done by the following divergence theorem, which involves the divergence of a vector function

$$\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}.$$

The following theorem is useful to finding the volume of a closed and bounded solid, is introduced by Carl Friedrich Gauss (1777 – 1855). This theorem transforms the surface integral into volume integral under the base calculation with divergence of a vector valued function.

6.4 GAUSS DIVERGENCE THEOREM

Statement: Let T be a closed and bounded region in space whose boundary is a piecewise smooth orientable surface S. Let $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ be a vector valued function that is continuous and has continuous first order partial derivatives in some domain containing T. Then

$$\iint_S \vec{F}\cdot\vec{n}\, dA = \iiint_T \text{div}\vec{F}\, dV$$

where $\vec{n}$ is the outer unit normal vector on S.

That is the flux of $\vec{F}$ over S equals to the triple integral of the divergence of $\vec{F}$ over T.

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Workout Examples

Example 1: Evaluate the integral $\iint_S \vec{F}\cdot\vec{n}\, ds$, where $\vec{F} = 4x\vec{i} + 2y^2\vec{j} + z^2\vec{k}$ and s is the surface of the cube $|x| \le 1, |y| \le 1, |z| \le 1$.

Solution: $\vec{F} = (4x, 2y^2, z^2)$

Given that and the surface is the cube $|x| \le 1, |y| \le 1, |z| \le 1$.

By Gauss divergence theorem we have,

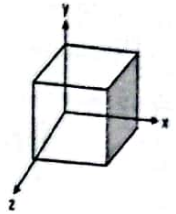
$$\iint_S \vec{F}\cdot\vec{n}\, ds = \iiint_V \text{div}\vec{F}\, dV \quad \ldots (i)$$

Here, $\text{div}\,\vec{F} = \nabla\cdot\vec{F} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)\cdot(4x\vec{i} + 2y^2\vec{j} + z^2\vec{k})$

$= 4 + 4y + 2z$

Since the surface is a cube with $|x| \le 1, |y| \le 1, |z| \le 1$.

So, $-1 \le x \le 1, -1 \le y \le 1, -1 \le z \le 1$.



Now, by (i)

$$\iint_S (\vec{F}\cdot\vec{n})\, ds = \int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} (4 + 4y + 2z)\, dx\, dy\, dz$$

$$= \int_{-1}^{1}\int_{-1}^{1} [4x + 4xy + 2xz]_{-1}^{1}\, dy\, dz$$

$$= \int_{-1}^{1}\int_{-1}^{1} [4(1 + 1) + 4y(1 + 1) + 2z(1 + 1)]\, dy\, dz$$

$$= 4\int_{-1}^{1}\int_{-1}^{1} (2 + 2y + z)\, dy\, dz$$

$$= 4\int_{-1}^{1} [2y + y^2 + zy]_{-1}^{1}\, dz$$

$$= 4\int_{-1}^{1} [2(1 + 1) + (1 - 1) + z(1 + 1)]\, dz$$

$$= 4\int_{-1}^{1} (4 + 2z)\, dz$$

$$= 4[4z + z^2]_{-1}^{1}$$

$$= 4[4(1 + 1) + (1 - 1)] = 32.$$

Thus, $\iint_S (\vec{F}\cdot\vec{n})\, ds = 32.$

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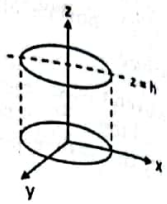
Example 2: Evaluate $\iint_S \vec{F}\cdot\vec{n}\, dA$ where $\vec{F} = (y^3, x^3, z^3)$, S: $x^2 + 4y^2 = 1$, $x \geq 0$, $y \geq 0$, $0 \leq z \leq h$.

Solution:

Given that $\vec{F} = (y^3, x^3, z^3)$

and surface is $x^2 + 4y^2 = 1$ for $x \geq 0, y \geq 0, 0 \leq z \leq h$.

Clearly, the surface is bounded, then by using Gauss divergence theorem,



$$\iint_S \vec{F}\cdot\vec{n}\, dA = \iiint_V \text{div}\vec{F}\, dV \quad \text{.....(i)}$$

Here,

$$\text{div}\vec{F} = \nabla\cdot\vec{F} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)\cdot(y^3\vec{i} + x^3\vec{j} + z^3\vec{k}) = 3z^2$$

Given surface is a ellipsoid having height h on first quadrant.

So, x varies on the region from $x = 0$ to $x = \sqrt{1-4y^2}$ and y moves from $y = 0$ to $y = \frac{1}{2}$. Also, the region moves from $z = 0$ to $z = h$.

Now, (i) becomes,

$$\iint_S \vec{F}\cdot\vec{n}\, dA = \int_0^{1/2}\int_0^{\sqrt{1-4y^2}}\int_0^h 3z^2\, dz dx dy$$

$$= \int_0^{1/2}\int_0^{\sqrt{1-4y^2}} [z^3]_0^h\, dx\, dy$$

$$= h^3 \int_0^{1/2} [x]_0^{\sqrt{1-4y^2}}\, dy$$

$$= h^3 \int_0^{1/2} \sqrt{1-4y^2}\, dy$$

$$= 2h^3 \int_0^{1/2} \sqrt{\frac{1}{4} - y^2}\, dy$$

$$= 2h^3 \left[\frac{y\sqrt{\frac{1}{4}-y^2}}{2} + \frac{1/4}{2}\sin^{-1}\frac{y}{1/2}\right]_0^{1/2}$$

$$= 2h^3\left[0 + \frac{1}{8}\sin^{-1}(1)\right]$$

$$= \frac{h^3}{4}\cdot\frac{\pi}{2} \quad [\because \sin^{-1}(1) = \frac{\pi}{2}].$$

$$= \frac{\pi h^3}{8}$$

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Example 3: Evaluate $\iint_S \vec{F}\cdot\vec{n}\, dA$ where $\vec{F} = (4xz, -y^2, yz)$, S is the surface of the cube bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

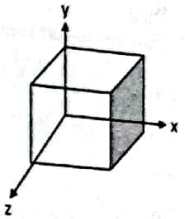
Solution:

Given that, $\vec{F} = (4xz, -y^2, yz)$.

And the surface is bounced by $x = 0, y = 0, y = 1, z = 0, z = 1$.

Clearly, the surface is bounded, then by using Gauss divergence theorem,

$$\iint_S \vec{F}\cdot\vec{n}\, dA = \iiint_V \text{div}\vec{F}\, dv \quad \text{... (i)}$$



Here,

$$\text{div}\vec{F} = \nabla\cdot\vec{F} = 4z - 2y + y = 4z - y.$$

Then (i) becomes,

$$\iint_S \vec{F}\cdot\vec{n}\, dA = \int_0^1\int_0^1\int_0^1 (4z - y)\, dz dy dx$$

$$= \int_0^1\int_0^1 [2z^2 - yz]_0^1\, dy dx$$

$$= \int_0^1\int_0^1 (2 - y) dy dx$$

$$= \int_0^1 \left[2y - \frac{y^2}{2}\right]_0^1 dx$$

$$= \int_0^1 \left(2 - \frac{1}{2}\right) dx$$

$$= \int_0^1 \frac{3}{2}\, dx$$

$$= \frac{3}{2}[x]_0^1$$

$$= \frac{3}{2}$$

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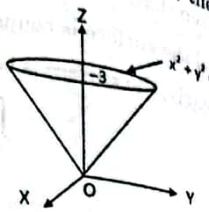
Example 4: Evaluate $\iint_S \vec{F}\cdot\vec{n}\, dA$ where $\vec{F} = (x, 3y, 6z)$ and S is the surface of the cone $\sqrt{x^2+y^2} \le z, 0 \le z \le 3$.

Solution:

The given surface is $\sqrt{x^2+y^2} \le z$, $0 \le z \le 3$. Clearly, the surface is bounded, then by using Gauss divergence theorem,

$$\iint_S \vec{F}\cdot\vec{n}\, dA = \iiint_V \text{div}\vec{F}\, dv \quad \ldots (i)$$



Given,

$$\vec{F} = x\vec{i} + 3y\vec{j} + 6z\vec{k}$$

Then, $\text{div}\vec{F} = 1 + 3 + 6 = 10.$

Then from equation (i), we get

$$\iint_S \vec{F}\cdot\vec{n}\, dA = \iiint_V 10\, dv$$

$$= 10\iiint_V dv$$

$$= 10 \times \text{volume of cone represented by the surface S}$$

$$= 10 \times \frac{1}{3}\pi(3)^2.3$$

$$= 90\pi.$$

Thus, $\iint_S \vec{F}\cdot\vec{n}\, dA = 90\pi$.

***Note:** If $\text{div}\vec{F}$ is not a constant value. Then we put the limits of x, y and z are as follows.*

$$0 \le z \le 3; -\sqrt{z} \le y \le \sqrt{z}; -\sqrt{z^2-y^2} \le x \le \sqrt{z^2-y^2}.$$

Example 5: Evaluate, $\iint_S \vec{F}\cdot\vec{n}\, dA$ where $\vec{F} = (x^2, e^y, 1)$; S: $x + y + z = 1$, $x \ge 0$, $y \ge 0, z \ge 0$.

Solution:

Given that

$$\vec{F} = (x^2, e^y, 1)$$

and the surface is, $x + y + z = 1$, for $x \ge 0, y \ge 0, z \ge 0$

By Gauss divergence theorem, we have

$$\iint_S \vec{F}\cdot\vec{n}\, dA = \iiint_V \text{div}\,\vec{F}\, dV \quad \ldots (i)$$

Here,

$$\text{div}\cdot\vec{F} = \nabla\cdot\vec{F} = 2x + e^y$$

The plane $x + y + z = 1$ with $x \ge 0, y \ge 0, z \ge 0$ is as shown in figure.

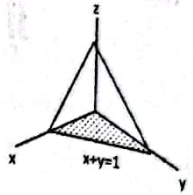
Clearly z varies from $z = 0$ to the plane $x + y + z = 1 \Rightarrow 1 - x - y$.

And, the variable y varies in xy–plane from $y = 0$ to $x + y = 1 \Rightarrow y = 1 - x$.

Also, on the region x moves from $x = 0$ to $x = 1$.

Then, (i) becomes,

$$\iint_S \vec{F}\cdot\vec{n}\, dA = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} (2x + e^y)\, dz\, dy\, dx$$



$$= \int_0^1 \int_0^{1-x} [2xz + ze^y]_0^{1-x-y}$$

$$= \int_0^1 \int_0^{1-x} [2x(1-x-y) + (1-x-y)e^y]\, dy\, dx$$

$$= \int_0^1 \int_0^{1-x} [2x - 2x^2 - 2xy + (1-x)e^y - ye^y]\, dy\, dx$$

$$= \int_0^1 [2xy - 2x^2y - xy^2 + (1-x)e^y - ye^y + e^y]_0^{1-x}\, dx$$

$$= \int_0^1 [(2x - x^2)(1-x) - x(1-x)^2 + (2-x)e^{1-x} -$$

$$(2-x)e^0 - (1-x)e^{1-x}]\, dx$$

$$= \int_0^1 [2x - x^2 - 2 + e\, e^{-x}]\, dx$$

$$= \left[x^2 - \frac{x^3}{3} - 2x + e.\frac{e^{-x}}{-1}\right]_0^1$$

$$= \left(1 - \frac{1}{3} - 2 - e.e^{-1}\right) + e$$

$$= -\frac{7}{3} + e \qquad [\because e.e^{-1} = 1]$$

Thus, $\iint_S \vec{F}\cdot\vec{n}\, dA = e - \frac{7}{3}$.

Example 6: Evaluate, $\iint_S \vec{F}\cdot\vec{n}\, dA$ where $\vec{F} = (4x, x^2y, -x^2z)$, S is the surface of the tetrahedron with vertices (0, 0, 0), (1, 0, 0), (0, 1, 0) and (0, 0, 1).

Solution:

Given that, $\vec{F} = (4x, x^2y, -x^2z)$.

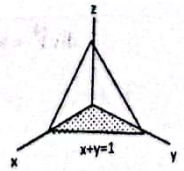
And the surface is the tetrahedron with vertices (0, 0, 0), (1, 0, 0), (0, 1, 0) and (0, 0, 1).

Thus, the region of integration is

$$0 \le x \le 1, 0 \le y \le 1 - x, 0 \le z \le 1 - y - x$$

By Gauss divergence theorem, we have,

$$\iint_S \vec{F}\cdot\vec{n}\, dA = \iiint_V \text{div}\vec{F}\, dv \quad \ldots (i)$$



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Here, $\text{div}\vec{F} = \nabla\cdot\vec{F} = 4 + x^2 - x^2 = 4.$

Then (i) becomes,

$$\iint_S \vec{F}\cdot\vec{n}\,dA = \int_0^1\int_0^{1-x}\int_0^{1-x-y} 4\,dz\,dy\,dx$$

$$= 4\int_0^1\int_0^{1-x}[z]_0^{1-x-y}\,dy\,dx$$

$$= 4\int_0^1\int_0^{1-x}(1-x-y)\,dy\,dx$$

$$= 4\int_0^1\left[y - xy - \frac{y^2}{2}\right]_0^{1-x}dx.$$

$$= \frac{4}{2}\int_0^1\left[1 - x - x(1-x) - \frac{(1-x)^2}{2}\right]dx$$

$$= 2\int_0^1(1 - 2x + x^2)\,dx$$

$$= 2\left[x - x^2 + \frac{x^3}{3}\right]_0^1$$

$$= 2\left(1 - 1 + \frac{1}{3}\right)$$

$$= \frac{2}{3}.$$

Example 7: Evaluate, $\iint_S \vec{F}\cdot\vec{n}\,dA$ where $\vec{F} = (x^3, y^3, z^3)$, S is the sphere $x^2 + y^2 + z^2 = 9$.

Solution:

Given that, $\vec{F} = (x^3, y^3, z^3)$.

And the surface is a sphere $x^2 + y^2 + z^2 = 9$, that has radius 3.

By Gauss divergence theorem, we have,

$$\iint_S \vec{F}\cdot\vec{n}\,dA = \iiint_V \text{div}\vec{F}\,dv \quad \ldots (i)$$

Here,

$$\text{div}\vec{F} = \nabla\cdot\vec{F} = 3x^2 + 3y^2 + 3z^2$$

$$= 3(x^2 + y^2 + z^2)$$

$$= 3 \times 9 \quad [\text{Being } x^2 + y^2 + z^2 = 9]$$

$$= 27.$$

Then (i) becomes,

$$\iint_S \vec{F}\cdot\vec{n}\,dA = \iiint_V 27\,dz\,dy\,dx$$

$$= 27\iiint_V dz\,dy\,dx$$

$$= 27 \times \text{volume of the sphere having radius 3}$$

$$= 27 \times \frac{4}{3}\pi(3)^3 \quad [\text{Volume of sphere, } V = \frac{4}{3}\pi(r)^3] = 972\pi$$

Thus, $\iint_S \vec{F}\cdot\vec{n}\,dA = 972\pi.$

Exercise 6 (B)

1. Evaluate $\iint_S \vec{F}\cdot\vec{n}\,dA$, by using Gauss divergence theorem of the following data:
   a. $\vec{F} = (x^2, 0, z^2)$, S is the box $|x| \le 1, |y| \le 3, |z| \le 2$.
   b. $\vec{F} = (4x, 2y^2, z^2)$ and s is the surface of the cube $|x| \le 1, |y| \le 1, |z| \le 1$.
   c. $\vec{F} = (4xz, -y^2, yz)$, S is the surface of the cube bounded by the planes $x = 0$, $x = 1, y = 0, y = 1, z = 0, z = 1$.
   d. $\vec{F} = (0, 20y, 2z^3)$, S: the surface of $0 \le x \le 6, 0 \le y \le 1, 0 \le z \le y$.
   e. $\vec{F} = (\cos y, \sin x, \cos z)$, S is the surface $x^2 + y^2 \le 4, |z| \le 2$.
   f. $\vec{F} = (1, 1, 1)$, S: $x^2 + y^2 + 4z^2 = 4, z \ge 0$.
   g. $\vec{F} = (x + z, y + z, x + y)$, S is the sphere $x^2 + y^2 + z^2 = 9$.
   h. $\vec{F} = (x^3, y^3, z^3)$, S is the sphere $x^2 + y^2 + z^2 = 9$.
   i. $\vec{F} = (9x, y\cosh^2 x, -z\sinh^2 x)$, S: the ellipsoid $4x^2 + y^2 + 9z^2 = 36$.
   j. $\vec{F} = (4x, x^2y, -x^2z)$, S is the surface of the tetrahedron with vertices (0, 0, 0), (1, 0, 0), (0, 1, 0) and (0, 0, 1).
2. Evaluate,
   a. $\iiint_S \phi\,dv$, where $\phi = 45x^2y$ and v is the closed region bounded by the planes $4x + 2y + z = 8, x = 0, y = 0, z = 0$.
   b. $\iiint_S (\nabla\times\vec{F})\,dv$ where $\vec{F} = (2x^2 - 3z, -2xy, -4x)$, and v is the closed region bounded by the planes $x = 0, y = 0, z = 0$ and $2x + 2y + z = 4$.

Answer Key

1. a. 0	b. 32	c. $\frac{3}{2}$	d. 63	e. 0
f. 0	g. 72π	h. 372π	i. 480π	j. $\frac{2}{3}$
2. a. 128	b. $\frac{8}{3}(\vec{j} - \vec{k})$			

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6.5 Relation between Line and Surface Integrals

The Stoke's theorem deals to the line integral and transforms it into surface integral. That is, the line integral around S (the boundary curve) of F's tangential component is equal to the surface integral of the normal component of the curl of F. The Stokes theorem is the generalized form of Green's theorem. The Stoke's theorem states that "the surface integral of the curl of a function over a surface bounded by a closed surface is equal to the line integral of the particular vector function around that surface." The Stokes theorem was first presented in 1854 by Irish mathematician George Gabriel Stokes (1819 – 1903). The most important application this is in Physics, Maxwell's laws.

6.6 Stoke's Theorem

Statement: Let S be a piecewise smooth oriented surface in space and the boundary of S be a piecewise smooth simple closed curve C. Let $\vec{F}(x, y, z)$ be a continuous vector valued function that has continuous first partial derivatives in a domain in space containing S. Then

$$\iint_S (\text{curl}\,\vec{F}\cdot\vec{n})\,ds = \oint_c \vec{F}\cdot d\vec{r}$$

where $\vec{n}$ is a unit normal vector of S and the integration around C is taken in anti-clockwise direction with respect to $\vec{n}$.

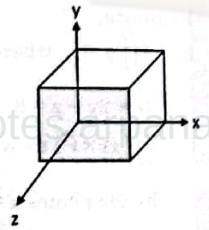
Workout Examples

Example 1: Evaluate $\oint_c \vec{F}\cdot d\vec{r}$ by using stocks theorem where $\vec{F} = (z^2, 5x, 0)$, and c is the boundary of the surface S: square $0 \le x \le 1, 0 \le y \le 1, z = 1$.

Solution:

Given that, $\vec{F} = (z^2, 5x, 0)$

and the surface is $0 \le x \le 1, 0 \le y \le 1, z = 1$.



By Stoke's theorem we have

$$\oint_c \vec{F}\cdot d\vec{r} = \iint_S (\nabla\times\vec{F})\cdot\vec{n}\,dx\,dy \quad \ldots (i)$$

where, $\vec{n} = \vec{r}_x \times \vec{r}_y$.

Here, $$\nabla\times\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ z^2 & 5x & 0 \end{vmatrix} = 2z\vec{j} + 5\vec{k} = (0, 2z, 5).$$

Since we have,

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k} = x\vec{i} + y\vec{j} + \vec{k} \quad \text{being } z = 1.$$

$$\vec{r}_x = \vec{i} \quad \text{and} \quad \vec{r}_y = \vec{j}.$$

$$\vec{n} = \vec{r}_x \times \vec{r}_y = \vec{i} \times \vec{j} = \vec{k} = (0, 0, 1).$$

Then,

$$(\nabla\times\vec{F})\cdot\vec{n} = (0, 2z, 5)\cdot(0, 0, 1) = 5.$$

Now,

Then (i) becomes

$$\oint_c \vec{F}\cdot d\vec{r} = \int_0^1\int_0^1 5\,dx\,dy = 5\int_0^1 [x]_0^1\,dy = 5\int_0^1 dy = 5[y]_0^1 = 5$$

Thus $\oint_c \vec{F}\cdot d\vec{r} = 5$.

Example 2: Evaluate $\oint_c \vec{F}\cdot d\vec{r}$ by using stocks theorem where $\vec{F} = (4z, -2x, 2x)$, C is the circle $x^2 + y^2 = 1, z = y + 1$.

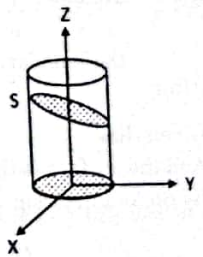
Solution:

Given that, $\vec{F} = (4z, -2x, 2x)$

and the surface is, $x^2 + y^2 = 1, z = y + 1$.

By Stoke's theorem we have

$$\iint_S (\nabla\times\vec{F})\cdot\vec{n}\,dx\,dy = \oint_c \vec{F}\cdot d\vec{r} \quad \ldots (i)$$



where, $\vec{n} = \vec{r}_x \times \vec{r}_y$.

Here,

$$\nabla\times\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 4z & -2x & 2x \end{vmatrix}$$

$$= 0\vec{i} + (4 - 2)\vec{j} + (-2)\vec{k}$$

$$= (0, 2, -2)$$

Since we have, $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k} = x\vec{i} + y\vec{j} + (y + 1)\vec{k}$. Then,

$$\vec{r}_x = \vec{i} = (1, 0, 0) \quad \text{and} \quad \vec{r}_y = \vec{j} + \vec{k} = (0, 1, 1).$$

So that,

$$\vec{n} = \vec{r}_x \times \vec{r}_y = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 0\vec{i} - \vec{j} + \vec{k} = (0, -1, 1).$$

Then,

$$(\nabla\times\vec{F})\cdot\vec{n} = (0, 2, -2)\cdot(0, -1, 1) = 0 - 2 - 2 = -4$$

Given surface on xy plane is $x^2 + y^2 = 1$ which is a circle in which y varies from $y = -\sqrt{1 - x^2}$ to $\sqrt{1 - x^2}$ and x moves on the region from $x = -1$ to $x = 1$.

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Therefore,

$$\iint_S (\nabla\times\vec{F})\cdot\vec{n}\,dx\,dy = \int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} (-4)\,dy\,dx$$

$$= -4\int_{-1}^{1} [y]_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\,dx$$

$$= -4\int_{-1}^{1} 2\sqrt{1-x^2}\,dx$$

$$= -8\left[\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}\left(\frac{x}{1}\right)\right]_{-1}^{1}$$

$$= -8\left[0 + \frac{1}{2}\sin^{-1}(1) - 0 - \frac{1}{2}\sin^{-1}(-1)\right]$$

$$= -8\sin^{-1}(1) \qquad [\because \sin^{-1}(-x) = -\sin^{-1}x]$$

$$= -8\times\frac{\pi}{2} \qquad [\because \sin^{-1}(1) = \frac{\pi}{2}]$$

$$= -4\pi$$

Thus, by (i), $\oint_c \vec{F}\cdot d\vec{r} = -4\pi$

Example 3: Evaluate, $\oint_c \vec{F}\cdot d\vec{r}$ by using stocks theorem where $\vec{F} = y^2\vec{i} + z^2\vec{j} + x^2\vec{k}$, and c is the boundary of the surface S in the first octant portion of the plane $x + y + z = 1$.

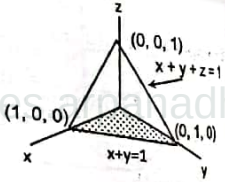
Solution:

Given that $\vec{F} = y^2\vec{i} + z^2\vec{j} + x^2\vec{k}$

And the surface is the portion of the plane $x + y + z = 1$ in the first octant.

By Stoke's theorem we have,

$$\iint_S (\nabla\times\vec{F})\cdot\vec{n}\,dx\,dy = \oint_c \vec{F}\cdot d\vec{r} \quad \ldots (i)$$



where $\vec{n} = \vec{r}_x \times \vec{r}_y$.

Here, $\nabla\times\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^2 & z^2 & x^2 \end{vmatrix} = -2z\vec{i} - 2x\vec{j} - 2y\vec{k}$

Since we have,

$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k} = x\vec{i} + y\vec{j} + (1 - x - y)\vec{k}$ $[\because x + y + z = 1]$

Then,

$\vec{r}_x = \vec{i} - \vec{k}$ and $\vec{r}_y = \vec{j} - \vec{k}$.

So that,

$$\vec{n} = \vec{r}_x \times \vec{r}_y = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{vmatrix} = \vec{i} + \vec{j} + \vec{k}.$$

Therefore,

$$(\nabla\times\vec{F})\cdot\vec{n} = (-2z\vec{i} - 2x\vec{j} - 2y\vec{k})\cdot(\vec{i} + \vec{j} + \vec{k})$$

$$= -2z - 2x - 2y$$

$$= -2(x + y + z)$$

$$= -2\,(1)$$

$$= -2.$$

The projection of the surface plane $x + y + z = 1$ on xy–plane is $x + y = 1, z = 0$.

In which y varies from $y = 0$ to $y = 1 - x$ and x moves from $x = 0$ t $x = 1$.

Therefore,

$$\iint_S (\nabla\times\vec{F})\cdot\vec{n}\,dx\,dy = \int_0^1\int_0^{1-x} (-2)\,dy\,dx$$

$$= -2\int_0^1 [y]_{-1}^{1-x}\,dx$$

$$= -2\int_0^1 (1 - x)\,dx$$

$$= -2\left[x - \frac{x^2}{2}\right]_0^1$$

$$= -2\left(1 - \frac{1}{2}\right)$$

$$= -1$$

Thus, by (i), $\oint_c \vec{F}\cdot d\vec{r} = -1$.

Example 4: Evaluate $\oint_c \vec{F}\cdot d\vec{r}$ by using stocks theorem where $\vec{F} = (y^3, 0, x^3)$, and c is the boundary of the triangle with vertices (1, 0, 0), (0, 1, 0), (0, 0, 1).

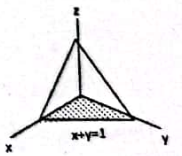
Solution:

Given that, $\vec{F} = (y^3, 0, x^3)$

And, the surface is a triangle having vertices (1, 0, 0), (0, 1, 0) and (0, 0, 1).

By Stoke's theorem we have,

$$\iint_S (\nabla\times\vec{F})\cdot\vec{n}\,dx\,dy = \oint_c \vec{F}\cdot d\vec{r} \quad \ldots (i)$$



where, $\vec{n} = \vec{r}_x \times \vec{r}_y$

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Here,

$$\nabla\times\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^3 & 0 & x^3 \end{vmatrix} = -3x^2\vec{j} - 3y^2\vec{k} = (0, -3x^2, -3y^2).$$

Since the equation of plane that passes through (1, 0, 0), (0, 1, 0) and (0, 0, 1) be

$$x + y + z = 1 \quad \ldots \text{(ii)}$$

Since we have,

$$\vec{r} = x\vec{i} + y\vec{j} + (1 - x - y)\vec{k} \qquad [\because \text{using (ii)}]$$

Then,

$$\vec{r_x} = \vec{i} - \vec{k} = (1, 0, -1) \text{ and } \vec{r_y} = \vec{j} - \vec{k} = (0, 1, -1).$$

So that,

$$\vec{n} = \vec{r_x}\times\vec{r_y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{vmatrix} = \vec{i} + \vec{j} + \vec{k} = (1, 1, 1)$$

Therefore,

$$(\nabla\times\vec{F})\cdot\vec{n} = (0, -3x^2, -3y^2)\cdot(1, 1, 1) = -3x^2 + 3y^2$$

Since the surface is the plane $x + y + z = 1$. On the xy–plane, the projection of the plane is $x + y = 1$ in which x varies from $x = 0$ to $x = 1 - y$ and y moves from $y = 0$ to $y = 1$.

Therefore,

$$\iint_S (\nabla\times\vec{F})\cdot\vec{n}\, dx\, dy = \int_0^1\int_0^{1-y} (-3x^2 - 3y^2)\, dx\, dy$$

$$= \int_0^1 [-x^3 - 3xy^2]_0^{1-y}\, dy$$

$$= \int_0^1 [-(1-y)^3 - 3y^2(1-y)]\, dy$$

$$= \int_0^1 [(y-1)^3 + 3y^2(y-1)]\, dy$$

$$= \int_0^1 [4y^3 - 6y^2 + 3y - 1]\, dy$$

$$= \left[y^4 - 2y^3 + 3\frac{y^2}{2} - y\right]_0^1$$

$$= \left[1 - 2 + \frac{3}{2} - 1\right]$$

$$= \frac{-1}{2}$$

Thus, by (i), $\oint_c \vec{F}\cdot d\vec{r} = \frac{-1}{2}$

Example 5: Evaluate, $\oint_c \vec{F}\cdot d\vec{r}$ by using stocks theorem where $\vec{F} = (2x - y, -yz^2, -y^2z)$ and c is the boundary of the surface S is the upper half surface of $x^2 + y^2 + z^2 = 1$, bounded by its projection on xy plane.

Solution:

Given that $\vec{F} = (2x - y, -yz^2, -y^2z)$.

And the region is the upper half of the sphere $x^2 + y^2 + z^2 = 1$ that is bounded by its projection on xy–plane.

By Stoke's theorem, we have,

$$\iint_S (\nabla\times\vec{F})\cdot\vec{n}\, dx\, dy = \oint_c \vec{F}\cdot d\vec{r} \quad \ldots \text{(i)}$$

where, $\vec{n} = \vec{r_x}\times\vec{r_y}$

Here,

$$\nabla\times\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2x-y & -yz^2 & -y^2z \end{vmatrix}$$

$$= (-2yz + 2yz)\vec{i} + (0 - 0)\vec{j} + (0 + 1))\vec{k} = (0, 0, 1).$$

Since we have,

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k} = (x, y, \sqrt{1 - x^2 - y^2})$$

Then,

$$\vec{r_x} = \left(1, 0, \frac{-x}{\sqrt{1 - x^2 - y^2}}\right) \text{ and } \vec{r_y} = \left(0, 1, \frac{-y}{\sqrt{1 - x^2 - y^2}}\right)$$

So that,

$$\vec{n} = \vec{r_x}\times\vec{r_y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -x/\sqrt{1=x^2-y^2} \\ 0 & 1 & -y/\sqrt{1-x^2-y^2} \end{vmatrix}$$

$$= \frac{x}{\sqrt{1-x^2-y^2}}\vec{i} + \frac{y}{\sqrt{1-x^2-y^2}}\vec{j} + \vec{k}.$$

Then,

$$(\nabla\times\vec{F})\cdot\vec{n} = 1.$$

Since the region is the projection of $x^2 + y^2 + z^2 = 1$ on xy–plane. So the region of integration is $x^2 + y^2 = 1$, $z = 0$.

This is a circle with radius $r = 1$.

Setting, $x = \cos\theta$ and $y = \sin\theta$ then $dx\, dy = r\, dr\, d\theta$. Also, θ varies from $\theta = 0$ to $\theta = 2\pi$.

Then,

$$\iint_S (\nabla\times\vec{F})\cdot\vec{n}\, dx\, dy = \int_0^{2\pi}\int_0^1 r\, dr\, d\theta = \left[\frac{r^2}{2}\right]_0^1 [\theta]_0^{2\pi} = \frac{1}{2}2\pi = \pi.$$

Then (i) gives, $\oint_c \vec{F}\cdot d\vec{r} = \pi$.

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Exercise 6 (C)

Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ by using stocks theorem:

1. $\vec{F} = (y^2, 2xy + \sin x, 0)$, C the boundary of $0 \le x \le \pi/2$, $0 \le y \le 2$, $z = 0$.
2. $\vec{F} = (\cos \pi y, \sin \pi x, \cos \pi x)$, C the boundary of $0 \le x \le \frac{1}{2}$, $0 \le y \le 4$, $z = x$.
3. $\vec{F} = (x^2 + y^2, -2xy, 0)$, C is the boundary of rectangle, $x = \pm a$, $y = 0$, $y = b$.
4. $\vec{F} = (e^z, e^z \sin y, e^z \cos y)$, S: $z = y^2$, $0 \le x \le 4$, $0 \le y \le 2$.
5. $\vec{F} = (e^x, e^y, e^z)$, C: $x = \log y$, $z = \log y$, $1 \le y \le 2$.
6. $\vec{F} = (y^2, z^2, x^2)$, S the portion of the parabolid $x^2 + y^2 = z$, $y \ge 0$, $z \le 1$.
7. $\vec{F} = [y, xz^3, -zy^3]$; C, the circle $x^2 + y^2 = 4$, $z = -3$.
8. $\vec{F} = (4z, -2x, 2x)$, C is the circle $x^2 + y^2 = 1$, $z = y + 1$.
9. $\vec{F} = \left(y, \frac{z}{2}, \frac{3y}{2}\right)$, C is the circle of $x^2 + y^2 + z^2 = 6z$, $z = x + 3$.
10. $\vec{F} = (x^3, e^{3y}, e^{-3z})$, C : $x^2 + 9y^2 = 9$, $z = x^2$.
11. $\vec{F} = (\sin \pi x, z, 0)$, C the boundary of the triangle with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(1, 1, 0)$.
12. Evaluate the surface integral is $\iint (\nabla \times \vec{F}) \cdot \vec{n}\, dA$, where $\vec{F} = [y^2, -x^2, 0]$, S the semi-circular disc $x^2 + y^2 \le 4$, $y \ge 0$ and $z = 0$.

Answer Key

1. 2	2. 4	3. $-4ab^2$	4. $4(1 - e^4)$
5. $2 + e^2 - e^1$	6. $-\frac{4}{3}$	7. -112π	8. -4π
9. $-9\sqrt{2}\,\pi$	10. 0	11. 0	12. $\frac{-32}{3}$

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Introduction

Fourier series is an important mathematical tool for the engineering field. It plays the important role in solving problems involve ordinary and partial differential equation. The Fourier series was introduced first time by French mathematician Baron Jean Baptiste Josheph Fourier (1768 – 1830) via his work *'Theorie Analytique de la Chaleur'*. Also, it is used for analyzing periodic functions, such as radio signals and alternating currents, for solving heat transfer problems.

7.1 Periodic Functions and Fundamental Period of Periodic Functions

Some functions have the nature that it repeats its value at regular interval; this length is the period of the function. For example, the functions sinx, cosx, secx and cosecx have repeated value at interval of 2π and the functions tanx and cotx are with period π.

Definition:

A function f(x) is said to be periodic if there exists a non zero positive number p such that f(x) = f(x + p) for all x.

This number p is said to be period of f(x). The smallest value of p is called the fundamental period or primitive period or basic period or prime period.

Example 1:

(i) The functions sinx, cosx are periodic with period 2π.

(ii) The functions tanx, cotx are periodic with period π.

(iii) The functions secx, cosecx are periodic with period 2π.

Example 2: Find the period of f(x) = sin nx.

Solution

Here,

$$f(x) = f(x + p)$$

$$\Rightarrow \quad \sin nx = \sin n(x + p) = \sin nx \cos np + \cos nx \sin np$$

Comparing the coefficient, we get

$$1 = \cos np$$

$$\Rightarrow \quad \cos 2\pi = \cos np$$

$$\Rightarrow \quad np = 2\pi$$

$$\Rightarrow \quad p = \frac{2\pi}{n}$$

Therefore, $p = \frac{2\pi}{n}$ is the period of f(x) = sin nx.

Example 3: Find the smallest period of $f(x) = \cos\left(\frac{2\pi x}{k}\right)$.

Solution

Here,

$$f(x) = f(x + p)$$

$$\Rightarrow \quad \cos\left(\frac{2\pi x}{k}\right) = \cos\left(\frac{2\pi(x + p)}{k}\right)$$

$$= \cos\left(\frac{2\pi x}{k}\right)\cos\left(\frac{2\pi p}{k}\right) - \sin\left(\frac{2\pi x}{k}\right)\sin\left(\frac{2\pi p}{k}\right)$$

Comparing the coefficient, we get

$$1 = \cos\left(\frac{2\pi p}{k}\right)$$

$$\Rightarrow \quad \cos 2\pi = \cos\left(\frac{2\pi p}{k}\right)$$

$$\Rightarrow \quad 2\pi = \frac{2\pi p}{k}$$

$$\Rightarrow \quad p = k$$

Therefore, p = k is the smallest period of f(x).

Example 4: Show that the fundamental period of sin2x is π.

Solution

Here,

$$f(x) = f(x + p)$$

$$\Rightarrow \quad \sin 2x = \sin 2(x + p) = \sin 2x \cos 2p + \cos 2x \sin 2p$$

Comparing the coefficient, we get

$$1 = \cos 2p$$

$$\Rightarrow \quad \cos 2\pi = \cos 2p$$

$$\Rightarrow \quad 2\pi = 2p$$

$$\Rightarrow \quad p = \pi$$

This means the period of f(x) is π. And, therefore the fundamental (i.e. smallest) period of f(x) is π.

Example 5: Find the fundamental period of $\tan\left(\frac{3x}{5}\right)$.

Solution

Here,

$$f(x) = f(x + p)$$

$$\Rightarrow \quad \tan\left(\frac{3x}{5}\right) = \tan\left(\frac{3(x + p)}{5}\right)$$

$$\Rightarrow \quad \tan\left(\pi + \frac{3x}{5}\right) = \tan\left(\frac{3(x + p)}{5}\right)$$

This implies

$$\pi = \frac{3p}{5}$$

$$\Rightarrow \quad p = \frac{5\pi}{3}$$

s, the smallest period of f(x) is $\frac{5\pi}{3}$.

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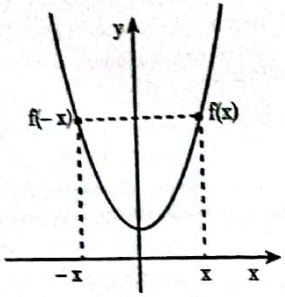
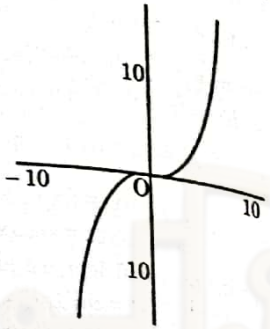
Note:

1. If f(x) and g(x) have same period p, then the function h(x) = a f(x) + b f(x), where a and b are constants also has period p.
2. If p_1 and p_2 are period function f(x) and g(x) respectively, then the period of f(x) + g(x) is a f(x) + b f(x) for a and b are constants) is the least common multiple of p_1 and p_2.

7.2 Even and Odd Function

If a function f(x) is symmetrical about the y-axis then it takes mirror shape in positive x value, which we know as an even function. On the other hand, if a function is symmetrical about the origin then f(x) is an odd function.





Definition

A function f(x) is said to be even function if f(– x) = f(x) for all x.

A function f(x) is said to be odd function if f(–x) = – f(x) for all x.

Example 1: Check $f(x) = x^2$ is even or odd function.

Solution

Let, $f(x) = x^2$. Replacing x by – x then

$$f(-x) = (-x)^2 = x^2 = f(x)$$

This means, $f(x) = x^2$ is an even function.

Example 2: Check $f(x) = x + x^2$ is even or odd function or neither.

Solution

Let, $f(x) = x + x^2$. Replacing x by – x then

$$f(-x) = -x + (-x)^2 = -x + x^2 \neq f(x) \text{ and } f(-x) \neq -f(x)$$

This means function f(x) is neither odd nor even.

Example 3: Check $f(x) = x^3$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ is even or odd function or neither.

Solution

Let, $f(x) = x^3$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

Replacing x by – x we get

$$f(-x) = (-x)^3 \text{ for } -\frac{\pi}{2} < -x < \frac{\pi}{2}$$

$$= -x^3 \text{ for } \frac{\pi}{2} > x > -\frac{\pi}{2}$$

$$= -f(x)$$

This means function f(x) is odd.

Example 4: Check even or odd or neither of the function, $f(x) = \begin{cases} -2x & \text{if } -\pi < x < 0 \\ 2x & \text{if } 0 < x < \pi \end{cases}$.

Solution

Let,

$$f(x) = \begin{cases} -2x & \text{if } -\pi < x < 0 \\ 2x & \text{if } 0 < x < \pi \end{cases}$$

Replacing x by – x then

$$f(-x) = \begin{cases} 2x & \text{if } -\pi < -x < 0 \\ -2x & \text{if } 0 < -x < \pi \end{cases}$$

$$= \begin{cases} 2x & \text{if } \pi > x > 0 \\ -2x & \text{if } 0 > x > -\pi \end{cases}$$

$$= \begin{cases} -2x & \text{if } -\pi < x < 0 \\ +2x & \text{if } 0 < x < \pi \end{cases}$$

$$= f(x)$$

This means f(x) is an even function.

Example 5: Check the function f(x) = x cosx + sinx is even or odd.

Solution

Let, f(x) = x cosx + sinx.

Replacing x by – x we get

$$f(-x) = (-x)\cos(-x) + \sin(-x) = -x\cos x - \sin x = -(x\cos x + \sin x) = -f(x)$$

This means, f(x) = (x cosx + sinx) is an odd function.

Example 6: Check the function f(x) = x|x| is even or odd or not.

Solution

Let, f(x) = x|x|

So, $f(-x) = (-x)|-x| = -x|x| = -f(-x)$.

Therefore, f(x) is an odd function.

Example 7: Check the function $f(x) = \frac{e^x + e^{-x}}{2}$ is even or odd or not.

Solution

Let, $f(x) = \frac{e^x + e^{-x}}{2} = \cosh x$

Here, $= \cosh(-x) = \cosh x = f(x)$

Therefore ion.

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Some Properties of Even and Odd Functions

(a) Product of two odd functions is even.

Proof:

Let f(x) and g(x) are two odd functions for all x.

Then

$$f(-x) = -f(x) \text{ and } g(-x) = -g(x).$$

Now,

$$f(-x) \cdot g(-x) = \{-f(x)\}\{-g(x)\} = f(x)\, g(x).$$

This shows the product of two odd functions is an even function.

(b) Product of odd and even functions is odd.

Proof:

Let f(x) is an odd and g(x) is an even function for all x. Then,

$$f(-x) = -f(x) \text{ and } g(-x) = g(x)$$

Now,

$$f(-x).\, g(-x) = \{-f(x)\} \cdot g(x) = -\{f(x).\, g(x)\}$$

This shows the product of odd and even functions is an odd function.

(c) Product of two even functions is even.

Proof:

Let f(x) and g(x) are even functions for all x. Then,

$$f(-x) = f(x) \text{ and } g(-x) = g(x)$$

Now,

$$f(-x) \cdot g(-x) = f(x).\, g(x)$$

This shows the product of two even functions is even.

(d) If f(x) is an even function then $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$.

Proof:

Let f(x) is an even function. Then f(−x) = f(x), for all x.

Now,

$$\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx$$

Take x = − t for first integral on the right,

$$= \int_{a}^{0} f(-t)\,(-dt) + \int_{0}^{a} f(x)\,dx$$

$$= \int_{0}^{a} f(-t)\,dt + \int_{0}^{a} f(x)\,dx \qquad \left[\because \int_{a}^{b} f(x)\,dx = -\int_{b}^{a} f(x)\,dx\right]$$

$$= \int_{0}^{a} f(-x)\,dx + \int_{0}^{a} f(x)\,dx \qquad \left[\because \int_{a}^{b} f(x)\,dx = \int_{a}^{b} f(t)\,dt\right]$$

$$= \int_{0}^{a} f(x)\,dx + \int_{0}^{a} f(x)\,dx \qquad [\because f(-x) = f(x)]$$

$$= 2\int_{0}^{a} f(x)\,dx.$$

Thus, $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$ whenever f(x) is an even.

(e) If f(x) is an odd function then $\int_{-a}^{a} f(x)\,dx = 0$.

Proof:

Let f(x) is an odd function. Then f(−x) = − f(x) for all x.

Then by (d),

$$\int_{-a}^{a} f(x)\,dx = \int_{0}^{a} f(-x)\,dx + \int_{0}^{a} f(x)\,dx = -\int_{0}^{a} f(x)\,dx + \int_{0}^{a} f(x)\,dx = 0$$

Thus, $\int_{-a}^{a} f(x)\,dx = 0$ whenever f(x) is an odd function.

7.3 TRIGONOMETRIC SERIES

From above, among the different trigonometric functions only sinx and cosx have similar nature that both have continuous on the real line R and both are of periodic with period 2π.

Definition

An infinite series of the form:

$$a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \ldots$$

$$a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where $a_0, a_1, a_2, \ldots b_1, b_2, \ldots$ are constants; is called a trigonometric series, where a_n and b_n are coefficients of the series.

After introducing the trigonometric series in mathematics, the uniqueness and the zeros of the series was the active research area in Europe. A German mathematician George Ferdinand Ludwig Philip Cantor first proved the trigonometric series is convergent to a function f(x) on the interval [0, 2π], which is identically zero and is non-zero at most finitely many points.

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7.4 FOURIER SERIES WITH PERIOD 2π

If the trigonometric series converges uniformly to f(x) on the interval $[0, 2\pi]$ and is identically zero, the series is known as Fourier series of f(x), which first introduced by Jean-Baptiste Joseph Fourier (1768 – 1830) in 1807 and 1822 in his book *Treatise on the propagation of heat in solid bodiesa* (in 1807) and *Analytical theory of heat* (in 1822).

Definition:

If f(x) be a periodic function of period 2π and is integrable over the period. Then the trigonometric series of the form

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \qquad \ldots (i)$$

is said to be Fourier series of f(x) if the series (i) converges to f(x) where a_0, a_n and b_n are said to be Euler's coefficients.

Note: The Euler's coefficients a_0, a_n and b_n are defined by Euler's in 1778 before the Fourier's work.

Determination of Euler's Coefficients (Euler's Formulae)

1. Determination of Constant a_0

We know Fourier series of f(x) with period $-\pi$ to π i.e. (2π) is:

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \qquad \ldots\ldots (i)$$

Integrating on both sides of (i) over $(-\pi, \pi)$, then we get

$$\int_{-\pi}^{\pi} f(x)\,dx = \int_{-\pi}^{\pi} \left[\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)\right] dx$$

$$= a_0 \int_{-\pi}^{\pi} dx + \sum_{n=1}^{\infty} \left[a_n \int_{-\pi}^{\pi} \cos x\,dx + b_n \int_{-\pi}^{\pi} \sin x\,dx\right]$$

$$= a_0 [x]_{-\pi}^{\pi} + \sum_{n=1}^{\infty} \left[a_n \left(\frac{\sin nx}{n}\right)_{-\pi}^{\pi} + b_n \left(\frac{-\cos nx}{n}\right)_{-\pi}^{\pi}\right]$$

$$= a_0 (\pi + \pi) + \sum_{n=1}^{\infty} \left[a_n (0) - b_n \left(\frac{\cos n\pi - \cos n\pi}{n}\right)\right]$$

$[\because \sin\pi = \sin(-\pi) = 0]$

$$= 2\pi a_0 + 0$$

$$= 2\pi a_0$$

$$\Rightarrow a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\,dx$$

2. Determination of Euler's Coefficients a_n (Euler's Formulae)

Multiply equation (i) by cosmx, where m is any fixed integer and then taking integration over $(-\pi, \pi)$, then we get

$$\int_{-\pi}^{\pi} f(x) \cos mx\,dx$$

$$= \int_{-\pi}^{\pi} \left[a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)\right] \cos mx\,dx$$

$$= a_0 \int_{-\pi}^{\pi} \cos mx\,dx + \sum_{n=1}^{\infty} \left[a_n \int_{-\pi}^{\pi} \cos nx \cos mx\,dx + b_n \int_{-\pi}^{\pi} \sin nx \cos mx\,dx\right] \ldots (ii)$$

Here, for m is an integer,

$$\int_{-\pi}^{\pi} \cos mx\,dx = \left[\frac{\sin mx}{m}\right]_{-\pi}^{\pi} = 0 \qquad [\text{Being } \sin n\pi = 0]$$

And,

$$\int_{-\pi}^{\pi} \cos nx \cos mx\,dx = 2\int_{0}^{\pi} \cos nx \cos mx\,dx \qquad [\text{Being } \cos nx \cos mx \text{ is even}]$$

$$= \int_{0}^{\pi} [\cos(n+m)x + \cos(n-m)x]\,dx$$

$$= \left[\frac{\sin(n+m)x}{n+m}\right]_0^{\pi} + \left[\frac{\sin(n-m)x}{n-m}\right]_0^{\pi}$$

$$= 0 + \left[\begin{cases} 0 & \text{for } n \neq m \\ \dfrac{x\cos(n-m)x}{1} & \text{for } n = m \end{cases}\right]_0^{\pi}$$

$$= \begin{cases} 0 & \text{for } n \neq m \\ \pi\cos(n-m)\pi & \text{for } n = m \end{cases}$$

$$= \begin{cases} 0 & \text{for } n \neq m \\ \pi & \text{for } n = m \end{cases}$$

Also,

$$\int_{-\pi}^{\pi} \sin nx \cos mx\,dx = 0 \qquad [\text{Being } \sin nx \cos mx \text{ is odd}]$$

Therefore, (ii) becomes

$$\int_{-\pi}^{\pi} f(x) \cos mx\,dx = 0 + \frac{a_n}{2}\pi + 0 \quad \text{for } n = m$$

$$\Rightarrow a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\,dx \qquad \text{for } n = 1, 2, 3, \ldots$$

3. Determination of Euler's Coefficients b_n (Euler's Formulae)

Multiplying equation (i) by sin mx and integrating over $(-\pi, \pi)$, then we get

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$$\int_{-\pi}^{\pi} f(x)\sin mx\,dx$$

$$= \int_{-\pi}^{\pi}\left[a_0 + \sum_{n=1}^{\infty}(a_n\cos nx + b_n\sin nx)\right]\sin mx\,dx$$

$$= a_0\int_{-\pi}^{\pi}\sin mx\,dx + \sum_{n=1}^{\infty}\left[a_n\int_{-\pi}^{\pi}\cos nx.\sin mx\,dx + b_n\int_{-\pi}^{\pi}\sin nx.\sin mx\,dx\right] \quad \text{...(i)}$$

Here, for m is integer,

$$\int_{-\pi}^{\pi}\sin mx\,dx = 0 \qquad \text{[Being sinmx is odd]}$$

And, $$\int_{-\pi}^{\pi}\cos nx\sin mx\,dx = 0 \quad \text{[Being cosnx sinmx is odd]}$$

Also,

$$\int_{-\pi}^{\pi}\sin nx\sin mx\,dx = 2\int_{0}^{\pi}\sin nx\sin mx\,dx \quad \text{[Being sinnx sinmx is even]}$$

$$= \int_{0}^{\pi}[\cos(n-m)x - \cos(n+m)x]\,dx$$

$$= \left[\frac{\sin(n-m)x}{n-m}\right]_0^{\pi} - \left[\frac{\sin(n+m)x}{n+m}\right]_0^{\pi}$$

$$= \left[\begin{cases} 0 & \text{for } n \neq m \\ \dfrac{x\cos(n-m)x}{1} & \text{for } n = m \end{cases}\right]_0^{\pi} - 0$$

$$= \begin{cases} 0 & \text{for } n \neq m \\ \pi\cos(n-m)\pi & \text{for } n = m \end{cases}$$

$$= \begin{cases} 0 & \text{for } n \neq m \\ \pi & \text{for } n = m \end{cases}$$

Therefore, (iii) becomes

$$\int_{-\pi}^{\pi} f(x)\sin mx\,dx = 0 + 0 + \frac{b_n}{2}\pi \qquad \text{for } n = m$$

$$\Rightarrow \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx \qquad \text{for } n = 1, 2, 3, \ldots$$

Thus the Fourier series of f(x) defined on $(-\pi, \pi)$ with period 2π is:

$$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n\cos nx + b_n\sin nx)$$

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where the Euler's formulae are

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx;\quad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx;\quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx.$$

Note: To integrate the product of two functions, we apply generalized rule of integration by parts, where dash denote differentiation and suffix denote integration

$$\int uv\,dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \ldots\ldots$$

where u and v are the functions of x.

Example 8: Find the Fourier series of $f(x) = x^2$ for $-\pi < x < \pi$.

Solution

Let,

$$f(x) = x^2 \qquad \text{for } -\pi < x < \pi$$

Clearly, the period of f(x) is

$$p = \pi - (-\pi) = 2\pi$$

Replacing x by $-x$ we get

$$f(-x) = (-x)^2 \qquad \text{for } -\pi < -x < x$$
$$= x^2 \qquad \text{for } \pi > x > -\pi$$
$$= f(x) \qquad \text{for } \pi > x > -\pi$$

This means f(x) is an even function. Therefore, $b_n = 0$ for Fourier series.

We know the Fourier series of period 2π is,

$$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n\cos nx + b_n\sin nx)$$

i.e. $$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n\cos nx) \qquad \text{..... (1)}$$

where

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{2}{2\pi}\int_{0}^{\pi} x^2\,dx = \frac{1}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi} = \frac{\pi^2}{3}$$

And,

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

$$= \frac{2}{\pi}\int_{0}^{\pi} x^2\cos nx\,dx \qquad \text{[Being } x^2\cos nx \text{ is even]}$$

$$= \frac{2}{\pi}\left[(x^2)\left(\frac{\sin nx}{n}\right) - (2x)\left(\frac{-\cos nx}{n^2}\right) + (2)\left(\frac{-\sin nx}{n^3}\right)\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{\pi^2\sin n\pi - 0}{n} + \frac{2\pi\cos n\pi - 0}{n^2} - \frac{2\sin n\pi - 0}{n^3}\right]$$

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$$= \frac{2}{\pi}\left[\frac{0-0}{n} + \frac{2\pi(-1)^n - 0}{n^2} - \frac{0-0}{n^3}\right] \quad [\because \sin n\pi = 0, \cos n\pi = (-1)^n]$$

$$= \frac{4}{n^2}(-1)^n$$

Now, substituting these values in (1) we get

$$f(x) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx$$

This is the Fourier expansion of f(x).

Example 9: Find the Fourier series of $f(x) = \begin{cases} 0 & -\pi < x < 0 \\ 1 & 0 \le x < \pi \end{cases}$

Solution

Let,

$$f(x) = \begin{cases} 0 & -\pi < x < 0 \\ 1 & 0 \le x < \pi \end{cases} \quad \ldots (1)$$

Clearly, the period of f(x) is

$$p = \pi - (-\pi) = 2\pi$$

We know the Fourier series of period 2π is,

$$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx) \quad \ldots\ldots (2)$$

where,

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{2\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi} 1\,dx\right] \left[\because \int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{-0}^{a} f(x)\,dx\right]$$

$$= \frac{1}{2\pi}\left[0 + [x]_0^{\pi}\right]$$

$$= \frac{1}{2\pi} \times \pi$$

$$= \frac{1}{2}$$

And,

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_{0}^{\pi} 1 \times \cos nx\,dx \quad [\text{Using (1)}]$$

$$= \frac{1}{\pi}\left[\frac{\sin nx}{n}\right]_0^{\pi}$$

$$= \frac{1}{n\pi}[\sin n\pi - \sin 0]$$

$$= 0 \quad [\because \sin n\pi = 0]$$

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Also,

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\int_{0}^{\pi} 1 \times \sin nx\,dx \quad [\text{Using (1)}]$$

$$= \frac{1}{\pi}\left[\frac{-\cos nx}{n}\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[\frac{-\cos n\pi + 1}{n}\right]$$

$$= \frac{1}{\pi}\left[\frac{1-(-1)^n}{n}\right] \quad [\because \cos n\pi = (-1)^n]$$

Now, substituting the values a_0, a_n and b_n in (2) we get,

$$f(x) = \frac{1}{2} + \frac{1}{\pi}\sum_{n=1}^{\infty}\left[\frac{1-(-1)^n}{n}\right]\sin nx$$

$$= \frac{1}{2} + \frac{1}{\pi}\left[2\sin x + \frac{2}{3}\sin 3x + \frac{2}{5}\sin 5x + \ldots\ldots\right]$$

$$= \frac{1}{2} + \frac{2}{\pi}\left[\sin x + \frac{\sin 3x}{3} + \frac{\sin 3x}{5} + \ldots..\right]$$

Example 10: Find the Fourier series of $f(x) = \frac{x^2}{2}$ for $-\pi < x < \pi$ and then using this show that $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \ldots\ldots = \frac{\pi^2}{6}$ and $1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \ldots\ldots = \frac{\pi^2}{12}$.

Solution

Given function is,

$$f(x) = \frac{x^2}{2} \quad \text{for } -\pi < x < \pi.$$

Clearly f(x) is 2π–periodic function.

Also, $f(-x) = \frac{(-x)^2}{2}$ for $-\pi < -x < \pi$

$$= \frac{x^2}{2} \quad \text{for } \pi > x > -\pi$$

$$= f(x)$$

This shows that f(x) is an even function. Therefore, $b_n = 0$ for Fourier series of f(x).

Now, the Fourier series of f(x) is,

$$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$$

i.e. $$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n \cos nx) \quad \ldots\ldots (1)$$

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where $a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$ and $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$

Here,

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi}\left(\frac{x^2}{2}\right)dx = \frac{2}{2\pi}\int_{0}^{\pi}\left(\frac{x^2}{2}\right)dx \quad \text{[Being f(x) is even]}$$

$$= \frac{1}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi} = \frac{\pi^2}{6}$$

And,

$$a_n = \frac{2}{\pi}\int_0^{\pi}\left(\frac{x^2}{2}\right)\cos nx\,dx \quad \text{[Being f(x) is even]}$$

$$= \frac{2}{\pi}\int_0^{\pi} x^2\cos nx\,dx$$

$$= \frac{1}{\pi}\left[(x^2)\frac{\sin nx}{n} - (2x)\frac{-\cos nx}{n^2} + (2)\frac{-\sin nx}{n^3}\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[0 + (2\pi)\frac{\cos n\pi}{n^2} + 0\right] \quad [\because \sin n\pi = 0]$$

$$= \frac{2\cos n\pi}{n^2}$$

Then (1) becomes,

$$f(x) = \frac{\pi^2}{6} + \sum_{n=1}^{\infty}\frac{2\cos n\pi}{n^2}\cos nx$$

i.e. $$f(x) = \frac{\pi^2}{6} + \sum_{n=1}^{\infty}\frac{2(-1)^n}{n^2}\cos nx \quad \text{.....(2)}$$

This is the Fourier series of f(x).

In particular if we take $x = 0$ then we get $f(0) = 0$. Then (2) gives,

$$f(0) = \frac{\pi^2}{6} + \sum_{n=1}^{\infty}\frac{2(-1)^n}{n^2} \quad [\because \cos 0 = 1]$$

$$\Rightarrow -\frac{\pi^2}{6} = 2\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2} = 2\left[\frac{-1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \frac{1}{4^2} - \frac{1}{5^2} + \ldots\right]$$

$$\Rightarrow \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots = \frac{\pi^2}{12}$$

And, if we take $x = \pi$ then $f(\pi) = \frac{\pi^2}{2}$. Then, (2) gives,

$$\frac{\pi^2}{2} = \frac{\pi^2}{6} + \sum_{n=1}^{\infty}\frac{2(-1)^n}{n^2}(-1)^n \quad [\because \cos n\pi = (-1)^n]$$

$$\Rightarrow \frac{2\pi^2}{6} = 2\sum_{n=1}^{\infty}\frac{1}{n^2} \quad [\because (-1)^{2n} = 1]$$

$$\Rightarrow \frac{\pi^2}{6} = \sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots\ldots$$

Example 11: Find Fourier series of $f(x) = |x|$ for $-\pi < x < \pi$ and then using the result show that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots = \frac{\pi^2}{8}$.

Solution:

Let $f(x) = |x|$ for $-\pi < x < \pi$

Clearly f(x) is 2π–periodic function.

Here,

$$f(-x) = |-x| \quad \text{for } -\pi < -x < \pi$$
$$= |x| \quad \text{for } \pi > x > -\pi$$
$$= f(x).$$

This shows that f(x) is an even function. Therefore, $b_n = 0$ for Fourier series of f(x).

Now, the Fourier series of f(x) is,

$$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n\cos nx + b_n\sin nx)$$

i.e. $$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n\cos nx) \quad \text{..... (1)}$$

where $a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$ and $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$.

Here,

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi}|x|\,dx = \frac{2}{2\pi}\int_0^{\pi} x\,dx \quad \text{[Being f(x) is even]}$$

$$= \frac{1}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi} = \frac{\pi}{2}$$

And,

$$a_n = \frac{2}{\pi}\int_0^{\pi} x\cos nx\,dx \quad \text{[Being f(x) is even]}$$

$$= \frac{2}{\pi}\left[(x)\frac{\sin nx}{n} - (1)\frac{-\cos nx}{n^2}\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[0 + \frac{\cos n\pi - 1}{n^2} + 0\right] \quad [\because \sin n\pi = 0]$$

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$$= \frac{2[\cos n\pi - 1]}{n\pi^2}$$

Then (1) becomes,

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2[\cos n\pi - 1]}{n\pi^2} \cos nx$$

i.e. $$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2[(-1)^n - 1]}{n\pi^2} \cos nx \quad \ldots\ldots(2)$$

This is the Fourier series of f(x).

In particular if we take $x = 0$ then we get $f(0) = 0$. And,

$$(-1)^n - 1 = \begin{cases} -2 & \text{if n is odd} \\ 0 & \text{if n is even} \end{cases}$$

Therefore, at $x = 0$ the result (2) gives

$$0 = \frac{\pi}{2} + \sum_{n=\text{odd}} \frac{2}{\pi n^2}(-2)\cos 0$$

i.e. $$\frac{\pi}{2} = \frac{4}{\pi} \sum_{n=\text{odd}} \frac{1}{n^2}$$

$$\Rightarrow \quad \frac{\pi^2}{8} = \sum_{n=\text{odd}} \frac{1}{n^2} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots$$

7.5 Fourier Series of f(x) in the Interval $(\alpha, \alpha + 2\pi)$

The Fourier series of a function f(x) defined on the interval $(\alpha, \alpha + 2\pi)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \ldots (1)$$

in which Fourier coefficients are a_0, a_n and b_n given by for $n = 1, 2, 3 \ldots$.

$$a_0 = \frac{1}{2\pi}\int_{\alpha}^{\alpha+2\pi} f(x)\,dx; \quad a_n = \frac{1}{\pi}\int_{\alpha}^{\alpha+2\pi} f(x)\cos nx\,dx; \quad b_n = \frac{1}{\pi}\int_{\alpha}^{\alpha+2\pi} f(x)\sin nx\,dx$$

Note: If we put $\alpha = 0$ then the formula (1) reduces to

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \ldots (2)$$

which is a Fourier series of a function f(x) defined on the interval $(0, 2\pi)$ where the Fourier coefficients are a_0, a_n and b_n given by for $n = 1, 2, 3 \ldots$.

$$a_0 = \frac{1}{2\pi}\int_{0}^{2\pi} f(x)\,dx; \quad a_n = \frac{1}{\pi}\int_{0}^{2\pi} f(x)\cos nx\,dx; \quad b_n = \frac{1}{\pi}\int_{0}^{2\pi} f(x)\sin nx\,dx$$

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Example 12: Find the Fourier series of $f(x) = x$ for $0 < x < 2\pi$.

Solution

Let, $f(x) = x$ for $0 < x < 2\pi$

The period of the function f(x) is

$$p = 2\pi - 0 = 2\pi$$

We know the Fourier series of f(x) of period 2π is

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \ldots\ldots (1)$$

where

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(x)\,dx = \frac{1}{2\pi}\int_0^{2\pi} x\,dx = \frac{1}{2\pi}\left[\frac{x^2}{2}\right]_0^{2\pi} = \frac{1}{2\pi}\left[\frac{4\pi^2 - 0}{2}\right] = \pi$$

And,

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} x\cos nx\,dx$$

$$= \frac{1}{\pi}\left[(x)\left(\frac{\sin nx}{n}\right) - (1)\left(\frac{-\cos nx}{n^2}\right)\right]_0^{2\pi}$$ [Applying integration by parts]

$$= \frac{1}{\pi}\left[0 - \frac{1-1}{n^2}\right]$$ [$\because \sin 2n\pi = 0, \cos 2n\pi = 1$]

$$= 0$$

Also,

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} x\sin nx\,dx$$

$$= \frac{1}{\pi}\left[(x)\left(\frac{-\cos nx}{n}\right) - (1)\left(\frac{-\sin nx}{n^2}\right)\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[\frac{-2\pi}{n} + \frac{0-0}{n^2}\right]$$ [$\because \cos 2n\pi = 1, \sin 2n\pi = 0$]

$$= \frac{-2}{n}$$

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Now, substituting these values in (1), we get

$$f(x) = \pi + \sum_{n=1}^{\infty}\left[0 + \left(\frac{-2}{n}\right)\sin nx\right]$$

i.e. $$f(x) = \pi - 2\sum_{n=1}^{\infty}\frac{\sin nx}{n}$$

This is the Fourier series of f(x).

Example 13: Find the Fourier series of $f(x) = \begin{cases} -1 & \text{if } 0 < x < \frac{\pi}{2} \\ 0 & \text{if } \frac{\pi}{2} < x < 2\pi \end{cases}$

Solution

Let,

$$f(x) = \begin{cases} -1 & \text{if } 0 < x < \frac{\pi}{2} \\ 0 & \text{if } \frac{\pi}{2} < x < 2\pi \end{cases} \quad \ldots (1)$$

The period of f(x) is

$$p = 2\pi - 0 = 2\pi$$

We know the Fourier series of f(x) of period 2π is

$$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx) \quad \ldots (2)$$

where

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(x)\,dx = \frac{1}{2\pi}\int_0^{\pi/2}(-1)\,dx + 0 \qquad \text{[Using (1)]}$$

$$= \frac{1}{2\pi}[-x]_0^{\pi/2}$$

$$= \frac{1}{2\pi}\left[\frac{-\pi}{2} + 0\right]$$

$$= \frac{-1}{4}$$

And,

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_0^{\pi/2}(-1)\cos nx\,dx + 0 \qquad \text{[Using (1)]}$$

$$= \frac{-1}{\pi}\left[\frac{\sin nx}{n}\right]_0^{\pi/2}$$

$$= \frac{-1}{n\pi}\left[\sin\frac{n\pi}{2}\right] \qquad [\because \sin 0 = 0]$$

Also,

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\int_0^{\pi/2}(-1)\sin nx\,dx + 0 \qquad \text{[Using (1)]}$$

$$= \frac{-1}{\pi}\left[\frac{-\cos nx}{n}\right]_0^{\pi/2}$$

$$= \frac{1}{n\pi}\left(\cos\frac{n\pi}{2} - 1\right) \qquad [\because \cos 0 = 1]$$

Now, substituting these values in (2) we get

$$f(x) = \frac{-1}{4} + \sum_{n=1}^{\infty}\frac{1}{n\pi}\left[-\sin\frac{n\pi}{2}\cos nx + \left(\cos\frac{n\pi}{2} - 1\right)\sin nx\right]$$

This is the Fourier series of f(x).

Example 14: Find its Fourier series of $f(x) = \begin{cases} x & \text{if } -\pi/2 < x < \pi/2 \\ \pi - x & \text{if } \pi/2 < x < 3\pi/2 \end{cases}$

Solution

Given function is

$$f(x) = \begin{cases} x & \text{for } -\pi/2 < x < \pi/2 \\ \pi - x & \text{for } \pi/2 < x < 3\pi/2 \end{cases}$$

Clearly f(x) is a periodic function with period 2π.

Here,

$$f(-x) = \begin{cases} -x & \text{for } -\pi/2 < -x < \pi/2 \\ \pi + x & \text{for } \pi/2 < -x < 3\pi/2 \end{cases}$$

$$= \begin{cases} -x & \text{for } \pi/2 > x > -\pi/2 \\ \pi + x & \text{for } -\pi/2 > x > -3\pi/2 \end{cases}$$

$$\neq f(x)$$

This shows that f(x) is neither even nor odd.

Now, Fourier series of f(x) is

$$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx) \quad \ldots (1)$$

where

$$a_0 = \frac{1}{2\pi}\left\{\int_{-\pi/2}^{\pi/2} x\,dx + \int_{\pi/2}^{3\pi/2}(\pi - x)\,dx\right\}$$

$$= \frac{1}{2\pi}\left\{\left[\frac{x^2}{2}\right]_{-\pi/2}^{\pi/2} + \left[\pi x - \frac{x^2}{2}\right]_{\pi/2}^{3\pi/2}\right\}$$

$$= \frac{1}{2\pi}\left[0 + \frac{3\pi^2}{2} - \frac{\pi^2}{2} - \frac{9\pi^2}{8} + \frac{\pi^2}{8}\right]$$

$$= \frac{\pi}{16}[12 - 4 - 9 + 1]$$

$$= 0$$

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and,

$$a_n = \frac{1}{\pi}\left\{\int_{-\pi/2}^{\pi/2} x\cos nx\,dx + \int_{\pi/2}^{3\pi/2} (\pi - x)\cos nx\,dx\right\}$$

Since x cosnx is odd therefore,

$$\int_{-\pi/2}^{\pi/2} x\cos nx\,dx = 0$$

Then

$$a_n = \frac{1}{\pi}\left\{\int_{\pi/2}^{3\pi/2} (\pi - x)\cos nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{\left[(\pi - x)\frac{\sin nx}{n} - (-1)\frac{-\cos nx}{n^2}\right]_{\pi/2}^{3\pi/2}\right\}$$

$$= \frac{1}{\pi}\left[\frac{-\pi}{2n}\sin\left(\frac{3n\pi}{2}\right) - \frac{\pi}{2n}\sin\left(\frac{n\pi}{2}\right) - \frac{1}{n^2}\cos\left(\frac{3n\pi}{2}\right) + \frac{1}{n^2}\cos\left(\frac{n\pi}{2}\right)\right]$$

Since we know

$$\sin\left(\frac{3n\pi}{2}\right) = \sin\left(2\pi - \frac{\pi}{2}\right)n = -\sin\left(\frac{\pi n}{2}\right)$$

and $$\cos\left(\frac{3n\pi}{2}\right) = \cos\left(2\pi - \frac{\pi}{2}\right)n = \cos\left(\frac{\pi n}{2}\right)$$

Then

$$a_n = \frac{1}{\pi}\left[\frac{\pi}{2n}\sin\left(\frac{\pi n}{2}\right) - \frac{\pi}{2n}\sin\left(\frac{n\pi}{2}\right) - \frac{1}{n^2}\cos\left(\frac{n\pi}{2}\right) + \frac{1}{n^2}\cos\left(\frac{n\pi}{2}\right)\right] = 0$$

Since x sinnx is even therefore,

$$\int_{-\pi/2}^{\pi/2} x\cos nx\,dx = 2\int_{0}^{\pi/2} x\cos nx\,dx$$

Also,

$$b_n = \frac{1}{\pi}\left\{\int_{-\pi/2}^{\pi/2} x\sin nx\,dx + \int_{\pi/2}^{3\pi/2} (\pi - x)\sin nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{2\int_{0}^{\pi/2} x\sin nx\,dx + \int_{\pi/2}^{3\pi/2} (\pi - x)\sin nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{2\left[x\left(\frac{-\cos nx}{n}\right) - (1)\frac{-\sin nx}{n^2}\right]_0^{\pi/2} + \left[(\pi - x)\left(\frac{-\cos nx}{n}\right) - (-1)\frac{-\sin nx}{n^2}\right]_{\pi/2}^{3\pi/2}\right\}$$

$$= \frac{1}{\pi}\left[-\frac{2\pi}{2n}\cos\left(\frac{nx}{2}\right) + \frac{2}{n^2}\sin\left(\frac{n\pi}{2}\right) + \frac{\pi}{2n}\cos\left(\frac{3n\pi}{2}\right) + \frac{\pi}{2n}\cos\left(\frac{n\pi}{2}\right) - \frac{1}{n^2}\sin\left(\frac{3n\pi}{2}\right) + \frac{1}{n^2}\sin\left(\frac{n\pi}{2}\right)\right]$$

$$= \frac{1}{\pi}\left[-\frac{\pi}{2n}\cos\left(\frac{nx}{2}\right) + \frac{2}{n^2}\sin\left(\frac{n\pi}{2}\right) + \frac{\pi}{2n}\cos\left(\frac{3n\pi}{2}\right) - \frac{1}{n^2}\sin\left(\frac{3n\pi}{2}\right)\right]$$

Since we know

$$\sin\left(\frac{3n\pi}{2}\right) = \sin\left(2\pi - \frac{\pi}{2}\right)n = -\sin\left(\frac{\pi n}{2}\right)$$

and $$\cos\left(\frac{3n\pi}{2}\right) = \cos\left(2\pi - \frac{\pi}{2}\right)n = \cos\left(\frac{\pi n}{2}\right)$$

Then

$$b_n = \frac{1}{\pi}\left[-\frac{\pi}{2n}\cos\left(\frac{nx}{2}\right) + \frac{2}{n^2}\sin\left(\frac{n\pi}{2}\right) + \frac{\pi}{2n}\cos\left(\frac{n\pi}{2}\right) + \frac{1}{n^2}\sin\left(\frac{n\pi}{2}\right)\right]$$

$$= \frac{4}{\pi n^2}\sin\left(\frac{n\pi}{2}\right)$$

Therefore (1) becomes,

$$f(x) = \sum_{n=1}^{\infty} \frac{4}{\pi n^2}\sin\left(\frac{n\pi}{2}\right)\sin nx \qquad \ldots\ldots (2)$$

Since $\sin n\pi = 0$. So, $\sin\left(\frac{n\pi}{2}\right) = 0$ for n is even.

Then (2) becomes

$$f(x) = \sum_{n=\text{odd}} \frac{4}{\pi n^2}\sin\left(\frac{n\pi}{2}\right)\sin nx \qquad \ldots\ldots (3)$$

$$\Rightarrow \quad f(x) = \frac{4}{\pi}\left[\sin x - \frac{\sin 3x}{32} + \frac{\sin 5x}{52} - \frac{\sin 7x}{72} + \ldots\ldots\right]$$

This is the Fourier series for f(x).

FOURIER SERIES IN ANY INTERVAL HAVING PERIOD 2ℓ

Let f(x) be a function with interval $(-\ell, \ell)$ with period $p = 2\ell$, let us suppose that its fourier series is

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi}{l} + b_n\sin\frac{n\pi}{l}\right) \qquad \ldots. (i)$$

Where,

$$a_0 = \frac{1}{2l}\int_{-l}^{l} f(x)\,dx \qquad \ldots.. (ii)$$

$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\left(\frac{n\pi x}{l}\right)dx \qquad \ldots.. (iii)$$

$$b_n = \frac{1}{l}\int_{-l}^{l} f(x)\sin\left(\frac{n\pi x}{l}\right)dx \qquad \ldots. (iv)$$

for all n = 1, 2, 3 .....

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Example 15: Find the Fourier series of $f(x) = \begin{cases} 1 & \text{for } -1 < x < 0 \\ -1 & \text{for } 0 < x < 1 \end{cases}$

Solution

Given periodic function

$$f(x) = \begin{cases} 1 & \text{for } -1 < x < 0 \\ -1 & \text{for } 0 < x < 1 \end{cases}$$

with period $2\ell = 1 - (-1) = 2 \Rightarrow \ell = 1$.

Then the Fourier series of $f(x)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{l} x + b_n \sin \frac{n\pi}{l} x \right) \text{ with } \ell = 1$$

$$\Rightarrow \quad f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\pi x + b_n \sin n\pi x) \qquad \ldots\ldots (1)$$

where

$$a_0 = \frac{1}{2l} \int_{-1}^{1} f(x)\, dx = \frac{1}{2} \left[\int_{-1}^{0} 1\, dx + \int_{0}^{1} (-1)\, dx \right] = \frac{1}{2} \left[[x]_{-1}^{0} - [x]_{0}^{1} \right] = 0$$

and

$$a_n = \frac{1}{l} \int_{-1}^{1} f(x) \cos \frac{n\pi x}{l}\, dx = \left[\int_{-1}^{0} 1 \cdot \cos n\pi x\, dx + \int_{0}^{1} (-1) \cos n\pi x\, dx \right]$$

$$\left[\frac{\sin n\pi x}{n\pi} \right]_{-1}^{0} - \left[\frac{\sin n\pi x}{n\pi} \right]_{0}^{1}$$

$$= 0 \qquad [\because \sin(-n\pi) = -\sin n\pi]$$

$$\Rightarrow \quad a_n = 0$$

also,

$$b_n = \frac{1}{l} \int_{-1}^{1} f(x) \sin \frac{n\pi}{l} x\, dx = \int_{-1}^{0} 1 \sin n\pi x\, dx + \int_{0}^{1} (-1) \sin n\pi x\, dx$$

$$= \left[\frac{-\cos n\pi x}{n\pi} \right]_{-1}^{0} - \left[\frac{-\cos n\pi x}{n\pi} \right]_{0}^{1}$$

$$= \frac{1}{n\pi} [-1 + \cos n\pi + \cos n\pi - 1]$$

$$= \frac{2}{n\pi} (\cos n\pi - 1)$$

Substituting these values in (1) we get;

$$f(x) = \sum_{n=1}^{\infty} \frac{2(\cos n\pi - 1)}{n\pi} \sin n\pi x = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(\cos n\pi - 1)}{n} \sin n\pi x$$

which is the Fourier series of $f(x)$.

Example 16: Obtain the Fourier series of $f(x)$ where $f(x) = \begin{cases} 1 + 2x/\pi & \text{for } -1 \le x \le 0 \\ 1 - 2x/\pi & \text{for } 0 \le x \le 1 \end{cases}$ and show that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \ldots = \frac{\pi^2}{8}$.

Solution

Let, $f(x) = \begin{cases} 1 + 2x/\pi & \text{for } -1 \le x \le 0 \\ 1 - 2x/\pi & \text{for } 0 \le x \le 1 \end{cases}$

Clearly $f(x)$ is 2–periodic function.

Here,

$$f(-x) = \begin{cases} 1 - 2x/\pi & \text{for } -1 \le -x \le 0 \\ 1 + 2x/\pi & \text{for } 0 \le -x \le 1 \end{cases}$$

$$= \begin{cases} 1 - 2x/\pi & \text{for } 1 \ge x \ge 0 \\ 1 + 2x/\pi & \text{for } 0 \ge x \ge -1 \end{cases}$$

$$= f(x)$$

This shows that $f(x)$ is an even function. So, the Fourier series of $f(x)$ is reduces to

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\pi x) \qquad \ldots\ldots (1)$$

where $a_0 = \frac{1}{2} \int_{0}^{1} f(x)\, dx$ and $a_n = \int_{0}^{1} f(x) \cos(n\pi x)\, dx$.

Here;

$$a_0 = \int_{0}^{1} \left(1 + \frac{2x}{\pi} \right) dx = \left[x + \frac{x^2}{\pi} \right]_{0}^{1} = \left(1 + \frac{1}{\pi} \right).$$

and,

$$a_n = 2 \int_{0}^{1} \left(1 + \frac{2x}{\pi} \right) \cos(n\pi x)\, dx$$

$$= 2 \left[\left(1 + \frac{2x}{\pi} \right) \left(\frac{\sin n\pi x}{n\pi} \right) - \frac{2}{\pi} \left(-\frac{\cos n\pi x}{n^2\pi^2} \right) \right]_{0}^{1}$$

$$= \frac{4}{n^2\pi^3} (\cos n\pi - 1), \text{ for } n = 1, 2, 3, \ldots \quad [\because \sin n\pi = 0]$$

$$= \frac{4}{n^2\pi^3} [(-1)^n - 1]$$

$$= \begin{cases} \frac{-8}{n^2\pi^3} & \text{for } n \text{ is odd} \\ 0 & \text{for } n \text{ is even} \end{cases}$$

Therefore, (1) becomes,

$$f(x) = \left(1 + \frac{1}{\pi} \right) - \frac{8}{\pi^3} \sum_{n\text{-odd}} \left(\frac{\cos n\pi x}{n^2} \right) \qquad \ldots\ldots (2)$$

This is required Fourier series of $f(x)$ on $[-1, 1]$.

Set $x = 0$ then $f(0) = 1$. Then (2) gives,

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$$1 = 1 + \frac{1}{\pi} - \frac{8}{\pi^3}\sum_{n-odd}\left(\frac{1}{n^2}\right)$$

$$\Rightarrow \sum_{n-odd}\left(\frac{1}{n^2}\right) = \frac{\pi^3}{8\pi} = \frac{\pi^2}{8}$$

$$\Rightarrow 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \ldots\ldots = \frac{\pi^2}{8}$$

Exercise 7 [A]

1. Find the smallest period (fundamental period) p of the following functions.
   a. $\sin x$ b. $\cos x$ c. $\sin 2x$ d. $\tan 5x$ e. $\sin 2\pi x$ f. $\cos 2\pi x$
2. Find out odd, even or neither function from the following:
   a. e^x b. x^4 c. x^7 d. $|x|$ e. $x \sin x$
   f. $\frac{\pi x^3}{2}$ for $-1 < x < 1$ g. $x \cos x$ h. $\sin x + \cos x$ i. $|\sin x|$
3. Find the Fourier series of f(x) on the given interval:

   a. $f(x) = \begin{cases} 1 & \text{if } -\pi < x < 0 \\ -1 & \text{if } 0 < x < \pi \end{cases}$

   b. $f(x) = \begin{cases} 1 & \text{if } \frac{-\pi}{2} < x < \frac{\pi}{2} \\ -1 & \text{if } \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$

   c. $f(x) = \begin{cases} 0 & \text{if } 0 < x < \pi \\ 1 & \text{if } \pi < x < 2\pi \end{cases}$

   d. $f(x) = \begin{cases} 1 & \text{for } 0 \le x < \pi \\ 0 & \text{for } \pi \le x < 2\pi \end{cases}$

   e. $f(x) = x$ (for $-\pi < x < \pi$)

   f. $f(x) = \begin{cases} x & \text{if } -\pi/2 < x < \pi/2 \\ 0 & \text{if } \pi/2 < x < 3\pi/2 \end{cases}$

   g. $f(x) = \begin{cases} x & \text{if } -\pi/2 < x < \pi/2 \\ \pi - x & \text{if } \pi/2 < x < 3\pi/2 \end{cases}$

   h. $f(x) = \begin{cases} x & \text{for } 0 \le x < \pi \\ 2\pi - x & \text{for } \pi \le x < 2\pi \end{cases}$

   i. $f(x) = \begin{cases} x & \text{if } -\pi/2 < x < \pi/2 \\ \pi - x & \text{if } \pi/2 < x < 3\pi/2 \end{cases}$

4. a. Find its Fourier series of $f(x) = \begin{cases} k & \text{if } -\pi/2 < x < \pi/2 \\ 0 & \text{if } \pi/2 < x < 3\pi/2 \end{cases}$ and show that:

   $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots\ldots = \frac{\pi}{4}$.

   b. Find the fourier series of $f(x) = \begin{cases} -k & \text{for } -\pi \le x \le 0 \\ k & \text{for } 0 < x \le \pi \end{cases}$ and show that:

   $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots\ldots$

   c. Find Fourier series of $f(x) = \begin{cases} 1 & \text{if } 0 \le x \le \pi \\ 2 & \text{if } \pi < x \le 2\pi \end{cases}$ and show that:

   $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots\ldots$

d. Find Fourier series of $f(x) = x - x^2$ if $-\pi < x < \pi$ and show that: $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots = \frac{\pi^2}{12}$.

e. Find Fourier series of $f(x) = |x|$ if $-\pi < x < \pi$ and show that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$.

f. Find the Fourier expansion of the function $f(x) = x + |x|$ $(-\pi < x < \pi)$.

g. Find Fourier series of $f(x) = e^{-x}$ if $0 < x < 2\pi$.

h. Expand the function $f(x) = x \sin x$ if $-\pi \le x \le \pi$ by using Fourier series.

Answer Key

1. a. 2π b. 2π c. π d. $\frac{\pi}{5}$ e. 1 f. 1
2. a. neither b. even c. odd d. even e. even f. odd g. odd
   h. neither i. even
3. a. $\frac{-4}{\pi}\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \ldots\ldots\right)$
   b. $\frac{4}{\pi}\left(\cos x - \frac{\cos 3x}{3} + \frac{\cos 5x}{5} - \ldots\ldots\right)$
   c. $\frac{1}{2} - \frac{2}{\pi}\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \ldots\ldots\right)$
   d. $\frac{1}{2} + \frac{2}{\pi}\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \ldots\ldots\right)$
   e. $2\left(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \frac{\sin 4x}{4} + \ldots\right)$
   f. $\frac{2}{\pi}\left[\sin x - \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} - \ldots\ldots\right]$
   g. $\frac{4}{\pi}\left[\sin x - \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} - \frac{\sin 7x}{7^2} + \ldots\ldots\right]$
   h. $\frac{\pi}{2} - \frac{4}{\pi}\sum_{n=0}^{\infty}\frac{\cos(2n+1)x}{(2n+1)^2}$
   i. $\frac{4}{\pi}\left[\sin x - \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} - \frac{\sin 7x}{7^2} + \ldots\ldots\right]$
4. a. $\frac{k}{2} + \frac{2k}{\pi}\cos x - \frac{2k}{3\pi}\cos 3 + \frac{2k}{5\pi}\cos 5x$
   b. $\frac{4k}{\pi}\left[\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \ldots\ldots\right]$
   c. $\frac{3}{2} + \sum_{n=1}^{\infty}\frac{1}{n\pi}(\cos n\pi - 1)\sin x$
   d. $\frac{-\pi^2}{3} - 2\sum_{n=1}^{\infty}\left(\frac{2(-1)^n}{n^2}\cos nx + \frac{(-1)^n}{n}\sin nx\right)$
   e. $\frac{\pi}{2} - \left[\frac{4}{\pi}\sum_{n-odd}\frac{1}{n^2}\cos(nx) + 2\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\sin nx\right]$
   f. $\frac{\pi}{2} - \frac{4}{\pi}\left[\cos x + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots.\right] + \left(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \frac{\sin 4x}{4} + \ldots.\right)$
   h. $1 - \frac{\cos x}{2} - 2\sum_{n=2}^{\infty}\frac{(-1)^{n+1}}{n^2-1}\cos nx$

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7.7 Fourier Cosine and Sine Series (The Fourier Series of Even and Odd Function)

Fourier Sine and Cosine Series in any interval having Period 2π

We know Fourier series of f(x) with period 2π is

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \ldots\ldots (1)$$

where,

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx,\ a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx \text{ and } b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

Case I:

If f(x) is an even function of period 2π, then $b_n = 0$ being f(x) sinnx is odd.

$$a_0 = \frac{2}{2\pi}\int_0^{\pi} f(x)\,dx = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx;\ a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx$$

Then (1) reduces to

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

with $a_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx;\ a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx$

which we called Fourier cosine series of f(x) of period 2π.

Case II:

If f(x) is an odd function of period 2π then $a_0 = 0, a_n = 0$ being f(x) cosnx is odd

And

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx$$

Then, (1) reduces to

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

with $b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx$

which we called Fourier sine series of f(x) of period 2π.

Fourier Sine and Cosine Series in any interval having Period 2ℓ

Let f(x) be a function with interval $(-\ell, \ell)$ with period $p = 2\ell$, let us suppose that its fourier series is

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi}{l} + b_n \sin\frac{n\pi}{l}\right) \quad \ldots (1)$$

where, for all n = 1, 2, 3, ...

$$a_0 = \frac{1}{2l}\int_{-l}^{l} f(x)\,dx;\ a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\left(\frac{n\pi x}{l}\right)dx;\ b_n = \frac{1}{l}\int_{-l}^{l} f(x)\sin\left(\frac{n\pi x}{l}\right)dx$$

Case I: When f(x) is odd function, then $f(x)\cos\left(\frac{n\pi x}{l}\right)$ is also odd. Then,

$$a_0 = \frac{1}{2l}\int_{-l}^{l} f(x)\,dx = 0;\ a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\left(\frac{n\pi x}{l}\right)dx = 0$$

And, $f(x)\sin\left(\frac{n\pi x}{l}\right)$ is even, therefore

$$b_n = \frac{1}{l}\int_{-l}^{l} f(x)\sin\left(\frac{n\pi x}{l}\right)dx = \frac{2}{l}\int_0^{l} f(x)\sin\left(\frac{n\pi x}{l}\right)dx$$

Then,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi}{l}$$

where $b_n = \frac{2}{l}\int_0^{l} f(x)\sin\left(\frac{n\pi x}{l}\right)dx$, which we called Fourier sine series of period 2ℓ.

Case II: When f(x) is even function, then $f(x)\sin\left(\frac{n\pi x}{l}\right)$ is odd and $f(x)\cos\left(\frac{n\pi x}{l}\right)$ is even.

Then,

$$a_0 = \frac{1}{2l}\int_{-l}^{l} f(x)\,dx = \frac{1}{l}\int_0^{l} f(x)\,dx;$$

$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\left(\frac{n\pi x}{l}\right)dx = \frac{2}{l}\int_0^{l} f(x)\cos\left(\frac{n\pi x}{l}\right)dx$$

$$b_n = \frac{1}{l}\int_{-l}^{l} f(x)\sin\left(\frac{n\pi x}{l}\right)dx = 0$$

Then,

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{l}\right)$$

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where $a_0 = \frac{1}{l}\int_0^l f(x)\,dx$, $a_n = \frac{2}{l}\int_0^l f(x)\cos\left(\frac{n\pi x}{l}\right)dx$, which we called Fourier cosine series of period 2ℓ.

Workout Examples

Example 1: Find the Fourier cosine series of $f(x) = x^2$ for $0 < x < \pi$.

Solution

Let,

$$f(x) = x^2 \text{ for } 0 < x < \pi$$

Clearly, the period of $f(x)$ is $p = 2\pi$.

Now, the Fourier cosine series of $f(x)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

with ..... (1)

$$a_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx = \frac{1}{\pi}\int_0^{\pi} x^2\,dx = \frac{1}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi} = \frac{\pi^2}{3}$$

and,

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos x\,dx$$

$$= \frac{2}{\pi}\int_0^{\pi} x^2 \cos nx\,dx$$

$$= \frac{2}{\pi}\left[(x^2)\left(\frac{\sin nx}{n}\right) - (2x)\left(\frac{-\cos nx}{n^2}\right) + (2)\left(\frac{-\sin nx}{n^3}\right)\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{0-0}{n} + \frac{2\pi\cos n\pi - 0}{n^2} - \frac{0-0}{n^3}\right] \quad [\because \sin n\pi = 0]$$

$$= \frac{4}{n^2}(-1)^n \quad [\because \cos n\pi = (-1)^n]$$

Now, substituting these values in (1) we get

$$f(x) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx$$

This is the Fourier cosine series of $f(x)$.

Example 2: Check whether the function: $f(x) = \begin{cases} -2x & \text{if } -\pi < x < 0 \\ 2x & \text{if } 0 < x < \pi \end{cases}$ is odd or even and hence obtain corresponding Fourier series.

Solution

Let, $f(x) = \begin{cases} -2x & \text{if } -\pi < x < 0 \\ 2x & \text{if } 0 < x < \pi \end{cases}$

Here, $f(-x) = \begin{cases} -2(-x) & \text{if } -\pi < -x < 0 \\ 2(-x) & \text{if } 0 < -x < \pi \end{cases} = \begin{cases} 2x & \text{if } \pi > x > 0 \\ -2x & \text{if } 0 > x > -\pi \end{cases} = f(x)$

This means $f(x)$ is an even function.

Being even function, the Fourier series of $f(x)$ is same as the Fourier cosine series of $f(x)$.

Clearly, the period of $f(x)$ is 2π.

Now, the Fourier cosine series of $f(x)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx \qquad \text{..... (1)}$$

with

$$a_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx = \frac{1}{\pi}\int_0^{\pi} 2x\,dx = \frac{1}{\pi}\left[2\cdot\frac{x^2}{2}\right]_0^{\pi} = \pi$$

and,

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx$$

$$= \frac{2}{\pi}\int_0^{\pi} 2x\cos nx\,dx$$

$$= \frac{4}{\pi}\left[(x)\left(\frac{\sin nx}{n}\right) - (1)\left(\frac{-\cos nx}{n^2}\right)\right]_0^{\pi}$$

$$= \frac{4}{\pi}\left[\frac{0-0}{n} + \frac{\cos n\pi - 1}{n^2}\right] \quad [\because \sin n\pi = 0, \cos 0 = 1]$$

$$= \frac{4}{\pi n^2}[(-1)^n - 1] \quad [\because \cos n\pi = (-1)^n]$$

Now, substituting these values in (1), we get

$$f(x) = \pi + \sum_{n=1}^{\infty}\frac{4}{\pi n^2}[(-1)^n - 1]\cos nx.$$

This is the Fourier cosine i.e. Fourier series of the given function $f(x)$.

Example 3: Find Fourier cosine series of $f(x) = x\sin x$ in $(0, \pi)$ and show that $\frac{1}{1\cdot3} - \frac{1}{3\cdot5} + \frac{1}{5\cdot7} - \frac{1}{7\cdot9} + \ldots = \frac{\pi - 2}{4}$.

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Solution

Given that,

$$f(x) = x \sin x \quad \text{for } 0 < x < \pi.$$

Clearly f(x) is of 2π–periodic function. The Fourier cosine series of f(x) is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) \quad \ldots\ldots (1)$$

with

$$a_0 = \frac{1}{\pi}\int_0^{\pi} x \sin x \, dx = \frac{1}{\pi}[x(-\cos x) - (1)(-\sin x)]_0^{\pi} = -\frac{\pi\cos\pi}{\pi} = 1.$$

and,

$$a_n = \frac{2}{\pi}\int_0^{\pi} x \sin x \cos nx \, dx$$

$$= \frac{1}{\pi}\int_0^{\pi} x\,(2\sin x \cos nx)\, dx$$

$$= \frac{1}{\pi}\int_0^{\pi} x\,[\sin(nx + x) - \sin(nx - x)]\, dx$$

$$= \frac{1}{\pi}\int_0^{\pi} x\,[\sin(n+1)x - \sin(n-1)x]\, dx$$

$$= \frac{1}{\pi}\left[x\left(\frac{-\cos(n+1)x}{n+1}\right) - (1)\left(\frac{-\sin(n+1)x}{(n+1)^2}\right) - x\left(\frac{-\cos(n-1)x}{n-1}\right) + (1)\left(\frac{-\sin(n-1)x}{(n-1)^2}\right)\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[\frac{-\pi\cos(n+1)\pi}{n+1} + \frac{\pi\cos(n-1)\pi}{n-1} - \frac{\sin(n-1)\pi}{(n-1)^2}\right] \quad [\text{Since } \sin n\pi = 0]$$

For $n \geq 2$,

$$a_n = \frac{1}{\pi}\left[\frac{-\pi\cos(n+1)\pi}{n+1} + \frac{\pi\cos(n-1)\pi}{n-1}\right] \quad [\text{Since } \sin n\pi = 0]$$

$$= \frac{1}{\pi}\left[\frac{-\pi(-1)^{n+1}}{n+1} + \frac{\pi(-1)^{n-1}}{n-1}\right] \quad \left[\begin{matrix}\text{Since } \cos(n-1)\pi = (-1)^{n-1} \\ \cos(n+1)\pi = (-1)^{n+1}\end{matrix}\right]$$

$$= \frac{\pi}{\pi}(-1)^n\left[\frac{-(-1)^1}{n+1} + \frac{(-1)^{-1}}{n-1}\right]$$

$$= (-1)^n\left[\frac{1}{n+1} - \frac{1}{n-1}\right]$$

$$= (-1)^n\left[\frac{-2}{n^2-1}\right]$$

At n = 1,

$$a_1 = \frac{2}{\pi}\int_0^{\pi} x \sin x \cos x \, dx$$

$$= \frac{1}{\pi}\int_0^{\pi} x \sin 2x \, dx$$

$$= \frac{1}{\pi}\left[x\frac{-\cos 2x}{2} - (1)\frac{-\sin 2x}{4}\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[\pi\left(\frac{-1}{2}\right)\right]$$

$$= \frac{-1}{2}.$$

Then (i) becomes,

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) = a_0 + a_1 \cos x + \sum_{n=2}^{\infty} a_n \cos(nx)$$

i.e. $$f(x) = 1 - \frac{\cos x}{2} + \sum_{n=2}^{\infty} (-1)^n\left(\frac{-2}{n^2-1}\right)\cos(nx), \ldots (2)$$

In particular if we take $x = \frac{\pi}{2}$ then (2) implies us,

$$f\left(\frac{\pi}{2}\right) = 1 - \frac{1}{2}\cos\left(\frac{\pi}{2}\right) + \sum_{n=2}^{\infty}\left(\frac{-2}{n^2-1}\right)\cos\left(\frac{n\pi}{2}\right)$$

$$\Rightarrow \frac{\pi}{2}\sin\left(\frac{\pi}{2}\right) = 1 - \sum_{n=2}^{\infty}\left(\frac{2}{n^2-1}\right)\cos\left(\frac{n\pi}{2}\right)$$

$$\Rightarrow \frac{\pi-2}{4} = -\sum_{n=2}^{\infty}\left[\frac{1}{(n-1)(n+1)}\right]\cos\left(\frac{n\pi}{2}\right)$$

Since, $\cos\left(\frac{n\pi}{2}\right) = 0$ for n is odd. Then $\cos\left(\frac{n\pi}{2}\right)$ gives alternative unity value with initial −1 at n = 2. Therefore,

$$\frac{\pi-2}{4} = -\sum_{n=\text{even with } n\geq 2}^{\infty}\left[\frac{1}{(n-1)(n+1)}\right]\cos\left(\frac{n\pi}{2}\right)$$

$$\Rightarrow \frac{\pi-2}{4} = -\left[-\frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} - \frac{1}{5\cdot 7} + \frac{1}{7\cdot 9} - \ldots\right]$$

$$\Rightarrow \frac{\pi-2}{4} = \frac{1}{1\cdot 3} - \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} - \frac{1}{7\cdot 9} + \ldots$$

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Example 4: A function f(x) is defined by $f(x) = \pi x - x^2$ for $0 \le x \le \pi$. Find the Fourier sine series of f(x).

Solution

Given that,

$$f(x) = \pi x - x^2 \quad \text{for } 0 \le x \le \pi.$$

Clearly, f(x) is half range periodic function with period 2π. So, the Fourier sine series of f(x) is,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx \qquad \ldots\ldots (1)$$

with

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx.$$

$$= \frac{2}{\pi}\int_0^{\pi} (\pi x - x^2)\sin nx\,dx$$

$$= \frac{2}{\pi}\left[(\pi x - x^2)\frac{-\cos nx}{n} - (\pi - 2x)\left(\frac{-\sin nx}{n^2}\right) + (-2)\left(\frac{\cos nx}{n^3}\right)\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{0-0}{n} + \left(\frac{0-0}{n^2}\right) - 2\left(\frac{(-1)^n - 1}{n^3}\right)\right] \qquad [\because \sin n\pi = 0, \cos 0 = 1, \cos n\pi = (-1)^n]$$

$$= \frac{4}{\pi}\left(\frac{1-(-1)^n}{n^3}\right)$$

Then (1) becomes,

$$f(x) = \sum_{n=1}^{\infty} \frac{4}{\pi}\left(\frac{1-(-1)^n}{n^3}\right)\sin nx$$

Example 5: Find the Fourier sine series of $f(x) = \pi \sin\pi x$ if $0 < x < 1$.

Solution

Given function is,

$$f(x) = \pi \sin\pi x \quad \text{for } 0 < x < 1.$$

Clearly f(x) is a periodic function with 2-period.

The Fourier sine series of f(x) is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(n\pi x) \qquad \ldots\ldots (1)$$

with

$$b_n = 2\int_0^1 f(x)\sin(n\pi x)\,dx$$

$$= 2\int_0^1 \pi \sin\pi x \sin(n\pi x)\,dx$$

$$= \pi\int_0^1 [\cos(n-1)\pi x - \cos(n+1)\pi x]\,dx$$

$$= \pi\left[\frac{\sin(n-1)\pi x}{(n-1)\pi} - \frac{\sin(n+1)\pi x}{(n+1)\pi}\right]_0^1$$

$$= \pi\frac{\sin(n-1)\pi}{(n-1)\pi} \qquad [\because \sin n\pi = 0 \text{ for n is integer}]$$

$$= \begin{cases} 0 & \text{for } n \ge 2 \\ \dfrac{\sin(n-1)\pi}{n-1} & \text{for } n = 1 \end{cases} = \begin{cases} 0 & \text{for } n \ge 2 \\ \dfrac{\pi\cos(n-1)\pi}{1} & \text{for } n = 1 \end{cases} = \begin{cases} 0 & \text{for } n \ge 2 \\ \pi & \text{for } n = 1 \end{cases}$$

Now, (1) becomes, $f(x) = \pi \sin\pi x$

This is the Fourier sine series of f(x).

Example 6: Find the Fourier cosine series as well as Fourier sine series of $f(x) = e^x$ if $0 < x < \ell$.

Solution

Let $f(x) = e^x \quad \text{for } 0 < x < \ell$

Clearly f(x) is a half range periodic function on $(0, \ell)$.

Now, the Fourier cosine series of f(x) is,

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) \qquad \ldots\ldots (1)$$

with, $$a_0 = \frac{1}{l}\int_0^l e^x\,dx = \frac{1}{l}[e^x]_0^{\ell} = \frac{e^l - 1}{l}$$

and $$a_n = \frac{2}{l}\int_0^l e^x \cos\left(\frac{n\pi x}{l}\right)dx$$

$$= \frac{2}{l}\left[\frac{e^x}{1+(n\pi/l)^2}\left(\cos\left(\frac{n\pi x}{l}\right) + \frac{n\pi}{l}\sin\left(\frac{n\pi x}{l}\right)\right)\right]_0^{\ell}$$

$$= \frac{2}{l}\left[\frac{e^l}{1+(n\pi/l)^2}(\cos(n\pi)) - \frac{1}{1+(n\pi/l)^2}(1)\right] = \frac{2l}{n^2\pi^2 + l^2}\left[(-1)^n e^l - 1\right]$$

Then (1) becomes,

$$f(x) = \frac{e^l - 1}{l} + \sum_{n=1}^{\infty} \frac{2l}{n^2\pi^2 + l^2}\left[(-1)^n e^l - 1\right]\cos\left(\frac{n\pi x}{l}\right)$$

which is Fourier cosine series of f(x).

And, the Fourier sine series of f(x) is,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \qquad \ldots\ldots (2)$$

with

$$b_n = \frac{2}{l}\int_0^l e^x \sin\left(\frac{n\pi x}{l}\right)dx = \frac{2}{l}\left[\frac{l^2 e^x}{l^2 + n^2\pi^2}\left\{\sin\left(\frac{n\pi x}{l}\right) - \frac{n\pi}{l}\cos\left(\frac{n\pi x}{l}\right)\right\}\right]_0^{\ell}$$

$$= \frac{2l}{n^2\pi^2 + l^2}\left[\frac{n\pi}{l}\left[1 - (-1)^n e^l\right]\right] = \frac{2n\pi}{n^2\pi^2 + l^2}\left[1 - (-1)^n e^l\right]$$

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Then (2) becomes,

$$f(x) = \sum_{n=1}^{\infty} \frac{2n\pi}{n^2\pi^2 + l^2}\left[1 - (-1)^n e^l\right] \sin(nx)$$

This is the Fourier sine series of f(x).

Exercise 7 [B]

1. Find the Fourier cosine series as well as sine series of the following functions:
   a. $f(x) = \pi - x$ if $0 < x < \pi$
   b. $f(x) = x$ for $0 < x < \pi$
   c. $f(x) = x$ if $0 < x < l$
2. a. Find the Fourier cosine series of $f(x) = \sin x$ for $0 < x < \pi$.
   b. Expand the function $f(x) = x^2$ for $0 \le x \le \pi$ in a Fourier cosine series and deduce that
      (i) $\sum_{n=1}^{\infty} \left(\frac{1}{n^2}\right) = \frac{\pi^2}{6}$
      (ii) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$
   c. Obtain the Fourier cosine series of $f(x) = \begin{cases} kx & \text{for } 0 \le x \le l/2 \\ k(l - x) & \text{for } l/2 \le x \le l \end{cases}$ and show that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots = \frac{\pi^2}{8}$.
3. a. Find the Fourier series for $f(x) = |x|, -\pi < x < \pi$ after testing odd and even.
   b. Find the Fourier series for $f(x) = x \sin x$ if $-\pi \le x \le \pi$ after testing odd and even.

Answer Key

1. a. $\frac{\pi}{2} + \frac{4}{\pi}\left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots.\right]$ and $2\left[\sin x + \frac{\sin 2x}{2} + \frac{\sin 3x}{3} + \ldots.\right]$
   b. $\frac{\pi}{2} - \frac{4}{\pi}\left(\cos x + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots\right)$ and $2\left(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \frac{\sin 4x}{4} + \ldots\right)$
   c. $\frac{l}{2} - \frac{4l}{\pi^2}\left(\cos\left(\frac{\pi x}{l}\right) + \frac{1}{9}\cos\left(\frac{3\pi x}{l}\right) + \frac{1}{25}\cos\left(\frac{5\pi x}{l}\right) + \ldots\ldots\right)$ and $\frac{2l}{\pi}\left[\sin\left(\frac{x\pi}{l}\right) - \frac{1}{2}\sin\left(\frac{\pi x}{l}\right) + \frac{1}{3}\sin\left(\frac{3\pi x}{l}\right) - \frac{1}{4}\sin\left(\frac{4\pi x}{l}\right) + \ldots.\right]$
2. a. $\frac{4}{\pi}\left[\frac{1}{2} - \frac{\cos 2x}{1.3} - \frac{\cos 4x}{3.5} - \frac{\cos 6x}{5.7} - \ldots.\right]$
   b. $\frac{2\pi^2}{6} + 2\sum_{n=1}^{\infty} \frac{(-2)^n}{n^2}\cos nx$
   c. $\frac{kl}{4} - \frac{4kl}{\pi^2}\left[\frac{1}{1^2}\cos\left(\frac{\pi x}{l}\right) + \frac{1}{3^2}\cos\left(\frac{3\pi x}{l}\right) + \frac{1}{5^2}\cos\left(\frac{5\pi x}{l}\right) + \ldots\right]$
3. a. $\frac{\pi}{2} + \frac{4}{\pi}\left(\cos x + \frac{1}{9}\cos 3x + \frac{1}{25}\cos 5x + \ldots.\right)$ and $2\left(\sin x + \frac{\sin 2x}{2} + \frac{\sin 3x}{3} + \ldots.\right)$
   b. Even function, $\frac{\pi}{2} \sum_{n=1}^{\infty} \frac{(\cos n\pi - 1)}{n^2}\cos nx$.

□□□

8 PARTIAL DIFFERENTIAL EQUATIONS

CHAPTER OUTLINE

After completion of this chapter, you will be able to gain the knowledge and understanding regarding:

8.1 Introduction

8.2 Linear constant coefficient equation

8.3 Applications in conservation laws, the breaking time, shock waves, nonlinear advection equations, and traffic flow.

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Introduction

Let's consider the linear first order constant coefficient partial differential equation

$$au_x + bu_y + cu = f(x, y) \qquad \ldots (i)$$

for a, b, and c constants with $a^2 + b^2 > 0$.

If b = 0, the equation (i) reduces to

$$au_x + cu = f \qquad \ldots (ii)$$

which is a first order linear (ordinary) differential equation with y a parameter. Recall that the solution of such equations (ii) can be obtained using an integrating factor.

If $b \neq 0$, we have to solve the equation

$$au_x + bu_y + cu = f$$

It would help if we could find a transformation which would eliminate one of the derivative terms reducing this problem to the previous case.

Example 1: Find the general solution: $u_x + u_y - u = 0$.

Solution:

We first identify a = 1, b = 1, and c = u. The relations between the differentials is

$$\frac{dx}{1} = \frac{dy}{1} = \frac{du}{u} \qquad \ldots (i)$$

From (i) we can pair the differentials in different ways.

Taking first differentials, we get

$$\frac{dx}{1} = \frac{dy}{1}$$

Integrating we get

$$x = y + A$$

And, taking last two differentials, we get

$$\frac{dy}{1} = \frac{du}{u}$$

Integrating we get

$$y = \ell n(u) + B$$

$$\Rightarrow \quad u = e^{y/B} = C_1 e^y \qquad \text{for } C_1 = e^{1/B} \text{ (constant)}$$

Since u = u(x, y), and form the above relation we observe that the constant with respect to y. So, let,

$$C_1 = f(A) = f(x - y)$$

Then the general solution of (i) is

$$u = u(x, y) = f(x - y)e^y$$

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8.1 LINEAR CONSTANT COEFFICIENT EQUATION

The n^{th} order ordinary differential equation is an equation for an unknown function y(x) n^{th} order ordinary differential equation that expresses a relationship between the unknown function and its first n derivatives. This can be written as for y = y(x),

$$F(y^n, y^{n-1}, \ldots \ldots, y', y, x) = 0 \qquad \ldots (1)$$

Here, y^n represents the n^{th} derivative of y = y(x).

And, the equation (1) satisfying the condition at fixed point x_0

$$y^n(x_0) = \alpha_1, y^{n-1}(x_0) = \alpha_2, \ldots \ldots, y'(x_0) = \alpha_{n-1}, y(x_0) = \alpha_n$$

is called initial value problem.

If u = u(x, y, ...), be a function, its general partial differential equation is

$$F\left(x, y, \ldots, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \ldots, \frac{\partial^2 u}{\partial x^2}, \ldots\right) = 0. \qquad \ldots (2)$$

The particular form of (2),

$$F\left(x, y, \ldots, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \ldots,\right) = 0$$

is known as first order ordinary differential equation.

A general first order partial differential equation for u = u(x, y) is given as

$$F(x, y, u, u_x, u_y) = 0.$$

And a linear first order partial linear first order partial differential equation is of the form equation,

$$a(x, y)u_x + b(x, y)u_y + c(x, y)u = f(x, y).$$

Example 2: Find the general solution of the equation $u_x + u_y - u = 0$ subject to u(x, 0) = 1.

Solution

Given equation is

$$u_x + u_y - u = 0$$

i.e. $$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} - u = 0$$

The characteristic equation is

$$\frac{\partial x}{1} = \frac{\partial y}{1} = \frac{\partial u}{u}$$

Its solution is

$$x - y = C_1 \text{ and } x = \ln(u) + C_2$$

That is,

$$u(x, y) = Ae^y$$

As given u(x, 0) = 1 then A = 1. Therefore,

$$u(x, y) = e^y$$

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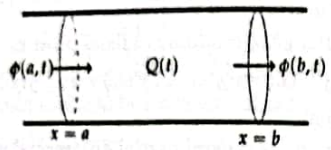
8.2 APPLICATIONS OF PDE

1. Conservation Laws (One Dimensional Conservation Laws)

The rate of change of some stuff Q(t), in a region, $a \le x \le b$, as depicted in as figure. The simple model is to think of fluid flowing in one dimension, such as water flowing in a stream.



The rate of change of Q(t) is given as

the rate of change of Q = Rate in − Rate Out + source term.

Here, the **Rate in** means the flow into the region x = a and **Rate out** means the flow out the region x = b. And, the **Source term** means some adding or removing Q from the region.

Let the flow in terms of the flux is $\phi(x, t)$ over the ends of the region. Then the boundary conditions of the flow are $\phi(a, t)$ and $\phi(b, t)$.

In terms of fluid flow, there could be a source of fluid inside the region such as a faucet adding more water.

Or, there could be a drain letting water escape.

Then, the total source over the interval is denoted by

$$\int_a^b f(x, t)\, dx.$$

Here, f(x, t) is the source density.

In summary, the rate of change of Q(x, t) can be written as

$$\frac{dQ}{dt} = \phi(a, t) - \phi(b, t) + \int_a^b f(x, y)\, dx \qquad \ldots (1)$$

From the Fundamental Theorem of Calculus, we have

$$\int_a^b \frac{\partial \phi(x, t)}{\partial x}\, dx = \phi(b, t) - \phi(a, t)$$

Then (1) can be written as

$$\frac{dQ}{dt} = -\int_a^b \frac{\partial \phi(x, t)}{\partial x}\, dx + \int_a^b f(x, t)\, dx \qquad \ldots (2)$$

This is the **integral form of the conservation law.**

We can rewrite the conservation law in differential form. First, we introduce the density function, u(x, t), so that the total amount of stuff at a given time t is

$$Q(t) = \int_a^b u(x, t)\, dx \qquad \ldots (3)$$

From (2), we have for all $x \in [a, b]$

$$\frac{d}{dt}\int_a^b u(x, t)\, dx = -\int_a^b \frac{\partial \phi(x, t)}{\partial x}\, dx + \int_a^b f(x, t)\, dx \qquad \ldots (4)$$

i.e. $$\int_a^b [u_t(x, t) + \phi_x(x, t) - f(x, t)]\, dx = 0$$

We cannot simply set the integrant to zero just because the integral vanishes. However, if this result holds for every region [a, b], then we can conclude the integrand vanishes. So,

$$u_t(x, t) + \phi_x(x, t) = f(x, t) \qquad \ldots (5)$$

This partial differential equation is actually an equation in terms of two unknown functions ϕ and f, assuming we know something about the source function. We would like to have a single unknown function. So, we need some additional information. This added information comes from the constitutive relation, a function relating the flux to the density function. Namely, we will assume that we can find the relationship $\phi = \phi(u)$. If so, then we can write

$$\frac{\partial \phi}{\partial x} = \frac{d\phi}{du}\frac{\partial u}{\partial x}$$

or, $$\phi_x = \phi'(u)\, u_x$$

Characteristics Equation

Consider a partial differential equation in xy is

$$a(x, y, u)\, u_x + b(x, y, u)\, u_y = c(x, y, u) \qquad \ldots (1)$$

Let, u = u (x, y) be solution of (1) then

$$f(x, y, u) = u(x, y) - u = 0 \qquad \ldots (2)$$

The normal of (2) is

$$\nabla f = (u_x, u_y, -u)$$

If v = (a, b, c) then

$$v \cdot \nabla f = a u_x + b u_y - c = 0 \quad [\because \text{using (1)}]$$

This means v is normal to ∇f and ∇f is normal to f.

Also, let $$p(t) = \frac{dp(t)}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{du}{dt}\right)$$

From (3) v(t) is the tangent to the surface associated with (1). Therefore,

$$\frac{dx}{dt} = a, \frac{dy}{dx} = b, \frac{du}{dt} = c$$

which is the characteristics curve for (1).

Alternatively,

Consider a PDE,

$$a u_x + b u_y = c$$

This implies

$$\frac{du}{dx} = \frac{c}{a}, \quad \frac{du}{dy} = \frac{c}{b}$$

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This gives

$$\frac{dx}{a} = \frac{dy}{b} = \frac{du}{c} = dt$$

$$\Rightarrow \frac{dx}{dt} = a, \quad \frac{dy}{dt} = b, \quad \frac{du}{dt} = c$$

Example 3: Find the general solution of $u_x + u_y - u = 0$

Solution:

The characteristics curve for (1) is

$$u_x + u_y - u = 0 \quad \ldots (1)$$

This implies

$$\frac{dx}{1} = \frac{dy}{1} = \frac{du}{u}$$

and

$$y = x + c_1$$

$$\ln u = x + \ln c_2 \quad \ldots (2)$$

$$\Rightarrow u = e^{x + \ln c_2} = e^x e^{\ln c_2} = c_2 e^x$$

Let, $u = u(x, y)$ be solution of (1).

Clearly, c_1 and c_2 associated with a relation

Then $$c_2 = G(c_1) = G(y - x) \quad [\because \text{Using (2)}]$$

$$u(x, y) = c_2 e^x = G(y - x)\, e^x,$$

2. Nonlinear Advection Equation

The equation of the form

$$u_t + c(u)u_x = 0 \quad \ldots (1)$$

with $c(u)$ is a constant function is called the nonlinear advection equation.

The characteristic equations of (1) is given by

$$\frac{dx}{dt} = c, \quad \frac{du}{dt} = 0$$

Solving we get

$$x = ct + x_0 \text{ and } u = x_0 + A$$

$= f(x_0)$

Therefore, the solution of (1) is

$u(x, t) =$ constant along the lines $x = ct + x_0$,

where x_0 is an arbitrary constant.

At $t = 0$, $x = x_0$

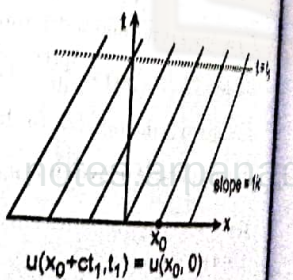
$$u(x, t) = u(x_0, 0) = f(x_0)$$

And, at $t = t_1$

$$u(x_0 + ct_1, t_1) = u(x_0, 0) = f(x_0)$$

Thus, the slope of (1) is

$$\text{Slope} = \frac{1}{c}$$



Example 4: Solve $u_t + uu_x = 0$, $u(x, 0) = e^{-x}$.

Solution

Given equation is

$$u_t + uu_x = 0, \quad \ldots (1)$$

satisfies $$u(x, 0) = e^{-x} \quad \ldots (2)$$

The characteristic equations of (1) is given by

$$\frac{dx}{dt} = u, \quad \frac{du}{dt} = 0$$

The equation $\frac{du}{dt} = 0$ gives u is constant.

And, $\frac{dx}{dt} = u$ implies us

$$x = u(x_0, 0)t + x_0 \quad \text{[Being u is constant]}$$

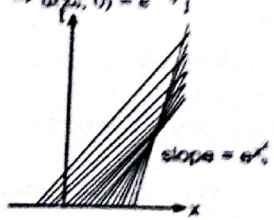
i.e. $x = t\, e^{-x_0} + x_0$ [Being $u(x, 0) = e^{-x} \Rightarrow u(x_0, 0) = e^{-x_0}$]

Thus, the solution of (1) is

$$u(x, t) = e^{-x_0} \text{ along } x = t\, e^{-x_0} + x_0$$

And, the slope of the line is

$$\text{Slope} = e^{x_0}$$



Example 5: Solve $u_t + uu_x = 0$, $u(x, 0) = \frac{1}{1 + x^2}$

Solution

Given equation is

$$u_t + uu_x = 0, \quad \ldots (1)$$

satisfies $$u(x, 0) = \frac{1}{1 + x^2} \quad \ldots (2)$$

The characteristic equations of (1) is given by

$$\frac{dx}{dt} = u, \quad \frac{du}{dt} = 0$$

The equation $\frac{du}{dt} = 0$ gives u is constant.

And, $\frac{dx}{dt} = u$ implies us

$$x = u(x_0, 0)t + x_0 \quad \text{[Being u is constant]}$$

i.e. $$x = \frac{t}{1 + x_0^2} + x_0$$

Thus, the solution of (1) is

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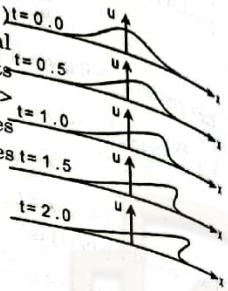
$$u(x, t) = \frac{1}{1 + x_0^2} \text{ along } x = \frac{t}{1 + x_0^2} + x_0$$

And, the slope of the line is

$$\text{Slope} = 1 + x_0^2$$

3. The Breaking Time

The development of the solution of above example 4 with the parametric plot of the point $(x_0 + e^{-x_0^2}, e^{-x_0^2})$ for different times is given in the figure. The initial profile propagates to the right with the higher points traveling faster than the lower points since $x'(t) = u > 0$. Around $t = 1.0$ the wave breaks and becomes multivalued. The time at which the function becomes multivalued is called the breaking time.



Consider,

$$u_t + c(u)u_x = 0 \quad \ldots (1)$$

Set $x_0 = \theta$, then the wave speed is

$$f(\theta) = c(u_0(\theta))$$

The characteristic line is given by

$$x = \theta + t\, f(\theta).$$

The solution of the equation (1) is

$$u(x, t) = u(\theta + t\, f(\theta), t)$$
$$= u_0(\theta) \text{ along } x = \theta + t\, f(\theta). \quad \ldots (2)$$

Here,

$$u_x = u_0'(\theta)\, \theta_x \text{ and } u_t = u_0'(\theta)\, \theta_t$$

Since,

$$x = \theta + t\, f(\theta)$$
$$\Rightarrow \quad \theta = x - t\, f(\theta)$$

Then,

$$\theta_x = 1 - t\, f'(\theta)\, \theta_x \quad \Rightarrow \theta_x = \frac{1}{1 + t\, f'(x)}$$

And,

$$\theta_t = \frac{\partial}{\partial t}(x - t\, f(\theta))$$
$$= -f(\theta) - t\, f'(\theta)\theta_t$$
$$\Rightarrow \theta_t (1 + t\, f'(\theta)) = -f(\theta)$$
$$\Rightarrow \theta_t = \frac{-f(\theta)}{1 + t\, f'(\theta)}$$

Clearly, θ_x and θ_t are undefined when

$$1 + t\, f'(\theta) = 0$$
$$\Rightarrow t = \frac{-1}{f'(\theta)}$$

The minimum time for the breaking time,

$$t_b = -\min\left\{\frac{1}{f'(\theta)}\right\}$$

Example 6: Find the breaking time for $u_t + uu_x = 0$, $u(x, 0) = e^{-x^2}$

Solution

Since

$$f(\theta) = c(u_0(\theta)) = e^{-\theta^2}$$

and

$$f'(\theta) = -2\theta\, e^{-\theta^2}$$

Therefore

$$t = \frac{-1}{f'(\theta)} = \frac{1}{2\theta\, e^{-\theta^2}} = \frac{e^{\theta^2}}{2\theta}$$

For min time, we set the derivative of time, is zero, so,

$$\frac{d}{d\theta}\left(\frac{e^{\theta^2}}{2\theta}\right) = 0$$

$$\Rightarrow \quad \frac{e^{\theta^2} \cdot 2\theta}{2\theta} + e^{\theta^2}\left(\frac{1}{2}\right) + \left(\frac{-1}{\theta^2}\right) = 0$$

$$\Rightarrow \quad e^{\theta^2}\left[1 - \frac{-1}{2\theta^2}\right] = 0$$

This gives $1 - \frac{1}{2\theta^2} = 0 \Rightarrow \theta = \frac{1}{\sqrt{2}}$

Thus, the main time for the breaking time is

$$t_b = \frac{e^{1/2}}{2\left(\frac{1}{\sqrt{2}}\right)} = \frac{e^{1/2}}{\sqrt{2}} = \sqrt{\frac{e}{2}} \approx 1.16$$

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4. Shock Waves

In the context of partial differential equations (PDEs), a shock wave refers to a sudden and discontinuous change in the solution of the equation. Shock waves are often associated with hyperbolic PDEs, which describe wave-like phenomena and exhibit characteristics such as wave propagation and conservation laws.

A shock wave occurs when there is a rapid transition from one state to another in the solution of a hyperbolic PDE. This abrupt change creates a shock front, where the solution undergoes a jump or discontinuity. Shock waves are common in various fields, including fluid dynamics, gas dynamics, and traffic flow modeling.

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The mathematical representation of a shock wave in a hyperbolic PDE involves introducing a shock condition or jump condition. The Rankine-Hugoniot jump conditions are often used to describe the conservation of mass, momentum, and energy across a shock wave.

For a scalar conservation law in one dimension, the Rankine-Hugoniot jump condition can be expressed as follows:

$$[u] = S[v],$$

where u and v are the values of the solution on either side of the shock, [u] is the jump in the solution across the shock, and S is the speed of the shock.

The initial value problem

$$u_t + \phi_x = 0,\ x \in R,\ t > 0,$$
$$u(x, 0) = u_0(x). \qquad \ldots (1)$$

The corresponding integral form of the conservation law is

$$\frac{d}{dt}\left(\int_a^b u(x, t)\, dx\right) = \phi(a, t) - \phi(b, t) \qquad \ldots (2)$$

We can find the shock path from equation (2)

Consider the solution of u(x, t), such that ϕ and their derivatives are continuous everywhere except on a line s = s(t) on xt –plane across which u(x, t) has a jump discontinuity. And a and b are choose so that a < s(t) < b.

Then the equation (2) can be written in [a, b] as

$$\frac{d}{dt}\left(\int_a^{s(t)} u(x, t)\, dx + \int_{s(t)}^b u(x, t)\, dx\right) = \phi(a, t) - \phi(b, t)$$

which implies that

$$\int_a^{s(t)} u_t(x, t)\, dx + \int_{s(t)}^b u_t(x, t)\, dx + [u(a, t) - u(b, t)]S = \phi(a, t) - \phi(b, t)$$

where, u(a, t) is a limit of u(s, t) as x→s(t) from left u(b, t) is a limit of u(s, t) as x→s(t) from right and $S = \frac{ds}{dt}$ is a speed of discontinuity.

So, $u_t(x, t)$ is bounded the integral and vanishes identically as s(t) and we obtain

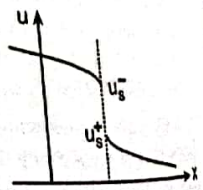
$$\phi(a, t) - \phi(b, t) = [u(a, t) - u(b, t)]\, S \qquad \ldots (3)$$

Thus by relating the jumps,

$$\Delta\phi = \phi(a, t) - \phi(b, t)$$
$$\Delta u = u(a, t) - u(b, t)$$

From (3) we get

$$\Delta\phi = \nabla u\, S$$
$$\Rightarrow \quad S = \frac{ds}{dt} = \frac{\Delta\phi}{\Delta u}$$



which is a discontinuity solution and we can find the shock wave with this path.

Example 7: Find the shock wave of the problem $u_t + uu_x = 0,\ |x| < \infty,\ t > 0$ satisfying the initial condition $u(x, 0) = \begin{cases} 1 & \text{for } x \le 0 \\ 0 & \text{for } x > 0 \end{cases}$.

Solution

Give that

$$u_t + uu_x = 0 \qquad \ldots (1)$$

with

$$u(x, 0) = \begin{cases} 1 & \text{for } x \le 0 \\ 0 & \text{for } x > 0 \end{cases} \qquad \ldots (2)$$

The characteristic equation of (1) is

$$\frac{dx}{dt} = u,\ \frac{du}{dt} = 0$$

This gives u is constant.

Also, if u = 0 then u is constant and if u = 1 along the characteristics curve with slope 1.

Therefore, $\frac{dx}{dt} = 1$ implies $x = t + x_0$

Thus,

$$x(t) = \begin{cases} x_0 & \text{for } x > 0 \\ t + x_0 & \text{for } x < 0 \end{cases}$$

Since $\phi_x = uu_x$ i.e. du = u du

Integrating we get,

$$\phi(u) = \frac{u^2}{2}$$

Now, the shock speed is,

$$S = \frac{ds}{dt} = \frac{\Delta\phi}{\Delta u} = \frac{\frac{1}{2}[(u^+)^2 - (u^-)^2]}{u^+ - u^-}$$

$$= \frac{1}{2}[u^+ - u^-]$$

$$= \frac{1}{2}[0 + 1] \qquad [\because \text{using (2)}]$$

$$= \frac{1}{2}$$

Therefore, integrating $S = \frac{1}{2}$ we get $s(t) = \frac{t}{2}$

Thus, the line $\frac{t}{2}$ separates the characteristics on the right and left of the shock solution. Thus the shock wave is

$$u(x, t) = \begin{cases} 1 & \text{for } x \le t/2 \\ 0 & \text{for } x > t/2 \end{cases}$$

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5. Traffic Flow (One-dimensional Traffic Flow)

The traffic flow problem is a classic topic in transportation engineering and applied mathematics. It is often modeled using partial differential equations (PDEs) to describe the evolution of traffic density or flow over time and space. The most commonly used PDE for modeling traffic flow is the Lighthill-Whitham-Richards (LWR) model.

The LWR model is a hyperbolic conservation law and is expressed as follows:

$$\frac{\partial u}{\partial t} + \frac{\partial \phi}{\partial x} = 0$$

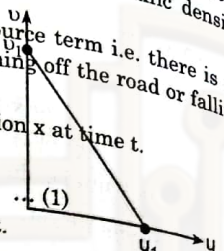
i.e. $u_t + \phi_x = 0$

where

u is the traffic density, t is time, x is space, and ϕ is flux function.

The flux function characterizes the relationship between traffic density and flow rate. A common choice for flux function includes linear, triangular, and piecewise linear functions. The solution of this PDE provides information about how traffic density evolves over time and space.

Let u(x, t) be the traffic density. Assume that there is no source term i.e. there is no way for a car to disappear from the flow by any means – turning off the road or falling into a sinkhole and also, there is no source of additional.



Let $\phi(x, t)$ denote the number of traffic per hour passing position x at time t.

Thus, the flux can be written as

$$\phi = uv$$

where v is the velocity of the traffic at position x and at time t.

Assume that the relation between velocity and density of traffic is continuous and simple straight line is given by

$$v = v_1 - \frac{v_1}{u_1} u \quad \ldots (1)$$

being the density is at $(u_1, 0)$ and velocity is at $(0, v_1)$.

Then the flux (from (1)) is,

$$\phi = uv = u\left(v_1 - \frac{v_1}{u_1} u\right) = v_1\left(u - \frac{u^2}{u_1}\right) \quad \ldots (2)$$

By conservation law the traffic density implies as

$$u_t + \phi_x = 0$$

i.e. $u_t + \phi' u_x = 0 \quad \ldots (3)$

$$\Rightarrow \quad u_t + v_1\left(1 - \frac{2u}{u_1}\right) u_x = 0 \quad \ldots (4)$$

with u_1 is maximum density and v_1 is initial velocity.

Let the traffic reached at u_1 when stopped and u_0 (minimum density) when incoming traffic.

Consider

$$u(x, 0) = \begin{cases} u_1, & x \geq 0 \\ u_0 & x < 0 \end{cases}$$

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The characteristic equation is

$$x = c(u(x_0, t)t + x_0 \quad \ldots (5)$$

Comparing (4) with the equation $u_t + c(u)\, u_x = 0$ we get

$$c(u(x_0, t)) = v_1\left(\frac{1 - 2u(x_0, 0)}{u_1}\right)$$

For $x \geq 0$,

$$c(u(x_0, t)) = c(u_1) = v_1\left(1 - \frac{2u_1}{u_1}\right) = -v_1$$

So, (5) becomes

$$x = -v_1 t + x_0 \quad \ldots (6)$$

and its slope is $\frac{-1}{v_1}$.

For $x < 0$,

$$c(u(x_0, t)) = c(u_0) = v_1\left(1 - \frac{2u_0}{u_1}\right)$$

So, (5) becomes,

$$x = v_1\left(1 - \frac{2u_0}{u_1}\right) t + x_0 \quad \ldots (7)$$

Clearly (6) and (7) meets when $t = 0$ and they cross each other for $t \neq 0$. So, the breaking time is 0.

Therefore,

$$x = \begin{cases} -v_1 t + x_0 & \text{for } x \geq 0 \\ v_1\left(1 - \frac{2u_0}{u_1}\right) t + x_0 & \text{for } x < 0 \end{cases}$$

From shock wave,

$$S = \frac{ds}{dt} = \frac{\phi^+ - \phi^-}{u^+ - u^-}$$

$$= \frac{\left[u_1 v_1 - \frac{v_1}{u_1} u_1^2\right] - \left(u_0 v_1 - \frac{v_1}{u_1} u_0^2\right)}{u_1 - u_0}$$

$$= \frac{(u_1 - u_0)v_1 - \frac{v_1}{u_1}\left(u_1^2 - u_0^2\right)}{u_1 - u_0}$$

$$= v_1 - \frac{v_1}{u_1}(u_1 + u_0)$$

$$= v_1 - v_1 - \frac{v_1 u_0}{u_1}$$

$$= \frac{-v_1 u_0}{u_1}$$

So, the shock path is $s(t) = -\frac{v_1 u_0}{u_1} t$

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Exercise 8 (A)

1. Convert the equation $u_t + cu_x = 0$ in conservation law form, $u_t + \phi_x = 0$ by finding the flux function $\phi(u)$.
2. Find the breaking time of the problem, $u_t + u^2u_x = 0$, $u(x, 0) = \frac{1}{1 + x^2}$.
3. Find the shock wave of the problem $u_t + uu_x = 0$, $|x| < \infty$, $t > 0$ satisfying the initial condition $u(x, 0) = \begin{cases} 2 & \text{for } x \le 0 \\ 1 & \text{for } x > 0 \end{cases}$.
4. Find the shock wave of the problem $u_t + uu_x = 0$, $|x| < \infty$, $t > 0$ satisfying the initial condition $u(x, 0) = \begin{cases} 1 & \text{for } x \le 0 \\ 2 & \text{for } x > 0 \end{cases}$.

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